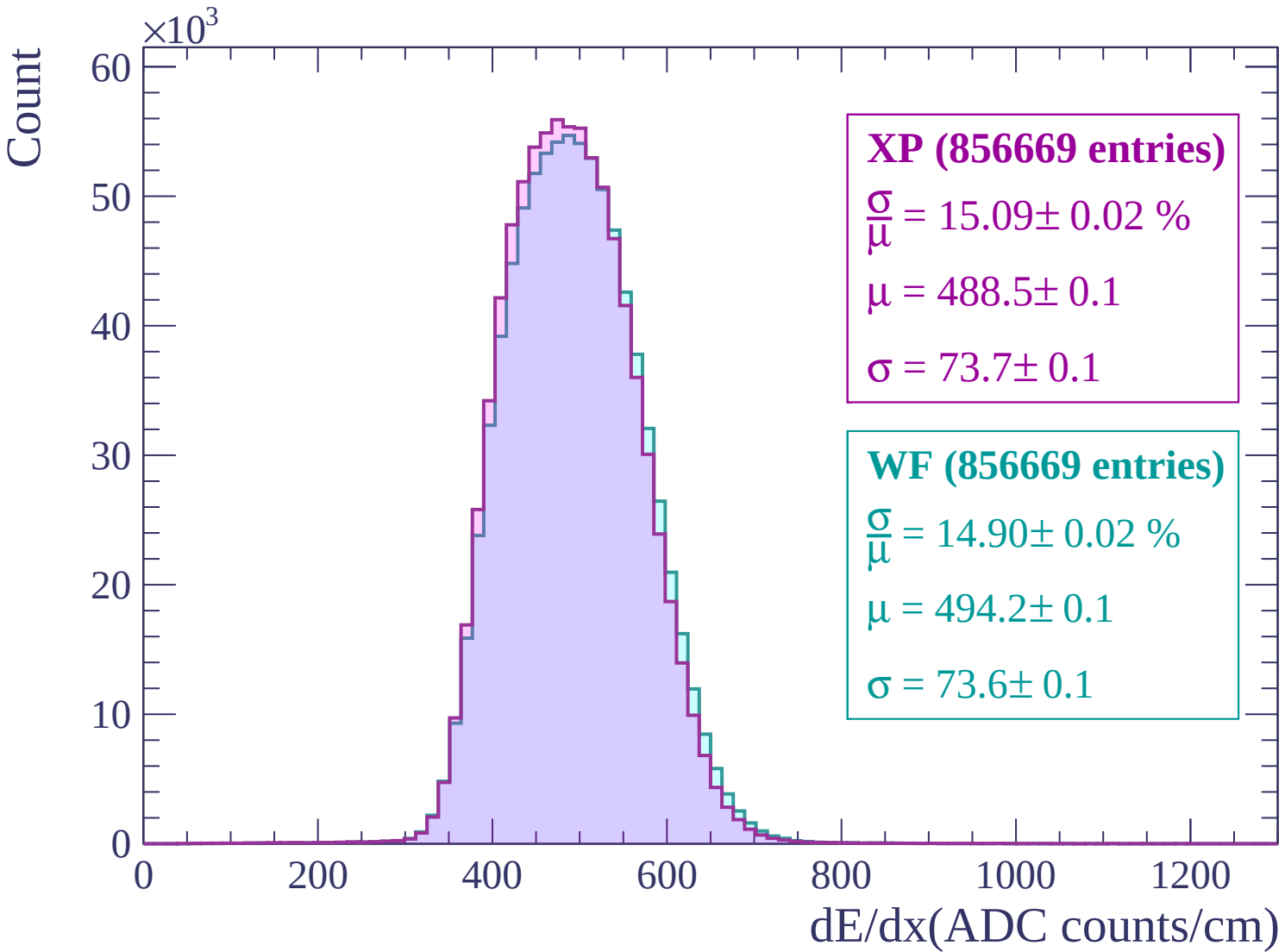
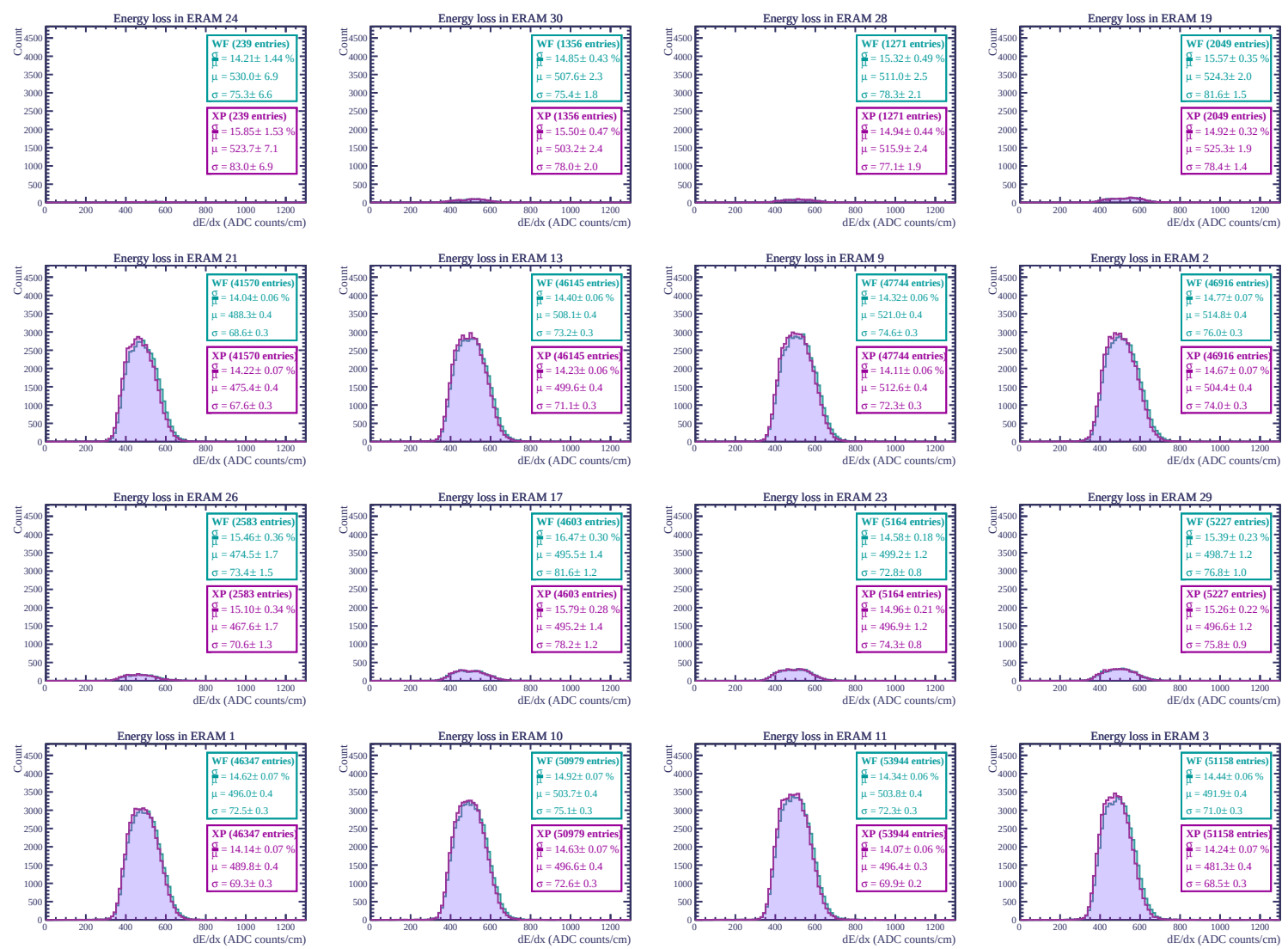
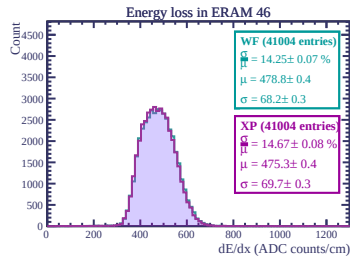
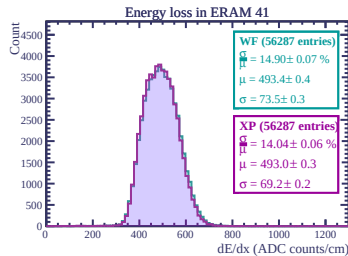
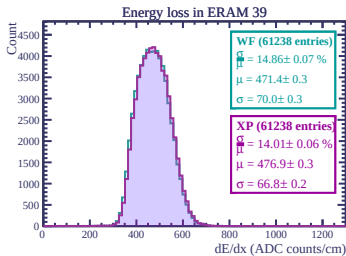
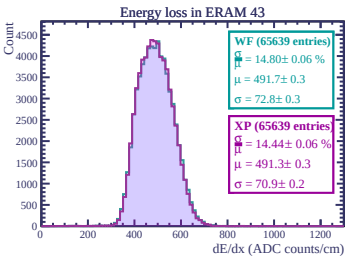
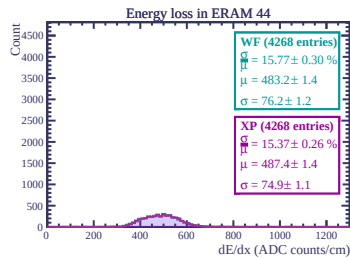
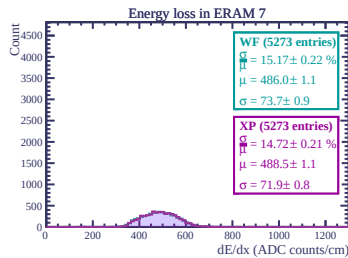
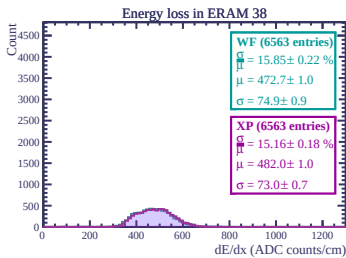
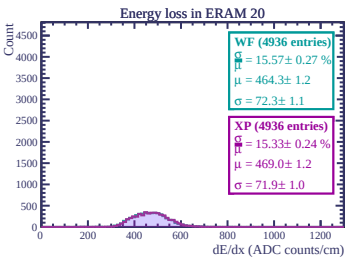
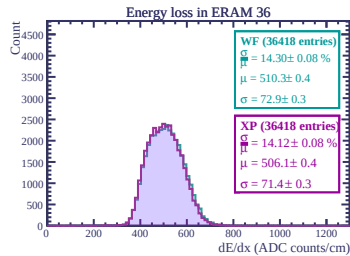
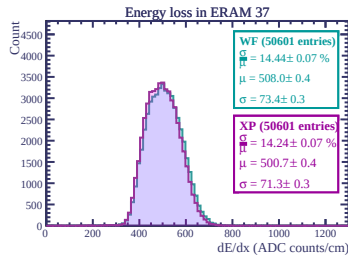
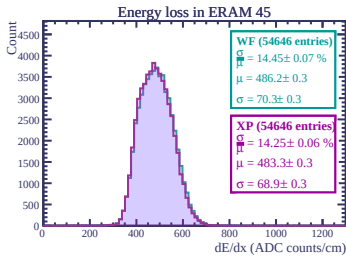
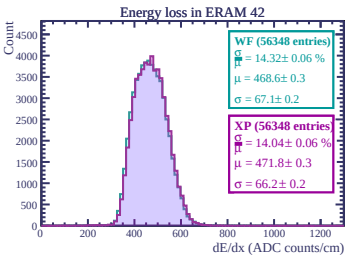
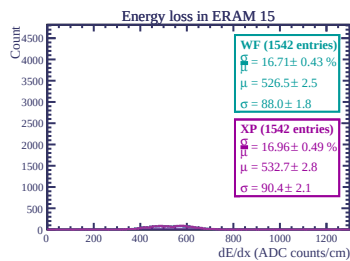
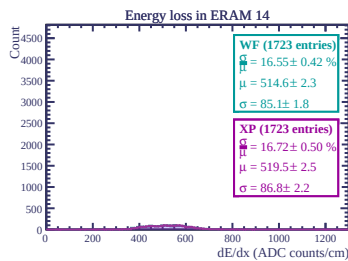
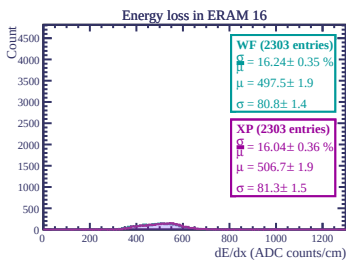
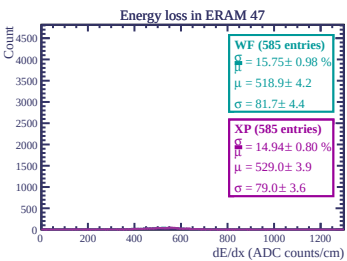
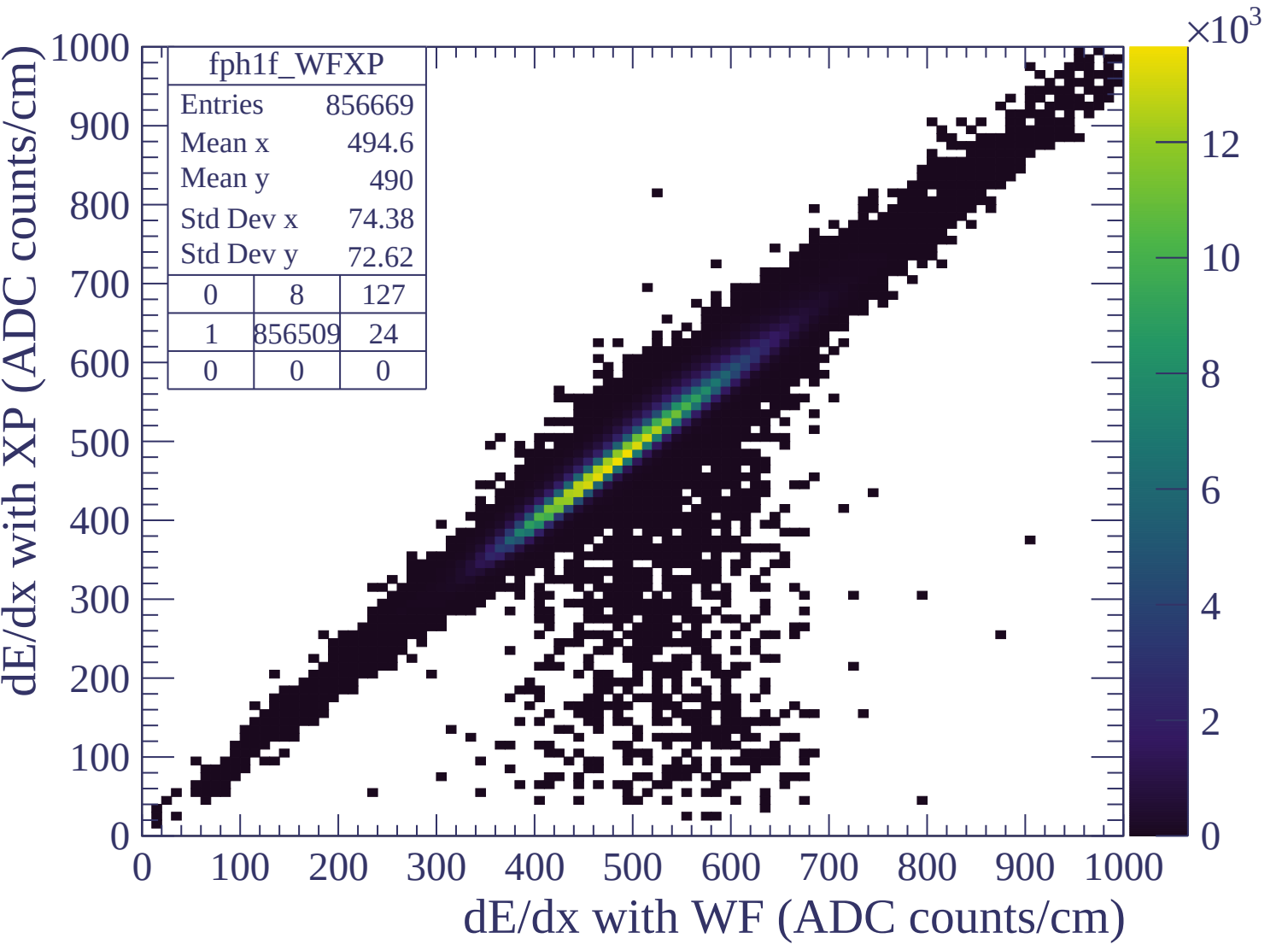
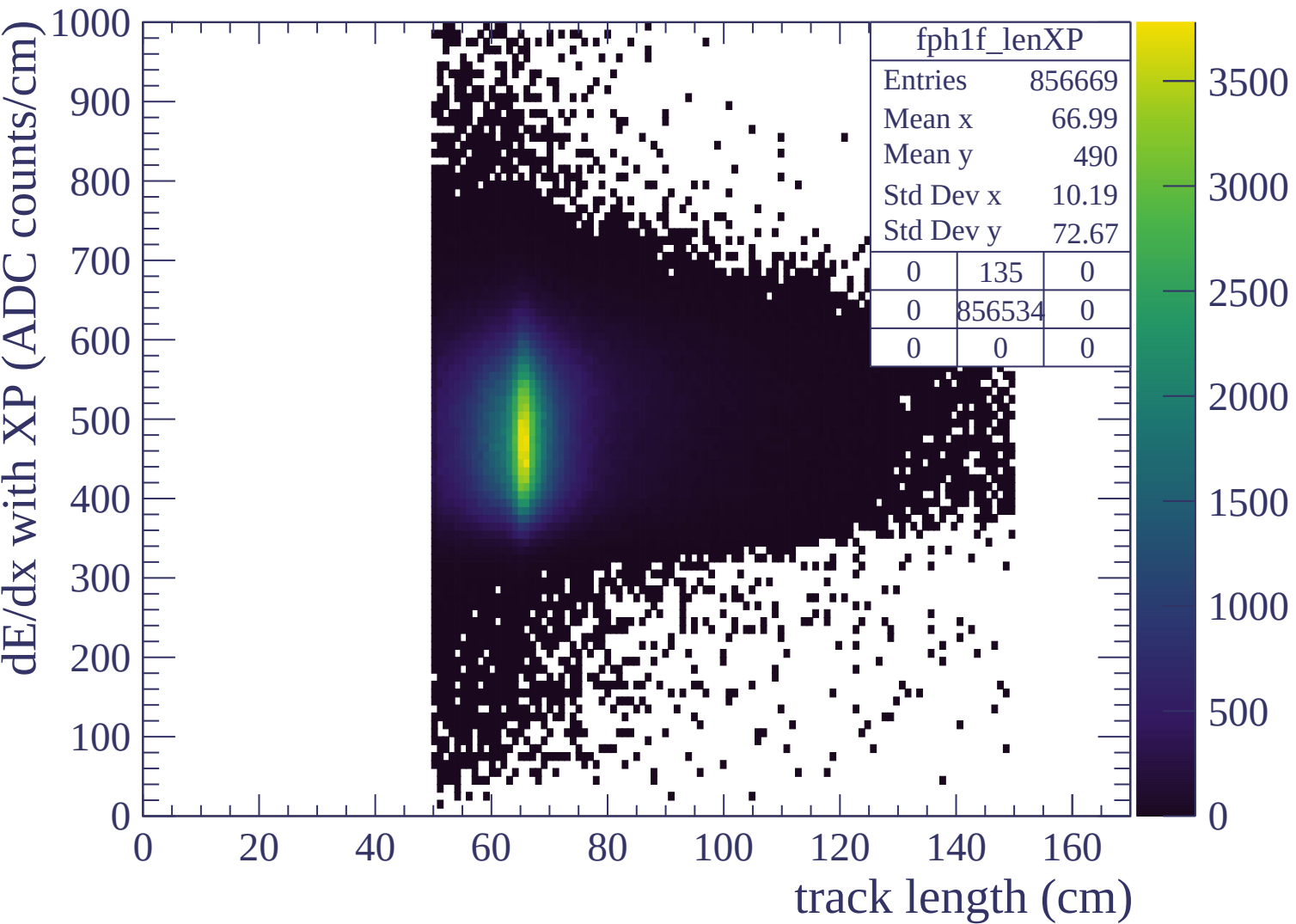


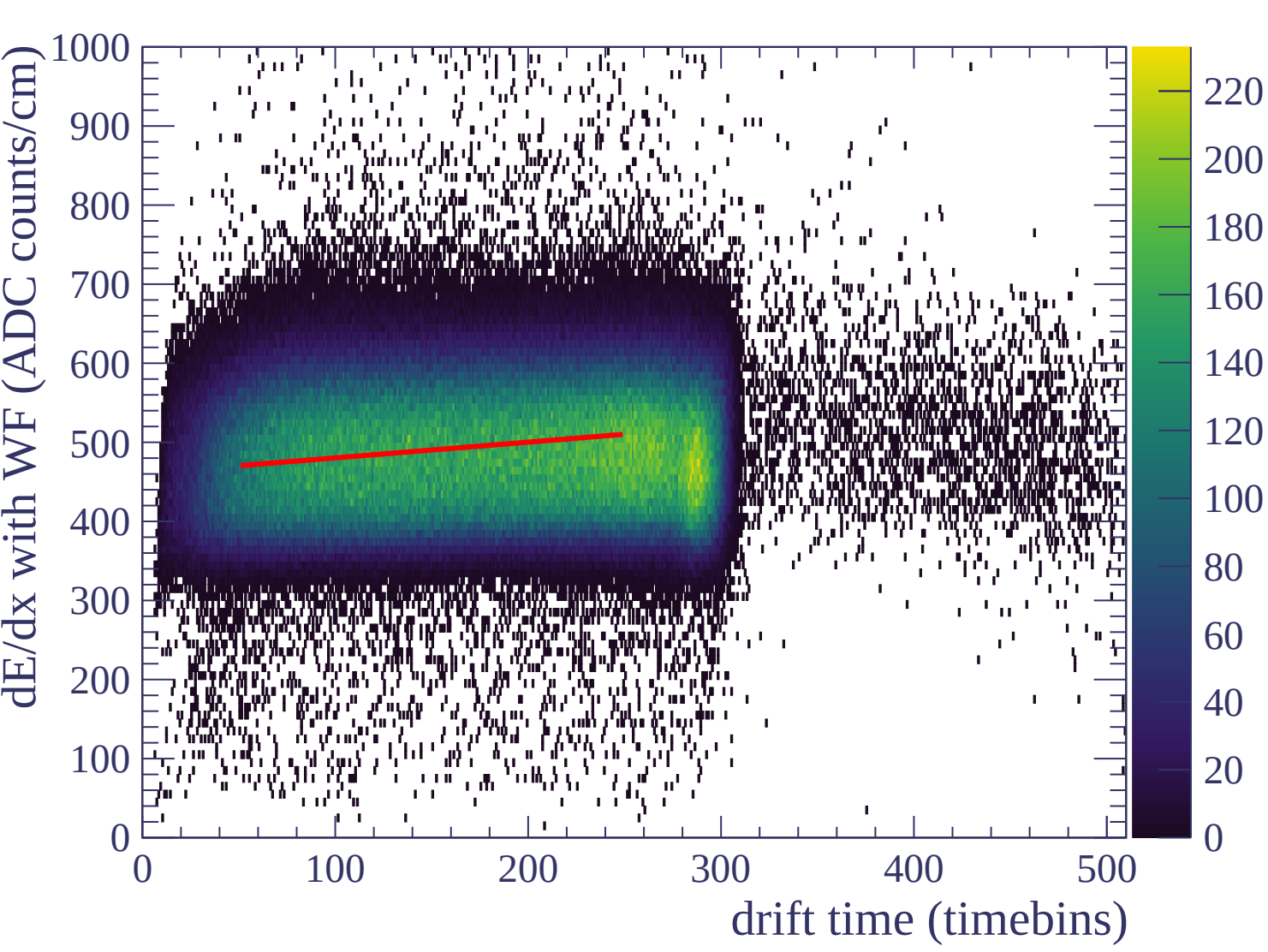


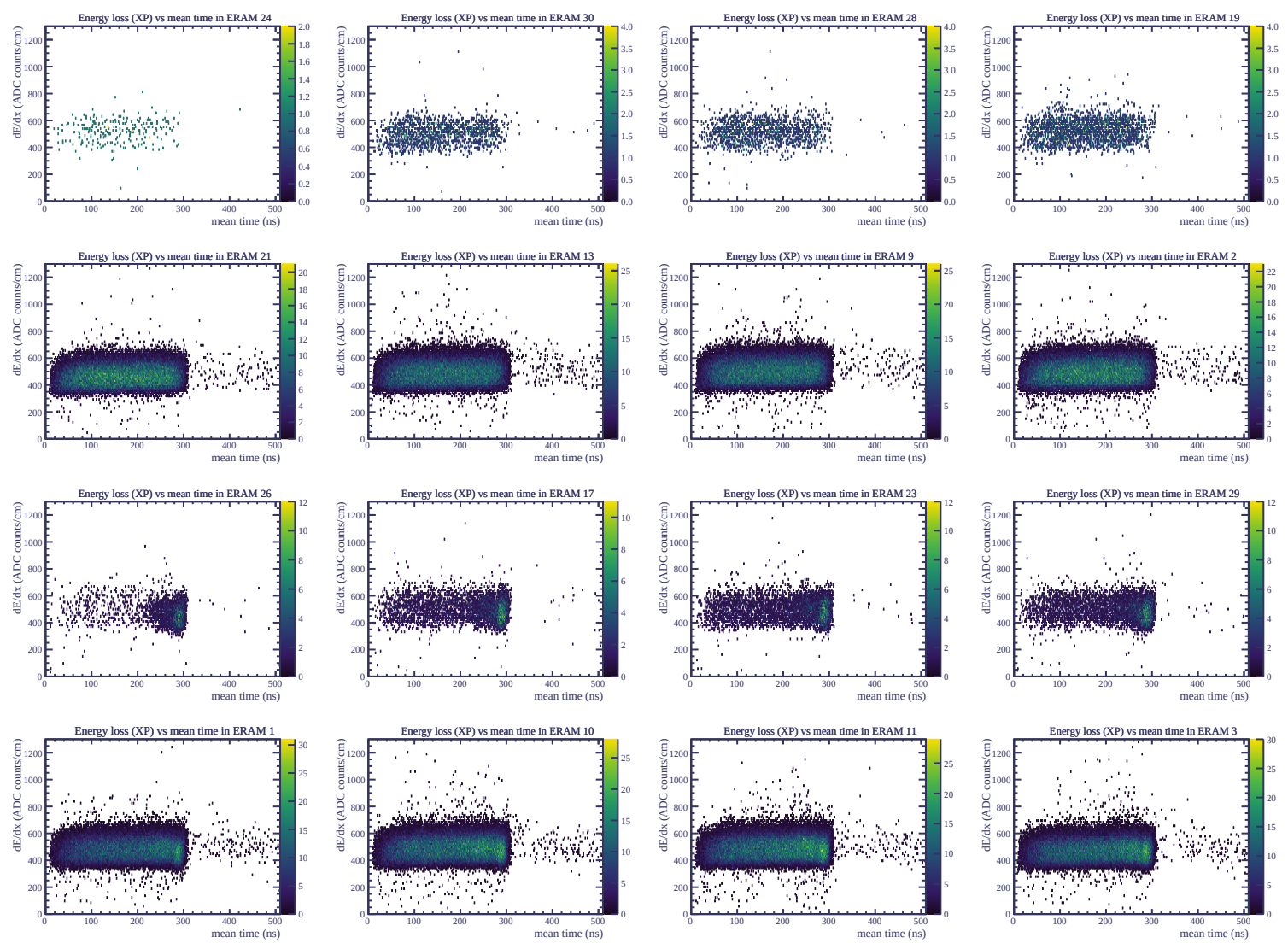


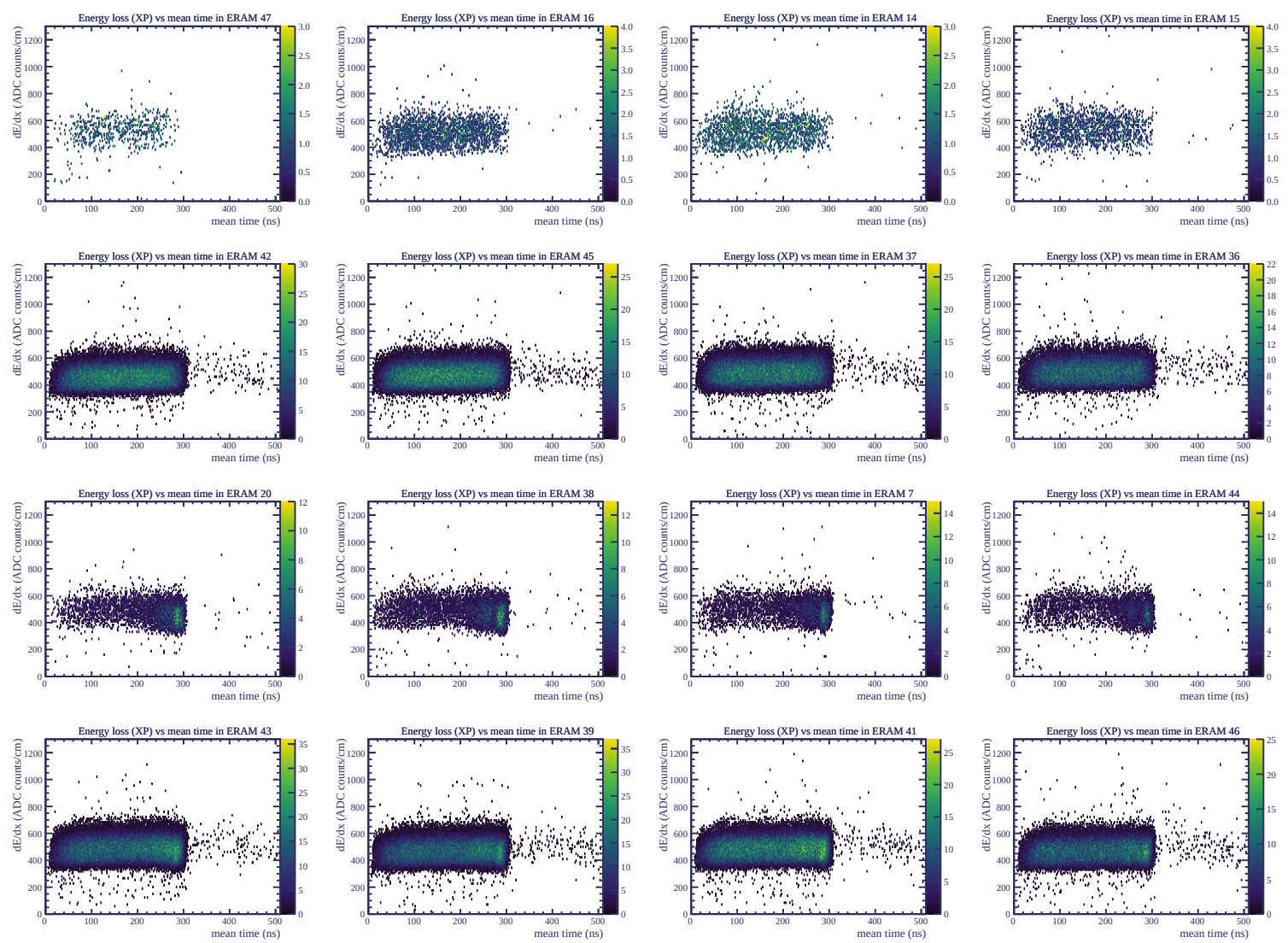






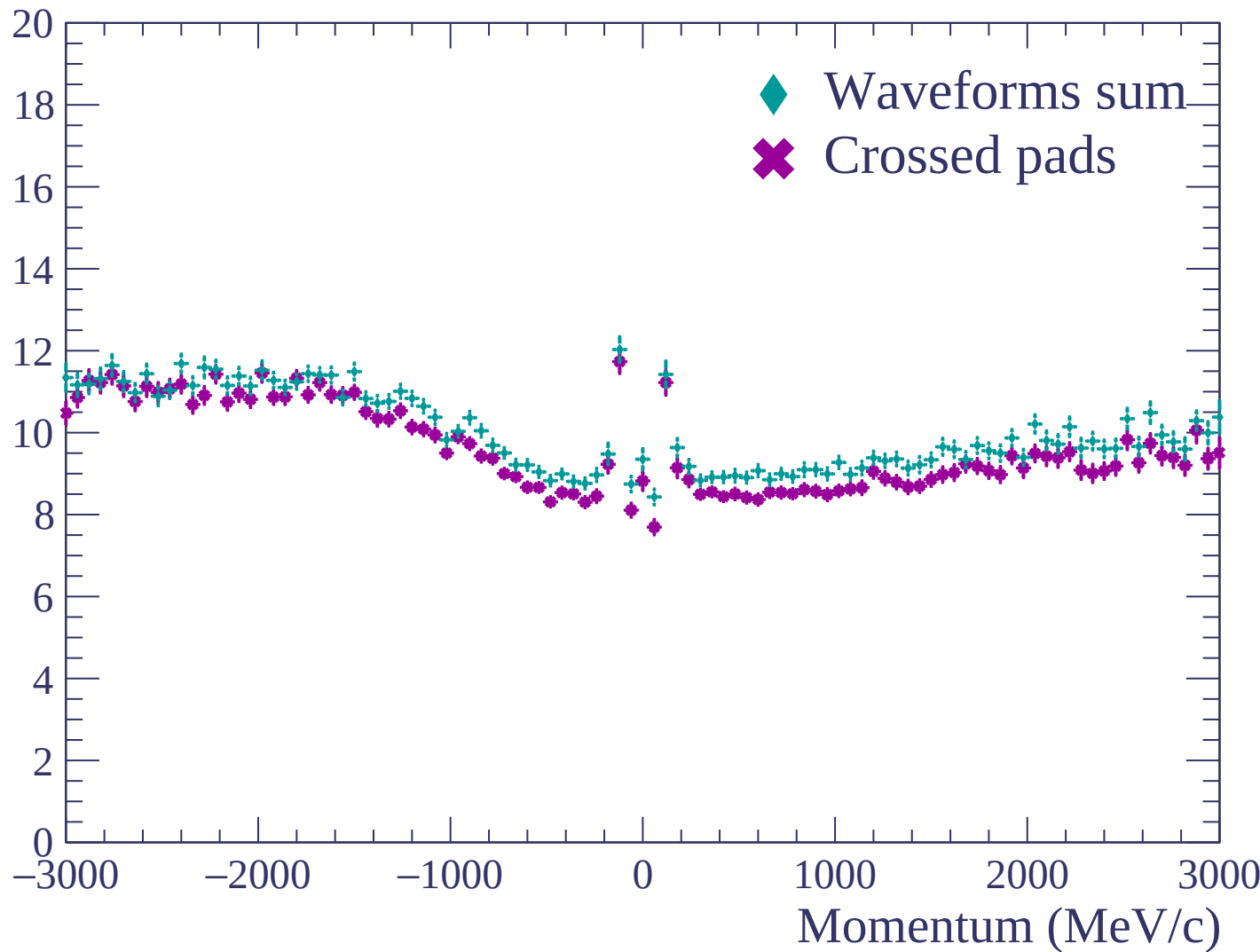


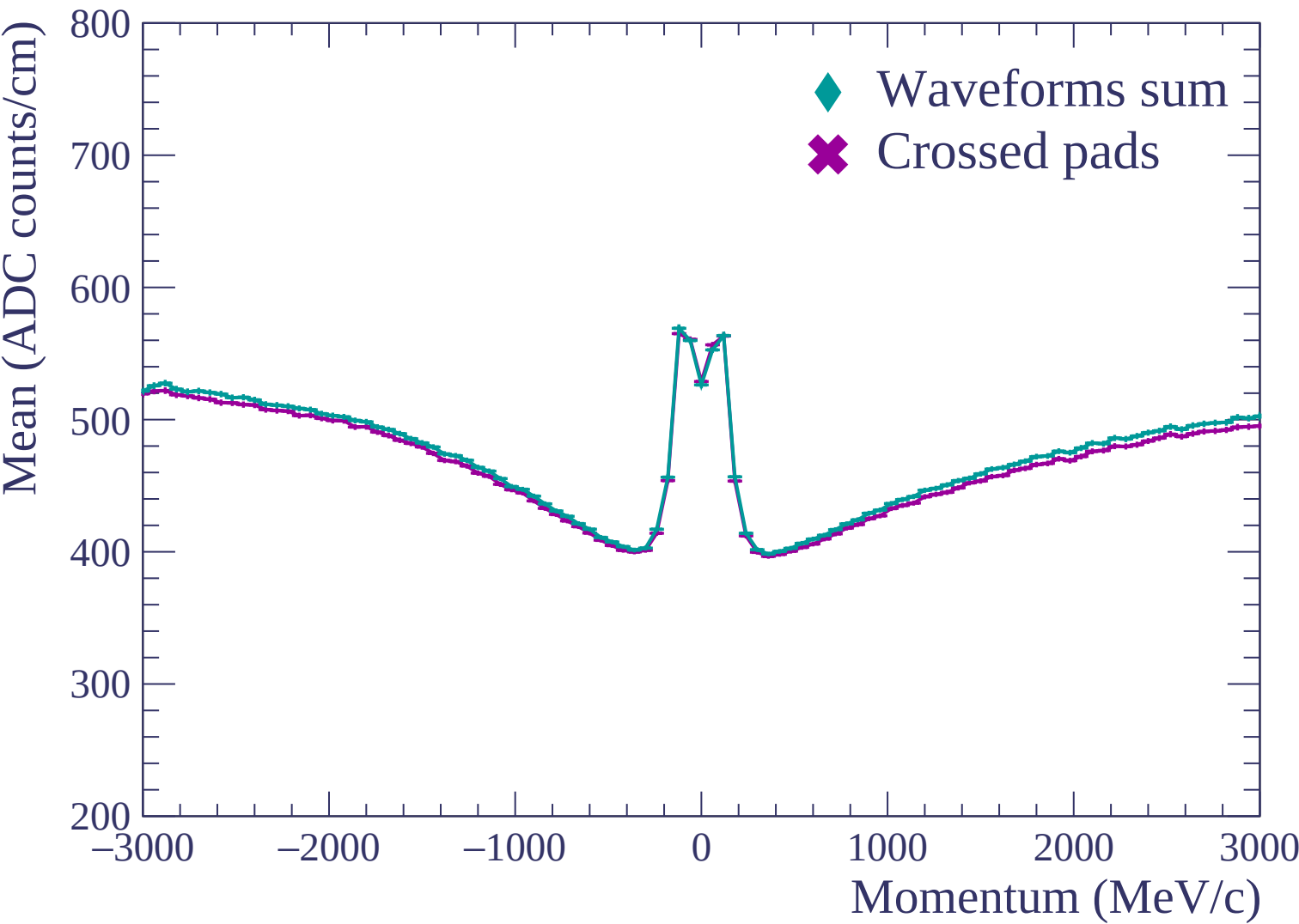


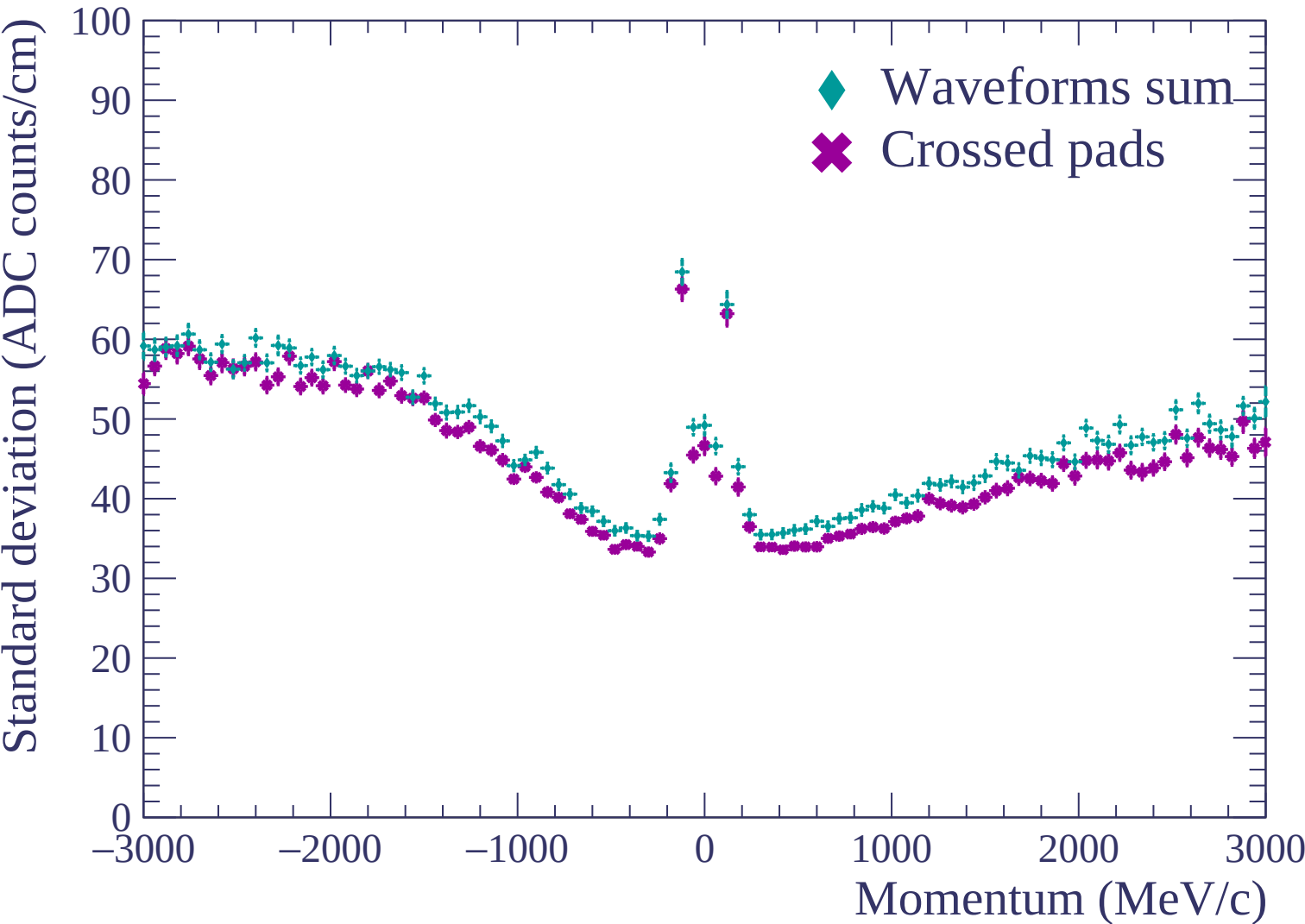


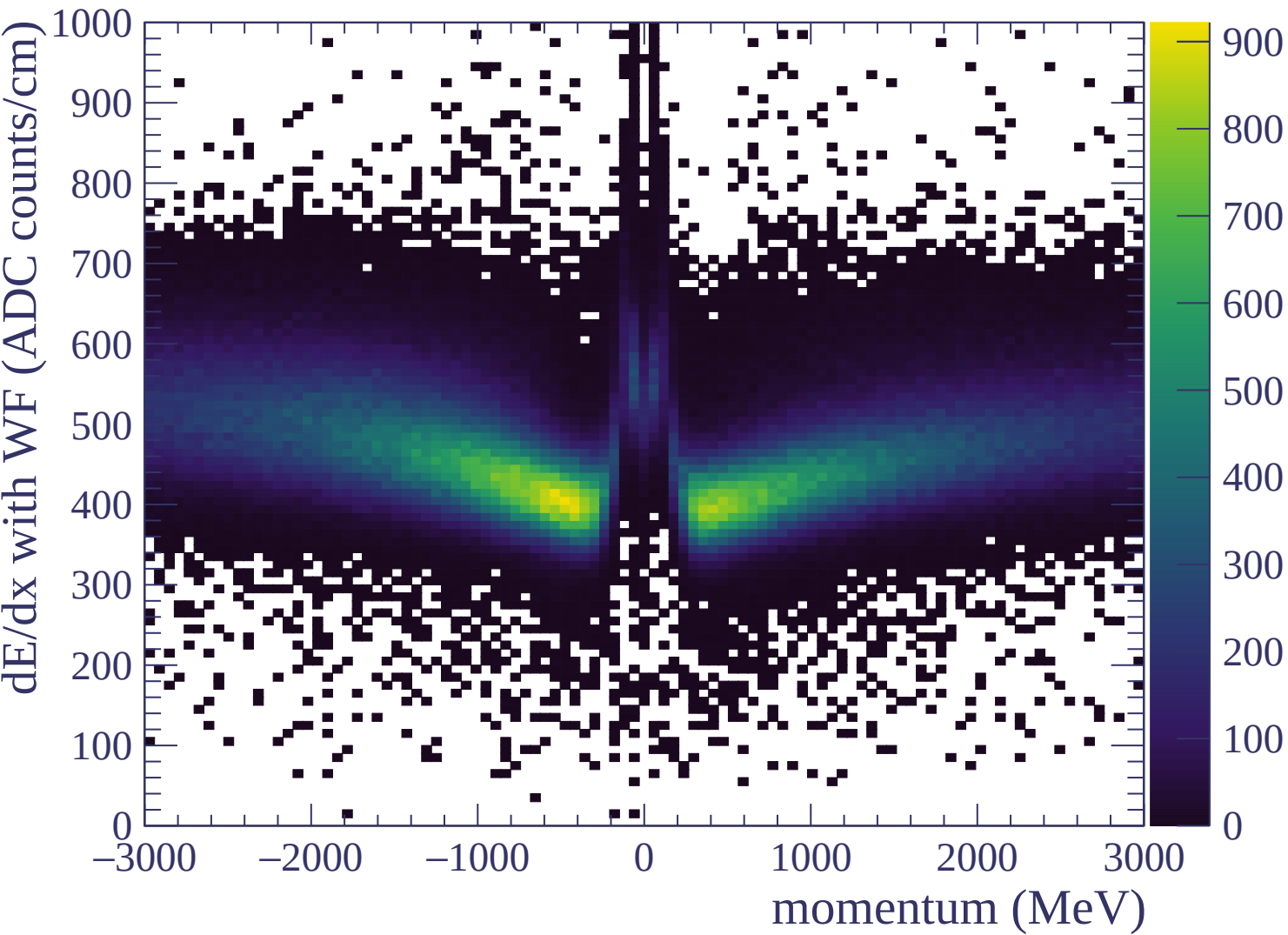
Resolution (%)

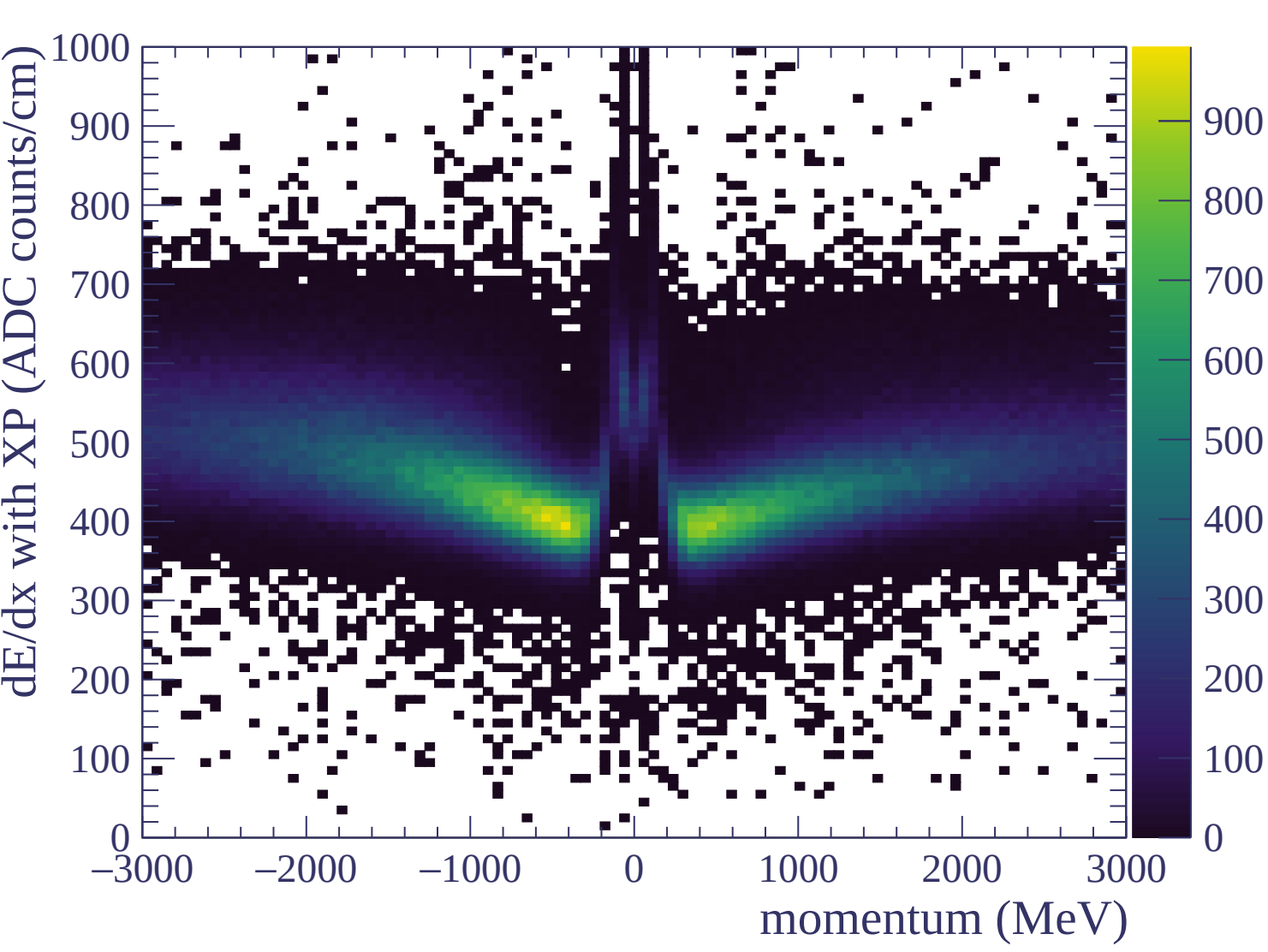
◆ Waveforms sum
✕ Crossed pads

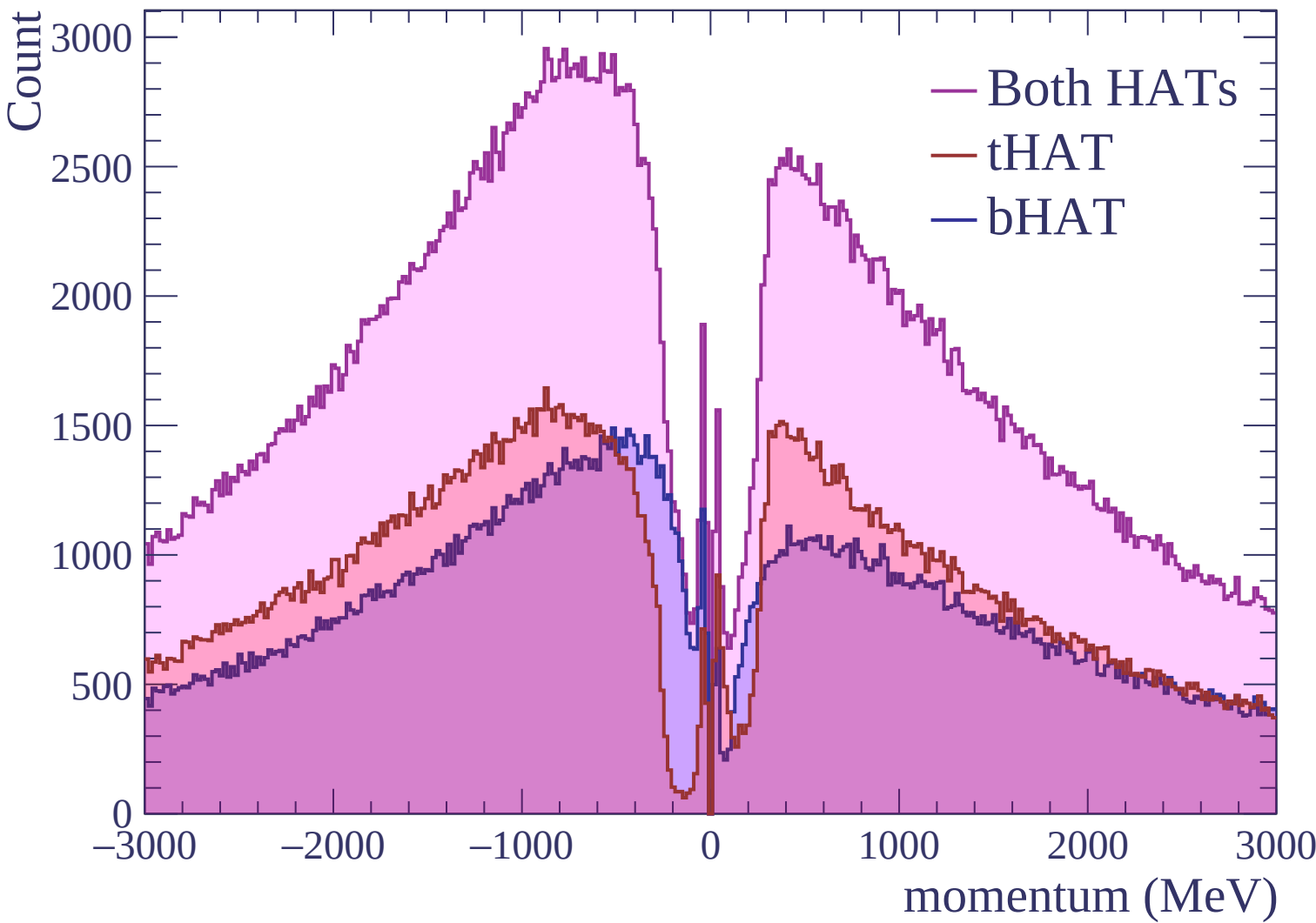






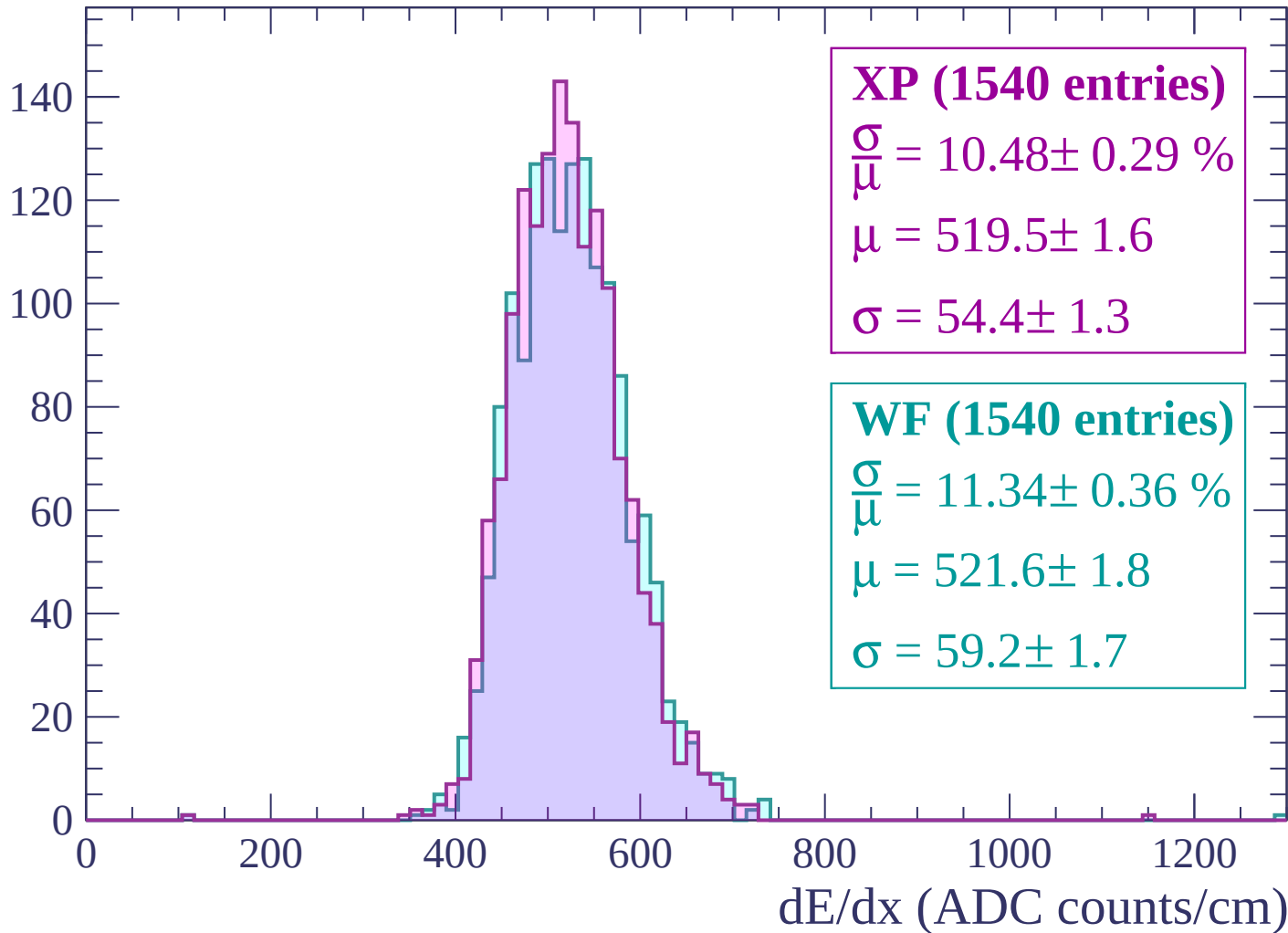






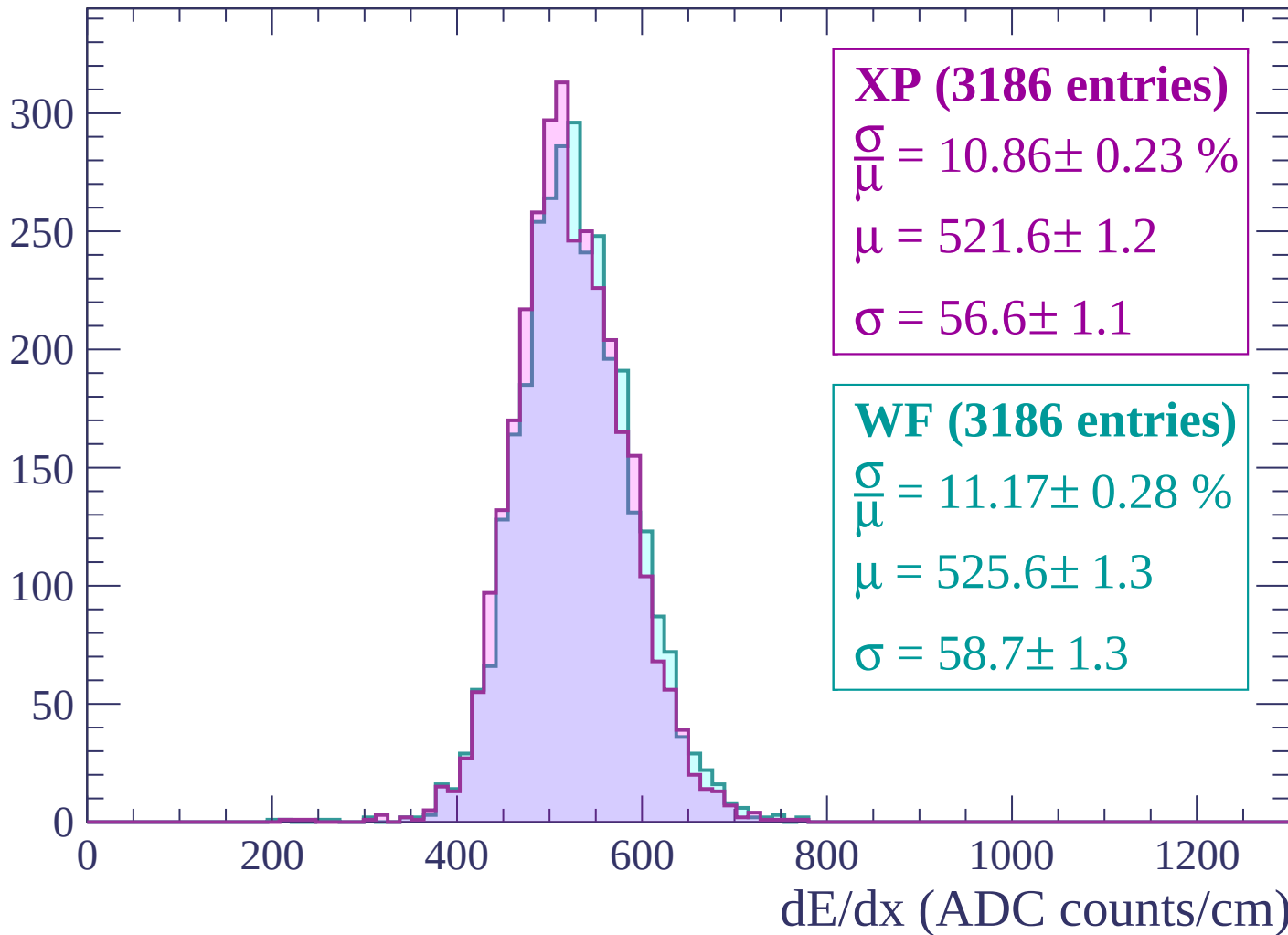
Energy loss | $-3000 < p < -2940$

Count



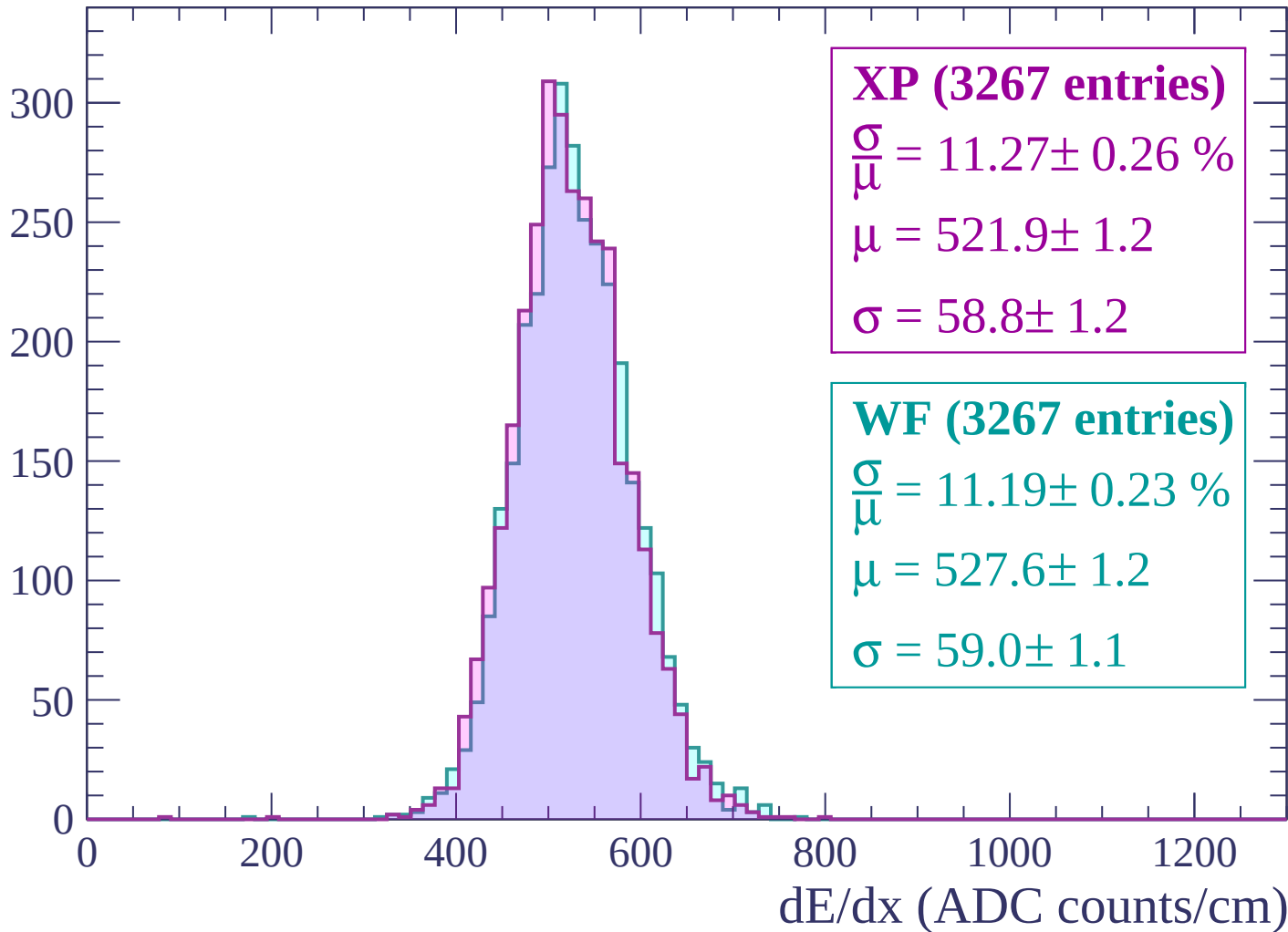
Energy loss | $-2940 < p < -2880$

Count



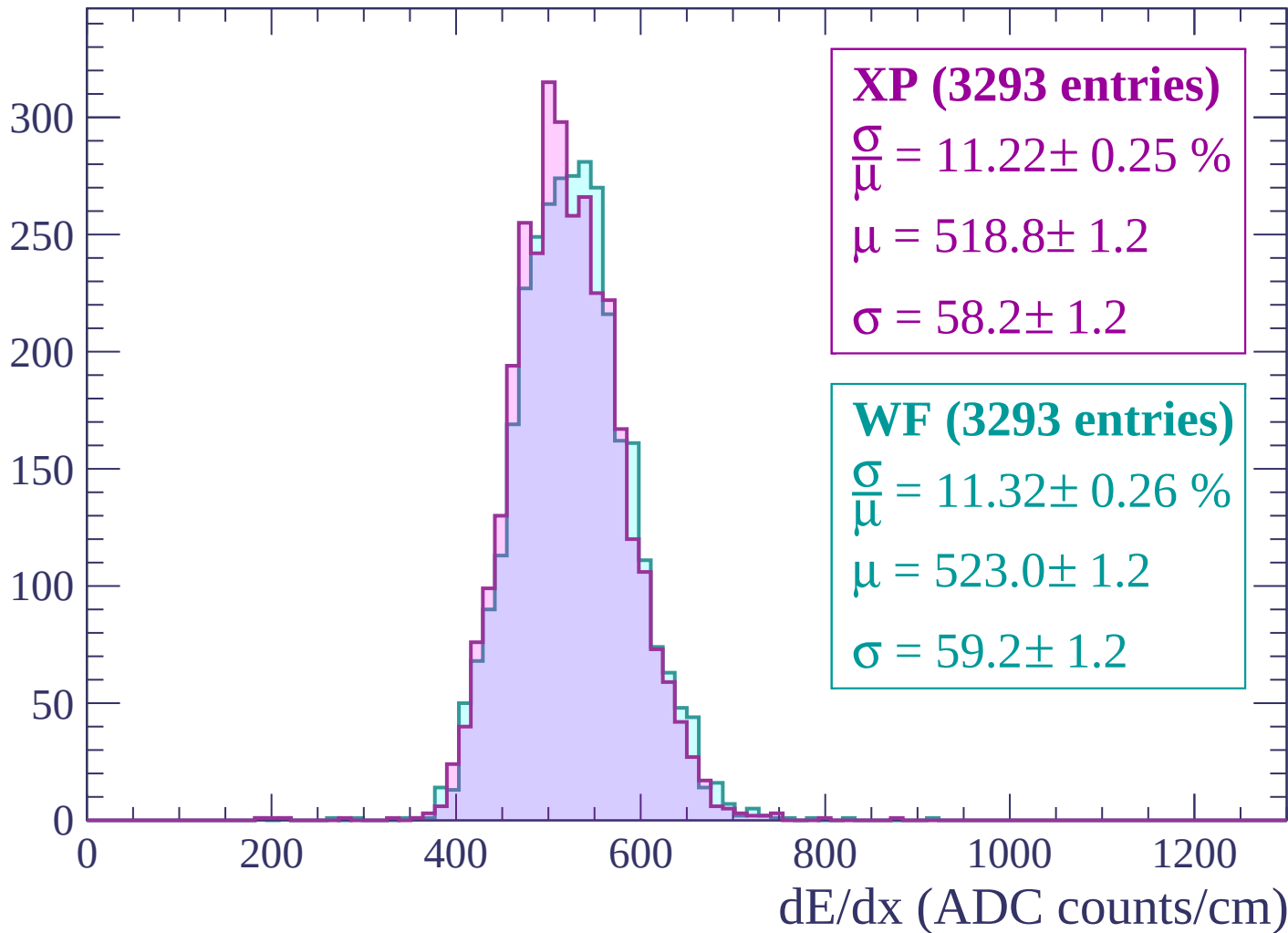
Energy loss | $-2880 < p < -2820$

Count



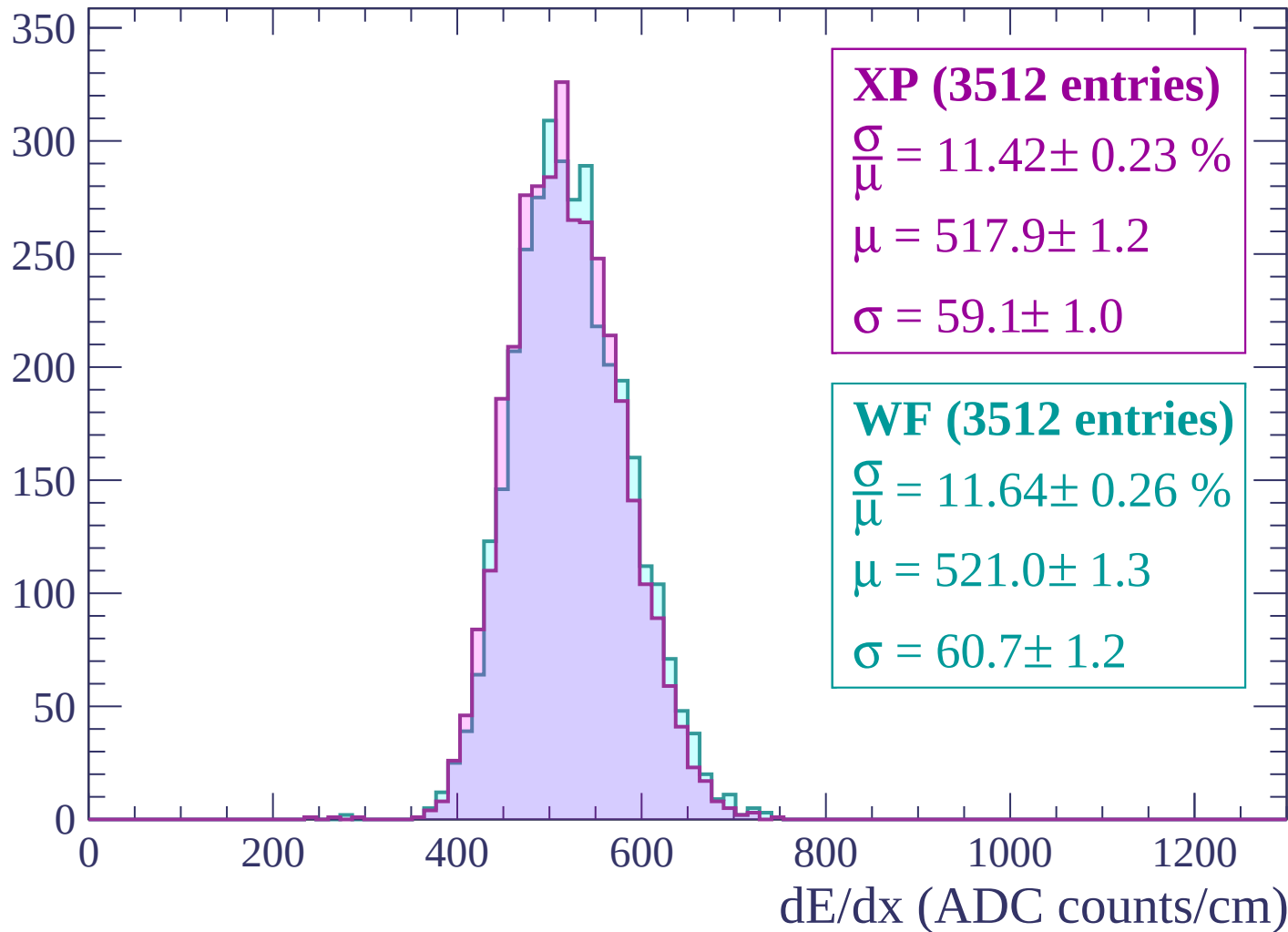
Energy loss | $-2820 < p < -2760$

Count



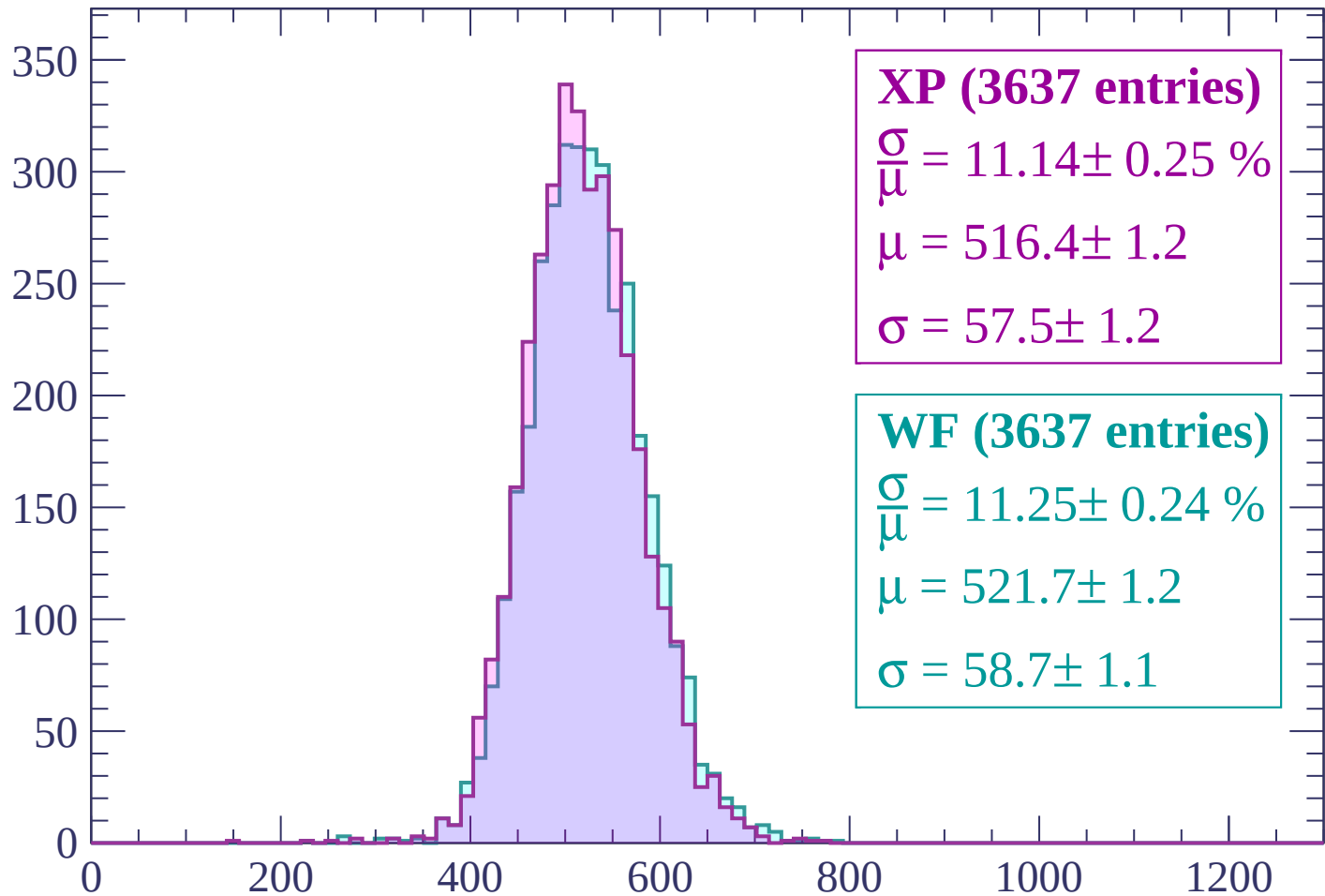
Energy loss | -2760 < p < -2700

Count



Energy loss | -2700 < p < -2640

Count



XP (3637 entries)

$$\frac{\sigma}{\mu} = 11.14 \pm 0.25 \%$$

$$\mu = 516.4 \pm 1.2$$

$$\sigma = 57.5 \pm 1.2$$

WF (3637 entries)

$$\frac{\sigma}{\mu} = 11.25 \pm 0.24 \%$$

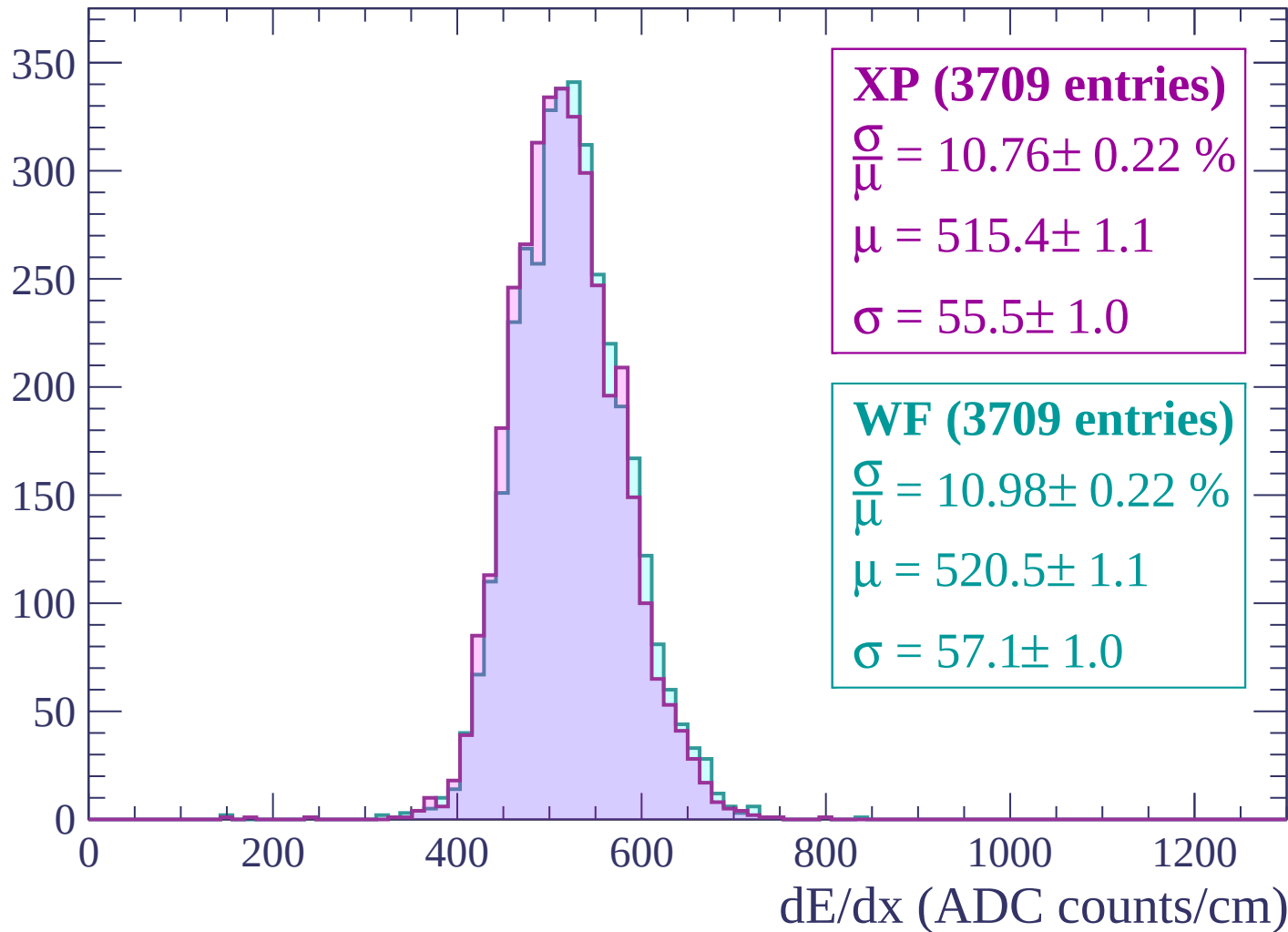
$$\mu = 521.7 \pm 1.2$$

$$\sigma = 58.7 \pm 1.1$$

dE/dx (ADC counts/cm)

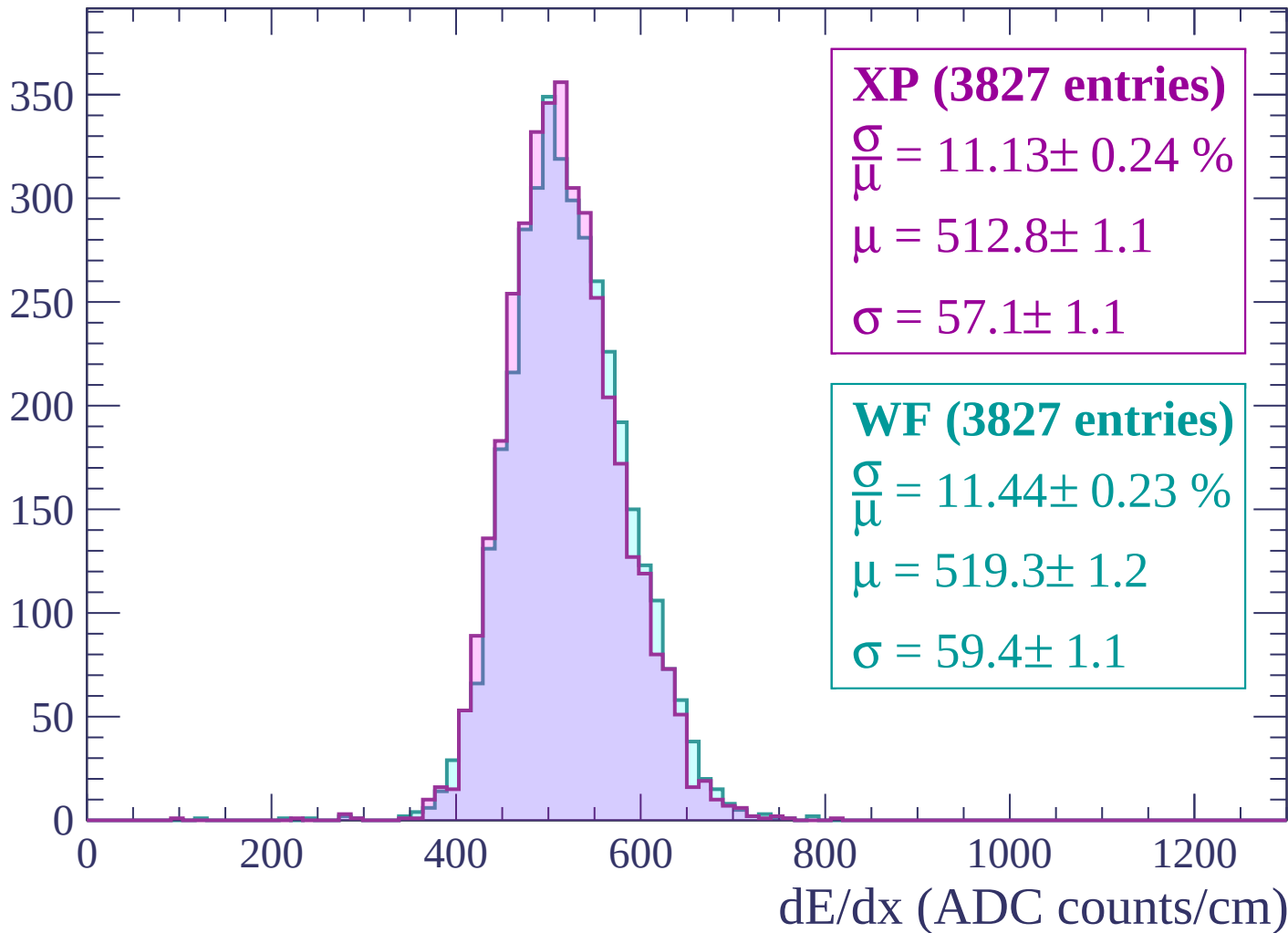
Energy loss | $-2640 < p < -2580$

Count



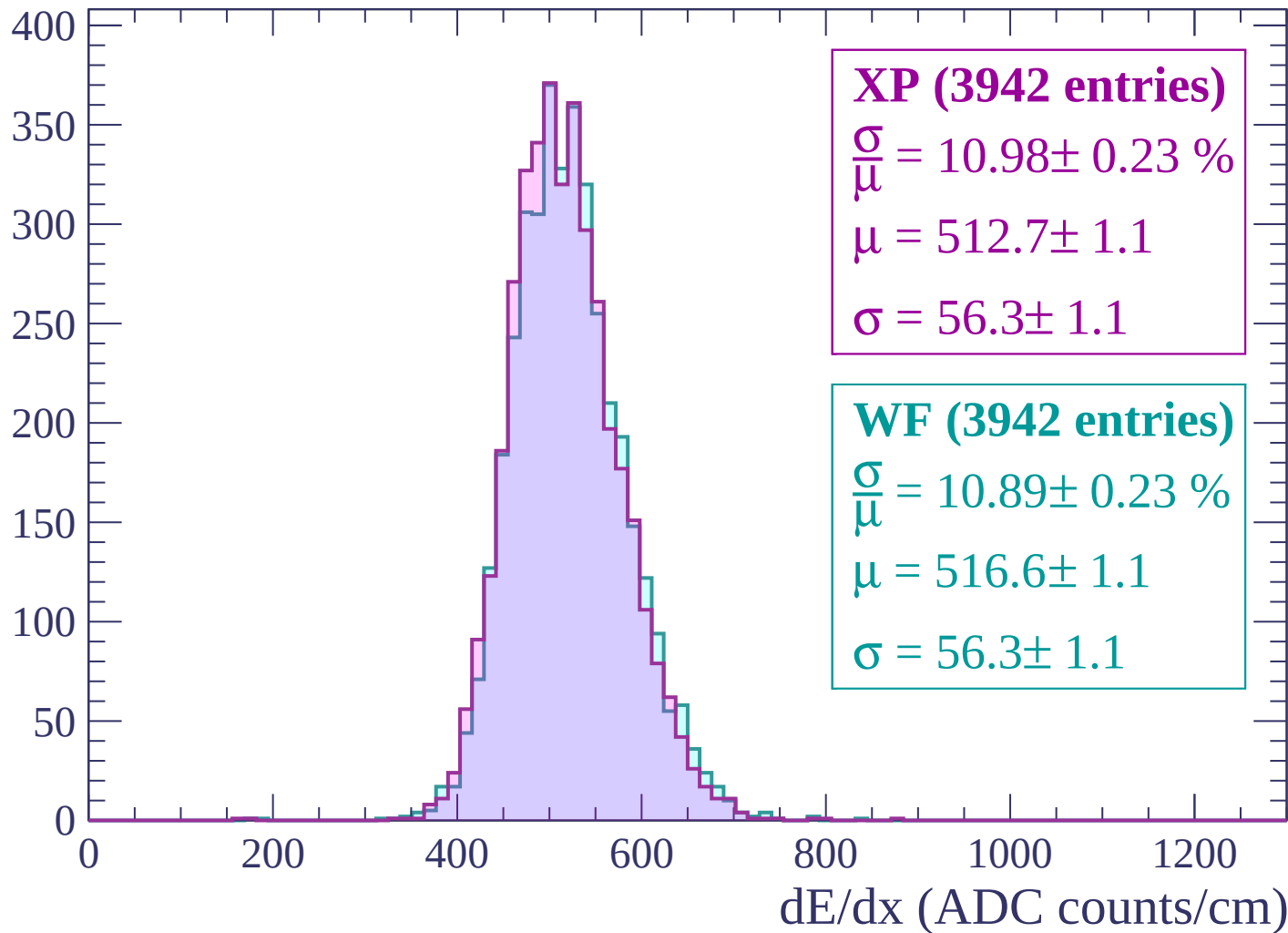
Energy loss | $-2580 < p < -2520$

Count



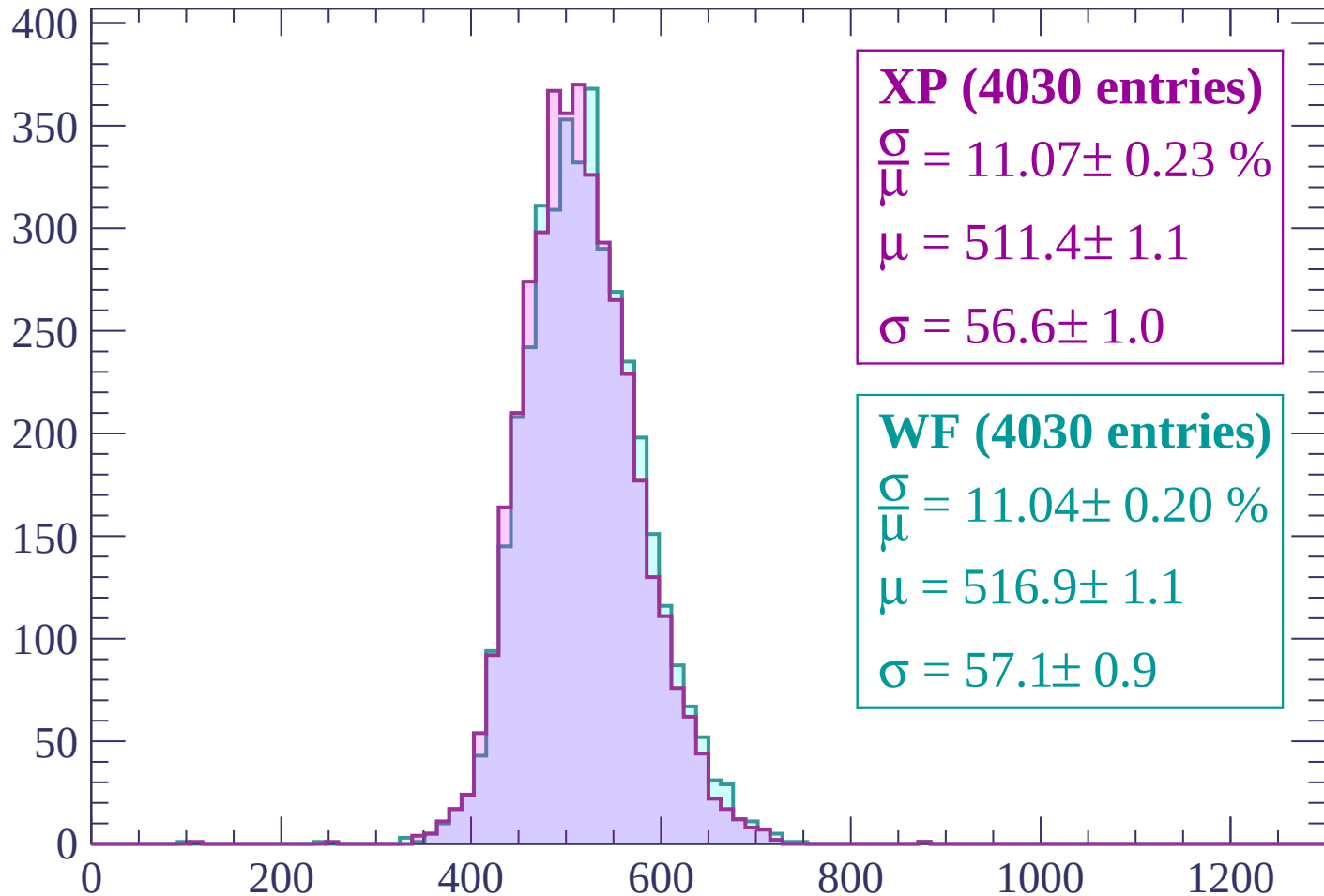
Energy loss | -2520 < p < -2460

Count



Energy loss | $-2460 < p < -2400$

Count



XP (4030 entries)

$$\frac{\sigma}{\mu} = 11.07 \pm 0.23 \%$$

$$\mu = 511.4 \pm 1.1$$

$$\sigma = 56.6 \pm 1.0$$

WF (4030 entries)

$$\frac{\sigma}{\mu} = 11.04 \pm 0.20 \%$$

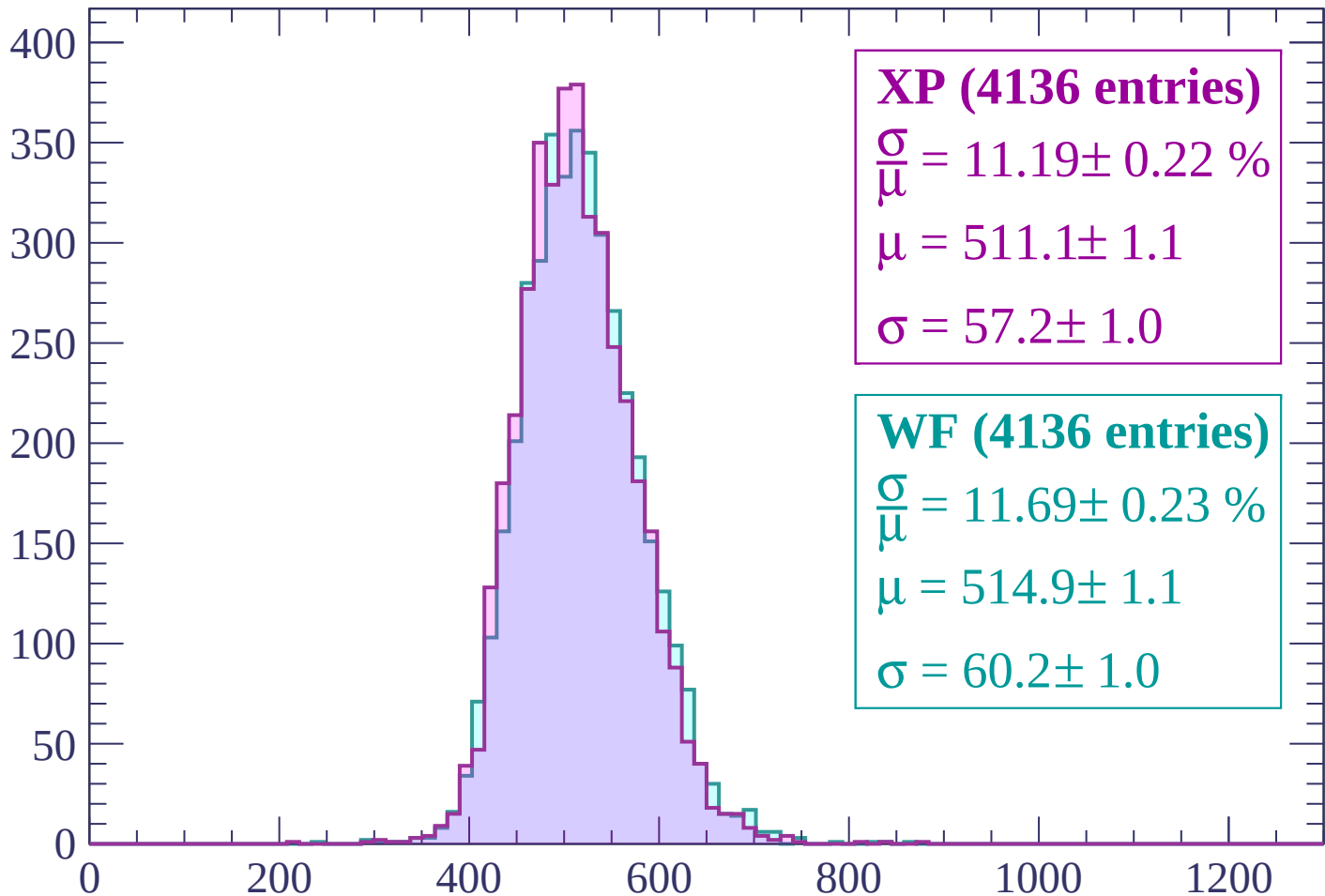
$$\mu = 516.9 \pm 1.1$$

$$\sigma = 57.1 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-2400 < p < -2340$

Count



XP (4136 entries)

$$\frac{\sigma}{\mu} = 11.19 \pm 0.22 \%$$

$$\mu = 511.1 \pm 1.1$$

$$\sigma = 57.2 \pm 1.0$$

WF (4136 entries)

$$\frac{\sigma}{\mu} = 11.69 \pm 0.23 \%$$

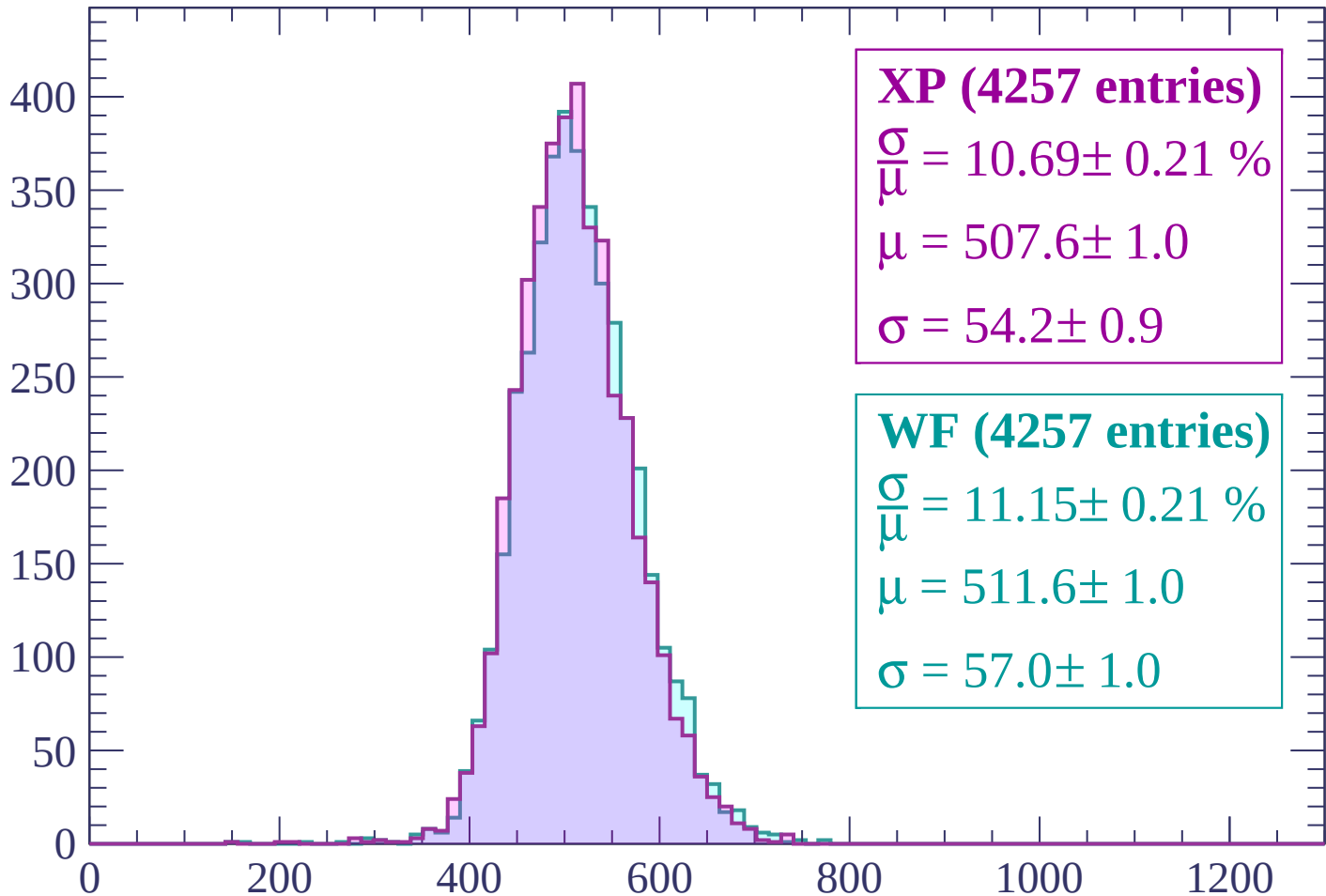
$$\mu = 514.9 \pm 1.1$$

$$\sigma = 60.2 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-2340 < p < -2280$

Count



XP (4257 entries)

$$\frac{\sigma}{\mu} = 10.69 \pm 0.21 \%$$

$$\mu = 507.6 \pm 1.0$$

$$\sigma = 54.2 \pm 0.9$$

WF (4257 entries)

$$\frac{\sigma}{\mu} = 11.15 \pm 0.21 \%$$

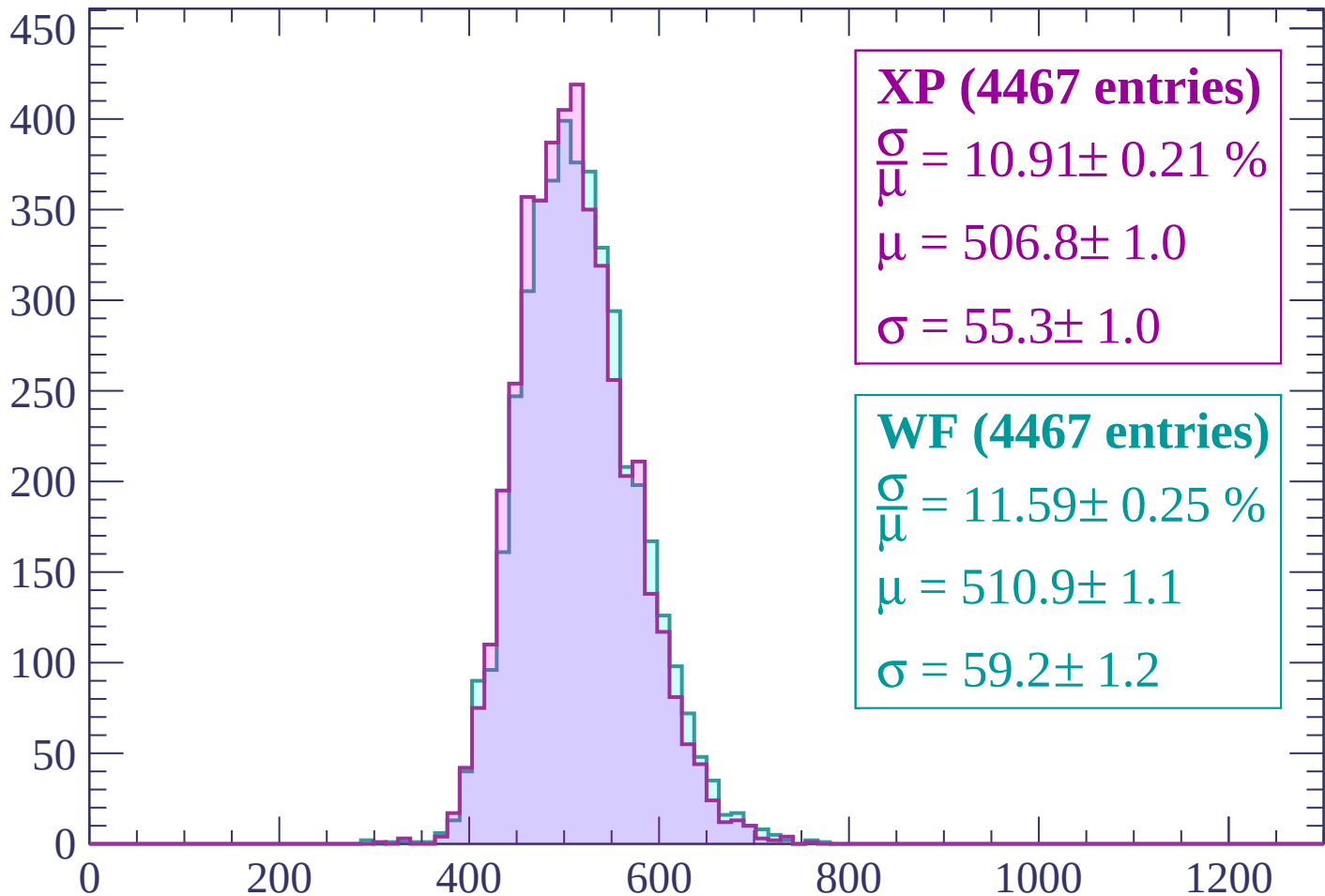
$$\mu = 511.6 \pm 1.0$$

$$\sigma = 57.0 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | -2280 < p < -2220

Count



XP (4467 entries)

$\frac{\sigma}{\mu} = 10.91 \pm 0.21 \%$

$\mu = 506.8 \pm 1.0$

$\sigma = 55.3 \pm 1.0$

WF (4467 entries)

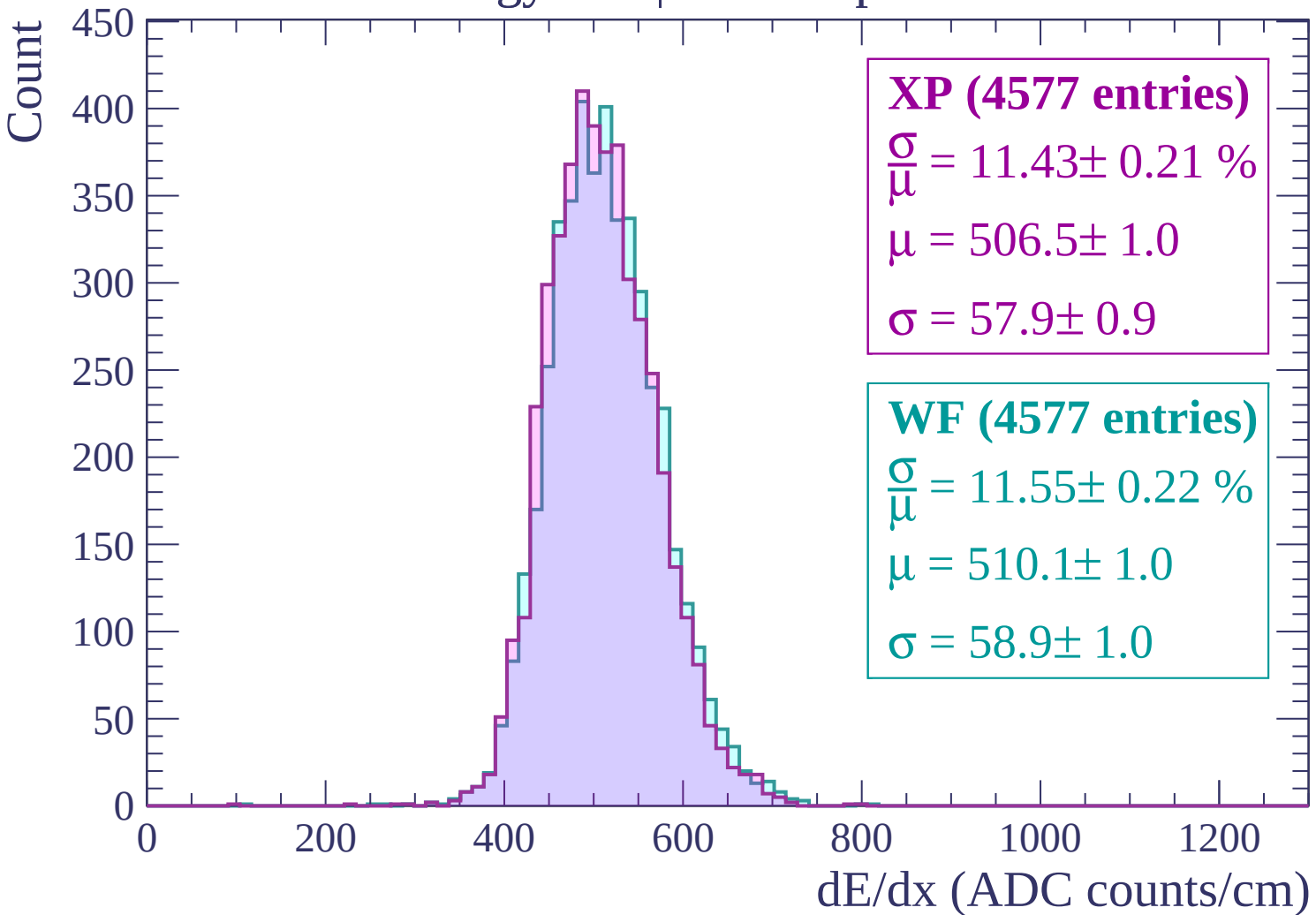
$\frac{\sigma}{\mu} = 11.59 \pm 0.25 \%$

$\mu = 510.9 \pm 1.1$

$\sigma = 59.2 \pm 1.2$

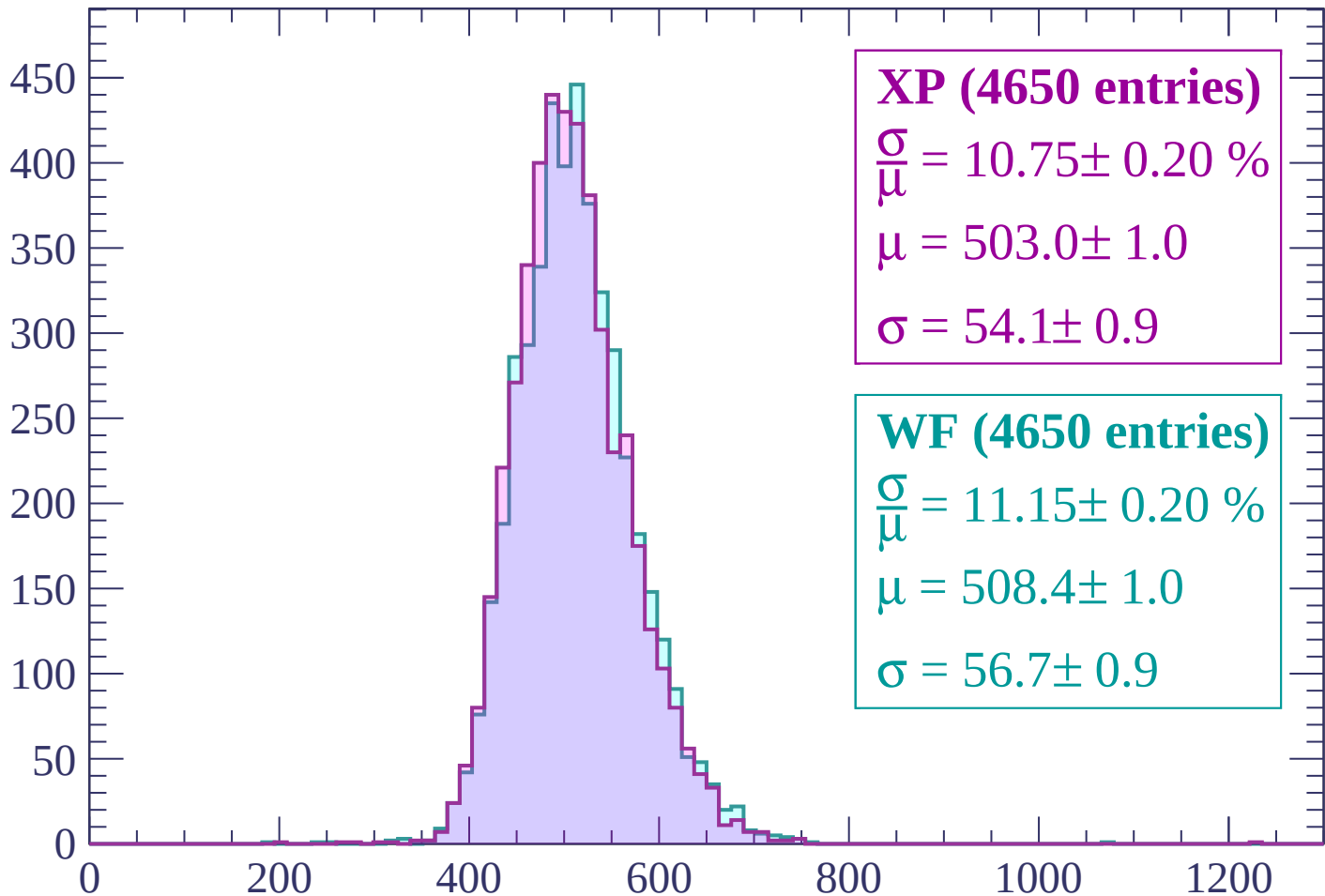
dE/dx (ADC counts/cm)

Energy loss | -2220 < p < -2160



Energy loss | -2160 < p < -2100

Count



XP (4650 entries)

$\frac{\sigma}{\mu} = 10.75 \pm 0.20 \%$

$\mu = 503.0 \pm 1.0$

$\sigma = 54.1 \pm 0.9$

WF (4650 entries)

$\frac{\sigma}{\mu} = 11.15 \pm 0.20 \%$

$\mu = 508.4 \pm 1.0$

$\sigma = 56.7 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | -2100 < p < -2040

Count

500

400

300

200

100

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (4887 entries)

$\frac{\sigma}{\mu} = 10.96 \pm 0.20 \%$

$\mu = 503.3 \pm 1.0$

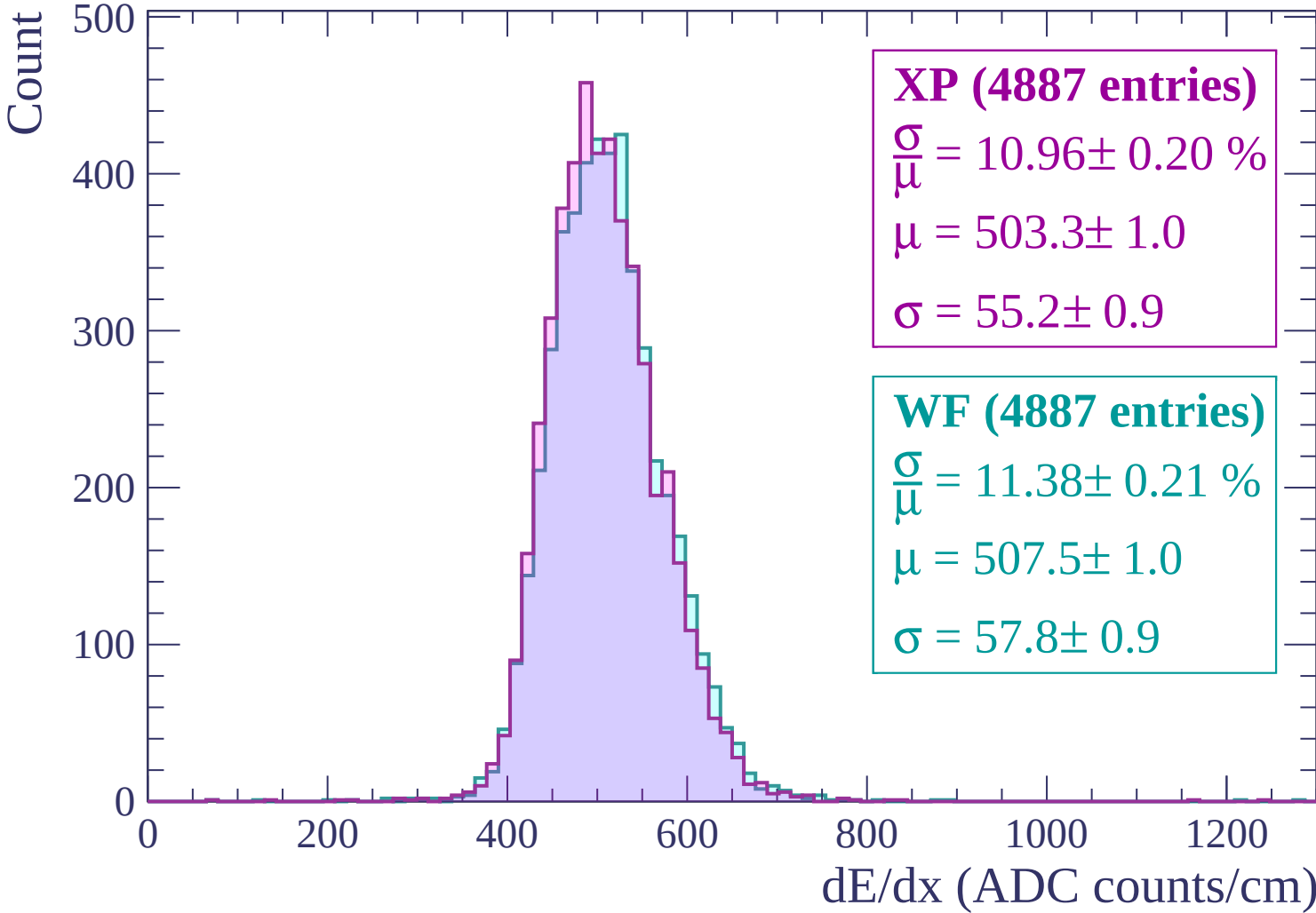
$\sigma = 55.2 \pm 0.9$

WF (4887 entries)

$\frac{\sigma}{\mu} = 11.38 \pm 0.21 \%$

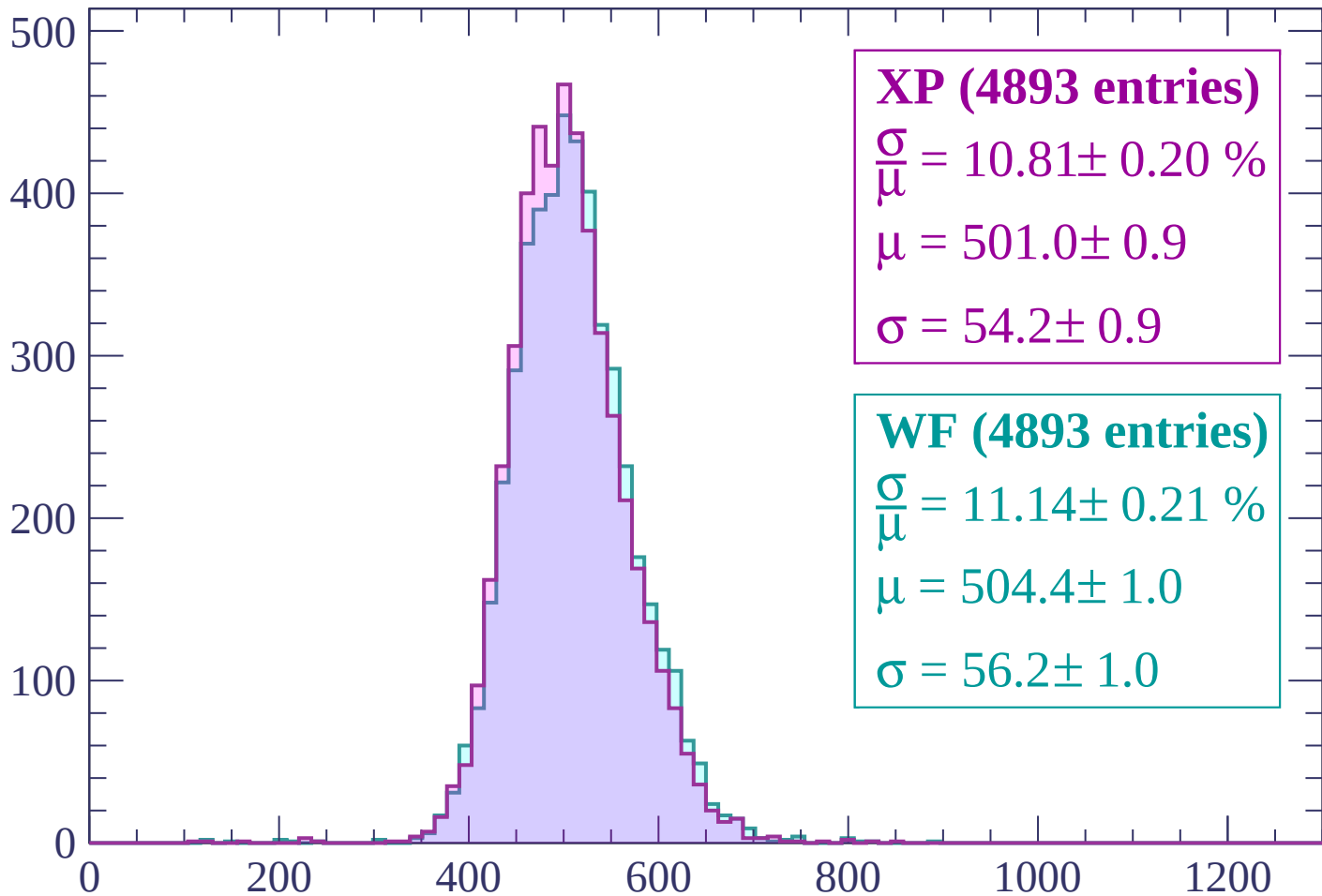
$\mu = 507.5 \pm 1.0$

$\sigma = 57.8 \pm 0.9$



Energy loss | -2040 < p < -1980

Count



XP (4893 entries)

$\frac{\sigma}{\mu} = 10.81 \pm 0.20 \%$

$\mu = 501.0 \pm 0.9$

$\sigma = 54.2 \pm 0.9$

WF (4893 entries)

$\frac{\sigma}{\mu} = 11.14 \pm 0.21 \%$

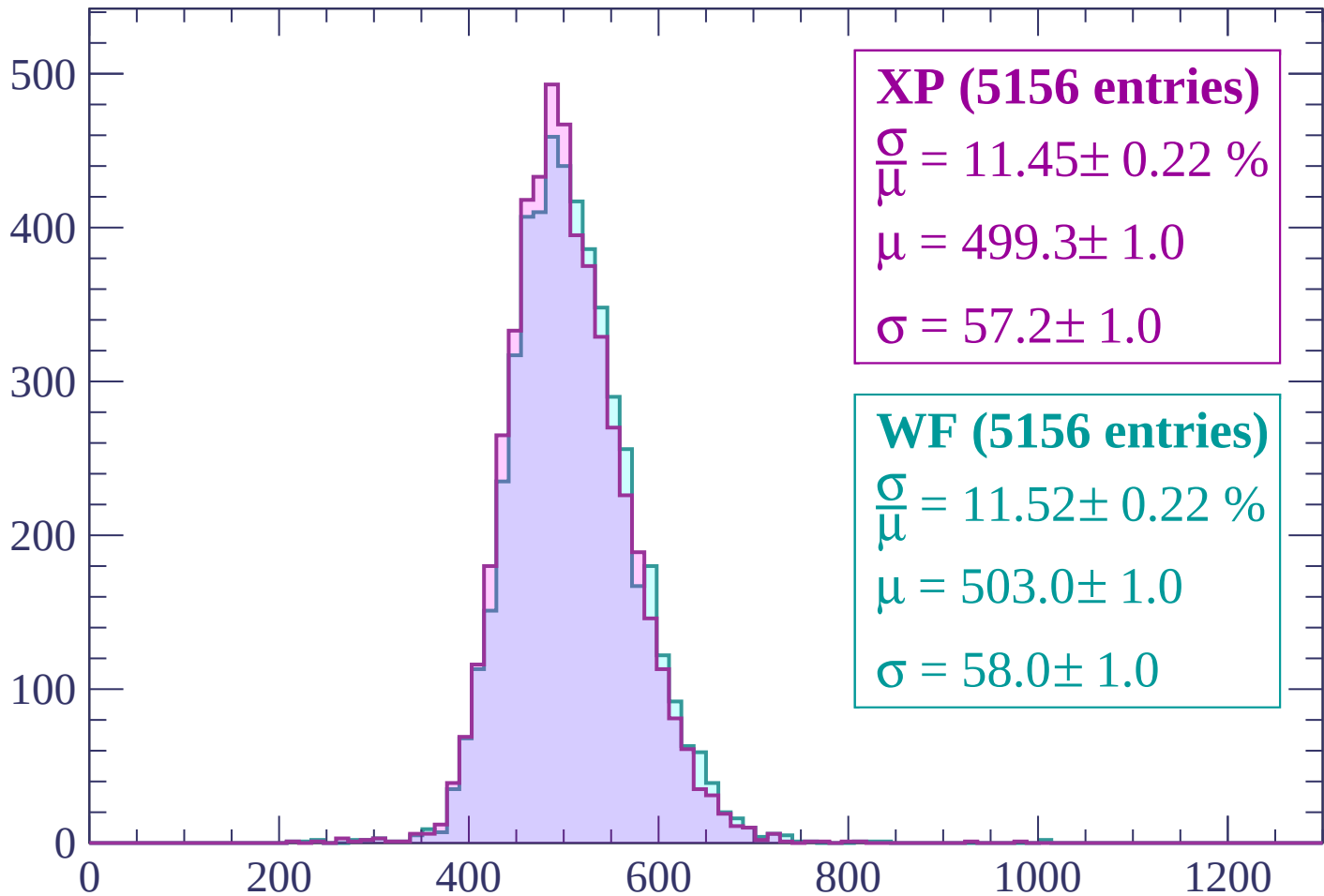
$\mu = 504.4 \pm 1.0$

$\sigma = 56.2 \pm 1.0$

dE/dx (ADC counts/cm)

Energy loss | -1980 < p < -1920

Count



XP (5156 entries)

$\frac{\sigma}{\mu} = 11.45 \pm 0.22 \%$

$\mu = 499.3 \pm 1.0$

$\sigma = 57.2 \pm 1.0$

WF (5156 entries)

$\frac{\sigma}{\mu} = 11.52 \pm 0.22 \%$

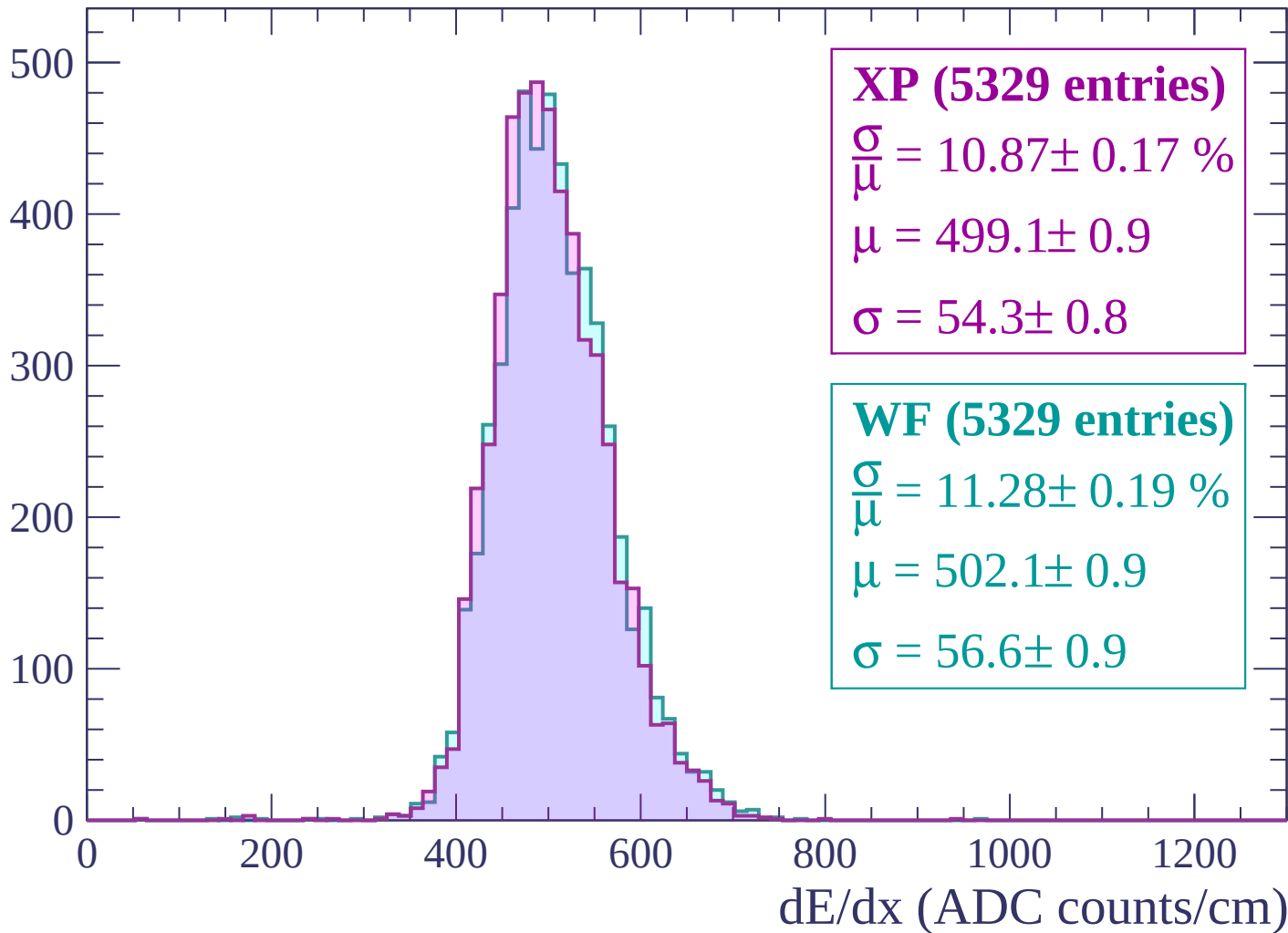
$\mu = 503.0 \pm 1.0$

$\sigma = 58.0 \pm 1.0$

dE/dx (ADC counts/cm)

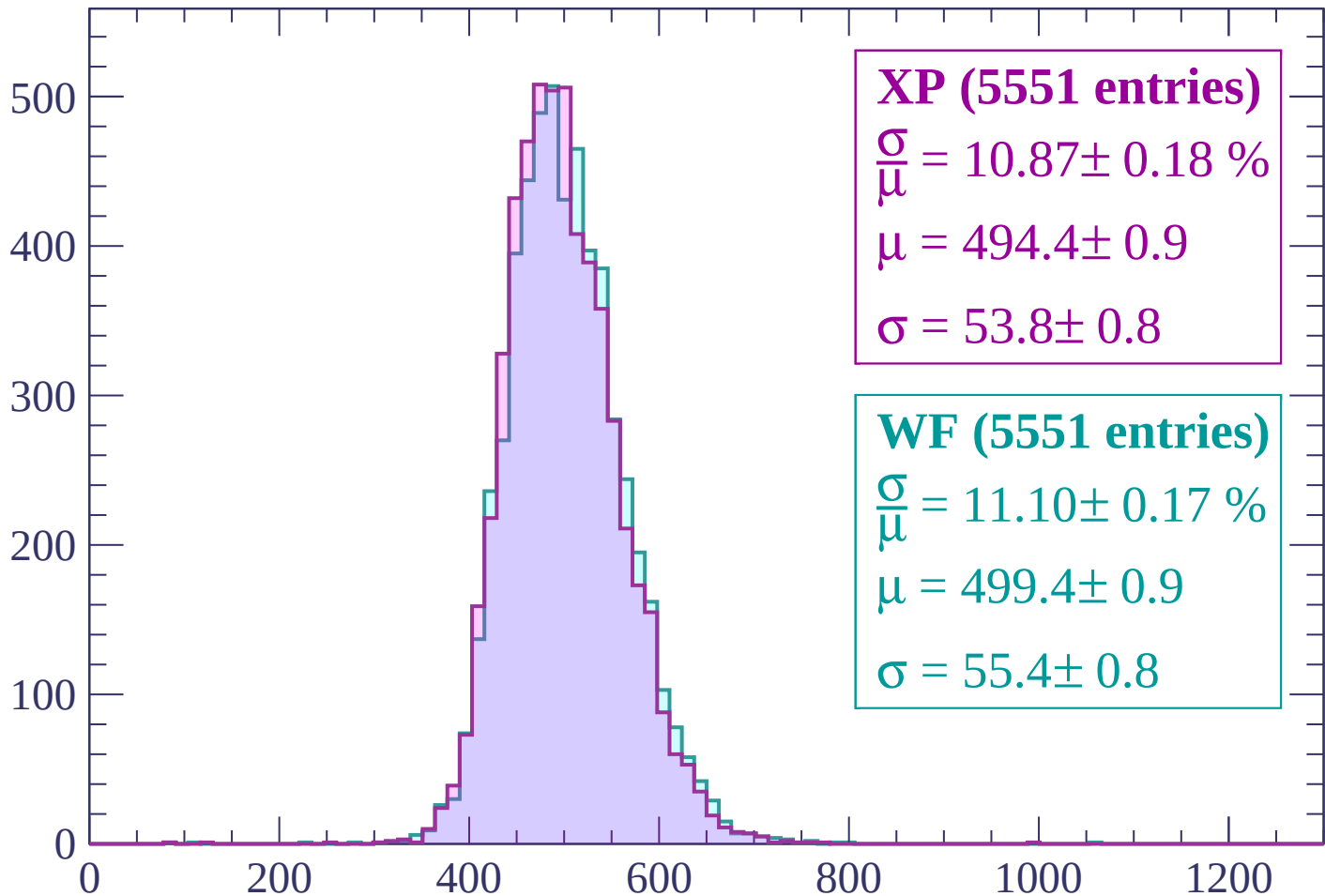
Energy loss | -1920 < p < -1860

Count



Energy loss | -1860 < p < -1800

Count



XP (5551 entries)

$\frac{\sigma}{\mu} = 10.87 \pm 0.18 \%$

$\mu = 494.4 \pm 0.9$

$\sigma = 53.8 \pm 0.8$

WF (5551 entries)

$\frac{\sigma}{\mu} = 11.10 \pm 0.17 \%$

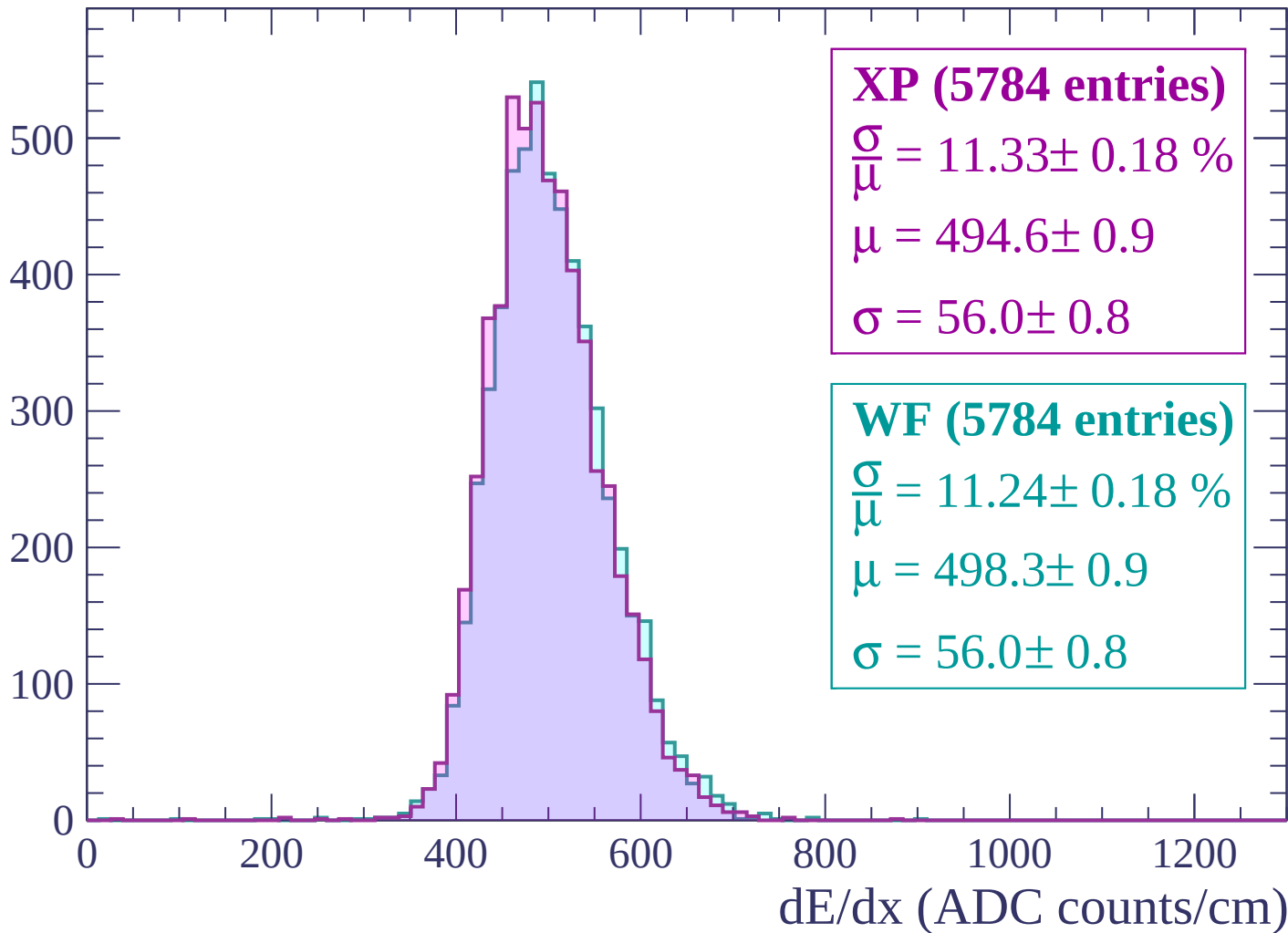
$\mu = 499.4 \pm 0.9$

$\sigma = 55.4 \pm 0.8$

dE/dx (ADC counts/cm)

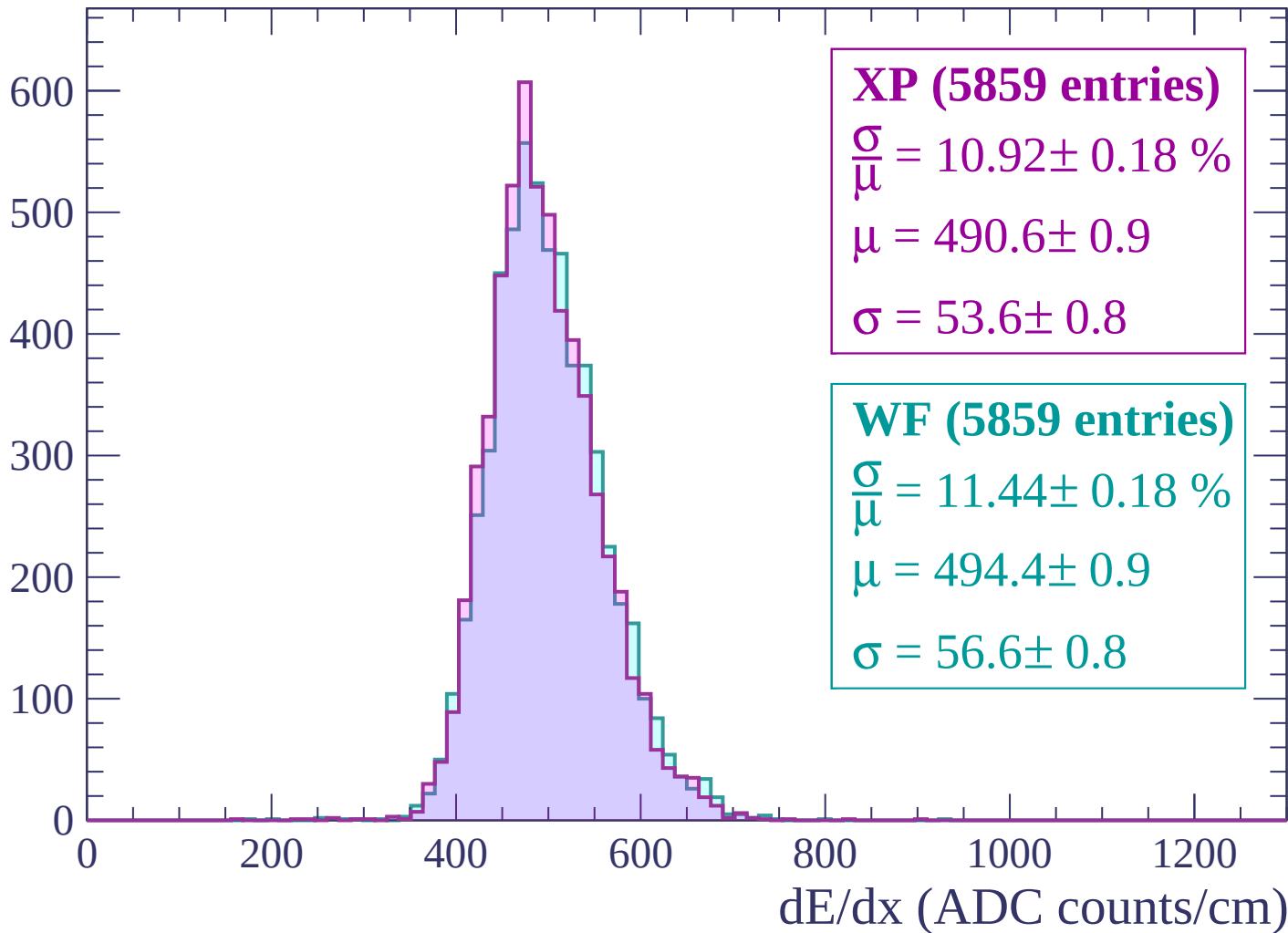
Energy loss | -1800 < p < -1740

Count



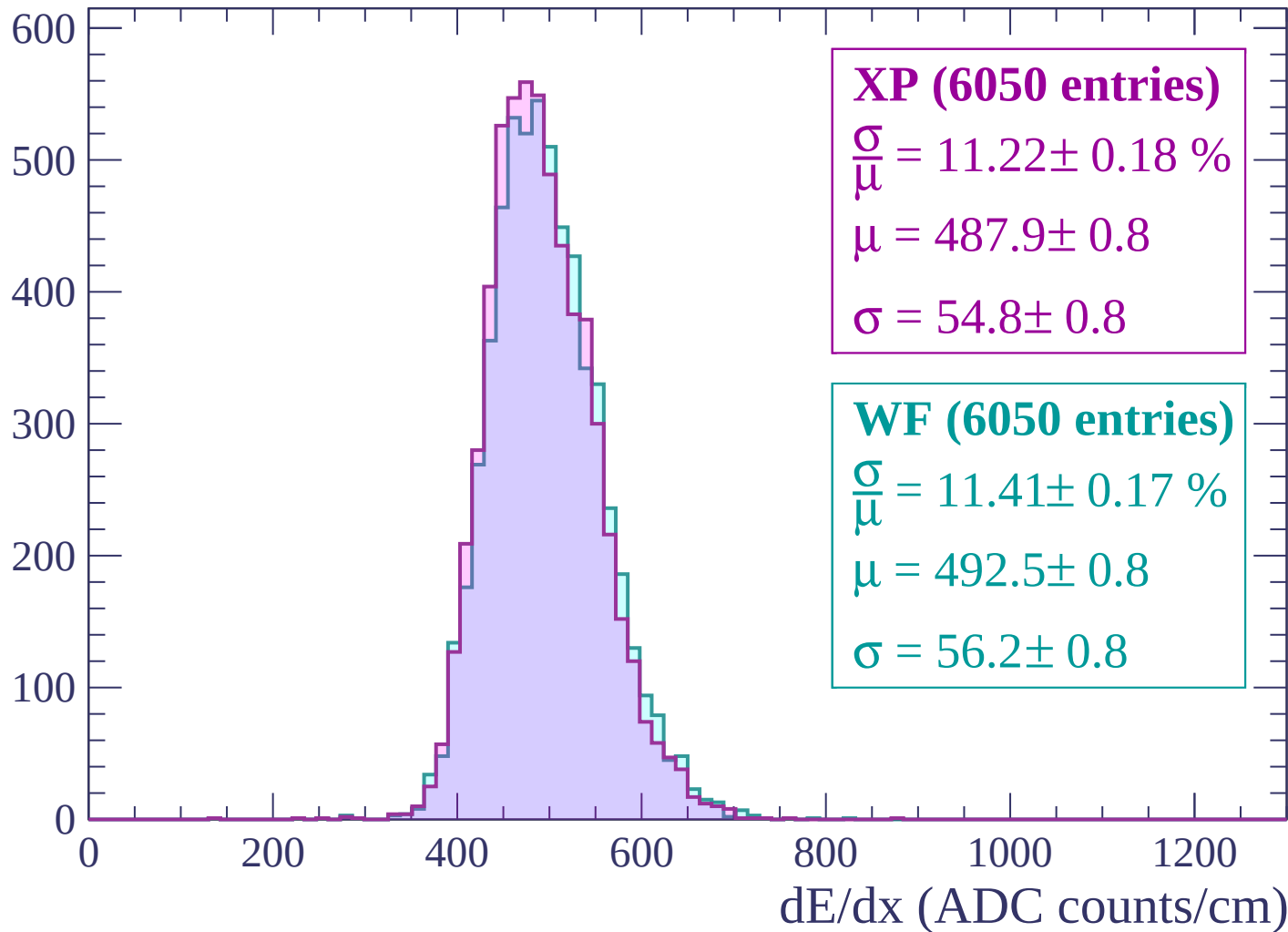
Energy loss | $-1740 < p < -1680$

Count



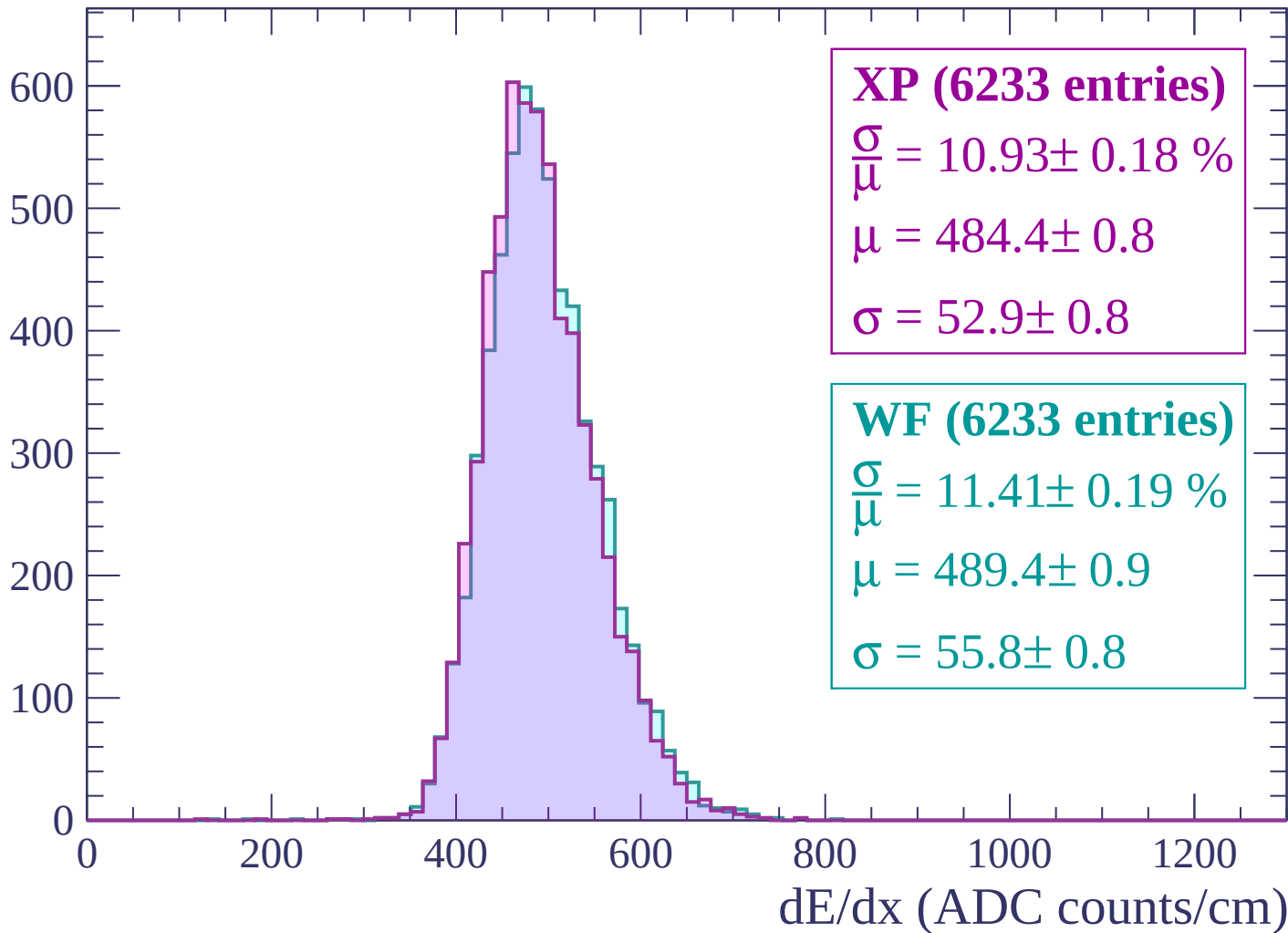
Energy loss | -1680 < p < -1620

Count



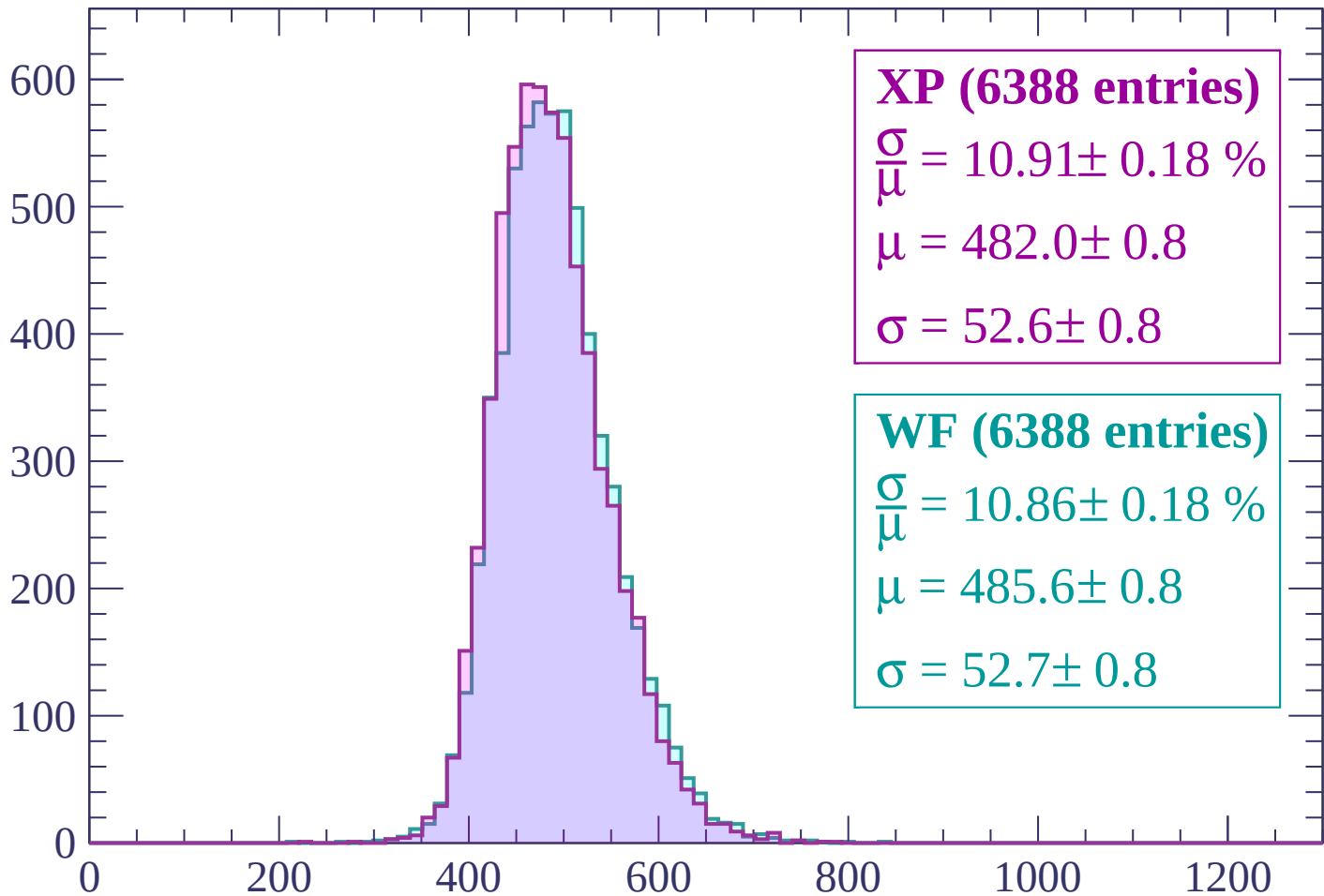
Energy loss | -1620 < p < -1560

Count



Energy loss | -1560 < p < -1500

Count



XP (6388 entries)

$$\frac{\sigma}{\mu} = 10.91 \pm 0.18 \%$$

$$\mu = 482.0 \pm 0.8$$

$$\sigma = 52.6 \pm 0.8$$

WF (6388 entries)

$$\frac{\sigma}{\mu} = 10.86 \pm 0.18 \%$$

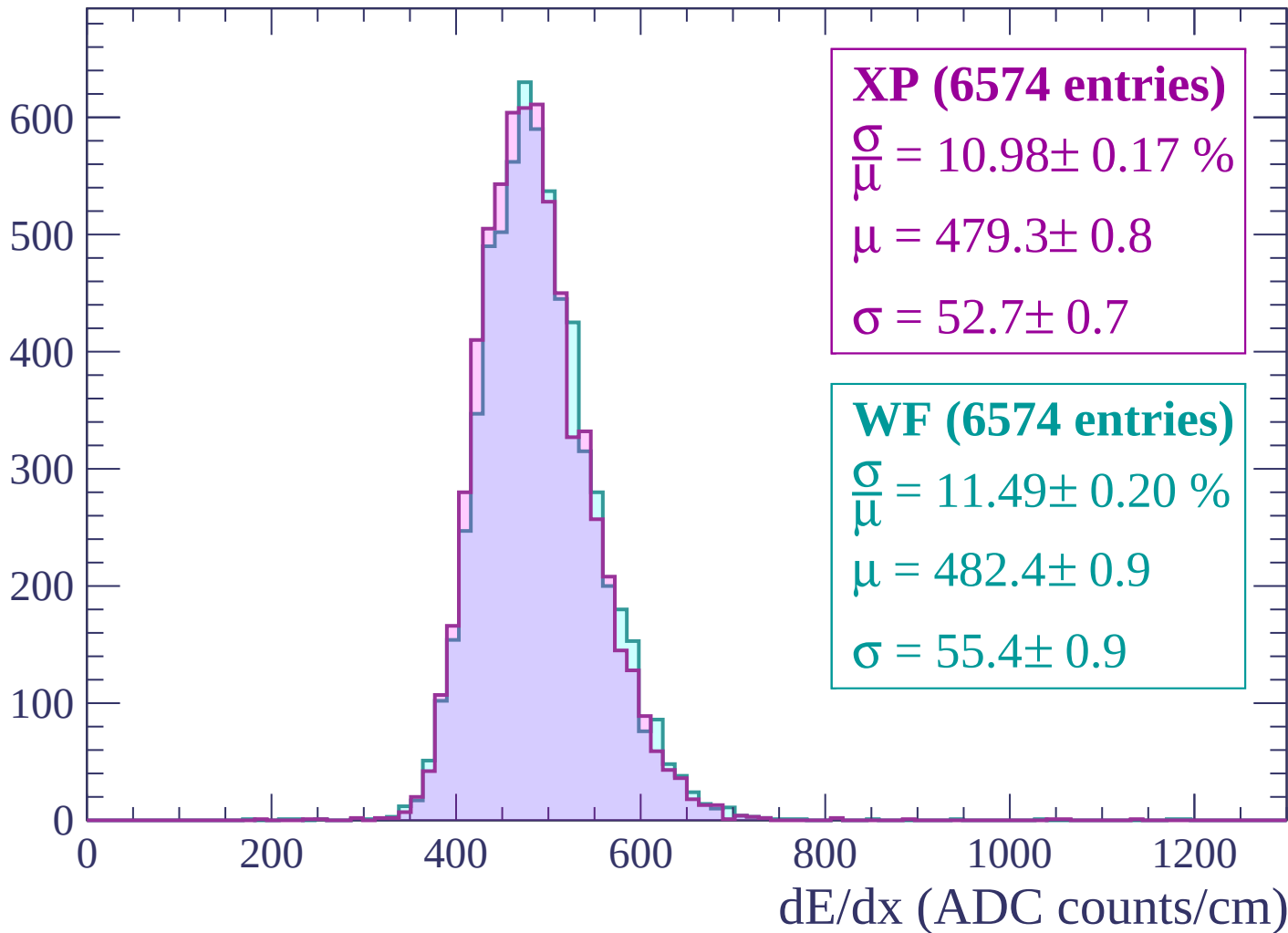
$$\mu = 485.6 \pm 0.8$$

$$\sigma = 52.7 \pm 0.8$$

dE/dx (ADC counts/cm)

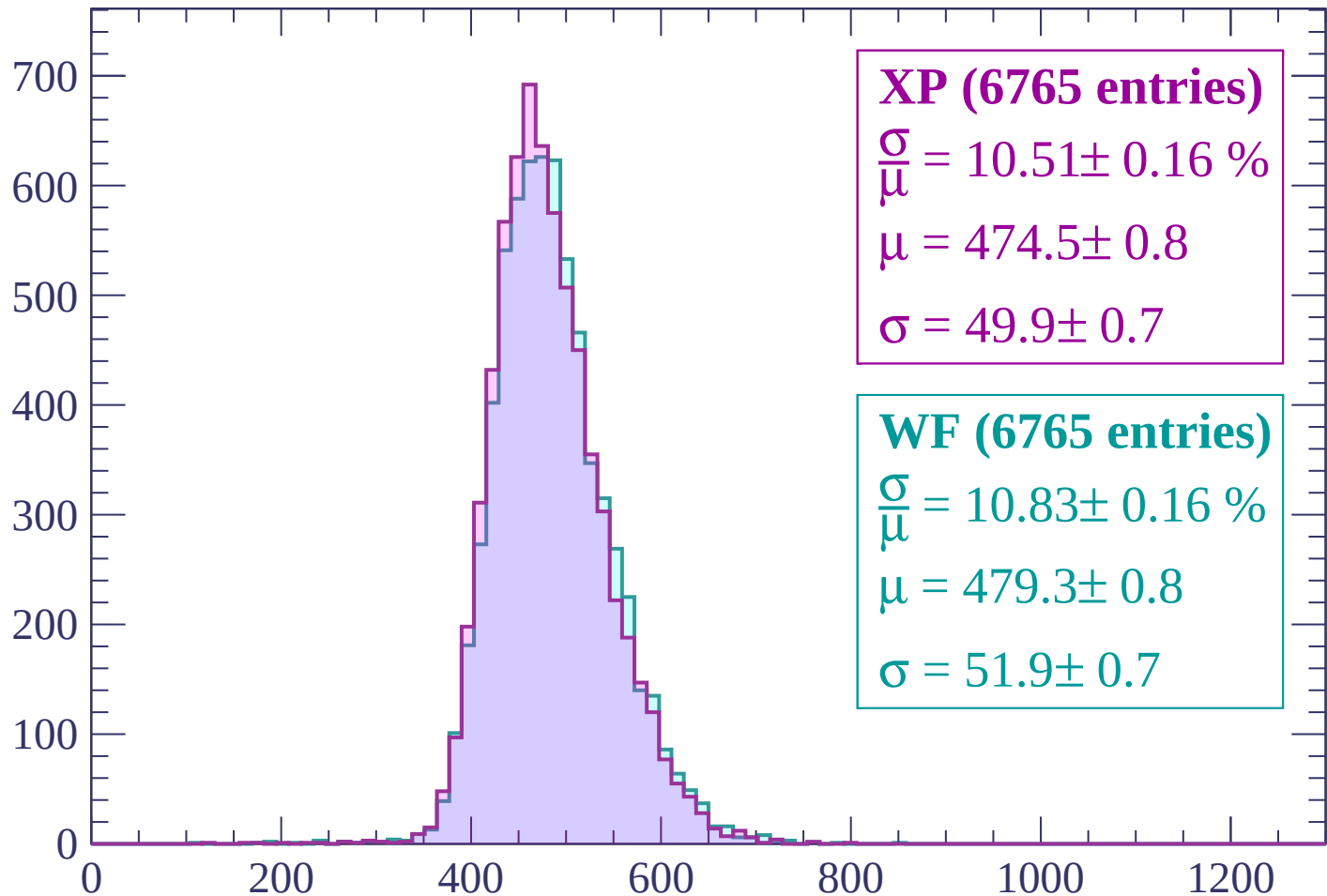
Energy loss | -1500 < p < -1440

Count



Energy loss | $-1440 < p < -1380$

Count



XP (6765 entries)

$\frac{\sigma}{\mu} = 10.51 \pm 0.16 \%$

$\mu = 474.5 \pm 0.8$

$\sigma = 49.9 \pm 0.7$

WF (6765 entries)

$\frac{\sigma}{\mu} = 10.83 \pm 0.16 \%$

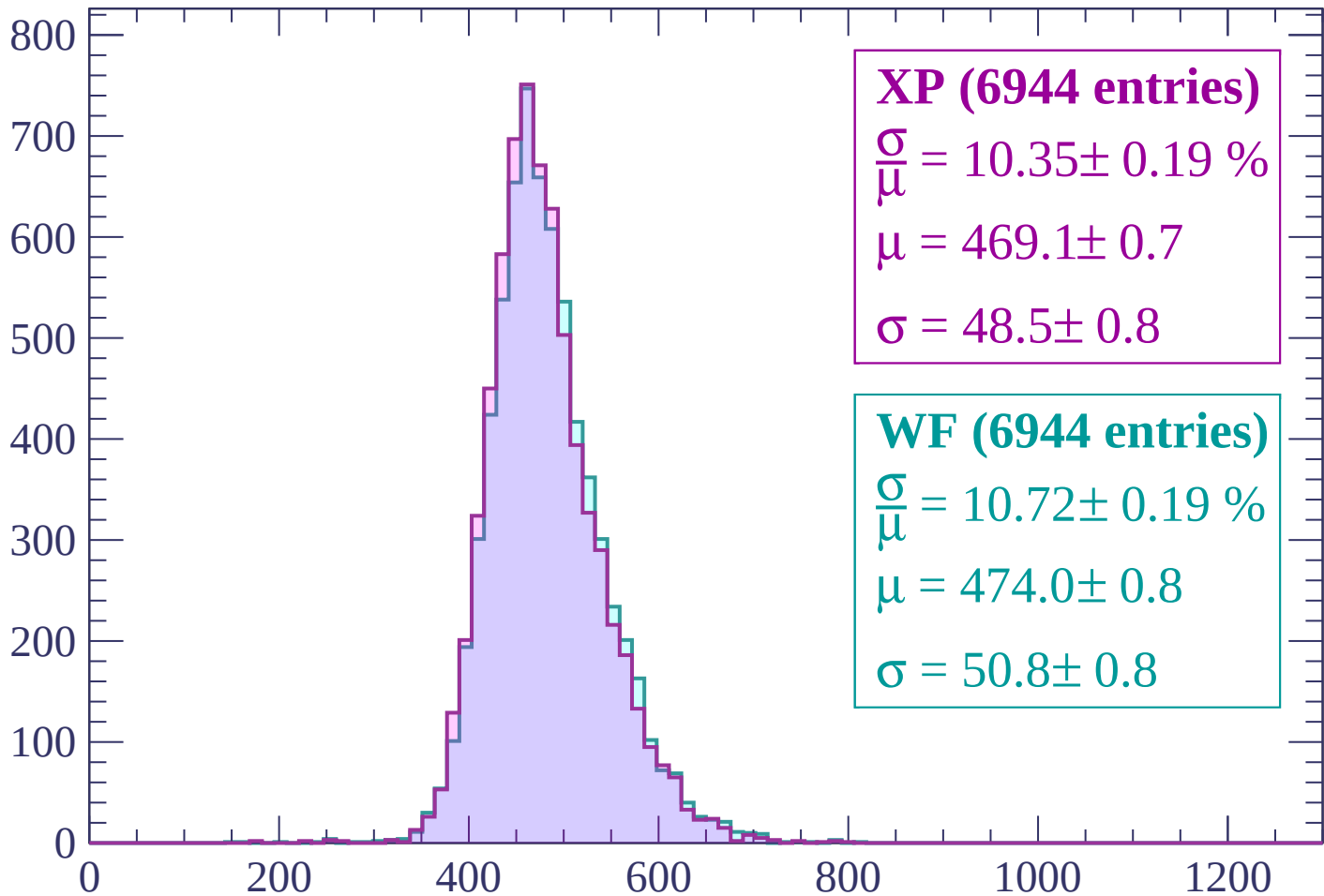
$\mu = 479.3 \pm 0.8$

$\sigma = 51.9 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $-1380 < p < -1320$

Count



XP (6944 entries)

$$\frac{\sigma}{\mu} = 10.35 \pm 0.19 \%$$

$$\mu = 469.1 \pm 0.7$$

$$\sigma = 48.5 \pm 0.8$$

WF (6944 entries)

$$\frac{\sigma}{\mu} = 10.72 \pm 0.19 \%$$

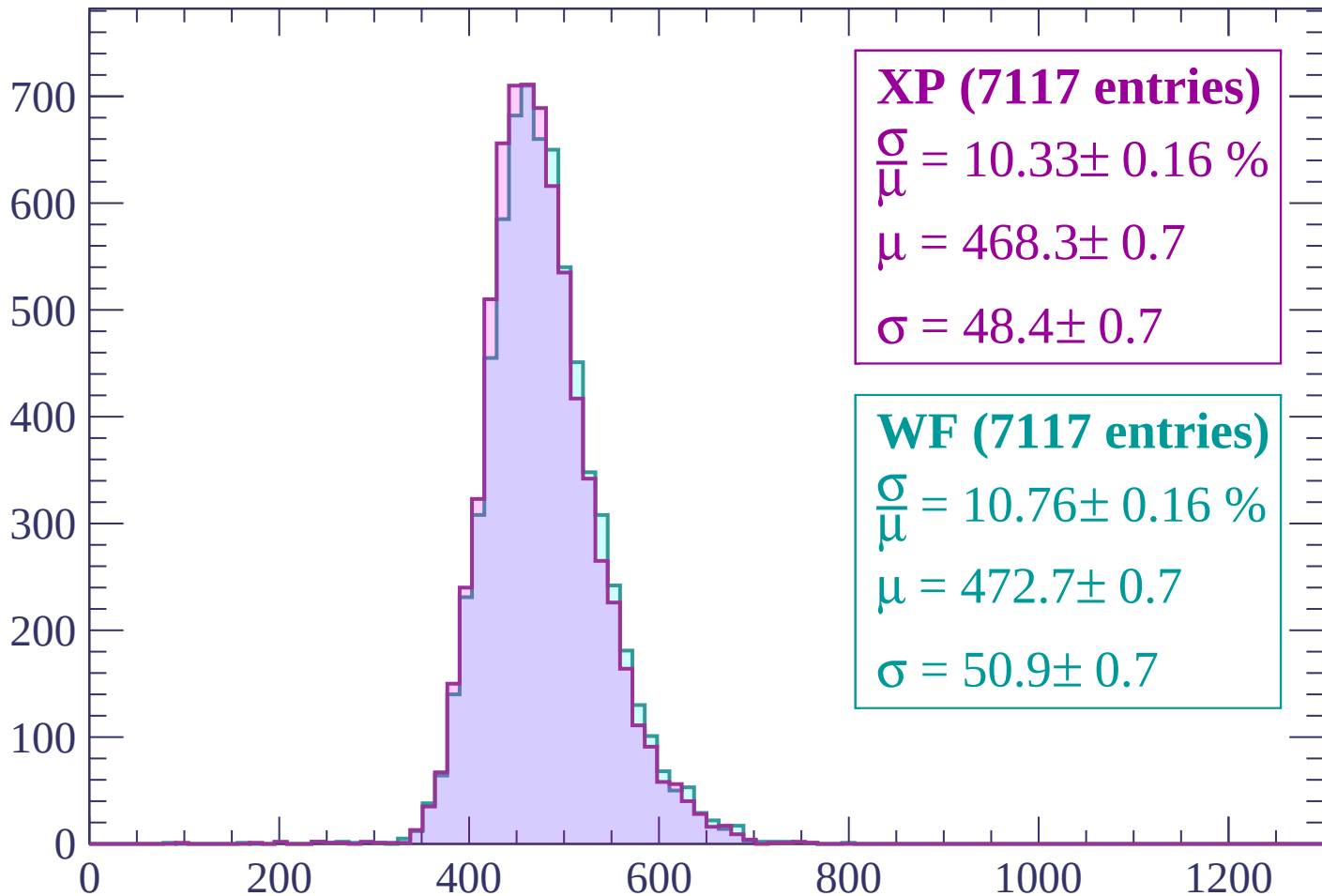
$$\mu = 474.0 \pm 0.8$$

$$\sigma = 50.8 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-1320 < p < -1260$

Count



XP (7117 entries)

$\frac{\sigma}{\mu} = 10.33 \pm 0.16 \%$

$\mu = 468.3 \pm 0.7$

$\sigma = 48.4 \pm 0.7$

WF (7117 entries)

$\frac{\sigma}{\mu} = 10.76 \pm 0.16 \%$

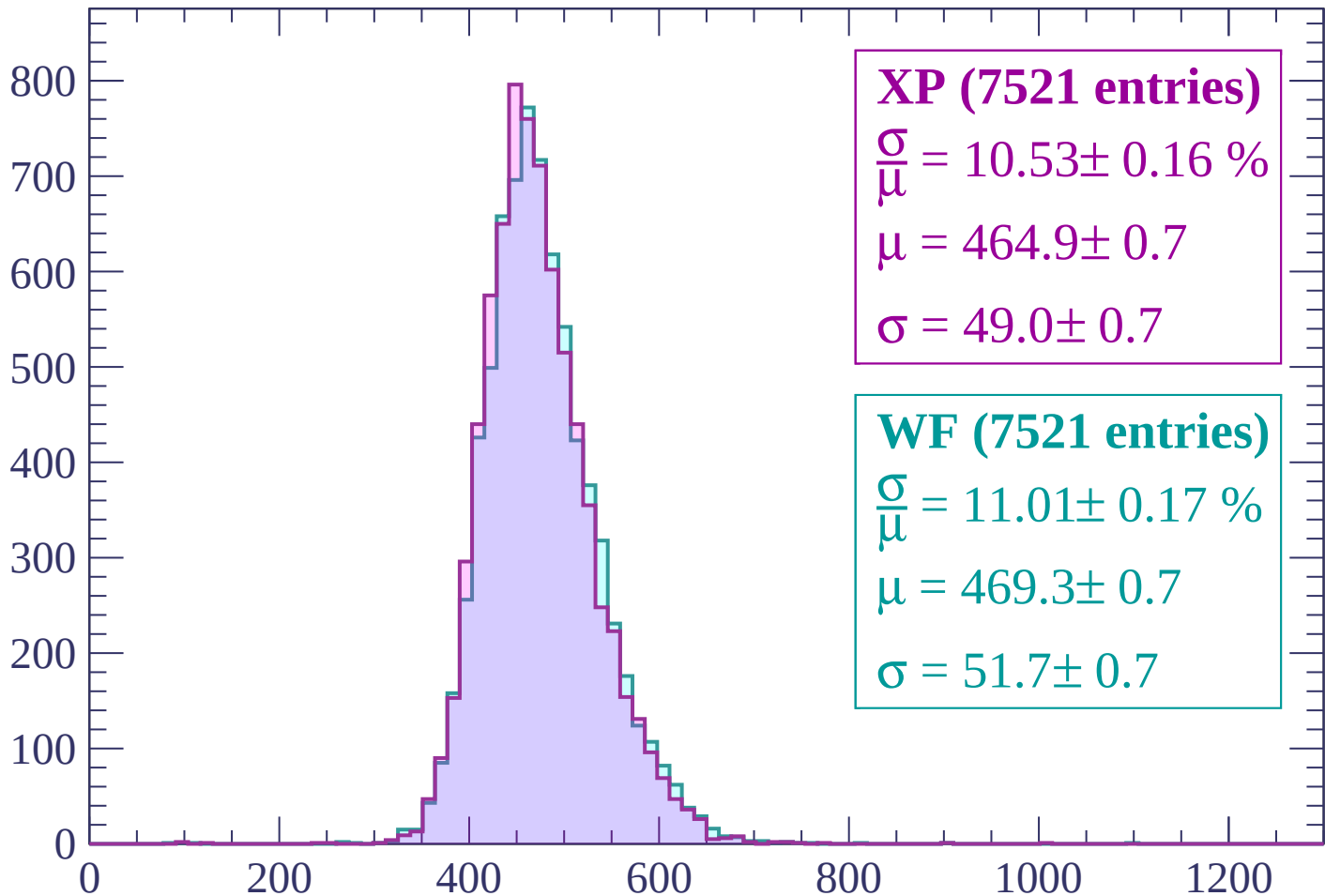
$\mu = 472.7 \pm 0.7$

$\sigma = 50.9 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | -1260 < p < -1200

Count



XP (7521 entries)

$\frac{\sigma}{\mu} = 10.53 \pm 0.16 \%$

$\mu = 464.9 \pm 0.7$

$\sigma = 49.0 \pm 0.7$

WF (7521 entries)

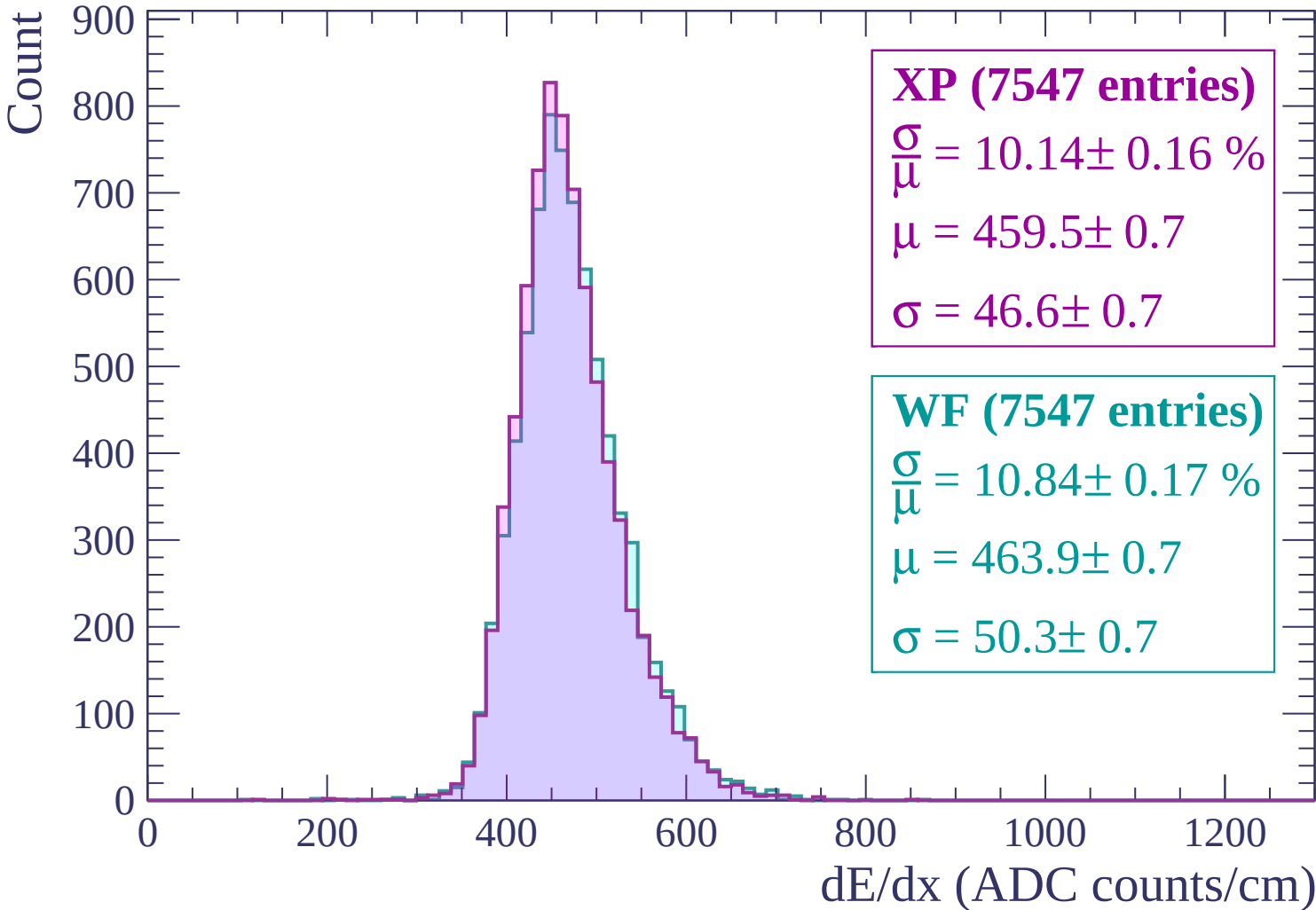
$\frac{\sigma}{\mu} = 11.01 \pm 0.17 \%$

$\mu = 469.3 \pm 0.7$

$\sigma = 51.7 \pm 0.7$

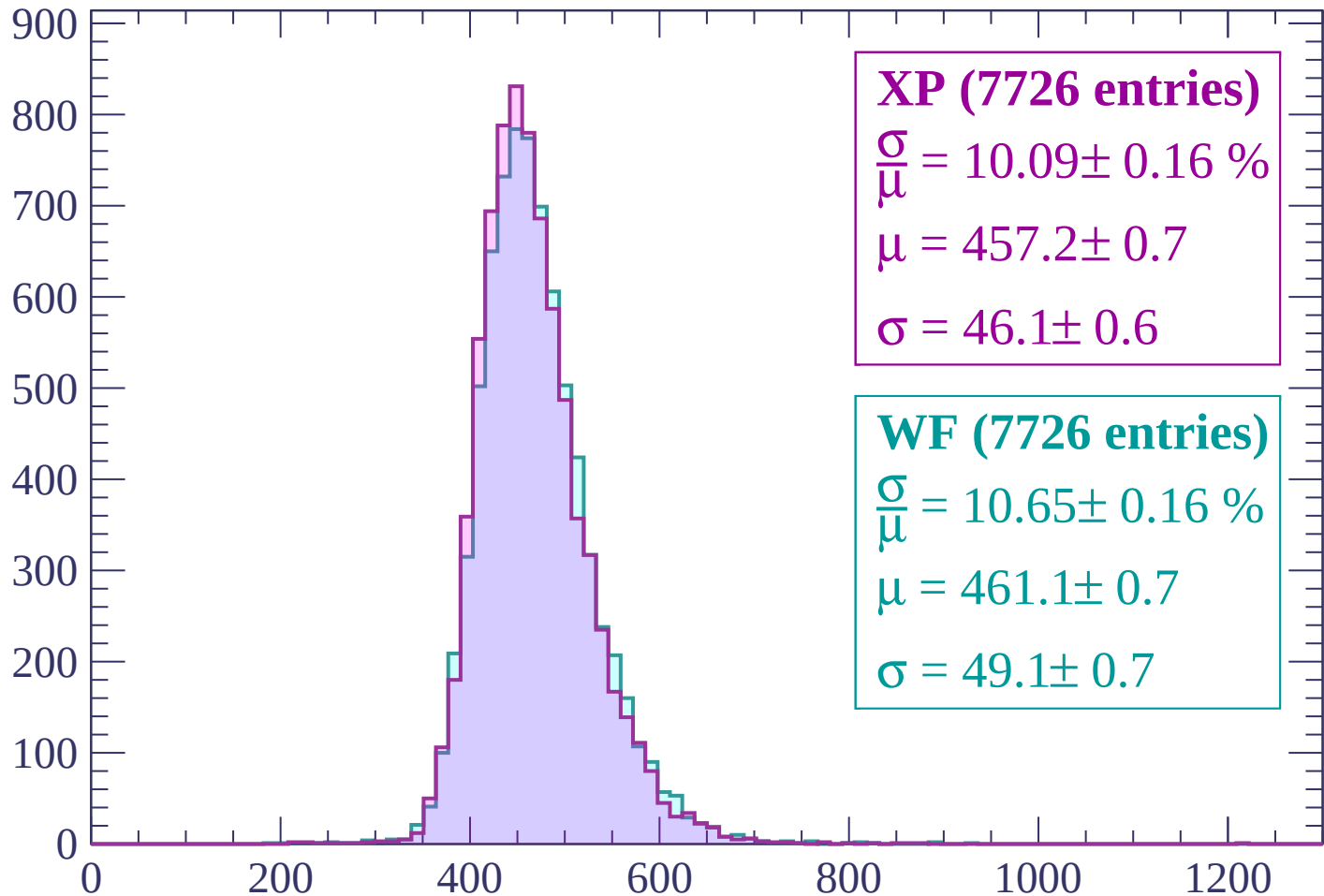
dE/dx (ADC counts/cm)

Energy loss | -1200 < p < -1140



Energy loss | -1140 < p < -1080

Count



XP (7726 entries)

$$\frac{\sigma}{\mu} = 10.09 \pm 0.16 \%$$

$$\mu = 457.2 \pm 0.7$$

$$\sigma = 46.1 \pm 0.6$$

WF (7726 entries)

$$\frac{\sigma}{\mu} = 10.65 \pm 0.16 \%$$

$$\mu = 461.1 \pm 0.7$$

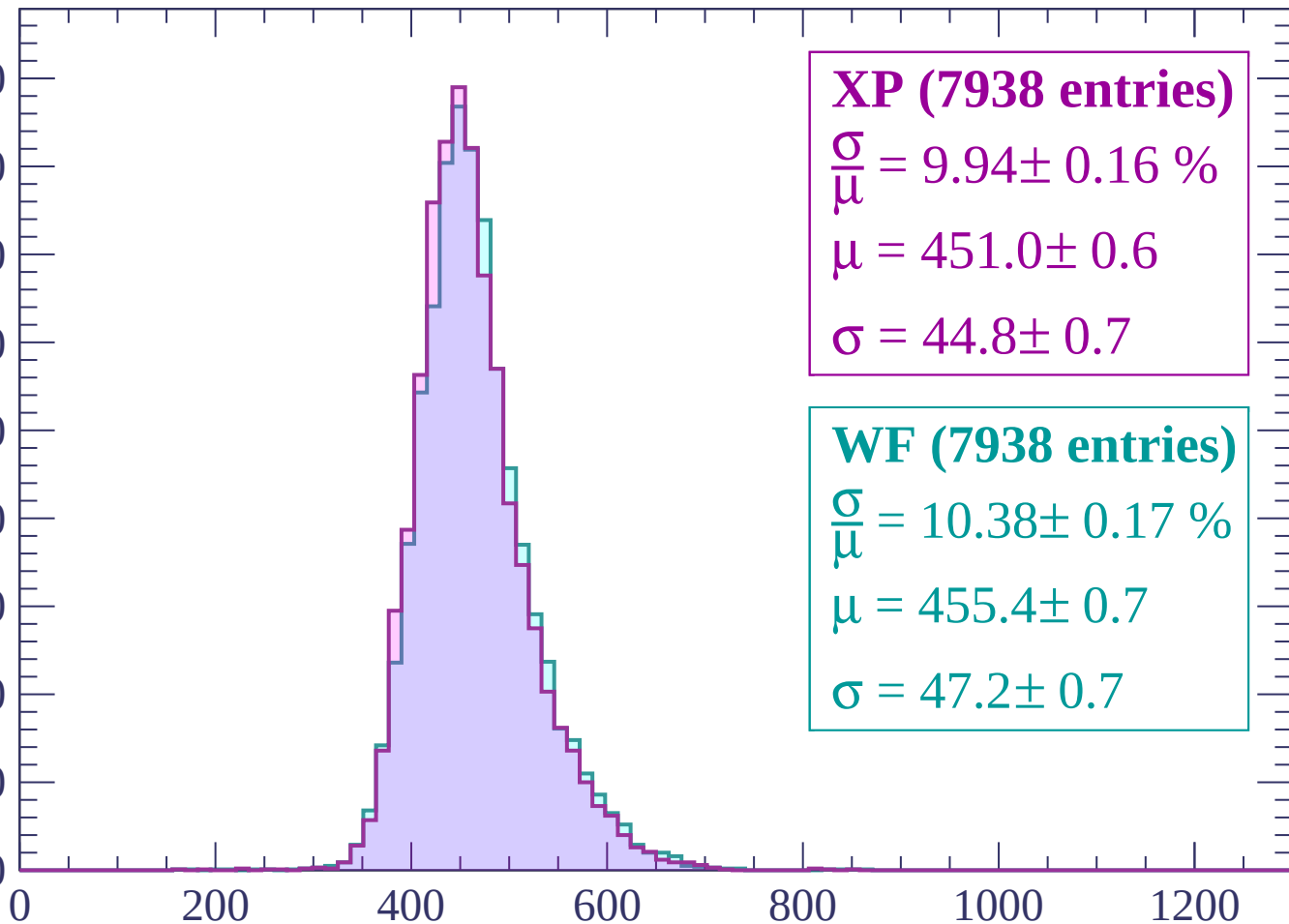
$$\sigma = 49.1 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1080 < p < -1020$

Count

900
800
700
600
500
400
300
200
100
0



XP (7938 entries)

$\frac{\sigma}{\mu} = 9.94 \pm 0.16 \%$

$\mu = 451.0 \pm 0.6$

$\sigma = 44.8 \pm 0.7$

WF (7938 entries)

$\frac{\sigma}{\mu} = 10.38 \pm 0.17 \%$

$\mu = 455.4 \pm 0.7$

$\sigma = 47.2 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $-1020 < p < -960$

Count

1000

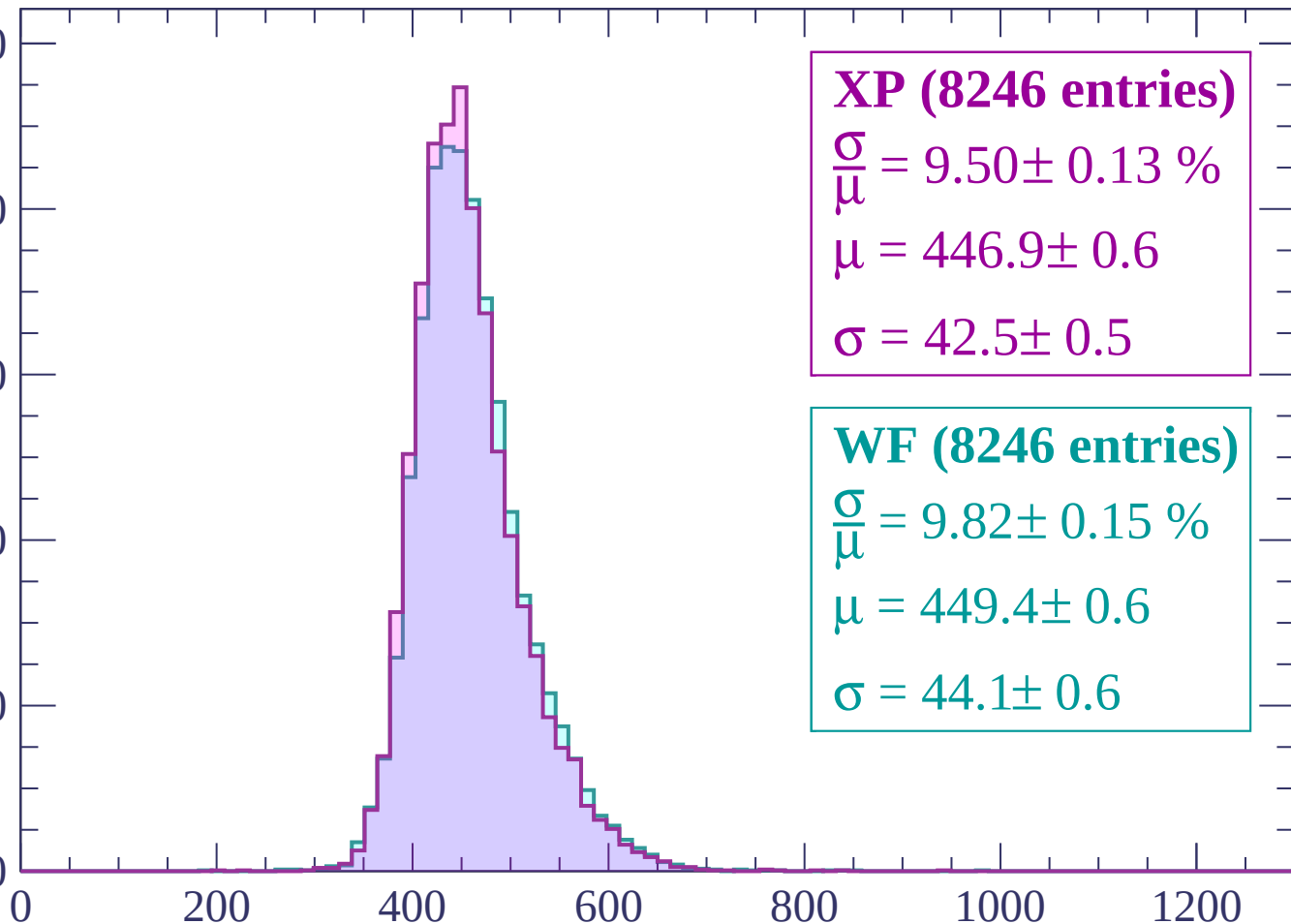
800

600

400

200

0



XP (8246 entries)

$\frac{\sigma}{\mu} = 9.50 \pm 0.13 \%$

$\mu = 446.9 \pm 0.6$

$\sigma = 42.5 \pm 0.5$

WF (8246 entries)

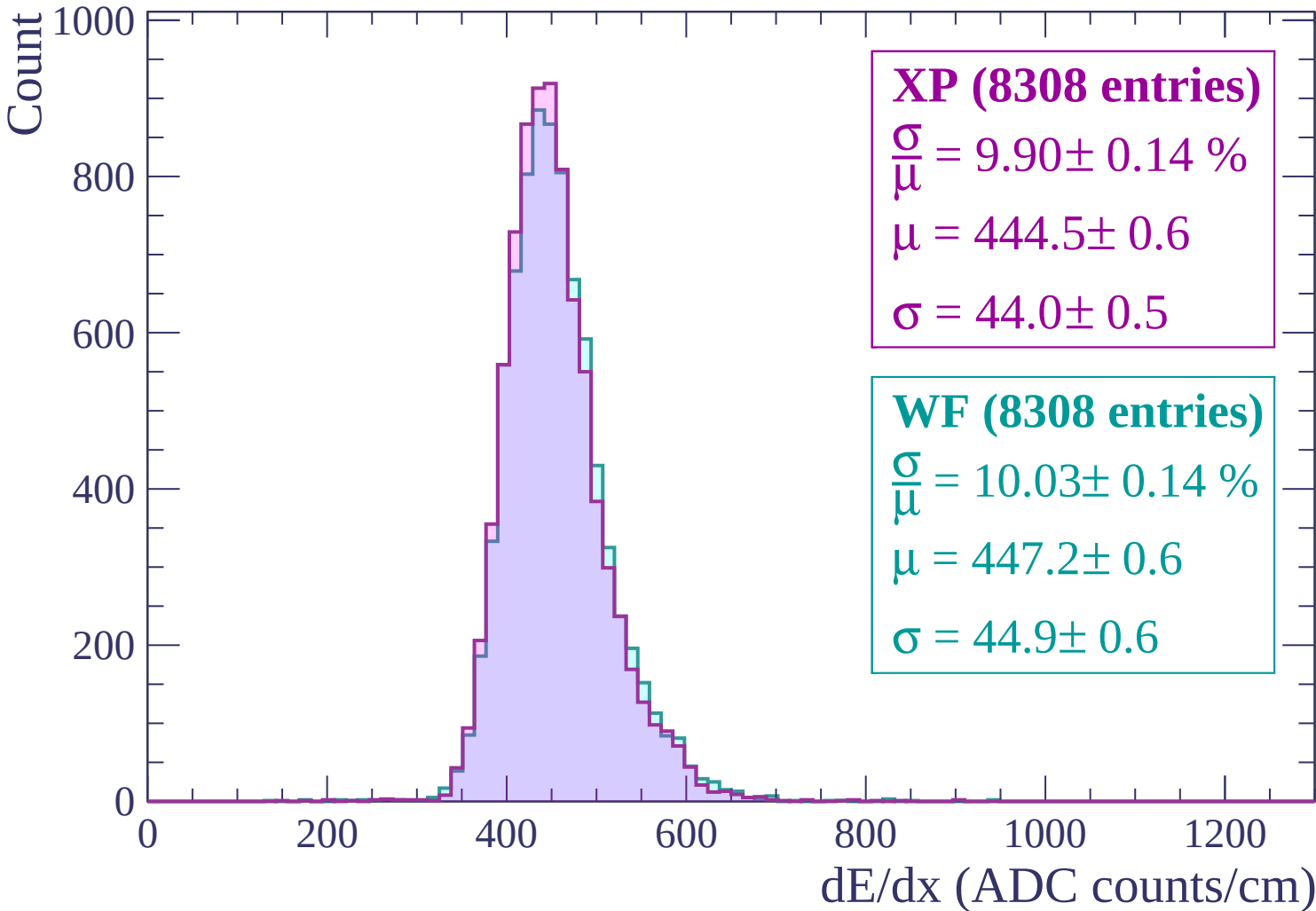
$\frac{\sigma}{\mu} = 9.82 \pm 0.15 \%$

$\mu = 449.4 \pm 0.6$

$\sigma = 44.1 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $-960 < p < -900$



Energy loss | $-900 < p < -840$

Count

1000

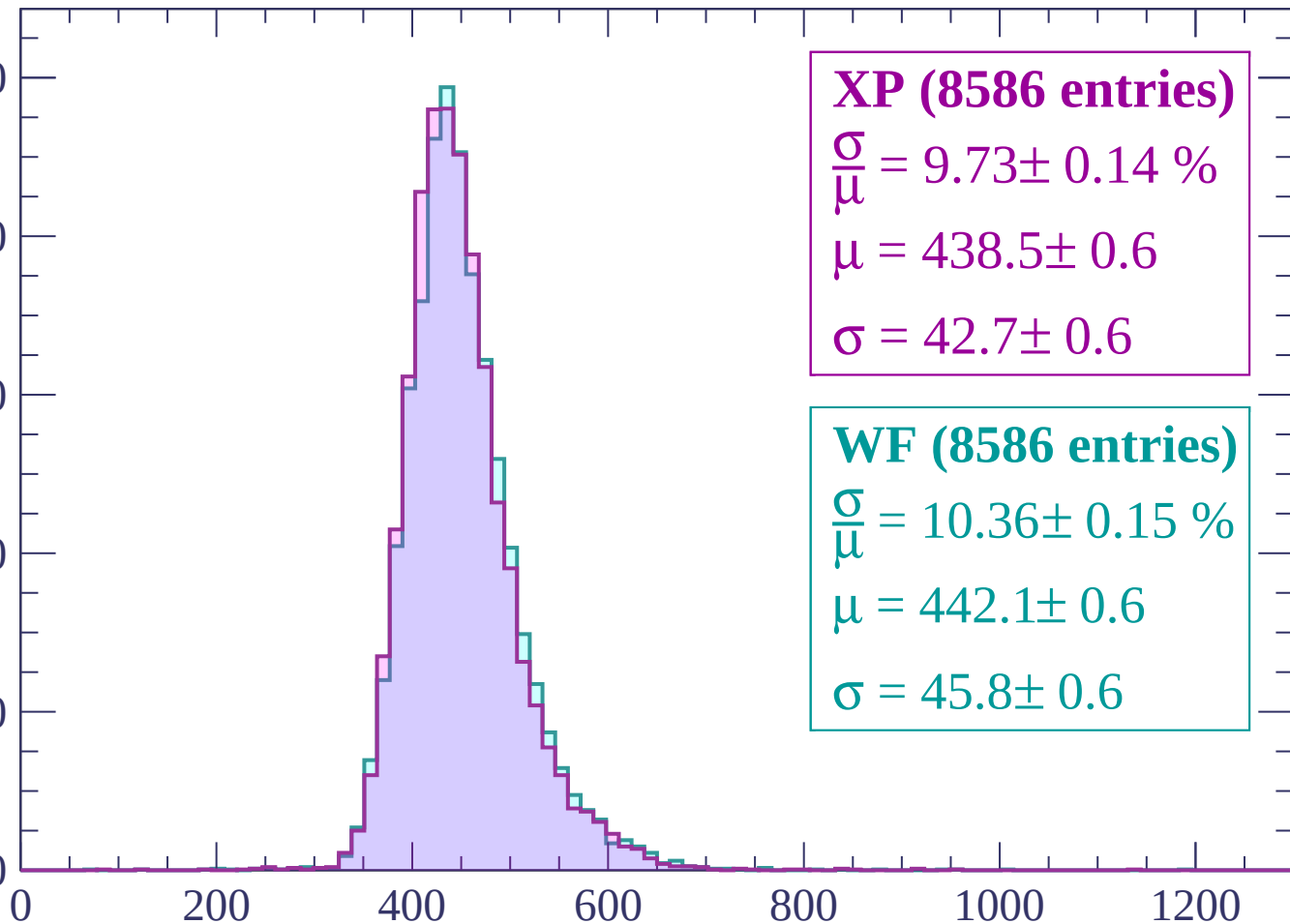
800

600

400

200

0



XP (8586 entries)

$\frac{\sigma}{\mu} = 9.73 \pm 0.14 \%$

$\mu = 438.5 \pm 0.6$

$\sigma = 42.7 \pm 0.6$

WF (8586 entries)

$\frac{\sigma}{\mu} = 10.36 \pm 0.15 \%$

$\mu = 442.1 \pm 0.6$

$\sigma = 45.8 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -780$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (8730 entries)

$\frac{\sigma}{\mu} = 9.43 \pm 0.14 \%$

$\mu = 432.9 \pm 0.6$

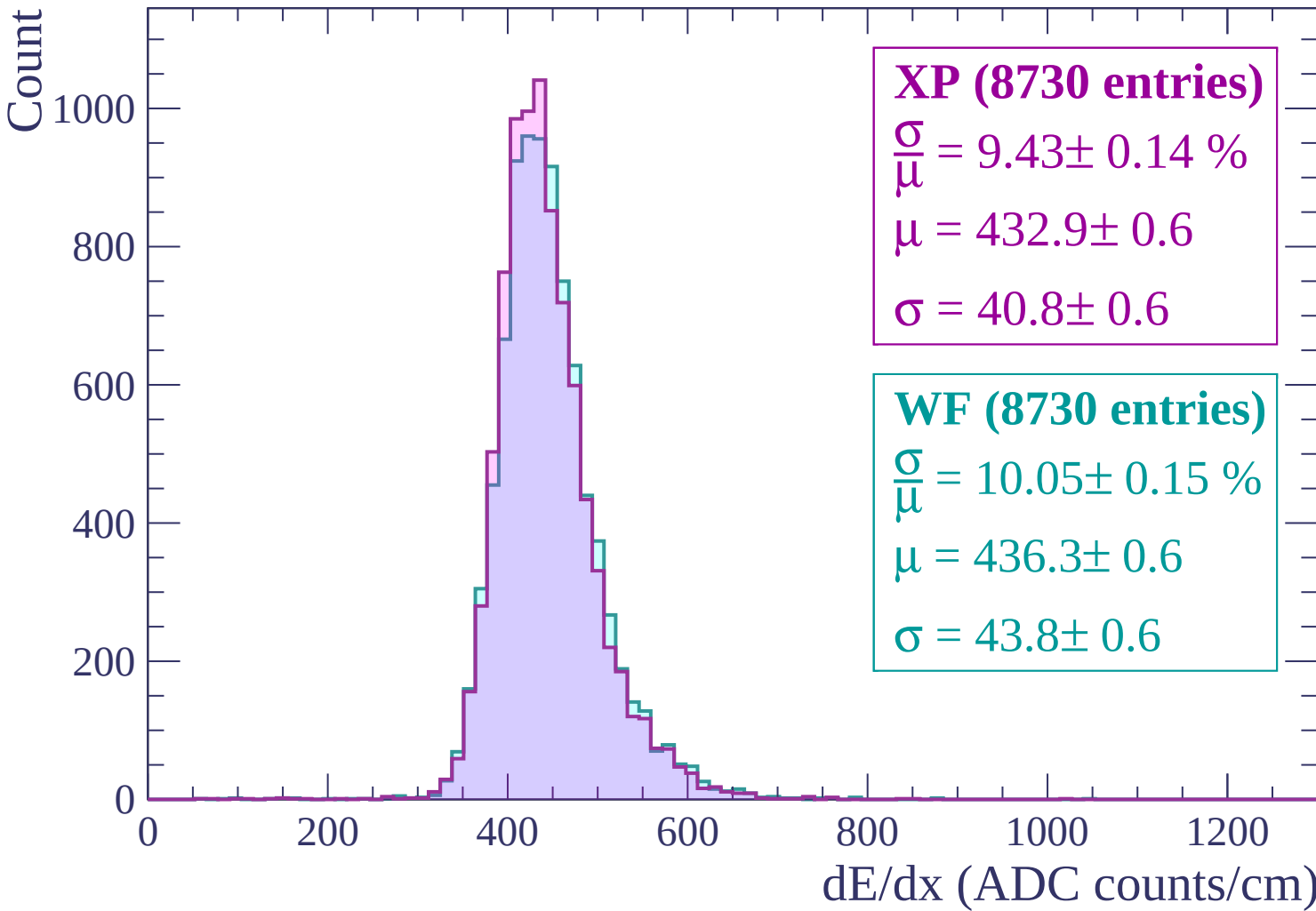
$\sigma = 40.8 \pm 0.6$

WF (8730 entries)

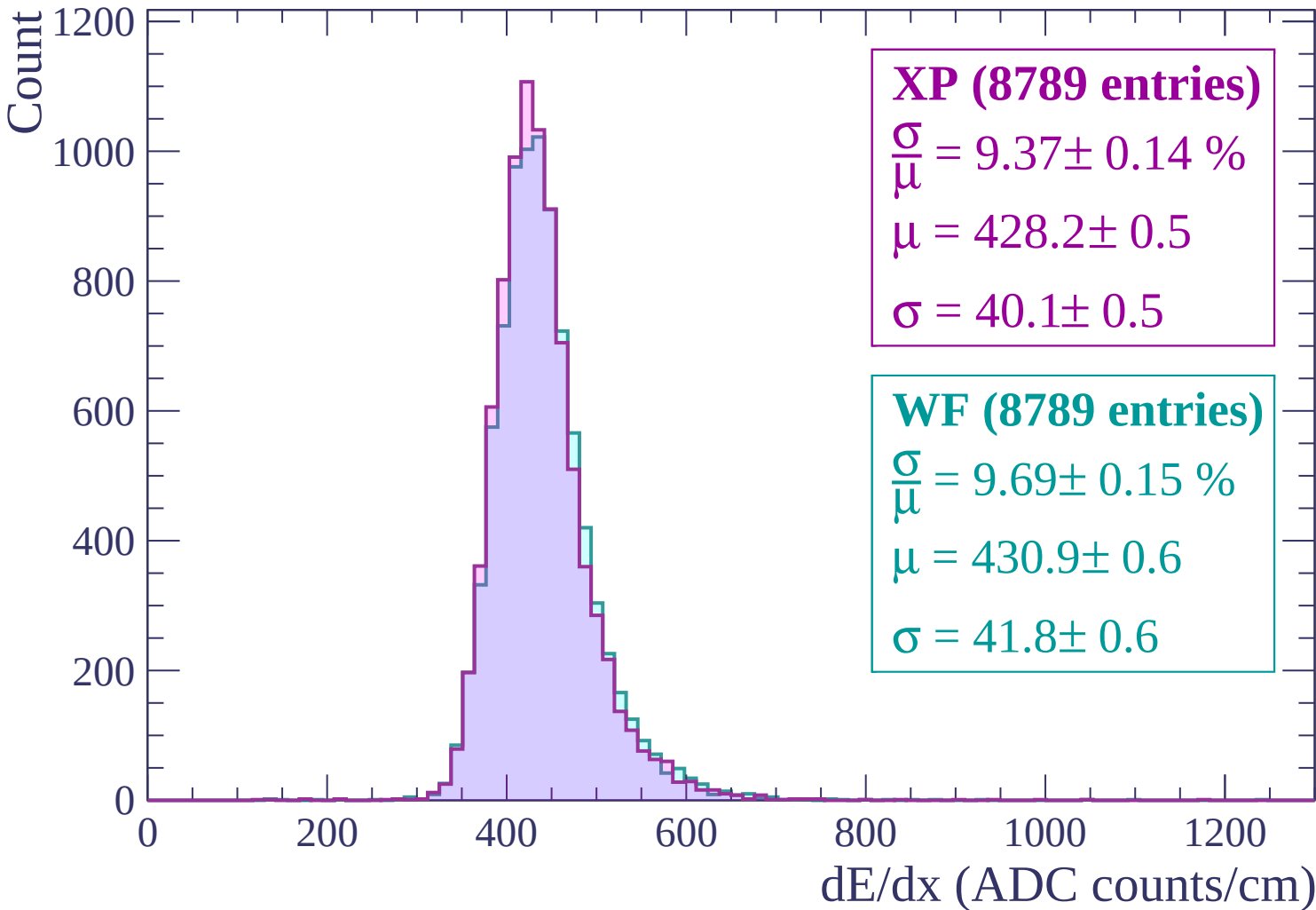
$\frac{\sigma}{\mu} = 10.05 \pm 0.15 \%$

$\mu = 436.3 \pm 0.6$

$\sigma = 43.8 \pm 0.6$

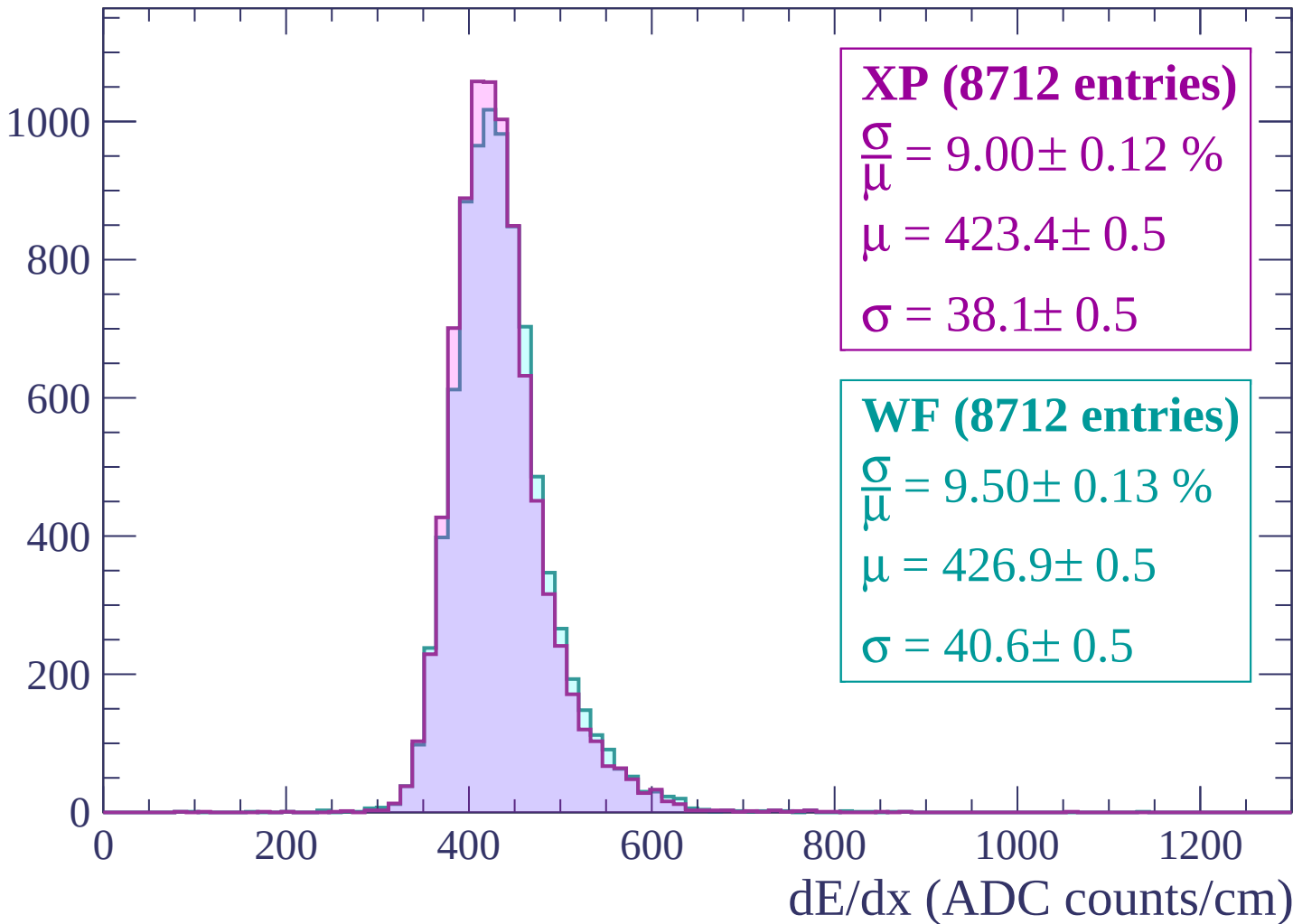


Energy loss | $-780 < p < -720$

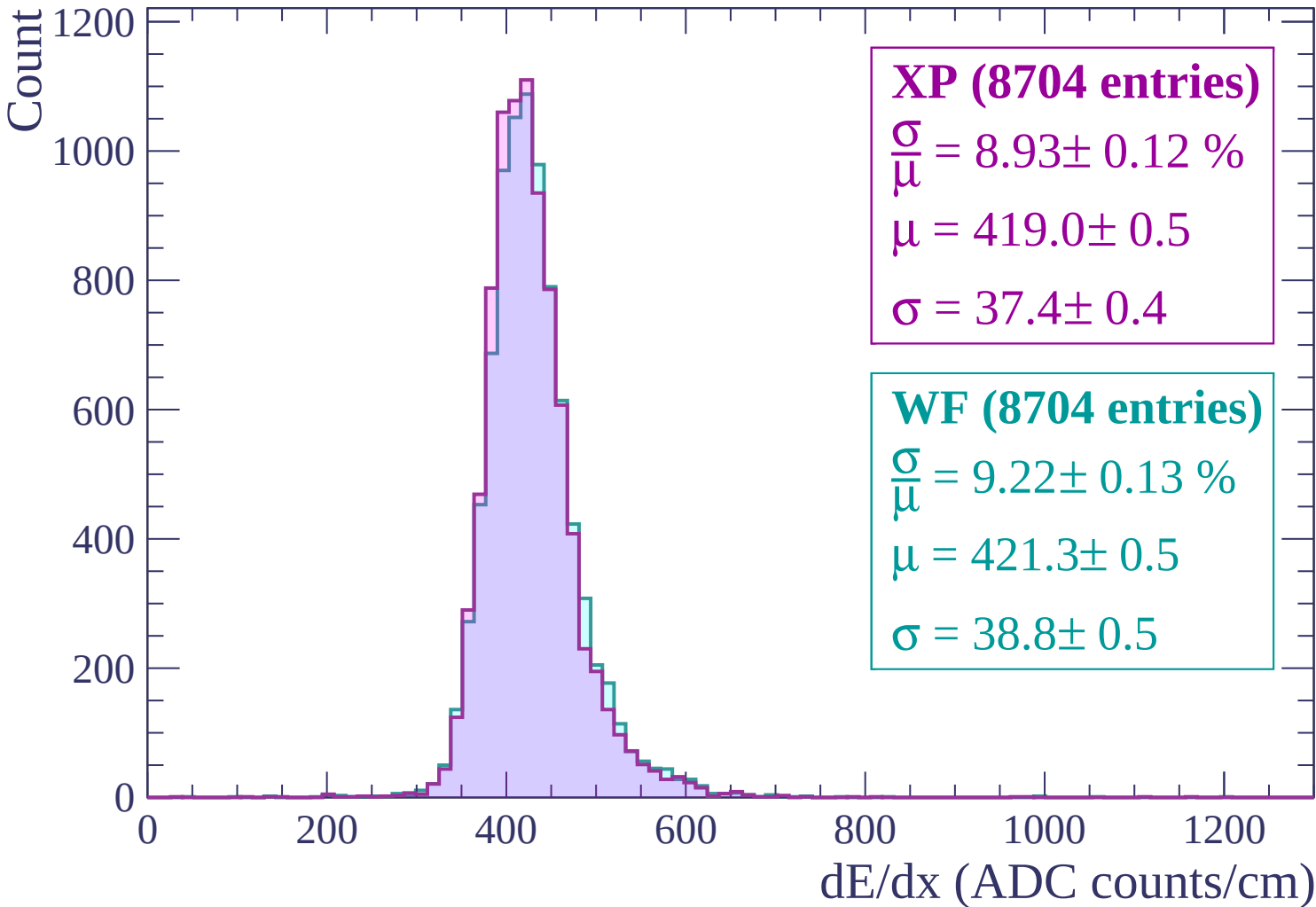


Energy loss | $-720 < p < -660$

Count



Energy loss | $-660 < p < -600$



Energy loss | $-600 < p < -540$

Count

1200
1000
800
600
400
200
0

XP (8642 entries)

$\frac{\sigma}{\mu} = 8.67 \pm 0.12 \%$

$\mu = 414.2 \pm 0.5$

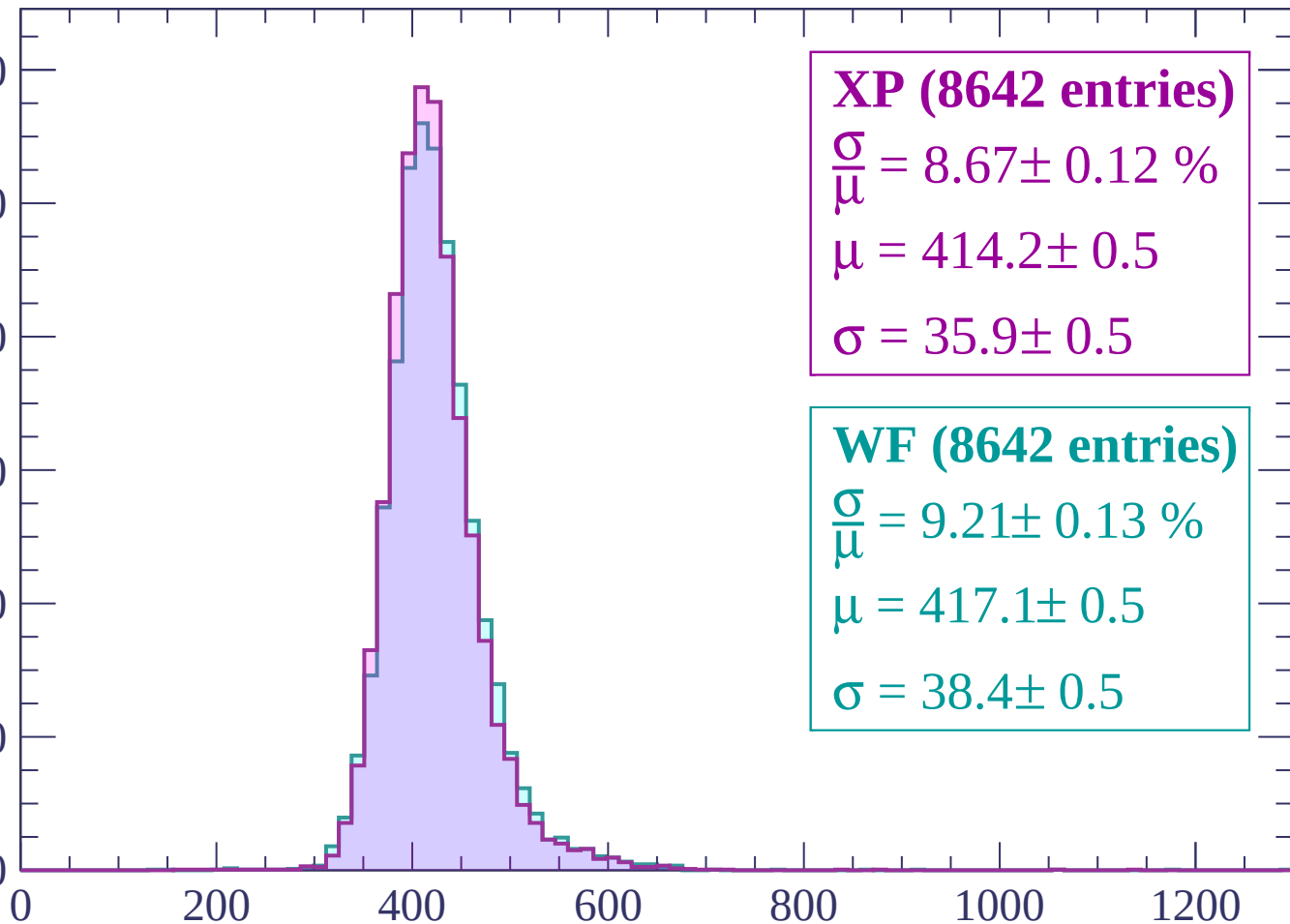
$\sigma = 35.9 \pm 0.5$

WF (8642 entries)

$\frac{\sigma}{\mu} = 9.21 \pm 0.13 \%$

$\mu = 417.1 \pm 0.5$

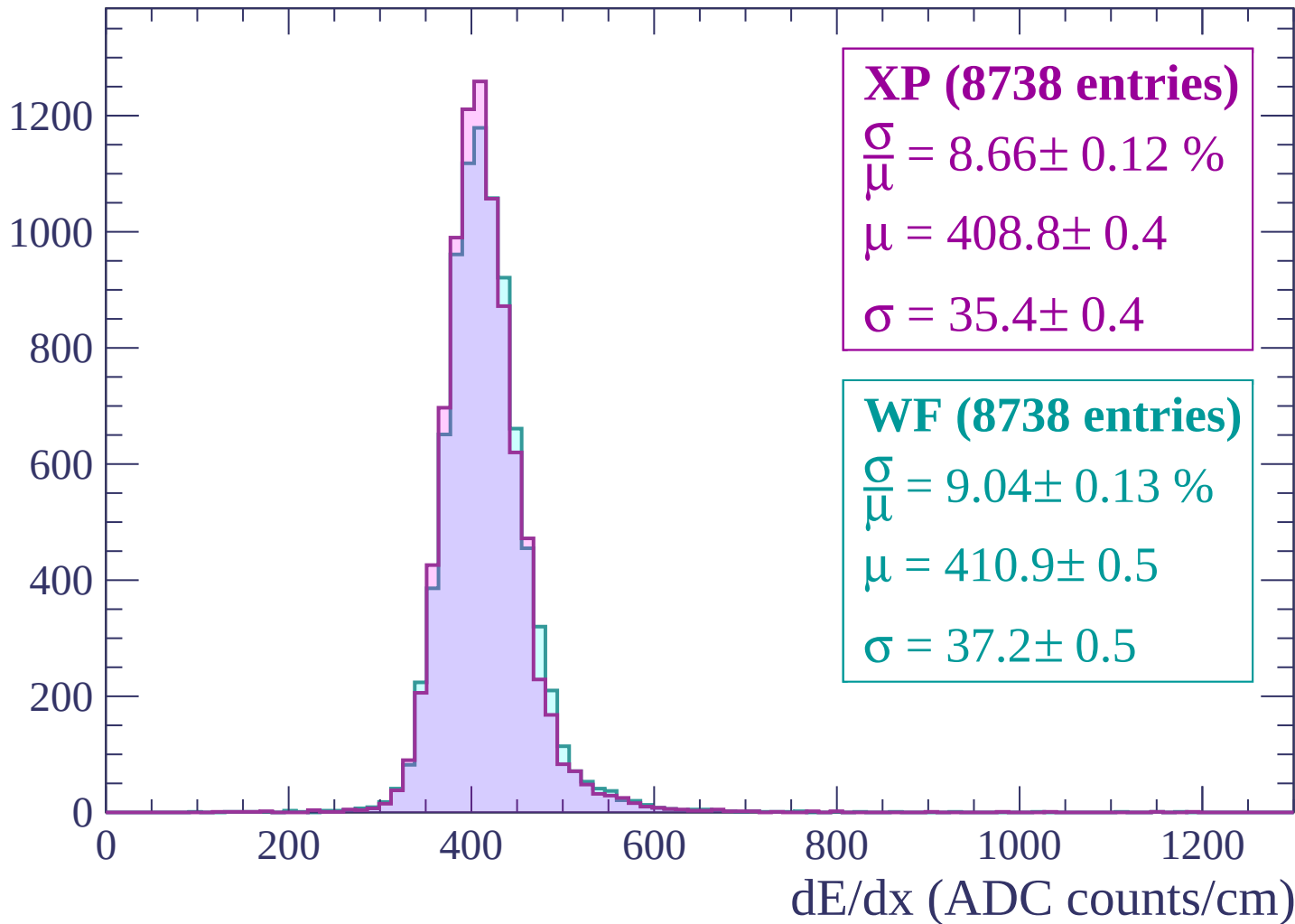
$\sigma = 38.4 \pm 0.5$



dE/dx (ADC counts/cm)

Energy loss | $-540 < p < -480$

Count



Energy loss | $-480 < p < -420$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (8500 entries)

$$\frac{\sigma}{\mu} = 8.31 \pm 0.12 \%$$

$$\mu = 404.8 \pm 0.4$$

$$\sigma = 33.6 \pm 0.4$$

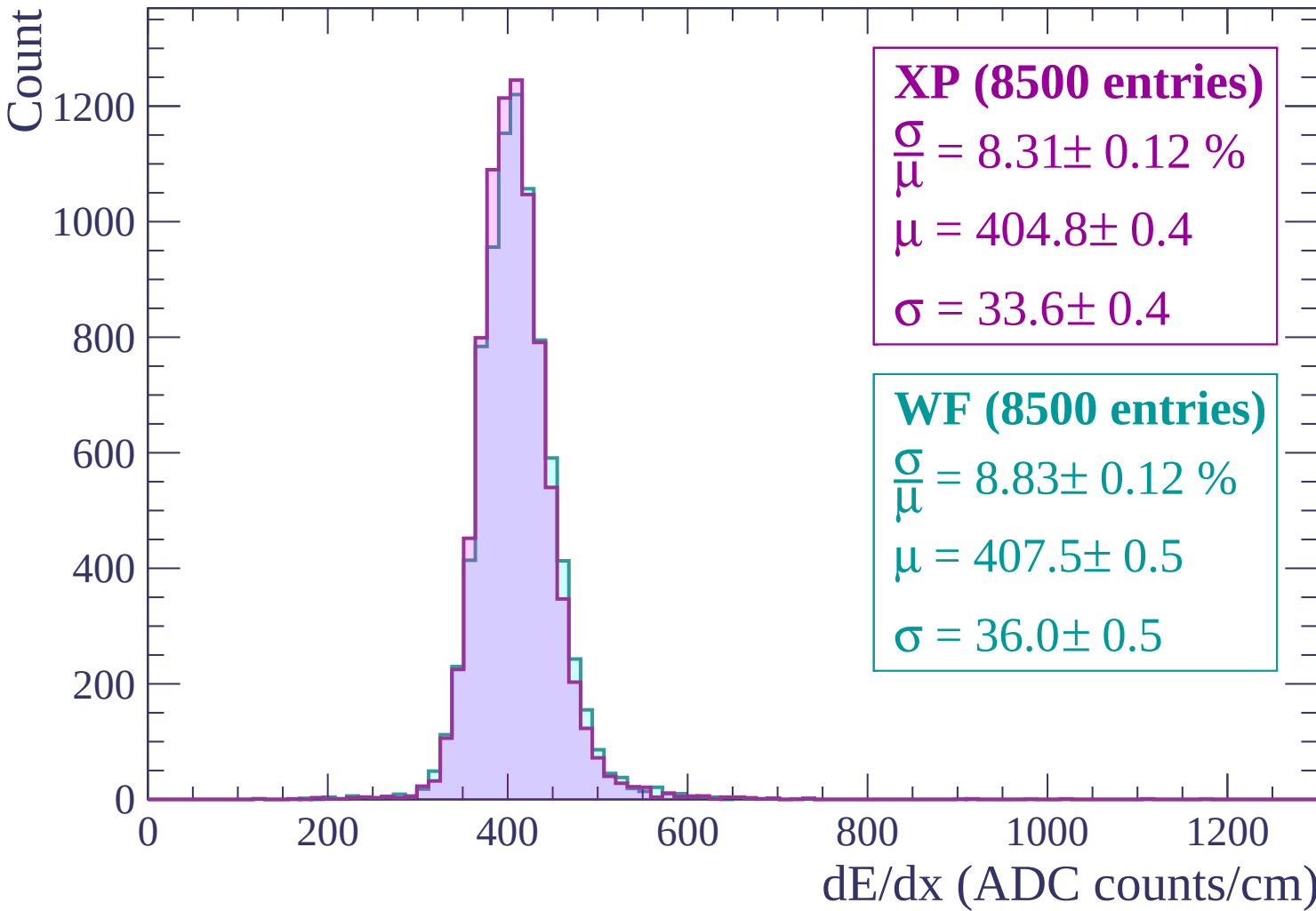
WF (8500 entries)

$$\frac{\sigma}{\mu} = 8.83 \pm 0.12 \%$$

$$\mu = 407.5 \pm 0.5$$

$$\sigma = 36.0 \pm 0.5$$

dE/dx (ADC counts/cm)



Energy loss | $-420 < p < -360$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (8409 entries)

$$\frac{\sigma}{\mu} = 8.54 \pm 0.13 \%$$

$$\mu = 401.1 \pm 0.5$$

$$\sigma = 34.2 \pm 0.5$$

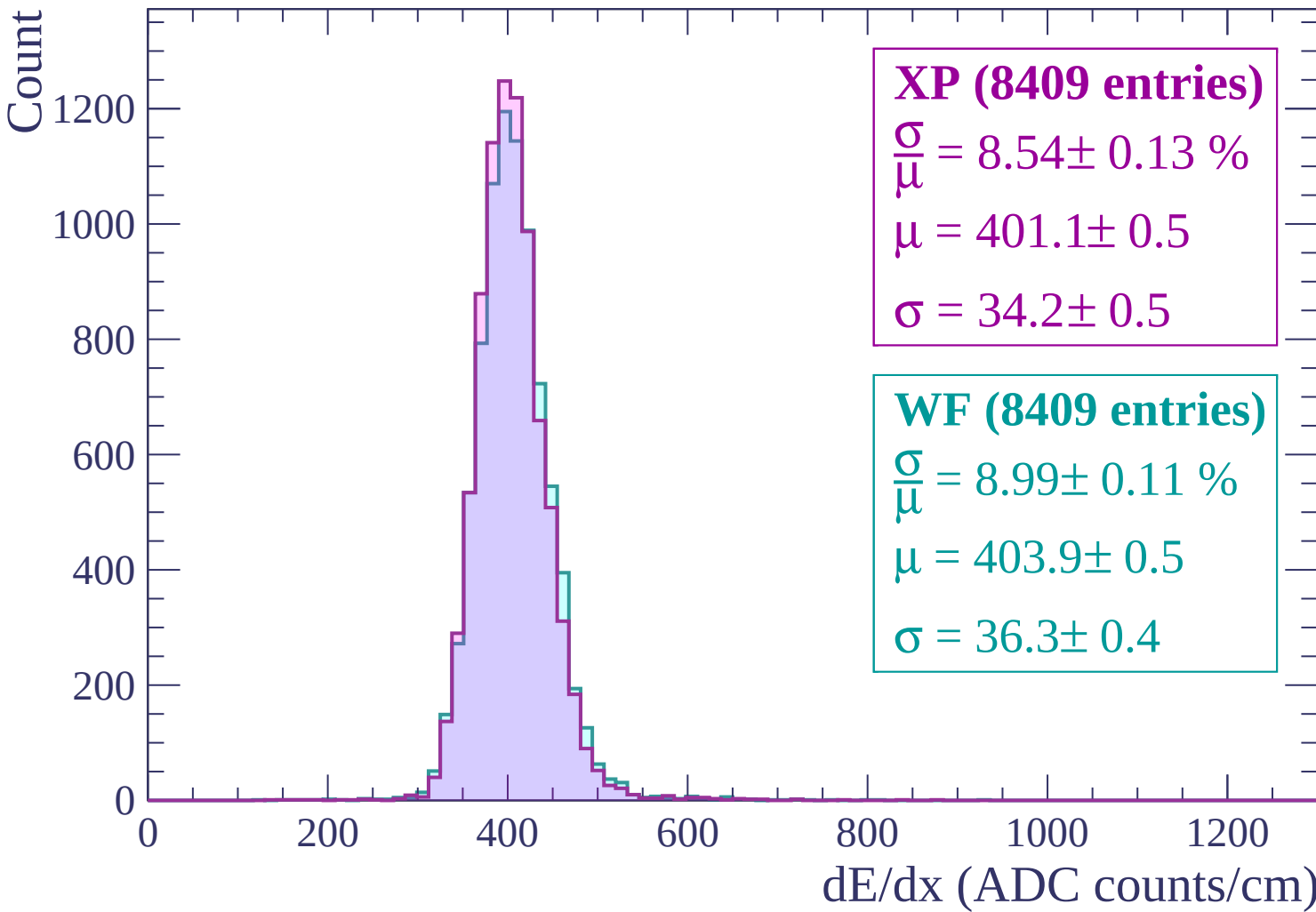
WF (8409 entries)

$$\frac{\sigma}{\mu} = 8.99 \pm 0.11 \%$$

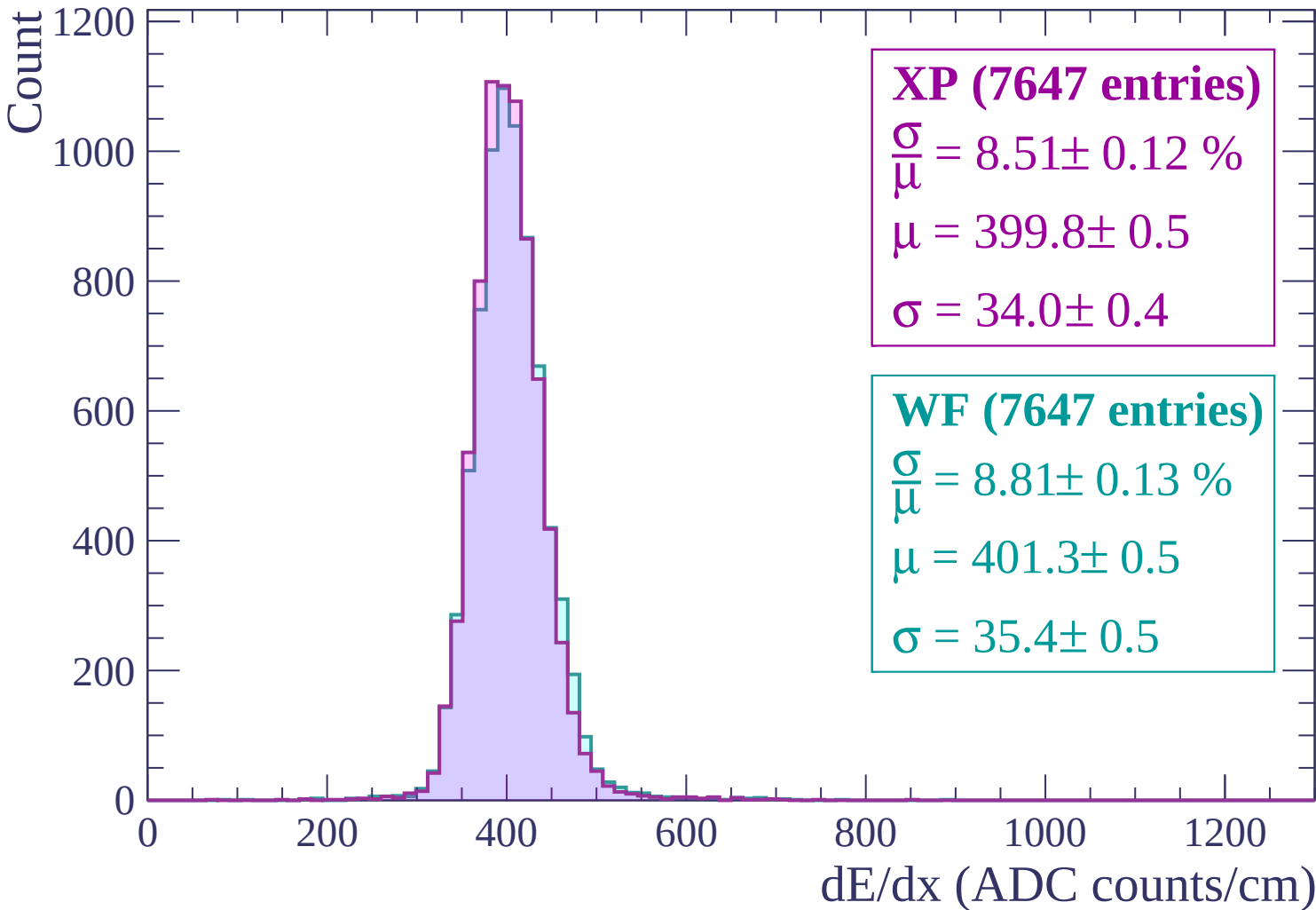
$$\mu = 403.9 \pm 0.5$$

$$\sigma = 36.3 \pm 0.4$$

dE/dx (ADC counts/cm)



Energy loss | $-360 < p < -300$



Energy loss | $-300 < p < -240$

Count

1000

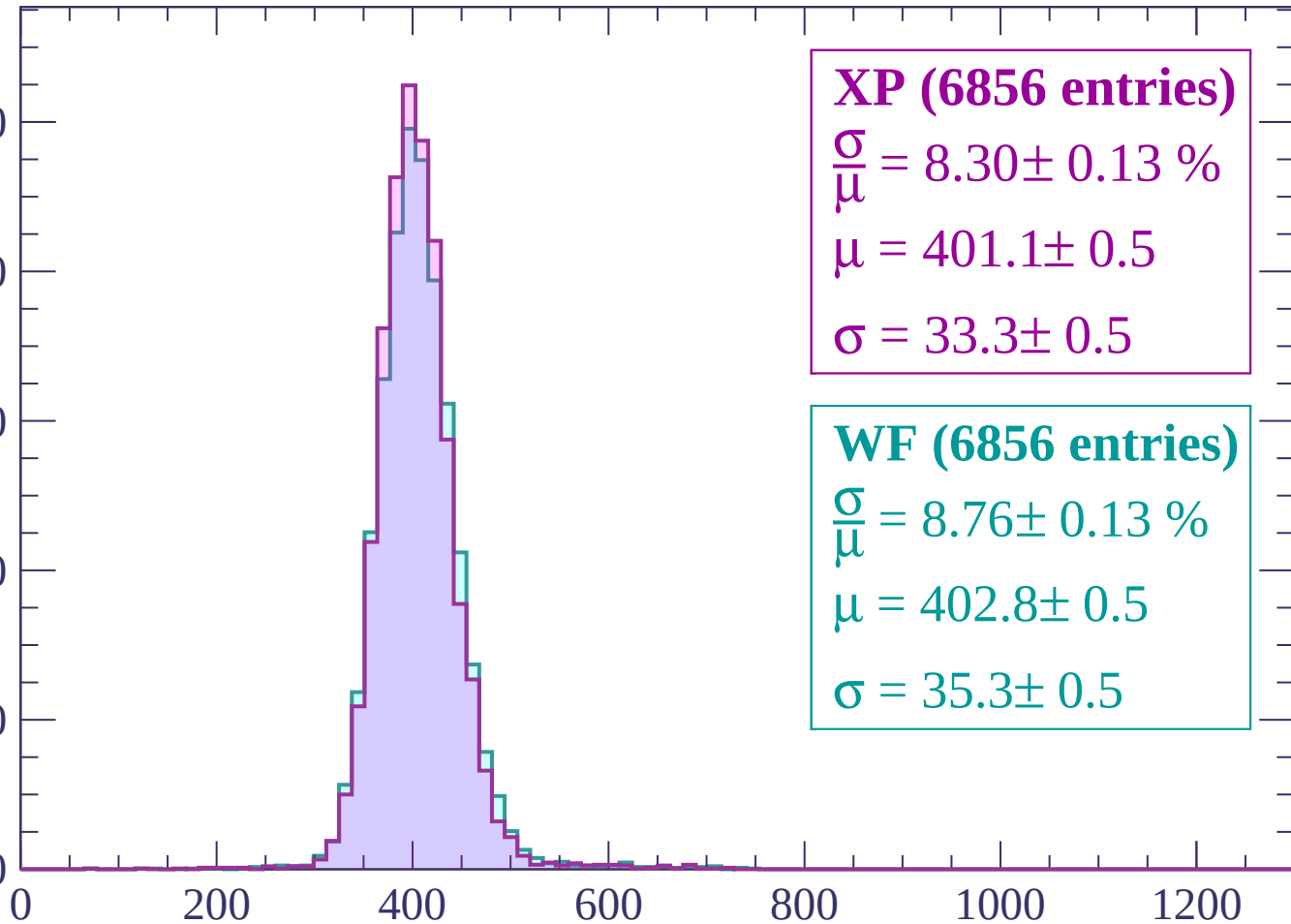
800

600

400

200

0



XP (6856 entries)

$\frac{\sigma}{\mu} = 8.30 \pm 0.13 \%$

$\mu = 401.1 \pm 0.5$

$\sigma = 33.3 \pm 0.5$

WF (6856 entries)

$\frac{\sigma}{\mu} = 8.76 \pm 0.13 \%$

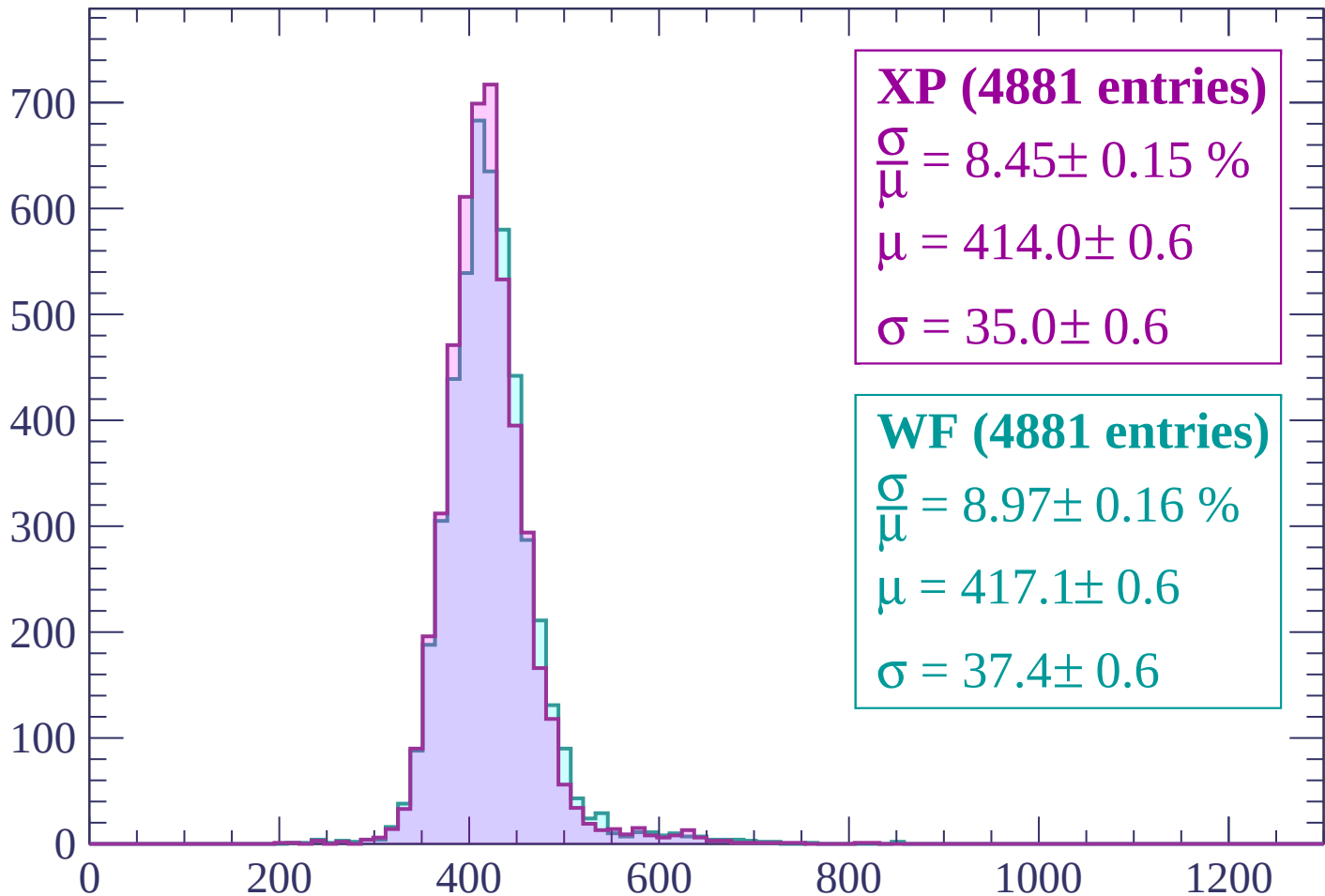
$\mu = 402.8 \pm 0.5$

$\sigma = 35.3 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -180$

Count



XP (4881 entries)

$\frac{\sigma}{\mu} = 8.45 \pm 0.15 \%$

$\mu = 414.0 \pm 0.6$

$\sigma = 35.0 \pm 0.6$

WF (4881 entries)

$\frac{\sigma}{\mu} = 8.97 \pm 0.16 \%$

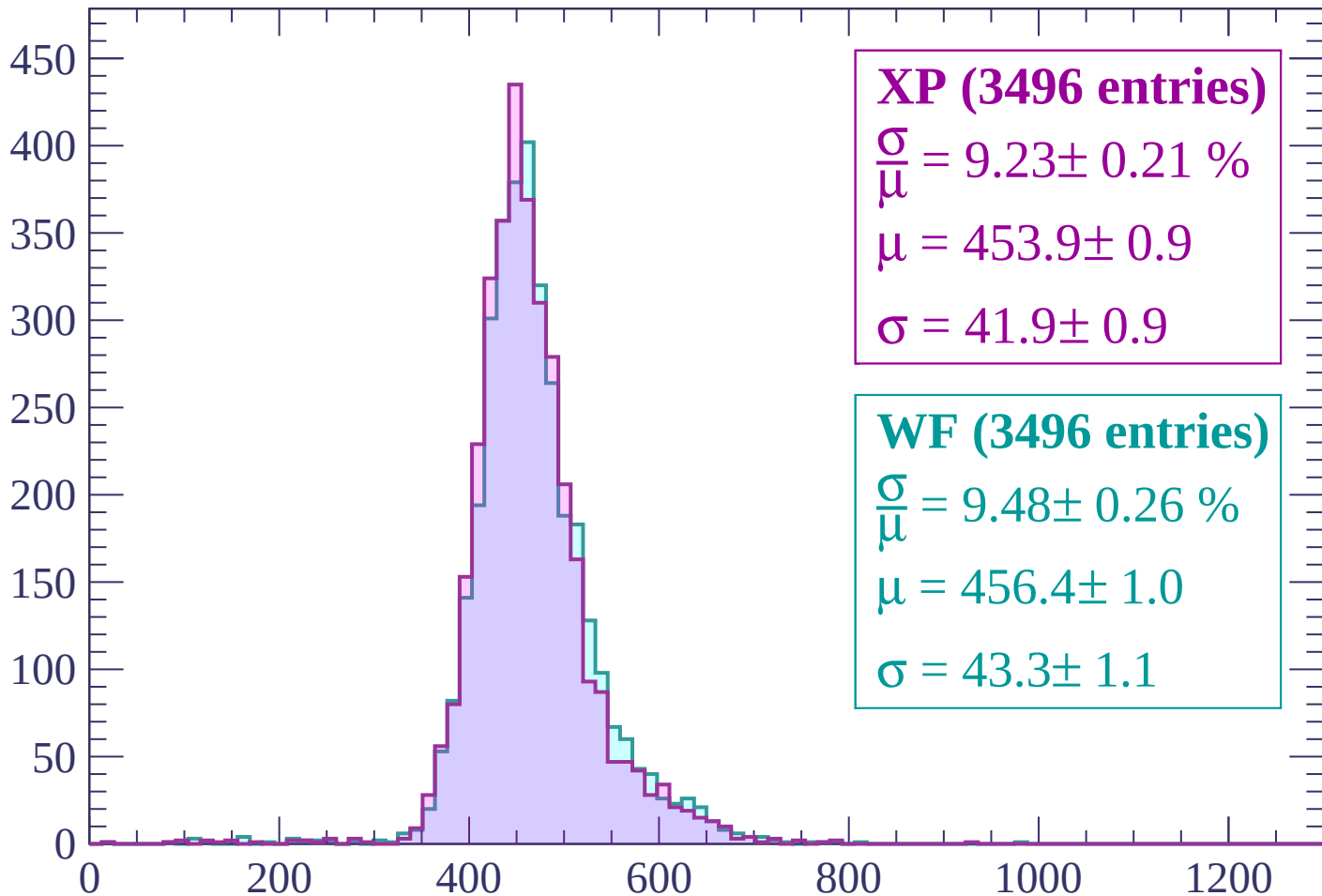
$\mu = 417.1 \pm 0.6$

$\sigma = 37.4 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $-180 < p < -120$

Count



XP (3496 entries)

$$\frac{\sigma}{\mu} = 9.23 \pm 0.21 \%$$

$$\mu = 453.9 \pm 0.9$$

$$\sigma = 41.9 \pm 0.9$$

WF (3496 entries)

$$\frac{\sigma}{\mu} = 9.48 \pm 0.26 \%$$

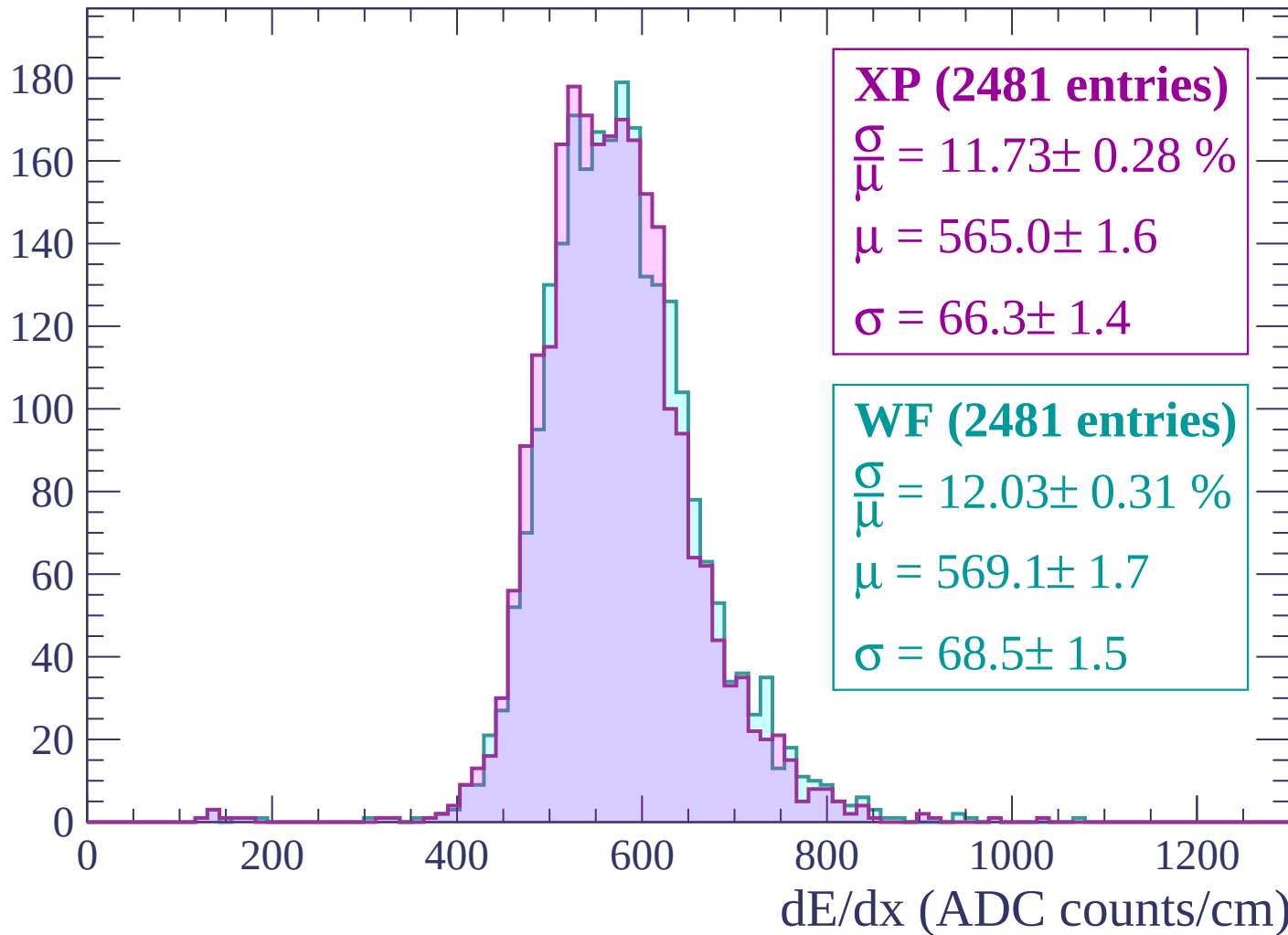
$$\mu = 456.4 \pm 1.0$$

$$\sigma = 43.3 \pm 1.1$$

dE/dx (ADC counts/cm)

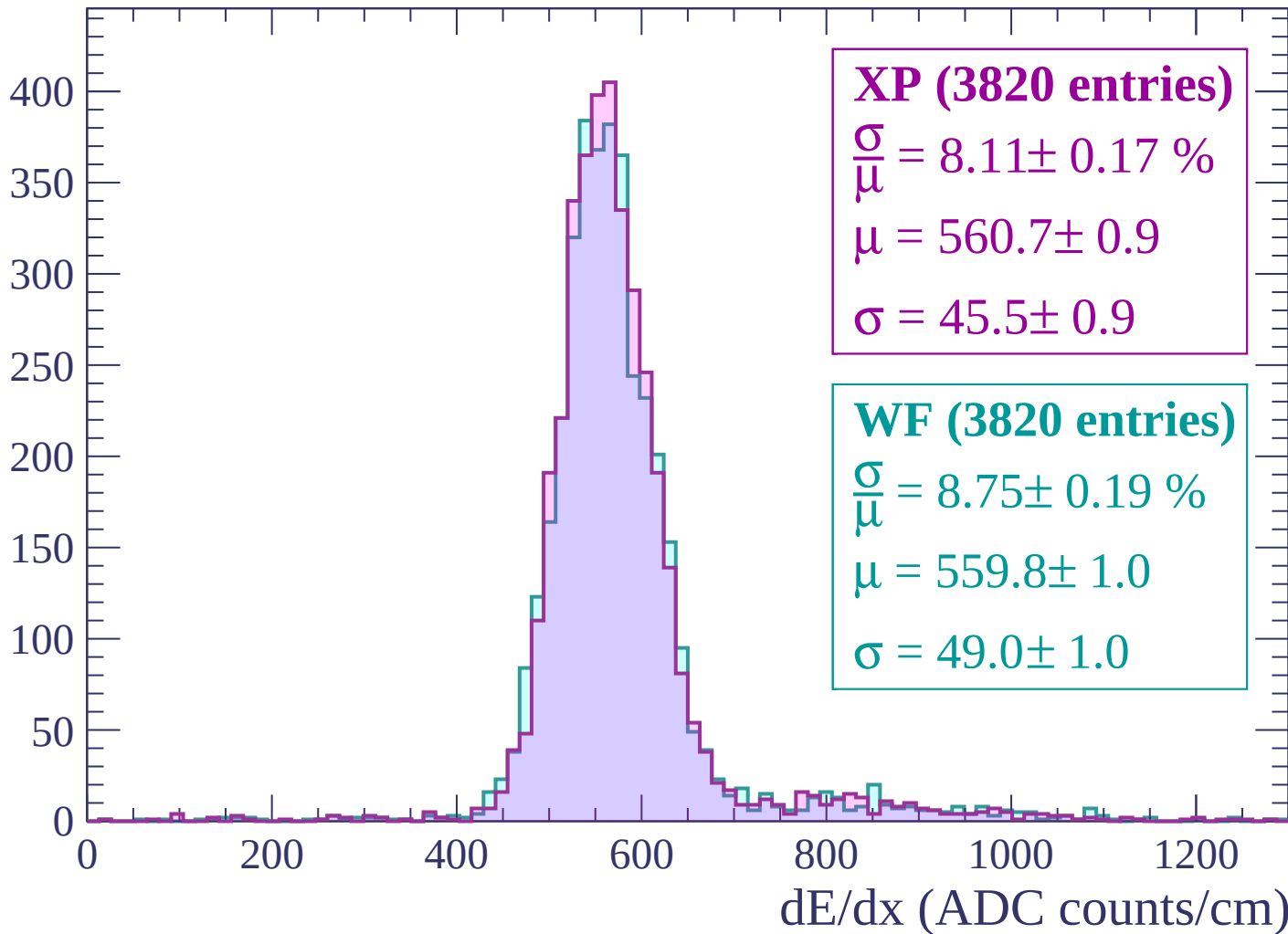
Energy loss | $-120 < p < -60$

Count



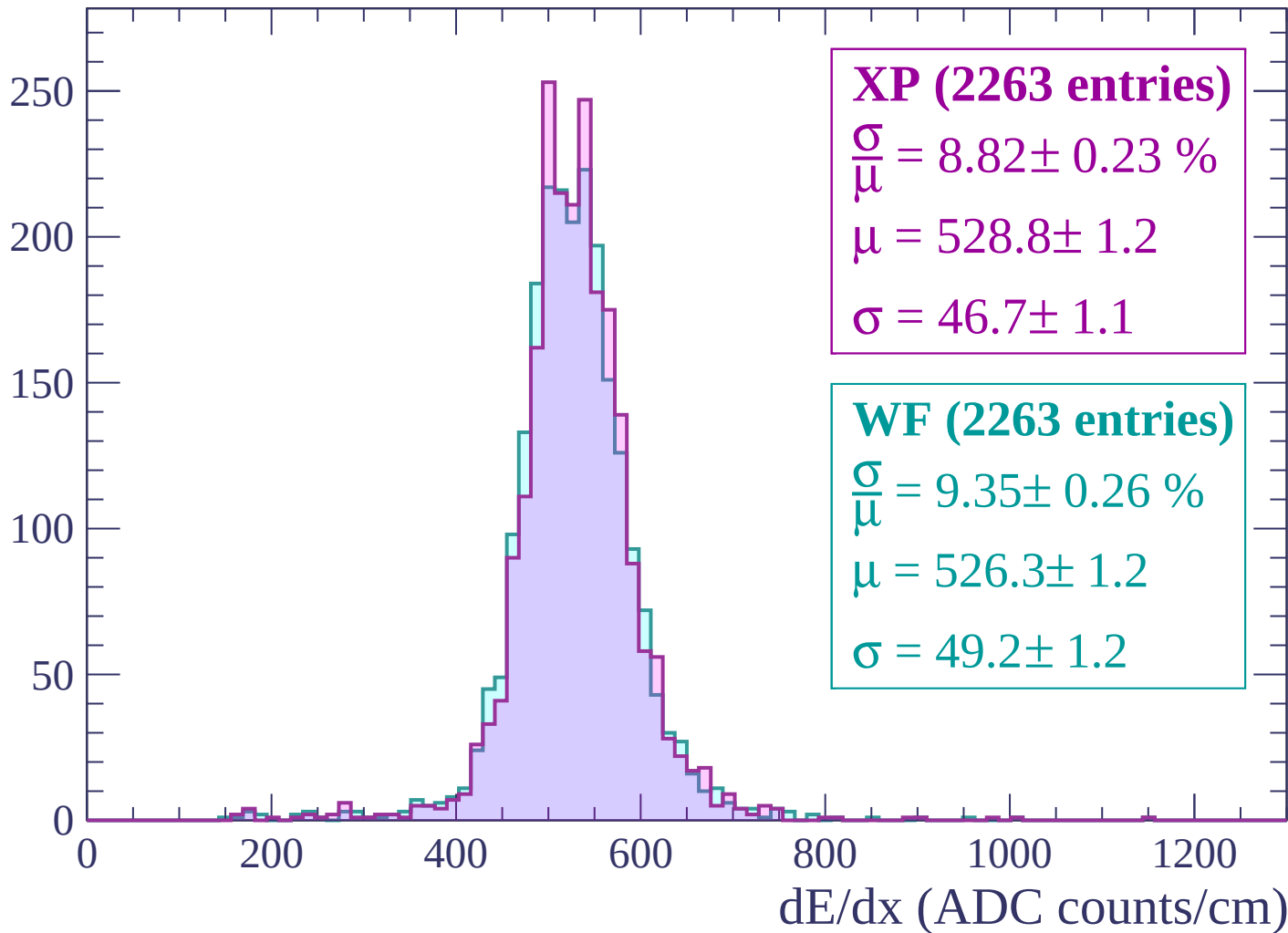
Energy loss | $-60 < p < 0$

Count



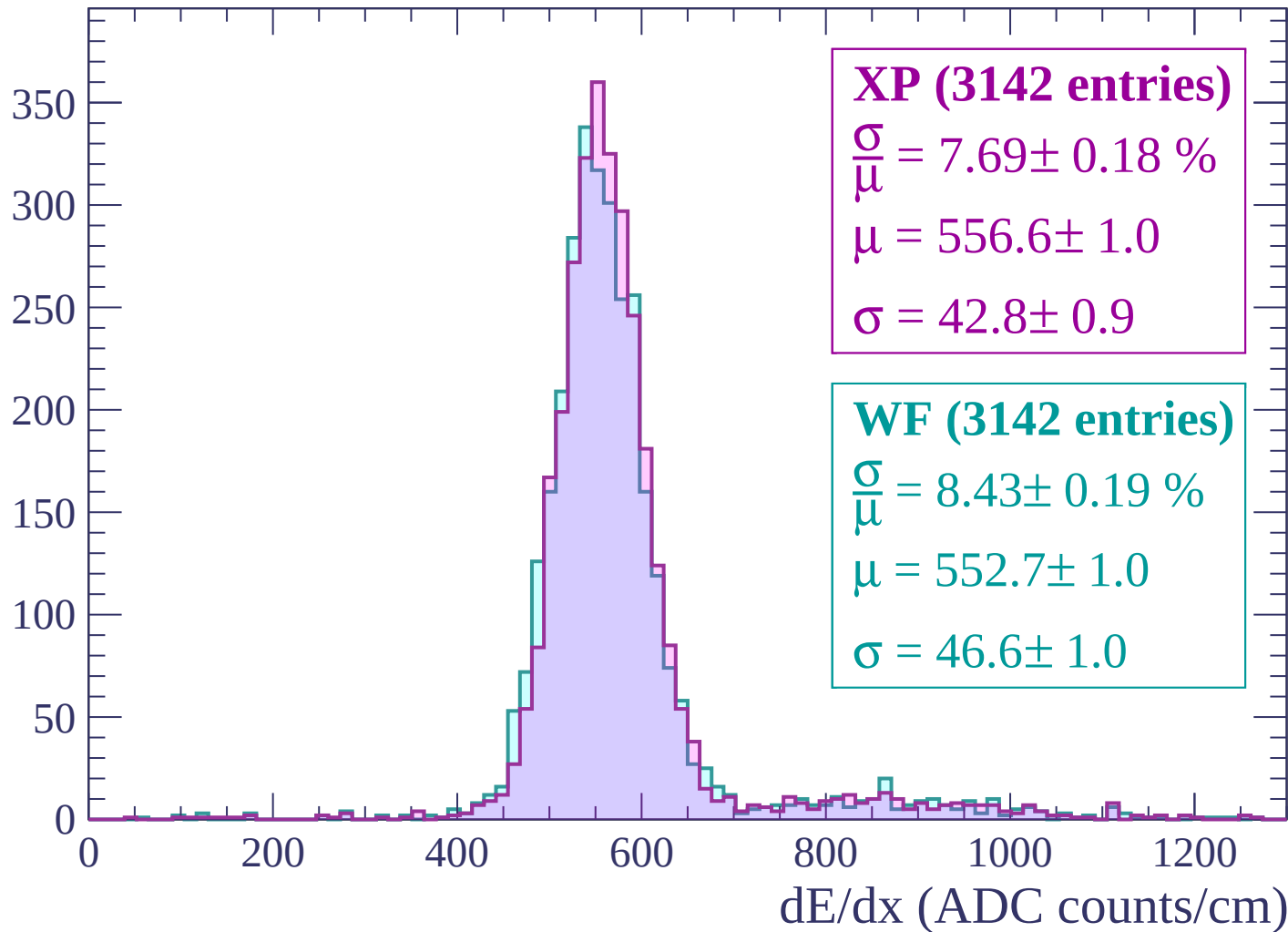
Energy loss | $0 < p < 60$

Count



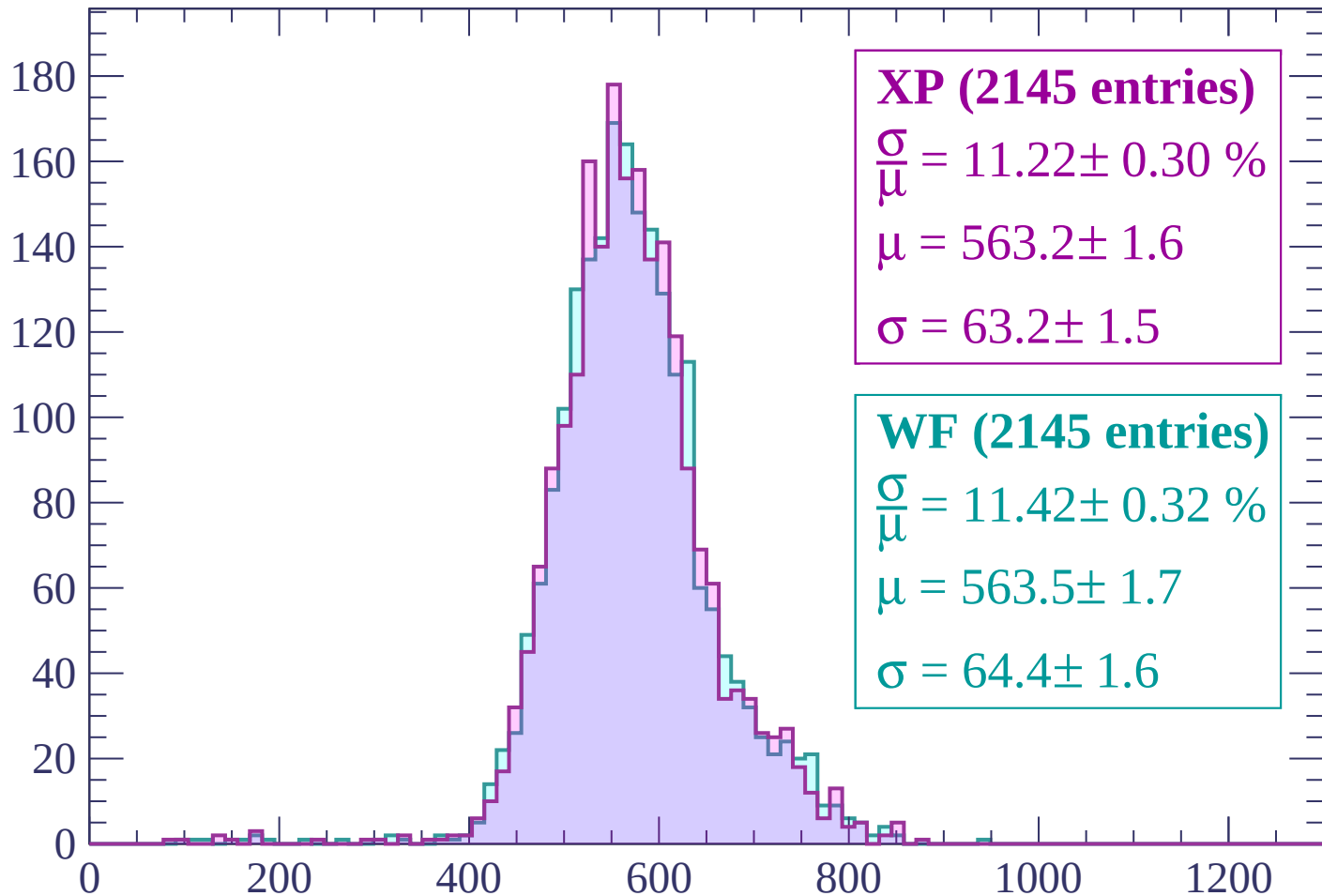
Energy loss | $60 < p < 120$

Count



Energy loss | $120 < p < 180$

Count



XP (2145 entries)

$\frac{\sigma}{\mu} = 11.22 \pm 0.30 \%$

$\mu = 563.2 \pm 1.6$

$\sigma = 63.2 \pm 1.5$

WF (2145 entries)

$\frac{\sigma}{\mu} = 11.42 \pm 0.32 \%$

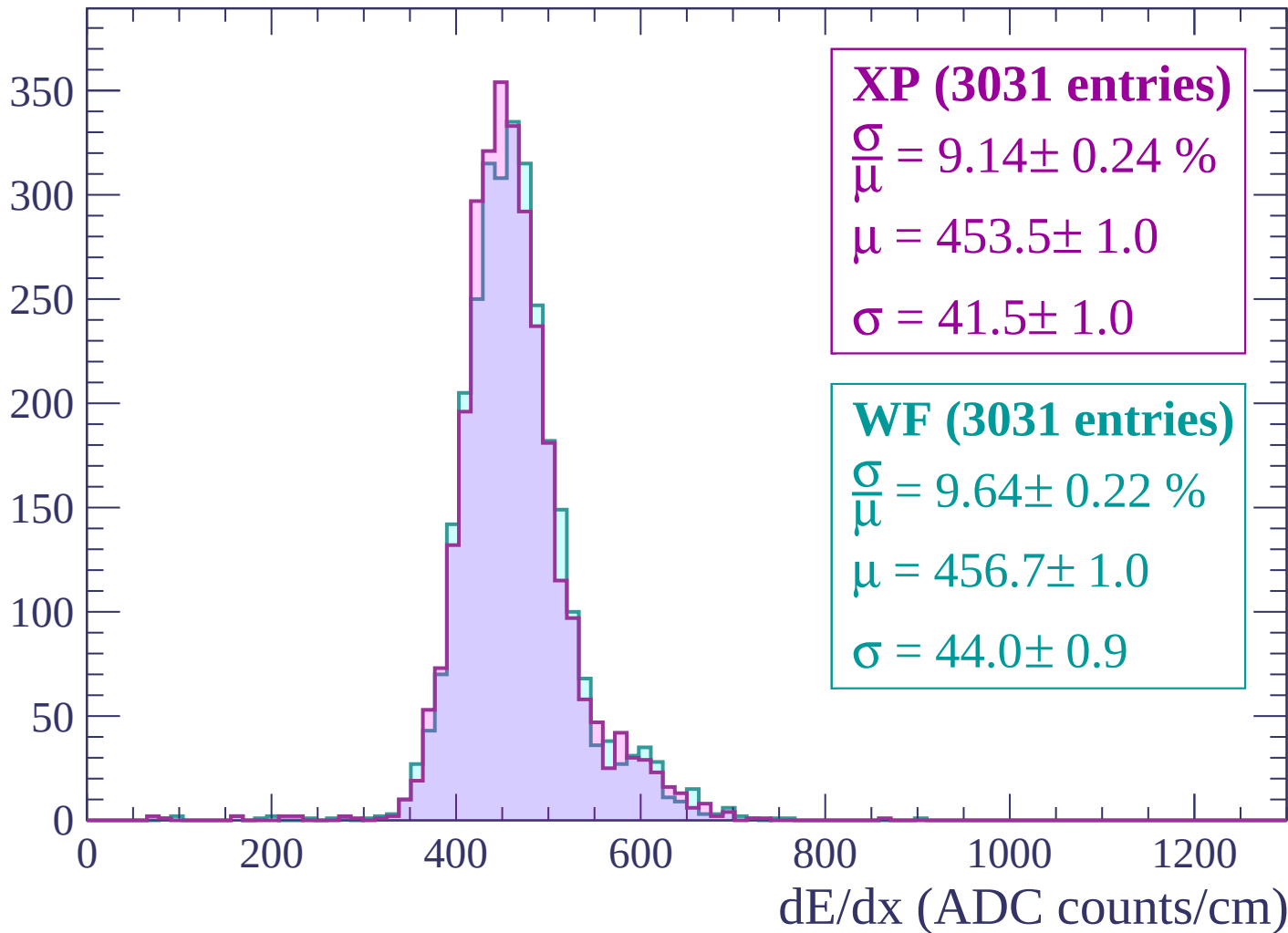
$\mu = 563.5 \pm 1.7$

$\sigma = 64.4 \pm 1.6$

dE/dx (ADC counts/cm)

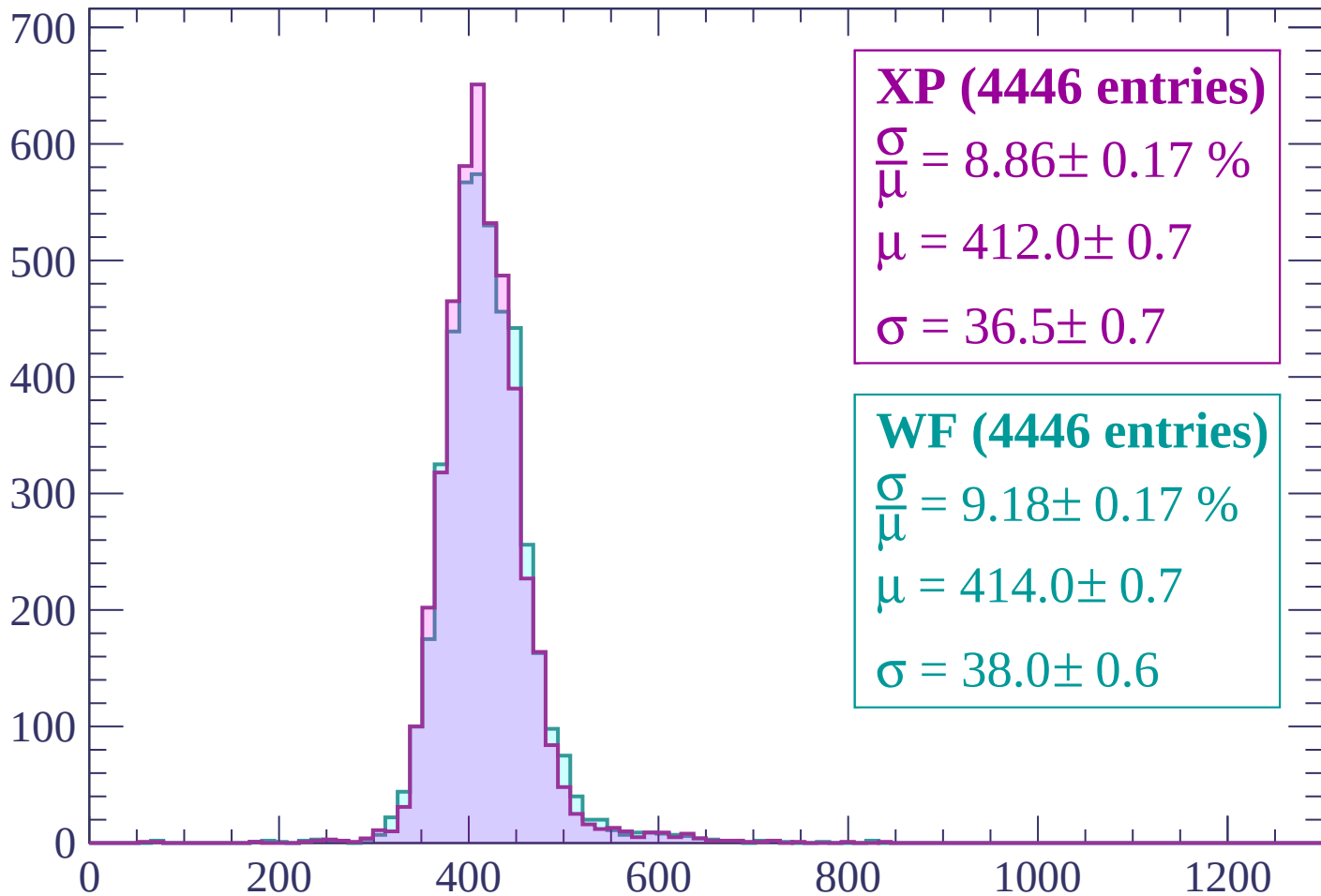
Energy loss | $180 < p < 240$

Count



Energy loss | $240 < p < 300$

Count



XP (4446 entries)

$$\frac{\sigma}{\mu} = 8.86 \pm 0.17 \%$$

$$\mu = 412.0 \pm 0.7$$

$$\sigma = 36.5 \pm 0.7$$

WF (4446 entries)

$$\frac{\sigma}{\mu} = 9.18 \pm 0.17 \%$$

$$\mu = 414.0 \pm 0.7$$

$$\sigma = 38.0 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $300 < p < 360$

Count

1000

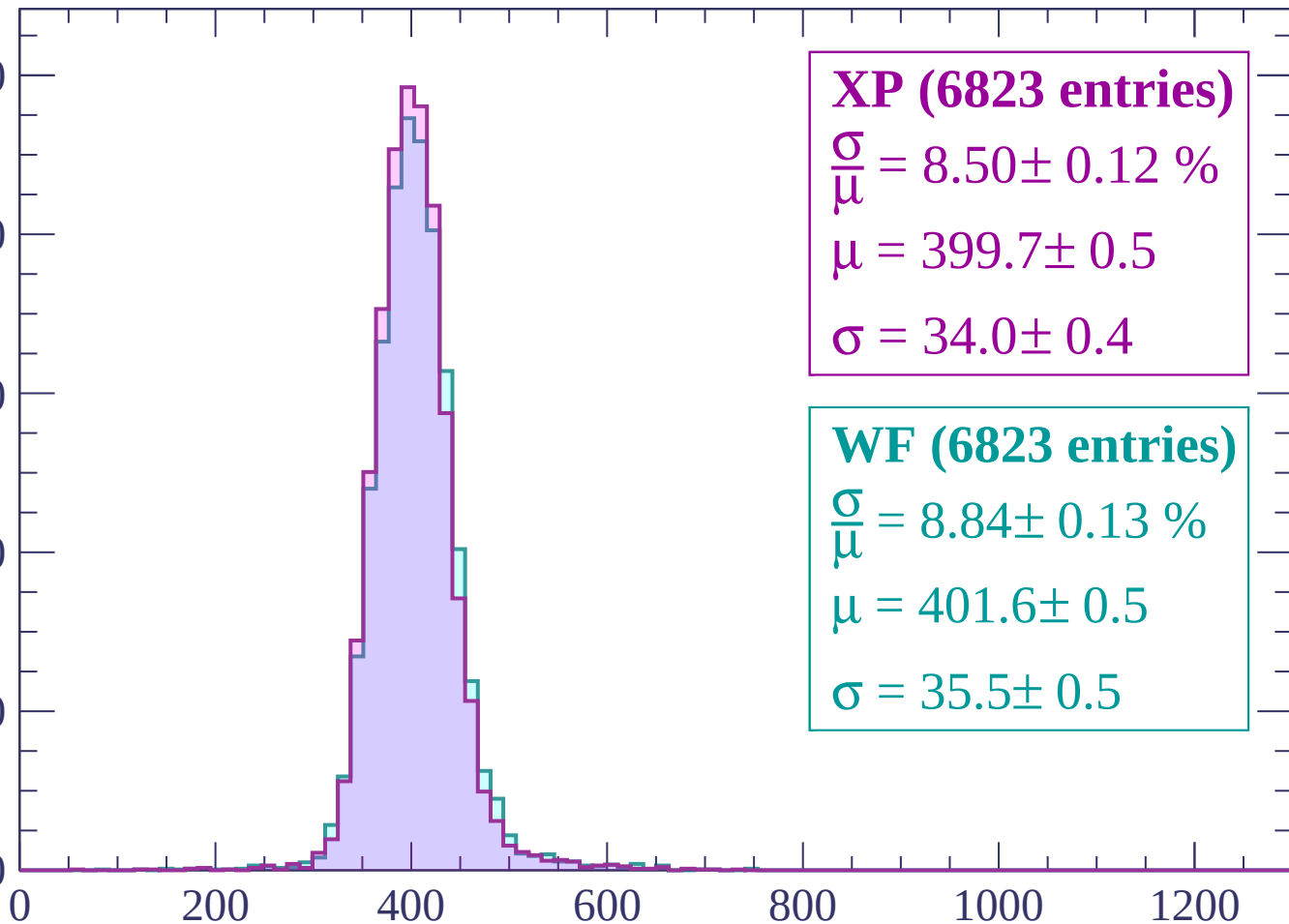
800

600

400

200

0



XP (6823 entries)

$\frac{\sigma}{\mu} = 8.50 \pm 0.12 \%$

$\mu = 399.7 \pm 0.5$

$\sigma = 34.0 \pm 0.4$

WF (6823 entries)

$\frac{\sigma}{\mu} = 8.84 \pm 0.13 \%$

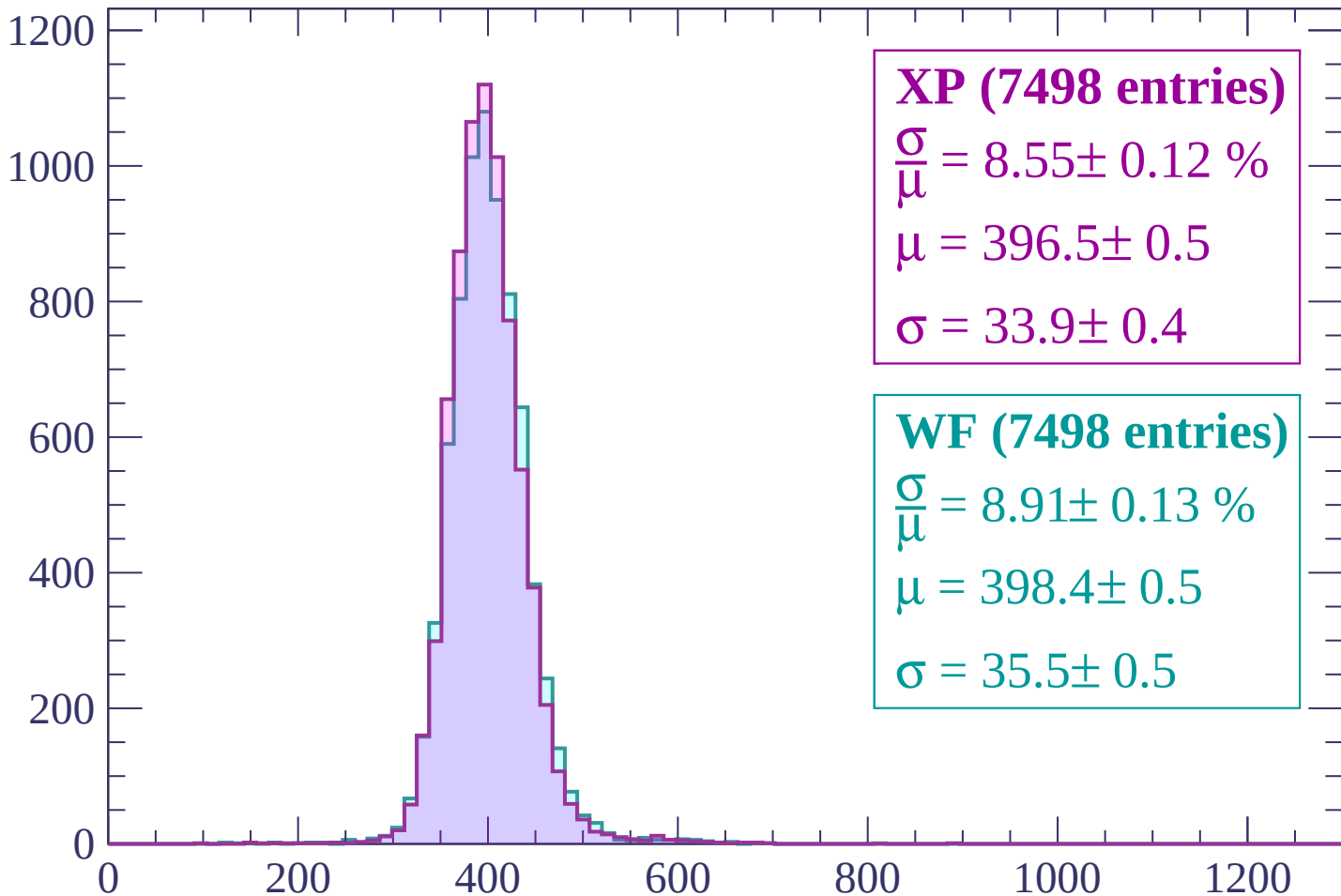
$\mu = 401.6 \pm 0.5$

$\sigma = 35.5 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $360 < p < 420$

Count



XP (7498 entries)

$$\frac{\sigma}{\mu} = 8.55 \pm 0.12 \%$$

$$\mu = 396.5 \pm 0.5$$

$$\sigma = 33.9 \pm 0.4$$

WF (7498 entries)

$$\frac{\sigma}{\mu} = 8.91 \pm 0.13 \%$$

$$\mu = 398.4 \pm 0.5$$

$$\sigma = 35.5 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $420 < p < 480$

Count

1200

1000

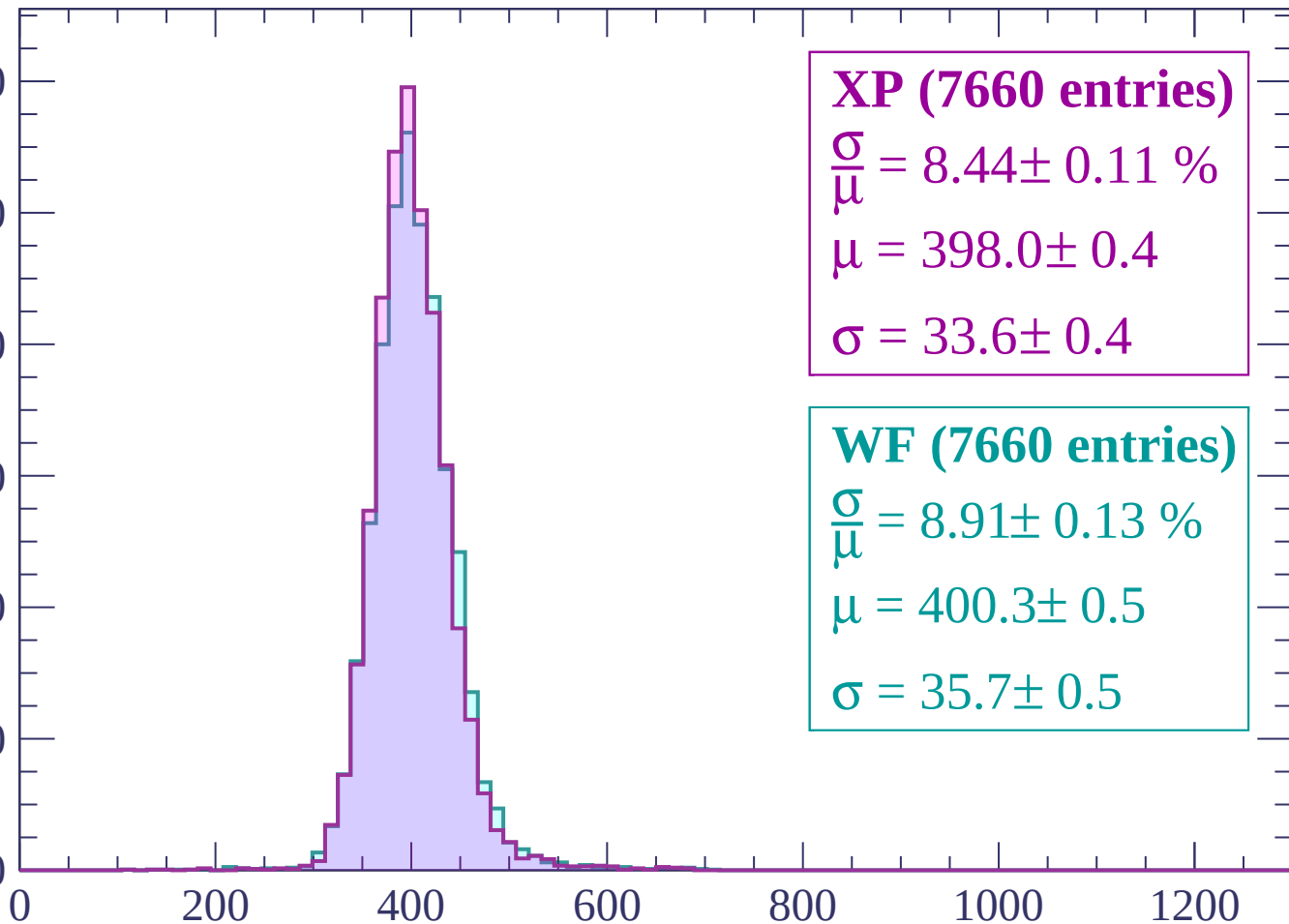
800

600

400

200

0



XP (7660 entries)

$\frac{\sigma}{\mu} = 8.44 \pm 0.11 \%$

$\mu = 398.0 \pm 0.4$

$\sigma = 33.6 \pm 0.4$

WF (7660 entries)

$\frac{\sigma}{\mu} = 8.91 \pm 0.13 \%$

$\mu = 400.3 \pm 0.5$

$\sigma = 35.7 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $480 < p < 540$

Count

1200
1000
800
600
400
200
0

XP (7538 entries)

$\frac{\sigma}{\mu} = 8.50 \pm 0.13 \%$

$\mu = 400.4 \pm 0.5$

$\sigma = 34.0 \pm 0.5$

WF (7538 entries)

$\frac{\sigma}{\mu} = 8.95 \pm 0.15 \%$

$\mu = 402.7 \pm 0.5$

$\sigma = 36.0 \pm 0.6$

0

0

200

400

600

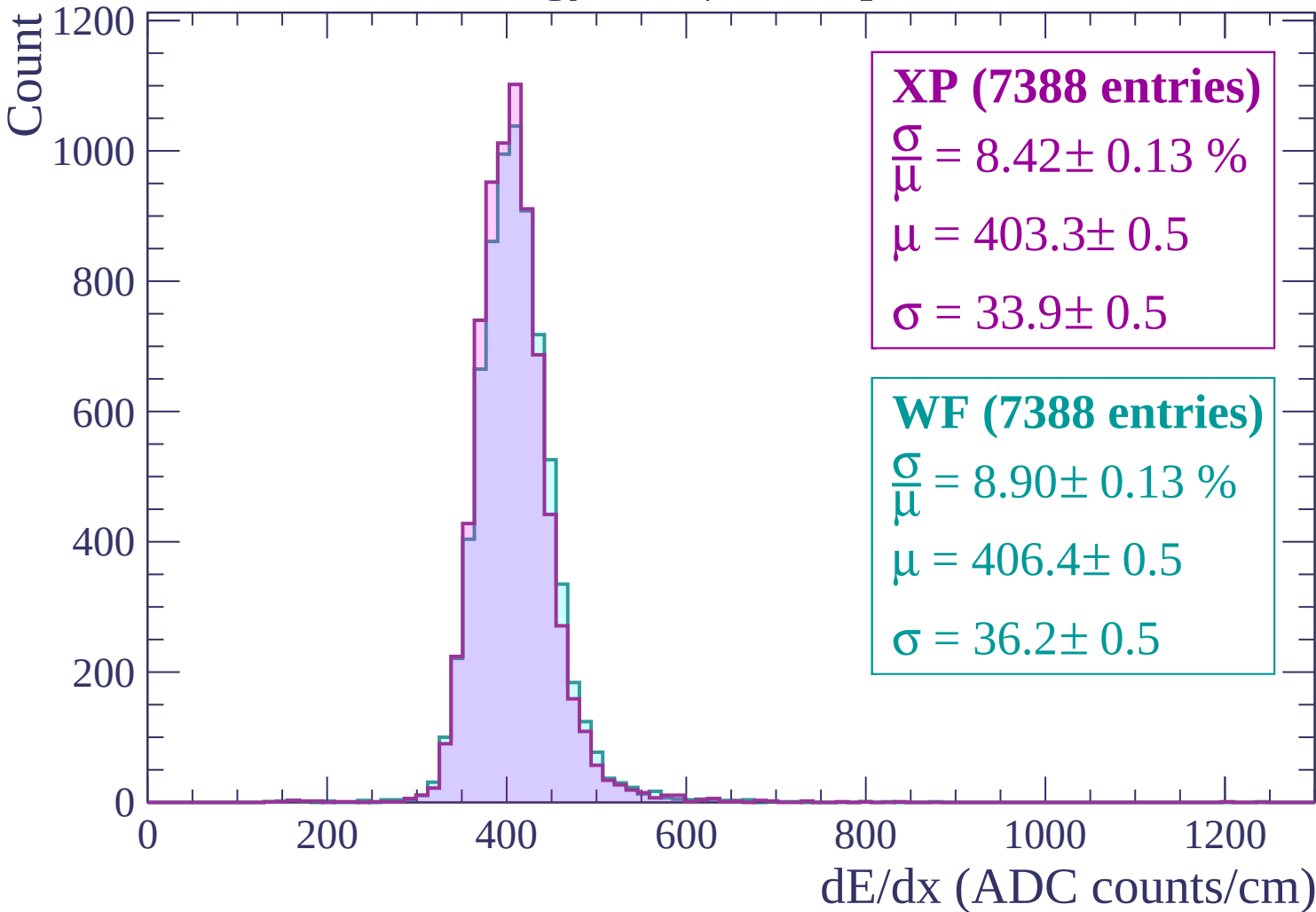
800

1000

1200

dE/dx (ADC counts/cm)

Energy loss | $540 < p < 600$



Energy loss | $600 < p < 660$

Count

1000

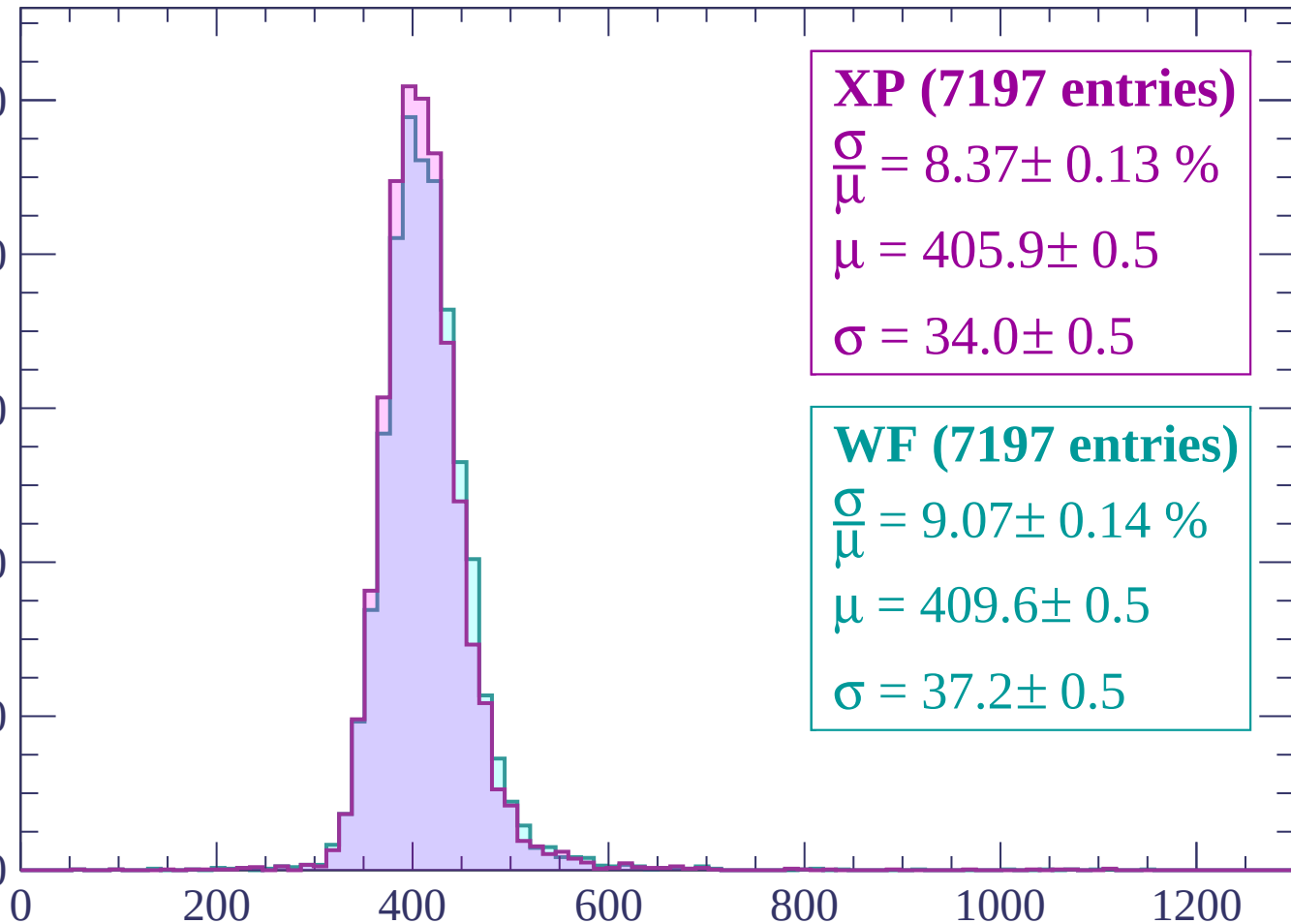
800

600

400

200

0



XP (7197 entries)

$\frac{\sigma}{\mu} = 8.37 \pm 0.13 \%$

$\mu = 405.9 \pm 0.5$

$\sigma = 34.0 \pm 0.5$

WF (7197 entries)

$\frac{\sigma}{\mu} = 9.07 \pm 0.14 \%$

$\mu = 409.6 \pm 0.5$

$\sigma = 37.2 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $660 < p < 720$

Count

1000

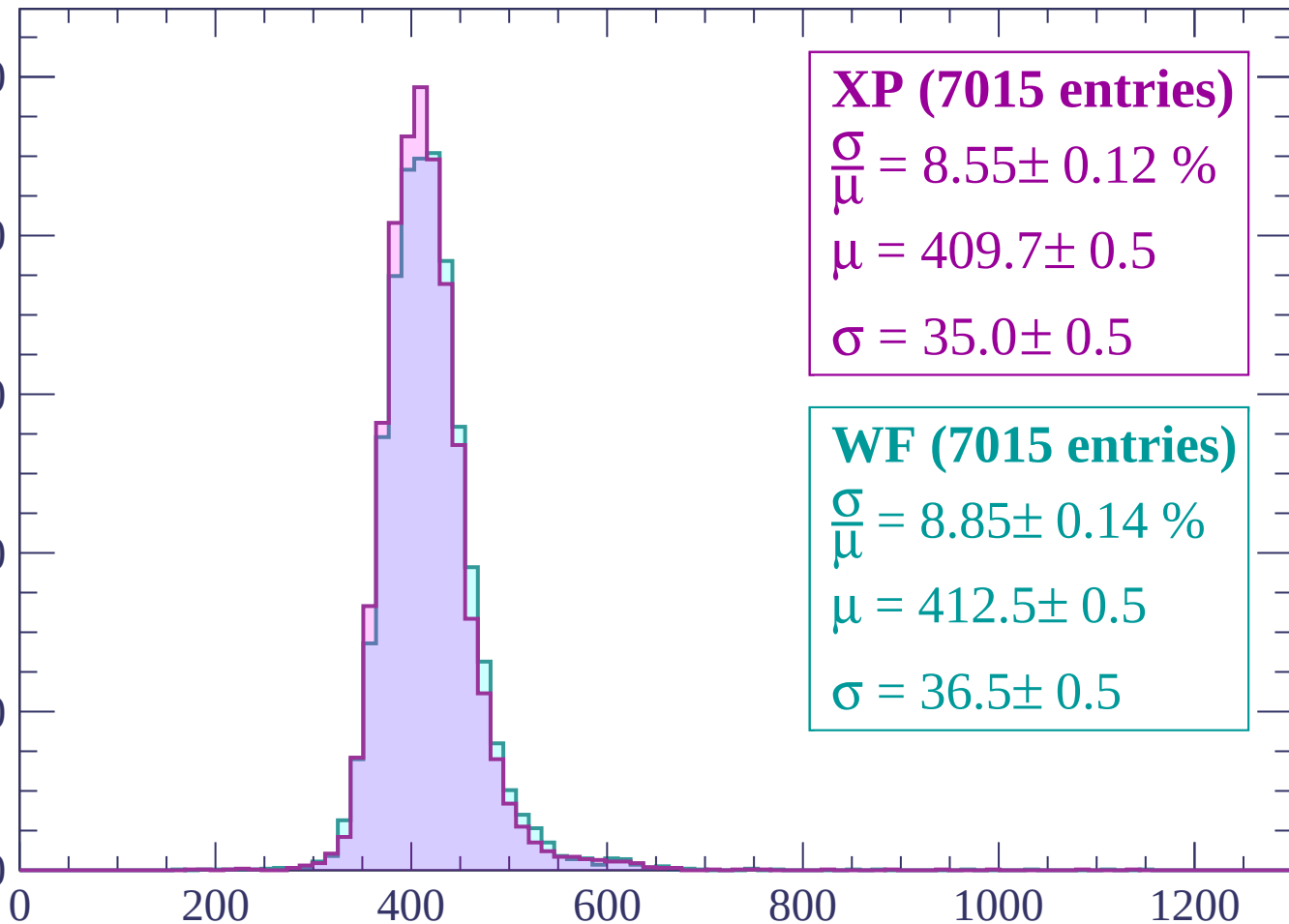
800

600

400

200

0



XP (7015 entries)

$\frac{\sigma}{\mu} = 8.55 \pm 0.12 \%$

$\mu = 409.7 \pm 0.5$

$\sigma = 35.0 \pm 0.5$

WF (7015 entries)

$\frac{\sigma}{\mu} = 8.85 \pm 0.14 \%$

$\mu = 412.5 \pm 0.5$

$\sigma = 36.5 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $720 < p < 780$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (7055 entries)

$\frac{\sigma}{\mu} = 8.54 \pm 0.13 \%$

$\mu = 413.5 \pm 0.5$

$\sigma = 35.3 \pm 0.5$

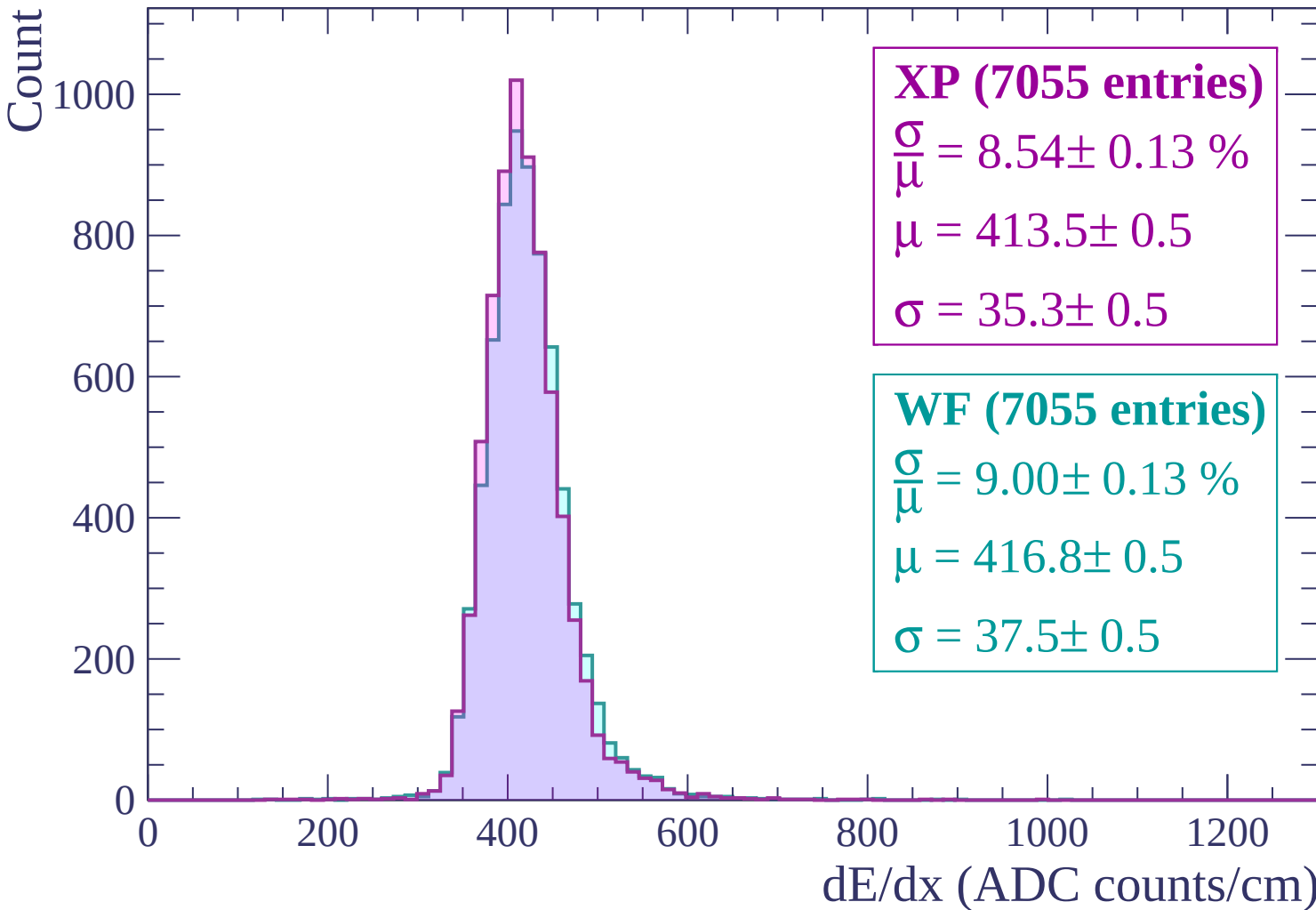
WF (7055 entries)

$\frac{\sigma}{\mu} = 9.00 \pm 0.13 \%$

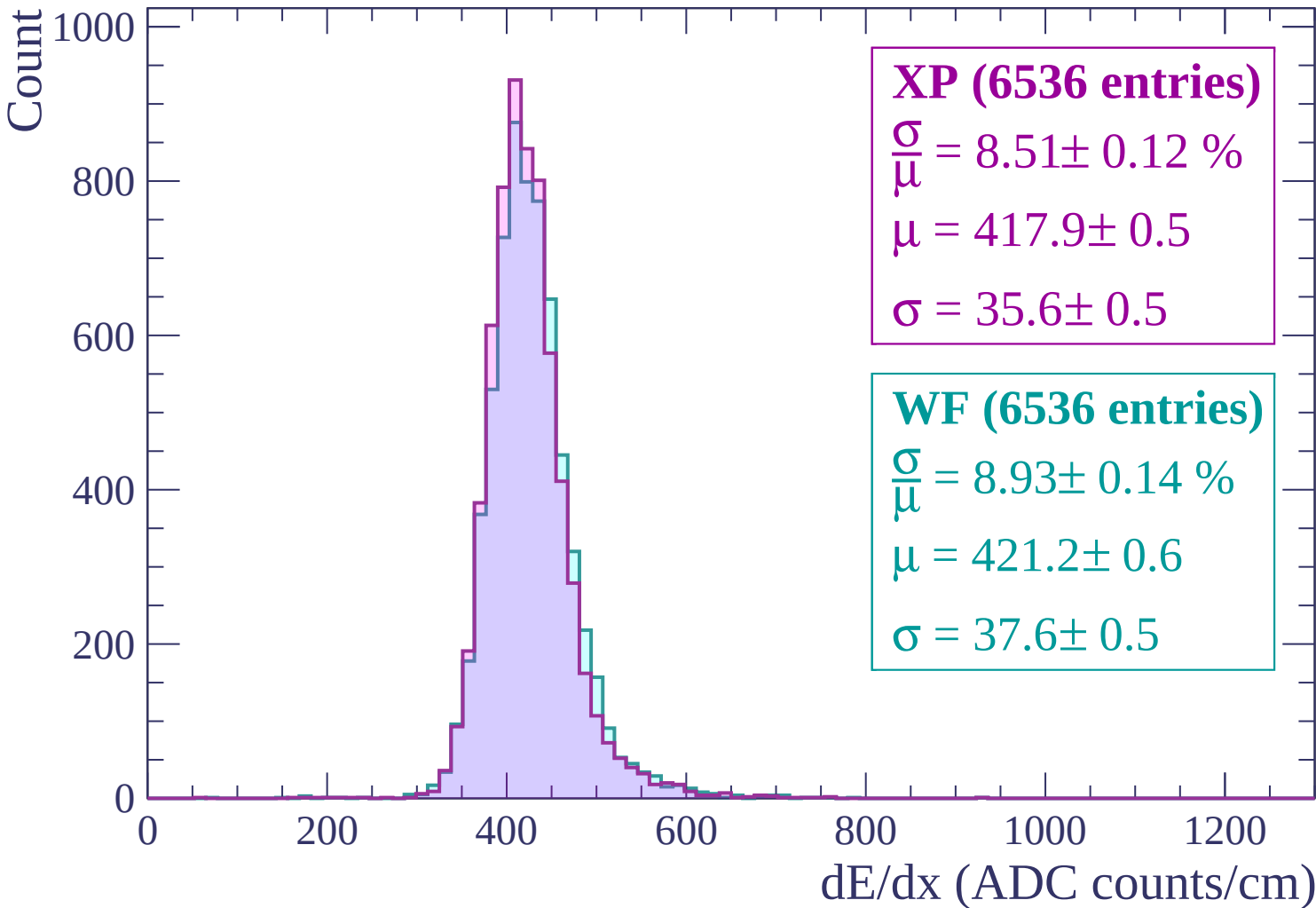
$\mu = 416.8 \pm 0.5$

$\sigma = 37.5 \pm 0.5$

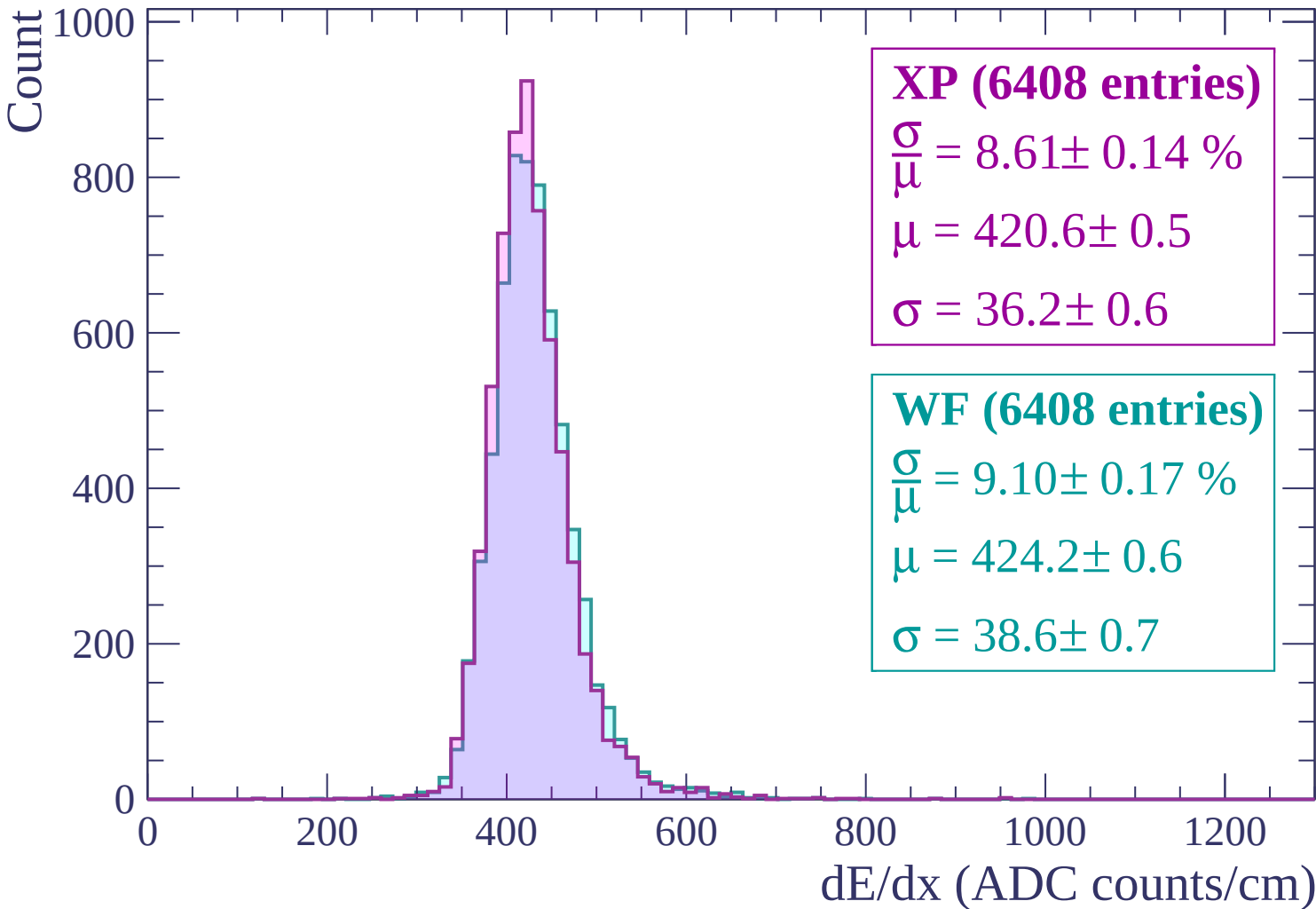
dE/dx (ADC counts/cm)



Energy loss | $780 < p < 840$

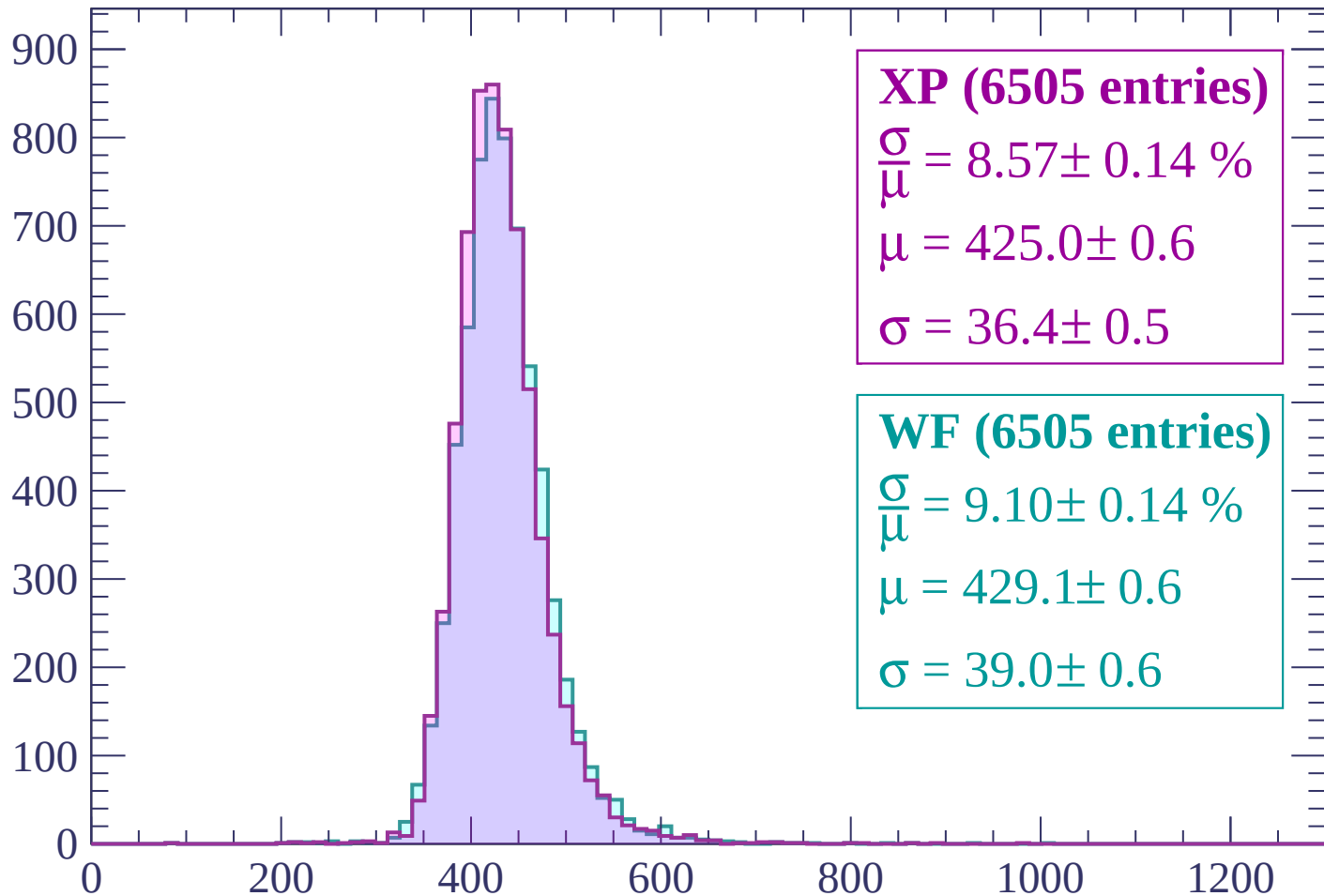


Energy loss | $840 < p < 900$



Energy loss | $900 < p < 960$

Count



XP (6505 entries)

$$\frac{\sigma}{\mu} = 8.57 \pm 0.14 \%$$

$$\mu = 425.0 \pm 0.6$$

$$\sigma = 36.4 \pm 0.5$$

WF (6505 entries)

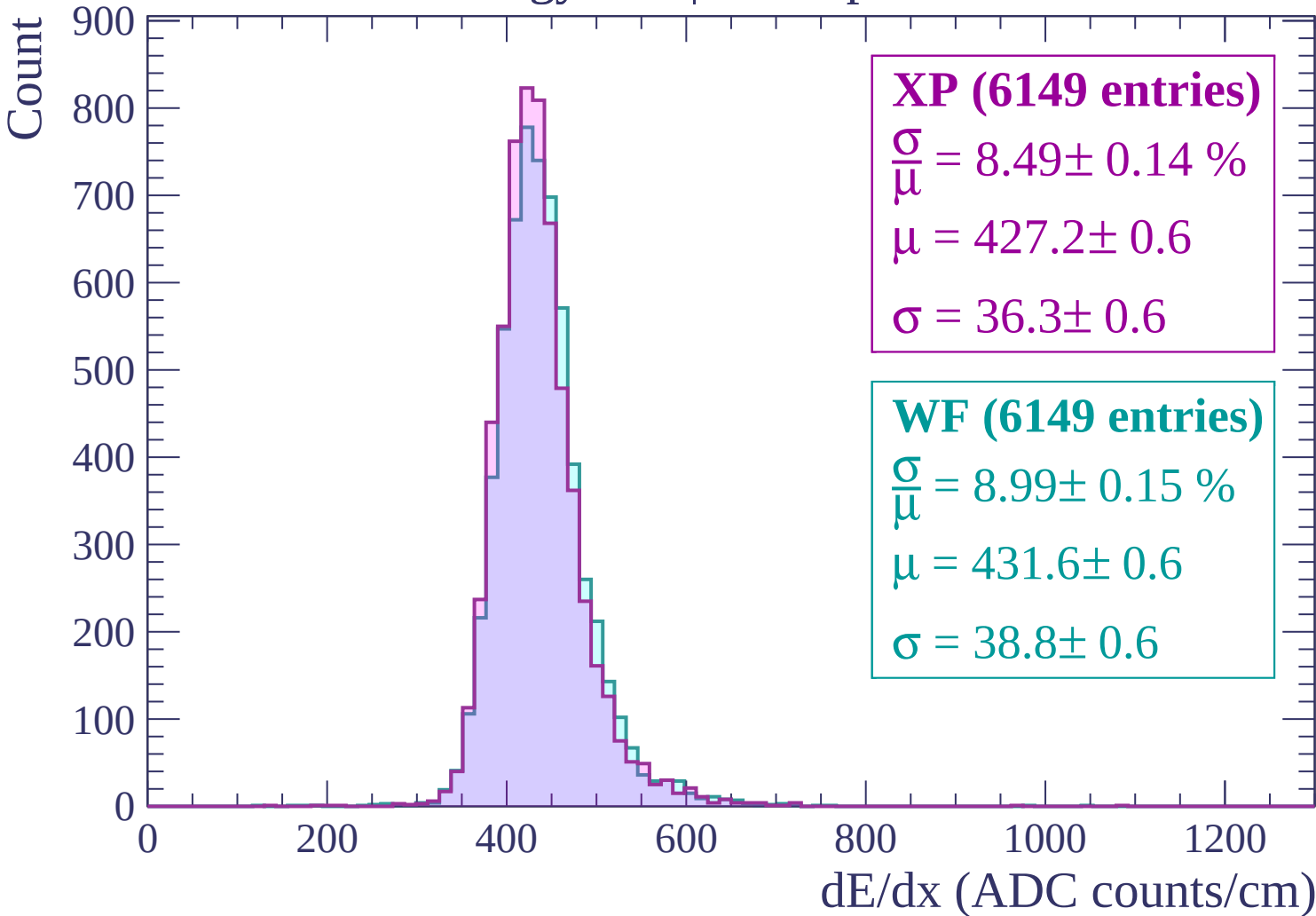
$$\frac{\sigma}{\mu} = 9.10 \pm 0.14 \%$$

$$\mu = 429.1 \pm 0.6$$

$$\sigma = 39.0 \pm 0.6$$

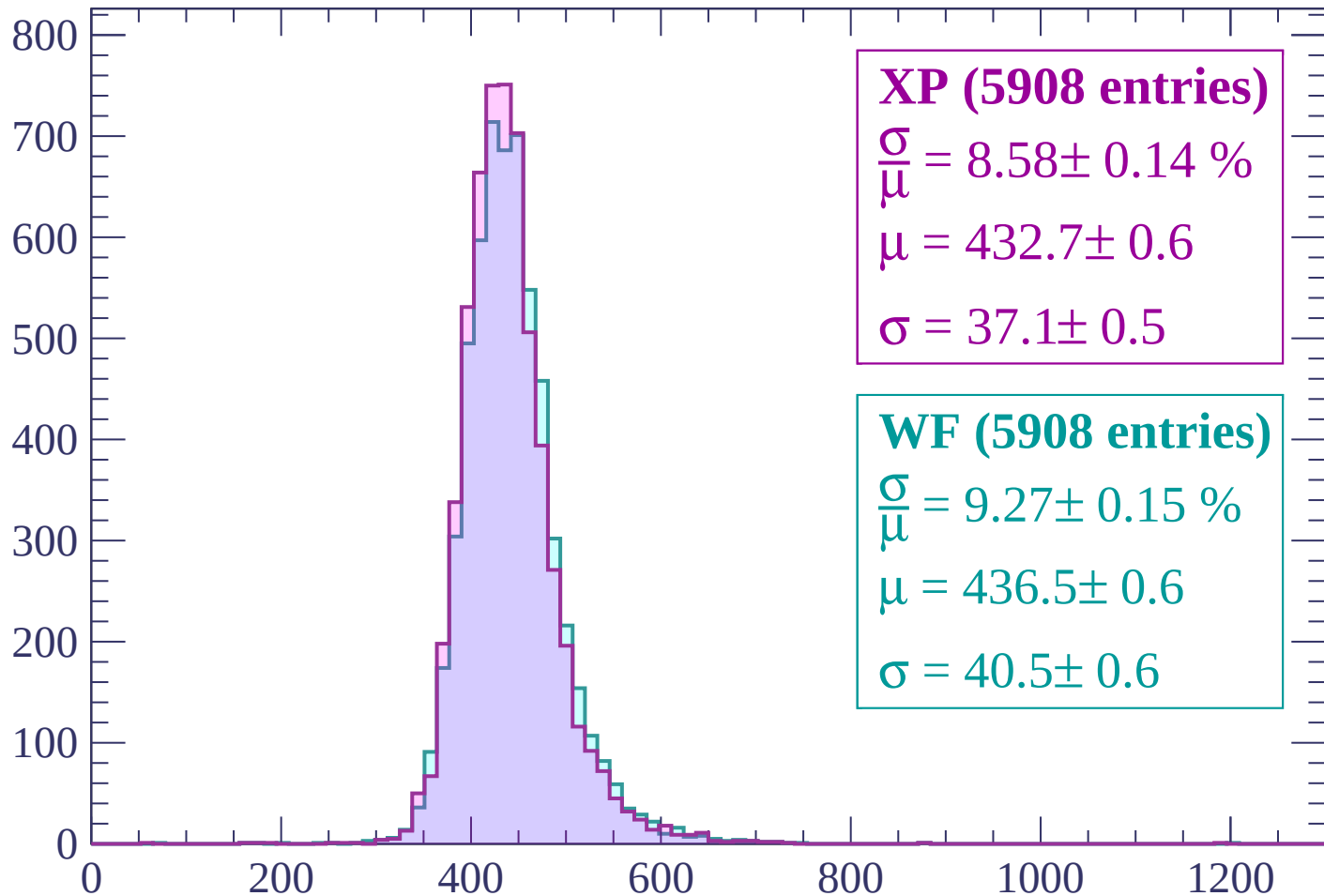
dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1020$



Energy loss | $1020 < p < 1080$

Count



XP (5908 entries)

$\frac{\sigma}{\mu} = 8.58 \pm 0.14 \%$

$\mu = 432.7 \pm 0.6$

$\sigma = 37.1 \pm 0.5$

WF (5908 entries)

$\frac{\sigma}{\mu} = 9.27 \pm 0.15 \%$

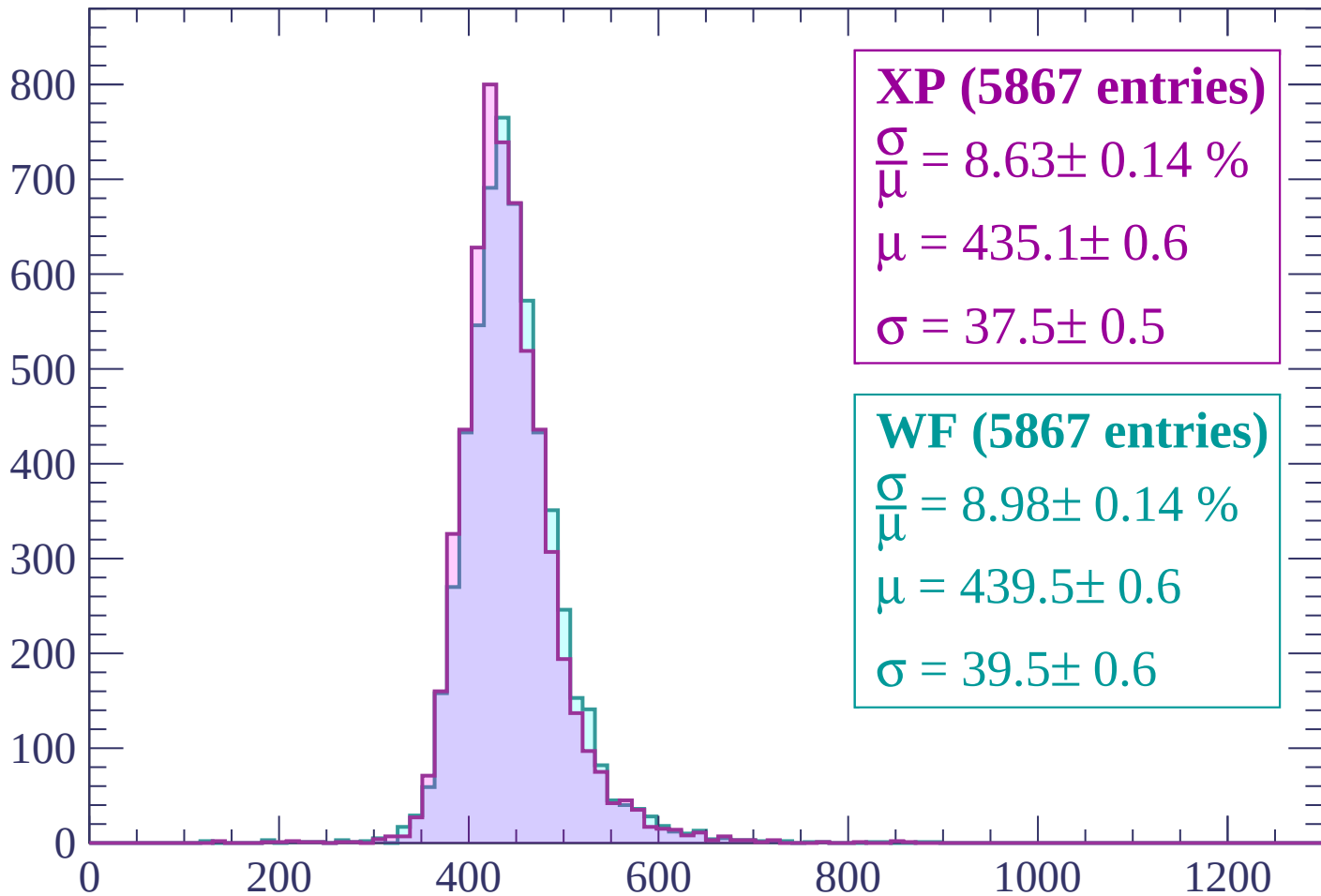
$\mu = 436.5 \pm 0.6$

$\sigma = 40.5 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $1080 < p < 1140$

Count



XP (5867 entries)

$\frac{\sigma}{\mu} = 8.63 \pm 0.14 \%$

$\mu = 435.1 \pm 0.6$

$\sigma = 37.5 \pm 0.5$

WF (5867 entries)

$\frac{\sigma}{\mu} = 8.98 \pm 0.14 \%$

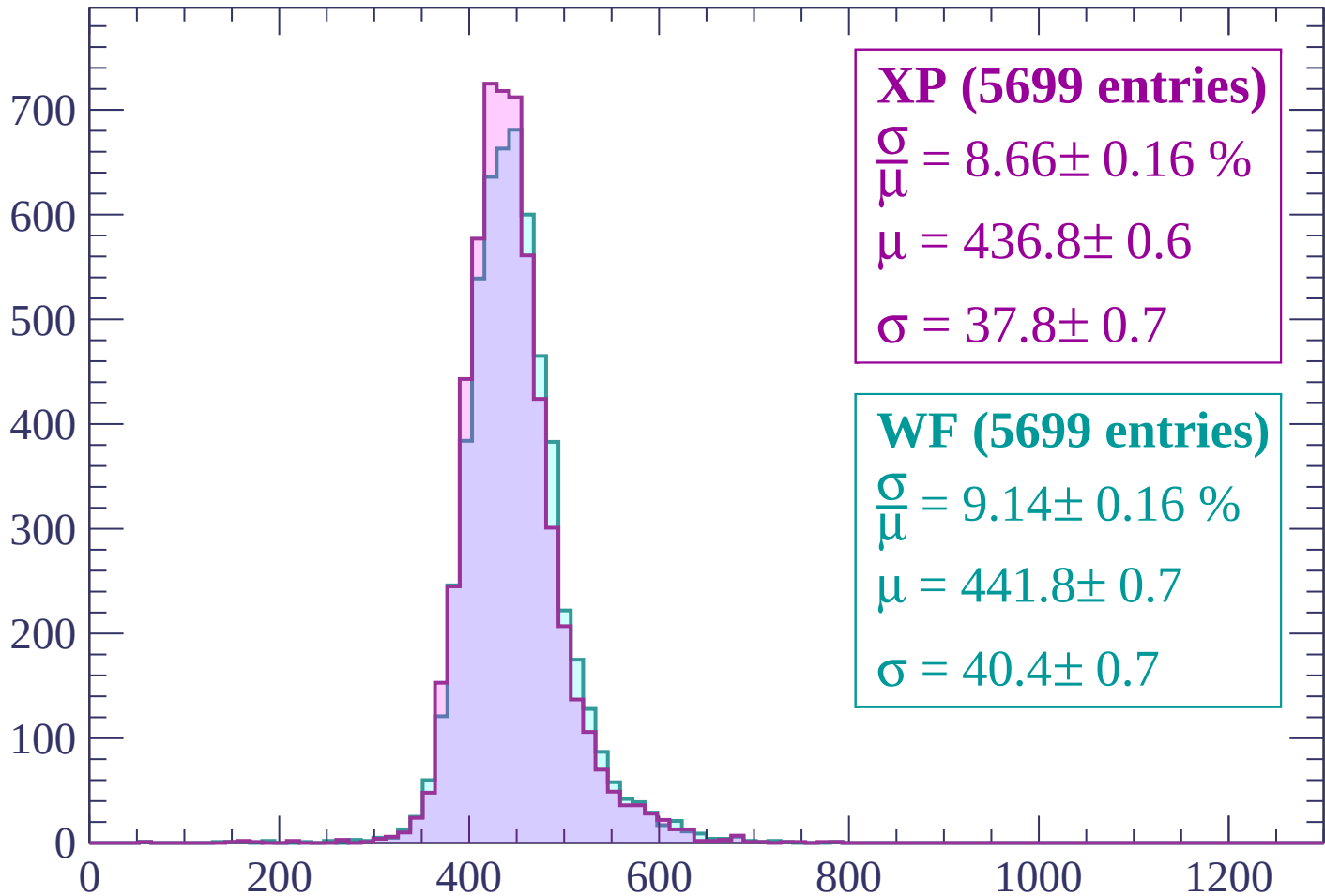
$\mu = 439.5 \pm 0.6$

$\sigma = 39.5 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $1140 < p < 1200$

Count



XP (5699 entries)

$$\frac{\sigma}{\mu} = 8.66 \pm 0.16 \%$$

$$\mu = 436.8 \pm 0.6$$

$$\sigma = 37.8 \pm 0.7$$

WF (5699 entries)

$$\frac{\sigma}{\mu} = 9.14 \pm 0.16 \%$$

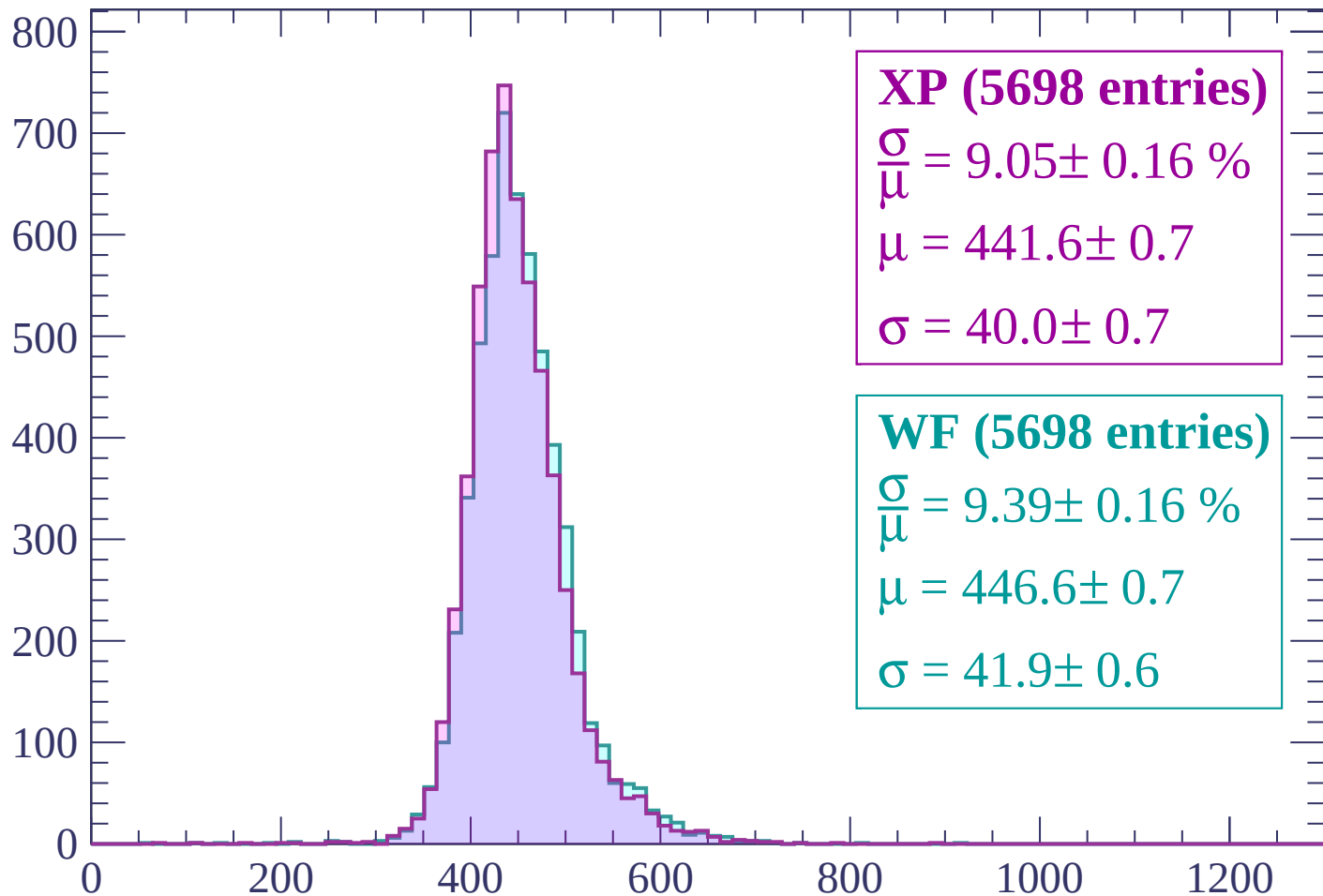
$$\mu = 441.8 \pm 0.7$$

$$\sigma = 40.4 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1200 < p < 1260$

Count



XP (5698 entries)

$\frac{\sigma}{\mu} = 9.05 \pm 0.16 \%$

$\mu = 441.6 \pm 0.7$

$\sigma = 40.0 \pm 0.7$

WF (5698 entries)

$\frac{\sigma}{\mu} = 9.39 \pm 0.16 \%$

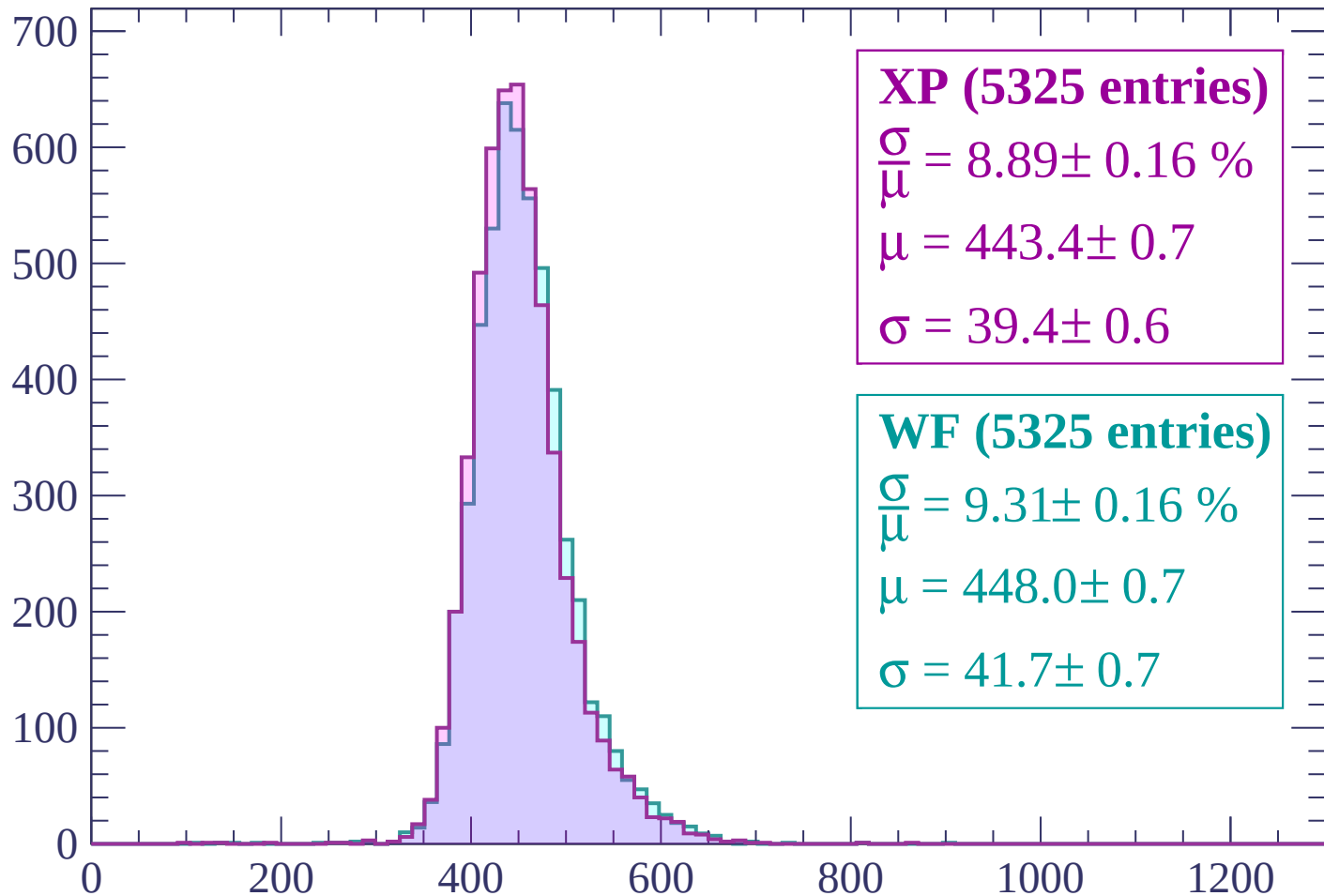
$\mu = 446.6 \pm 0.7$

$\sigma = 41.9 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $1260 < p < 1320$

Count



XP (5325 entries)

$$\frac{\sigma}{\mu} = 8.89 \pm 0.16 \%$$

$$\mu = 443.4 \pm 0.7$$

$$\sigma = 39.4 \pm 0.6$$

WF (5325 entries)

$$\frac{\sigma}{\mu} = 9.31 \pm 0.16 \%$$

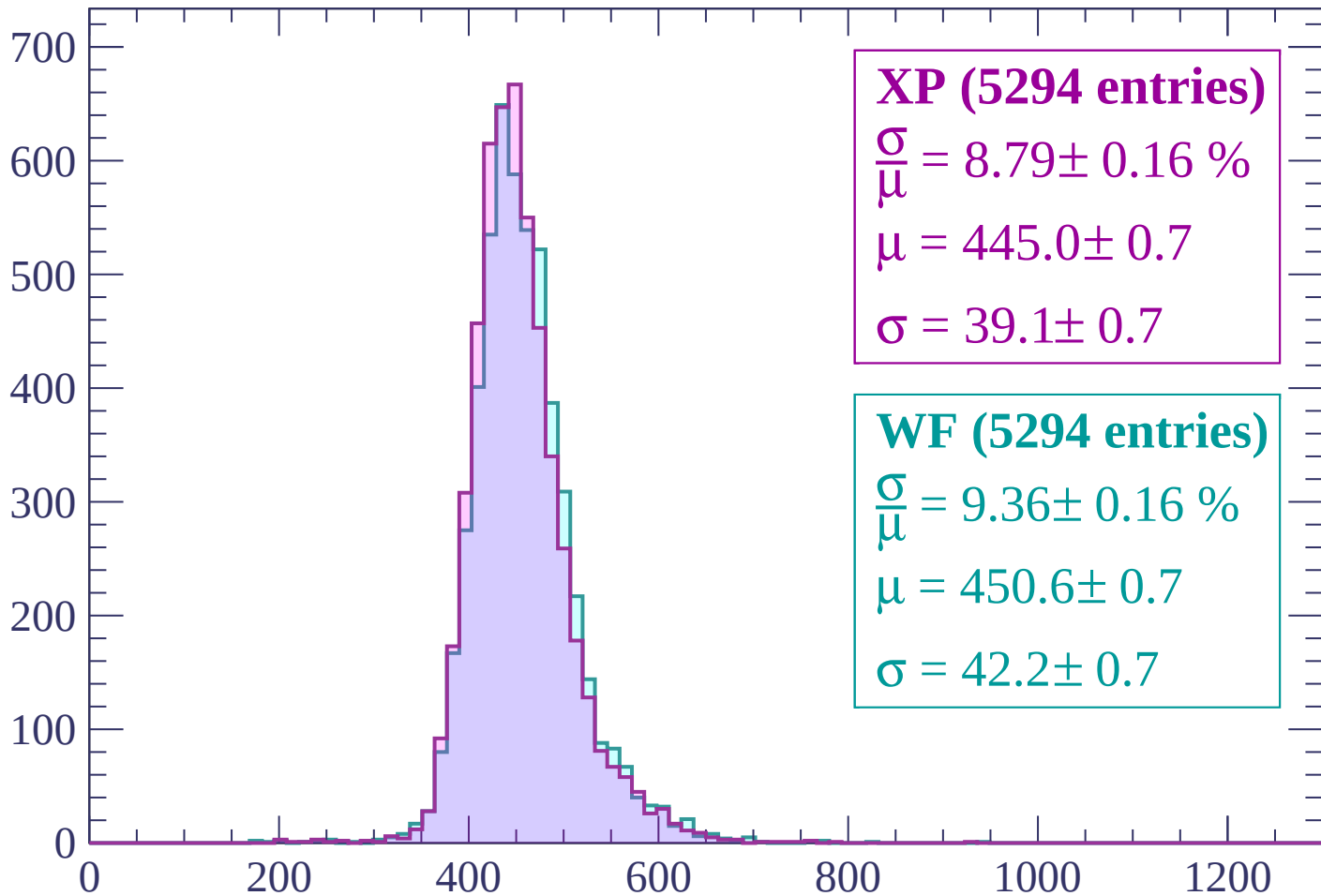
$$\mu = 448.0 \pm 0.7$$

$$\sigma = 41.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1320 < p < 1380$

Count



XP (5294 entries)

$$\frac{\sigma}{\mu} = 8.79 \pm 0.16 \%$$

$$\mu = 445.0 \pm 0.7$$

$$\sigma = 39.1 \pm 0.7$$

WF (5294 entries)

$$\frac{\sigma}{\mu} = 9.36 \pm 0.16 \%$$

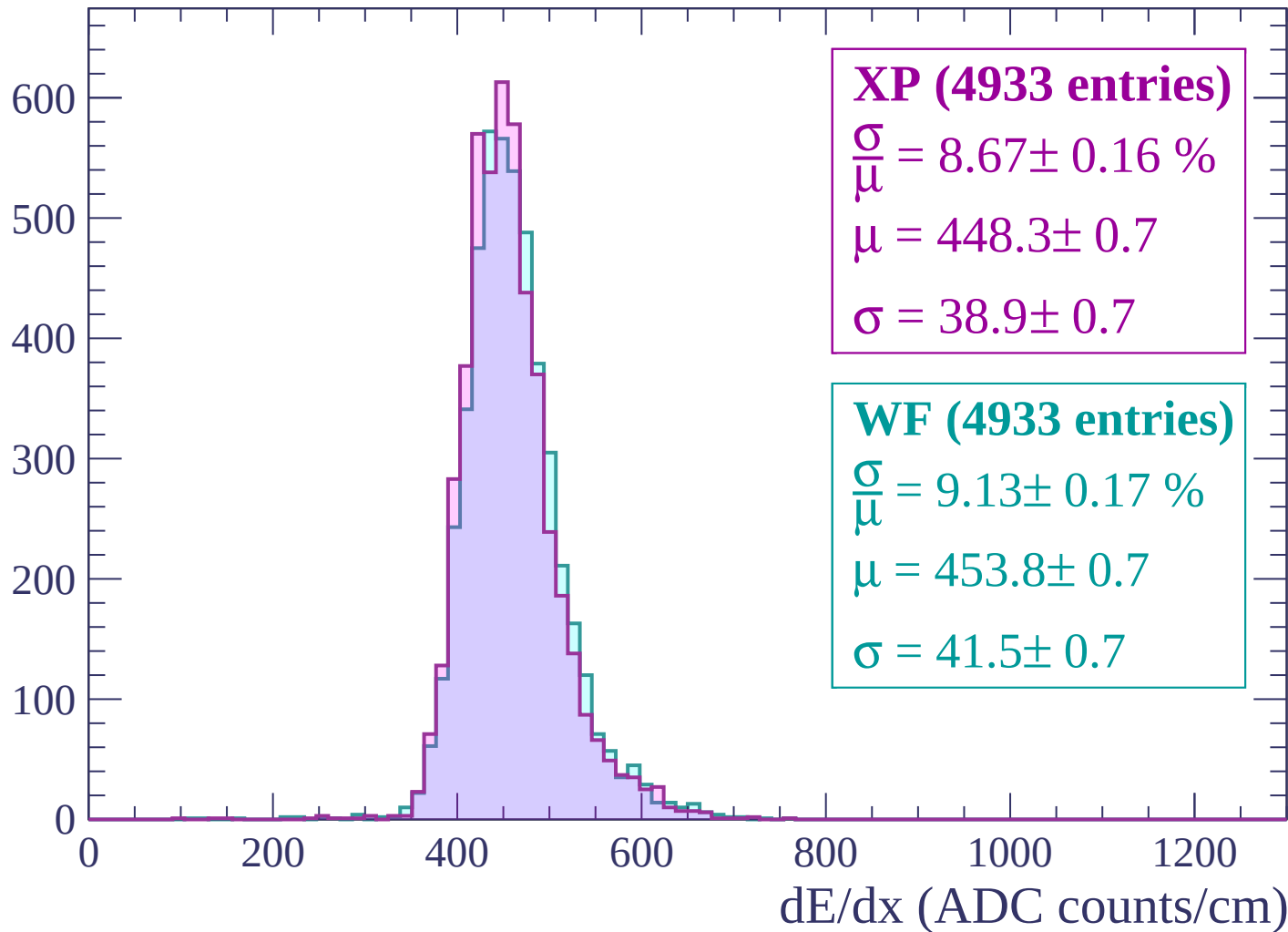
$$\mu = 450.6 \pm 0.7$$

$$\sigma = 42.2 \pm 0.7$$

dE/dx (ADC counts/cm)

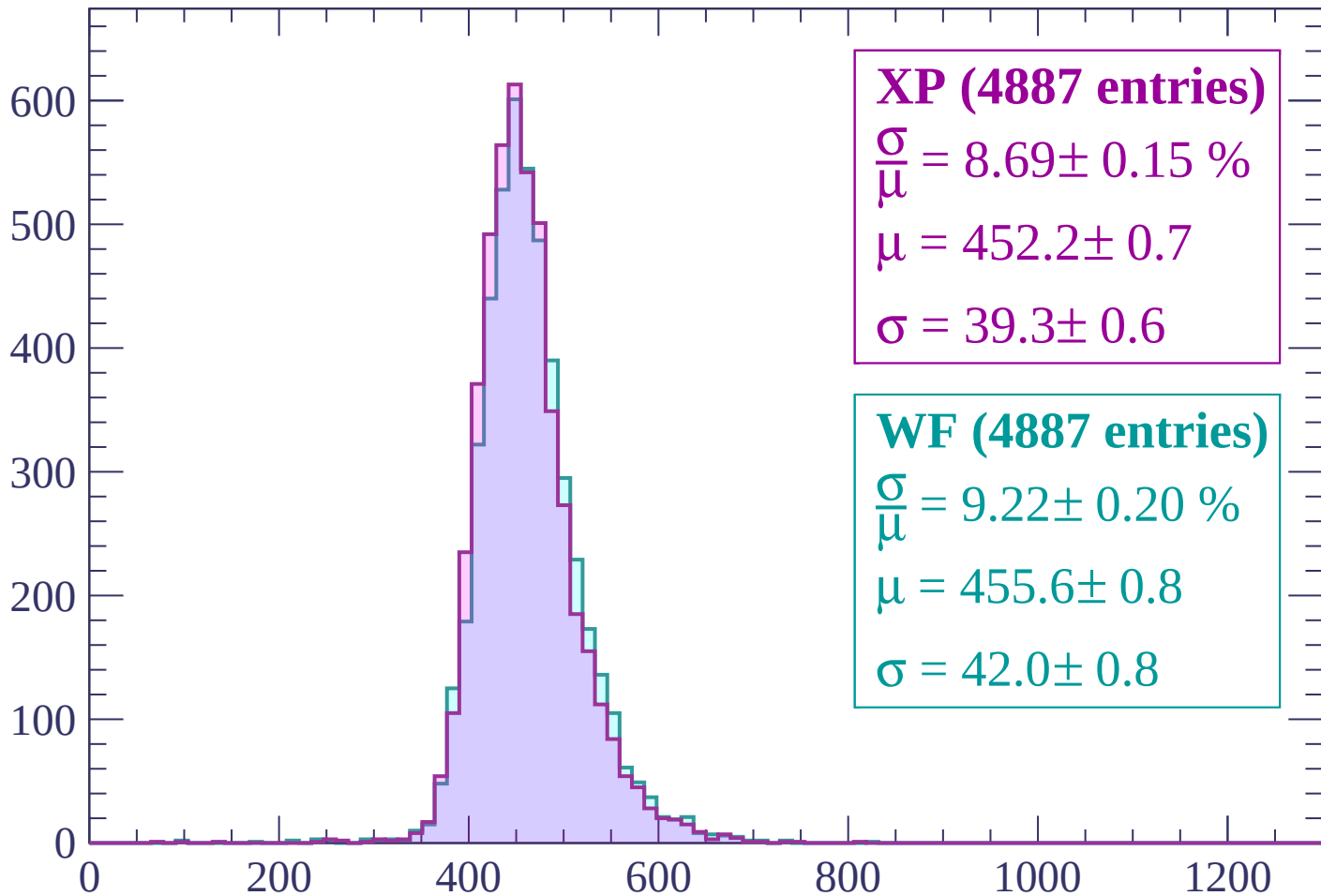
Energy loss | $1380 < p < 1440$

Count



Energy loss | $1440 < p < 1500$

Count



XP (4887 entries)

$$\frac{\sigma}{\mu} = 8.69 \pm 0.15 \%$$

$$\mu = 452.2 \pm 0.7$$

$$\sigma = 39.3 \pm 0.6$$

WF (4887 entries)

$$\frac{\sigma}{\mu} = 9.22 \pm 0.20 \%$$

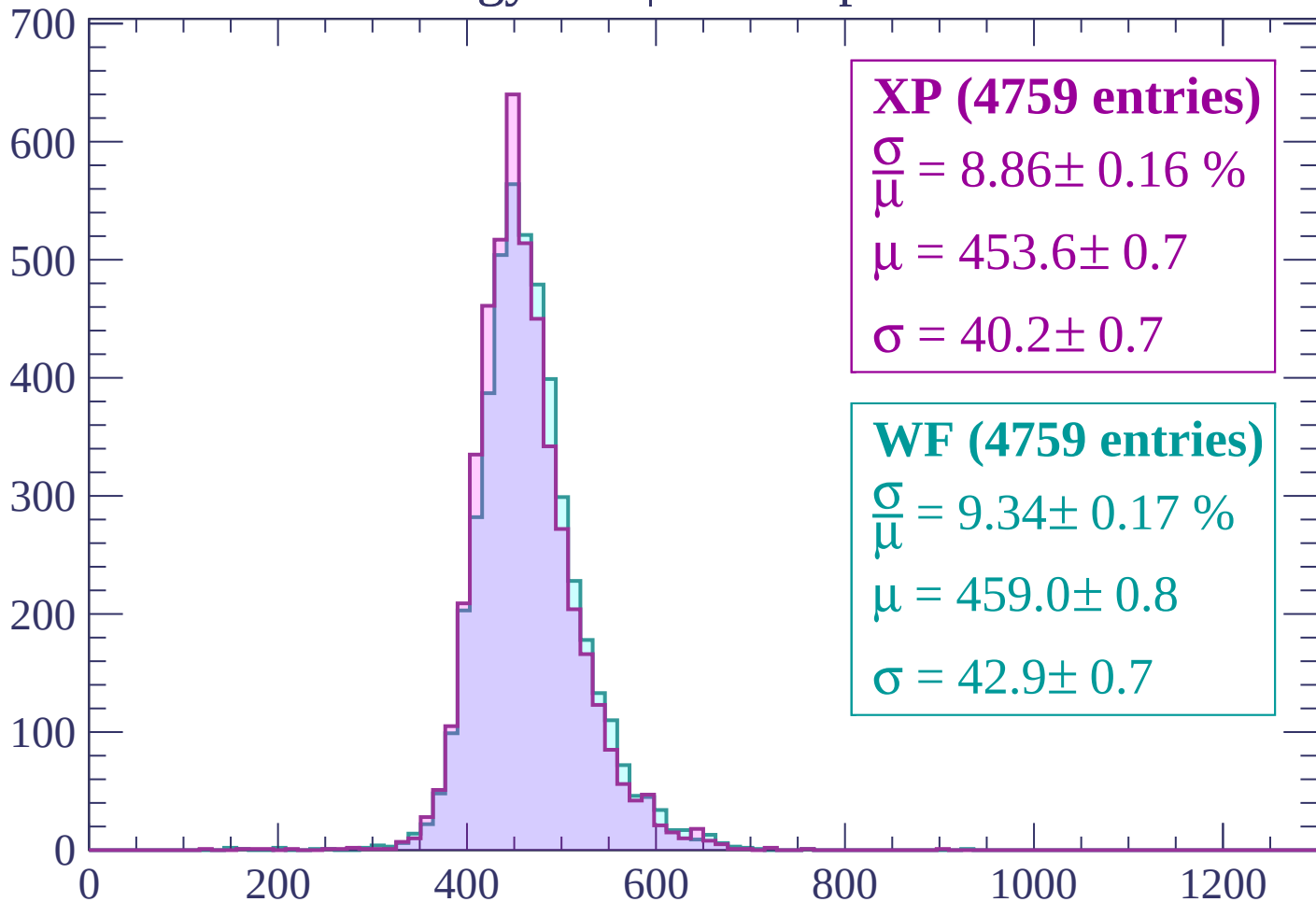
$$\mu = 455.6 \pm 0.8$$

$$\sigma = 42.0 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1500 < p < 1560$

Count



XP (4759 entries)

$$\frac{\sigma}{\mu} = 8.86 \pm 0.16 \%$$

$$\mu = 453.6 \pm 0.7$$

$$\sigma = 40.2 \pm 0.7$$

WF (4759 entries)

$$\frac{\sigma}{\mu} = 9.34 \pm 0.17 \%$$

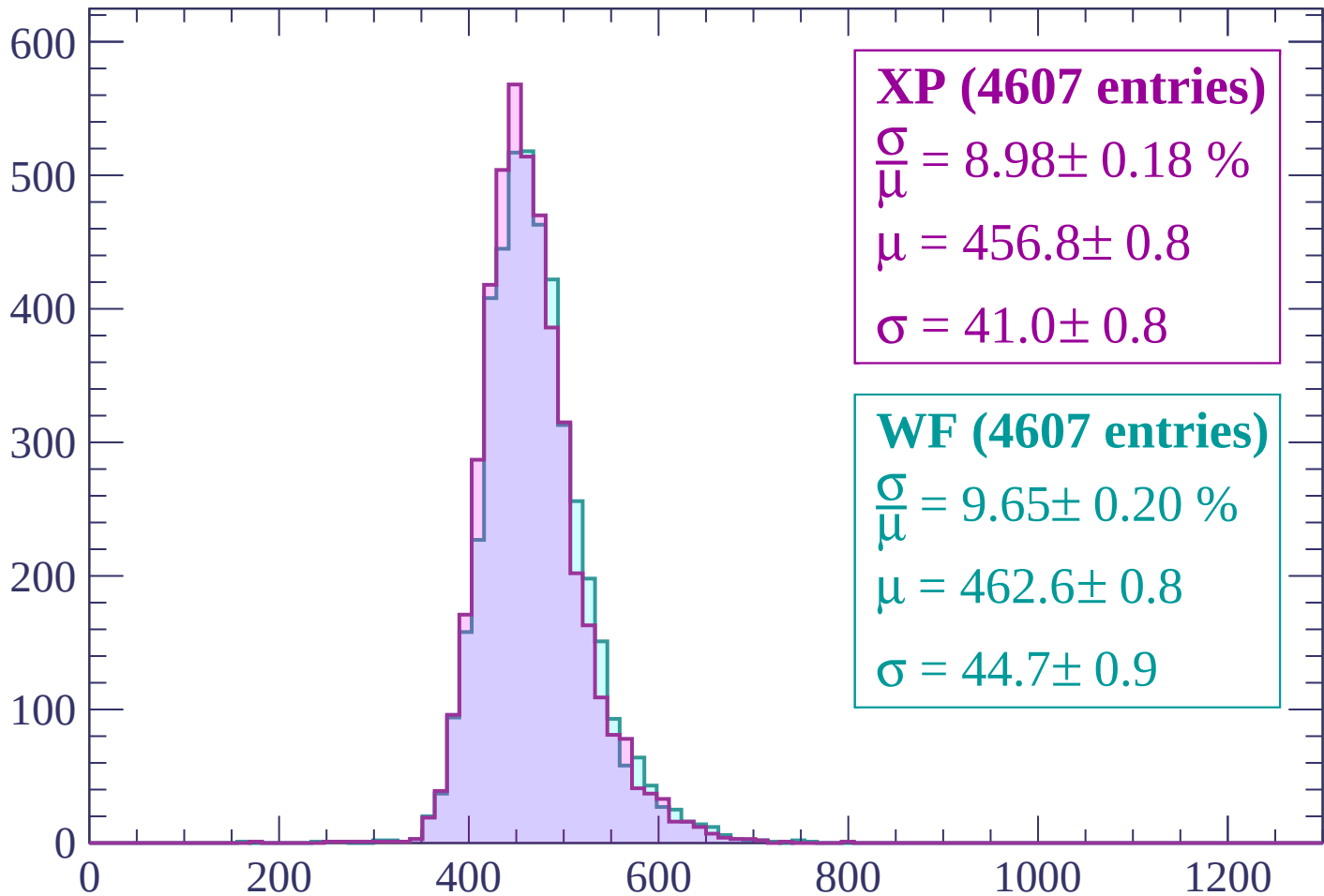
$$\mu = 459.0 \pm 0.8$$

$$\sigma = 42.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1560 < p < 1620$

Count



XP (4607 entries)

$$\frac{\sigma}{\mu} = 8.98 \pm 0.18 \%$$

$$\mu = 456.8 \pm 0.8$$

$$\sigma = 41.0 \pm 0.8$$

WF (4607 entries)

$$\frac{\sigma}{\mu} = 9.65 \pm 0.20 \%$$

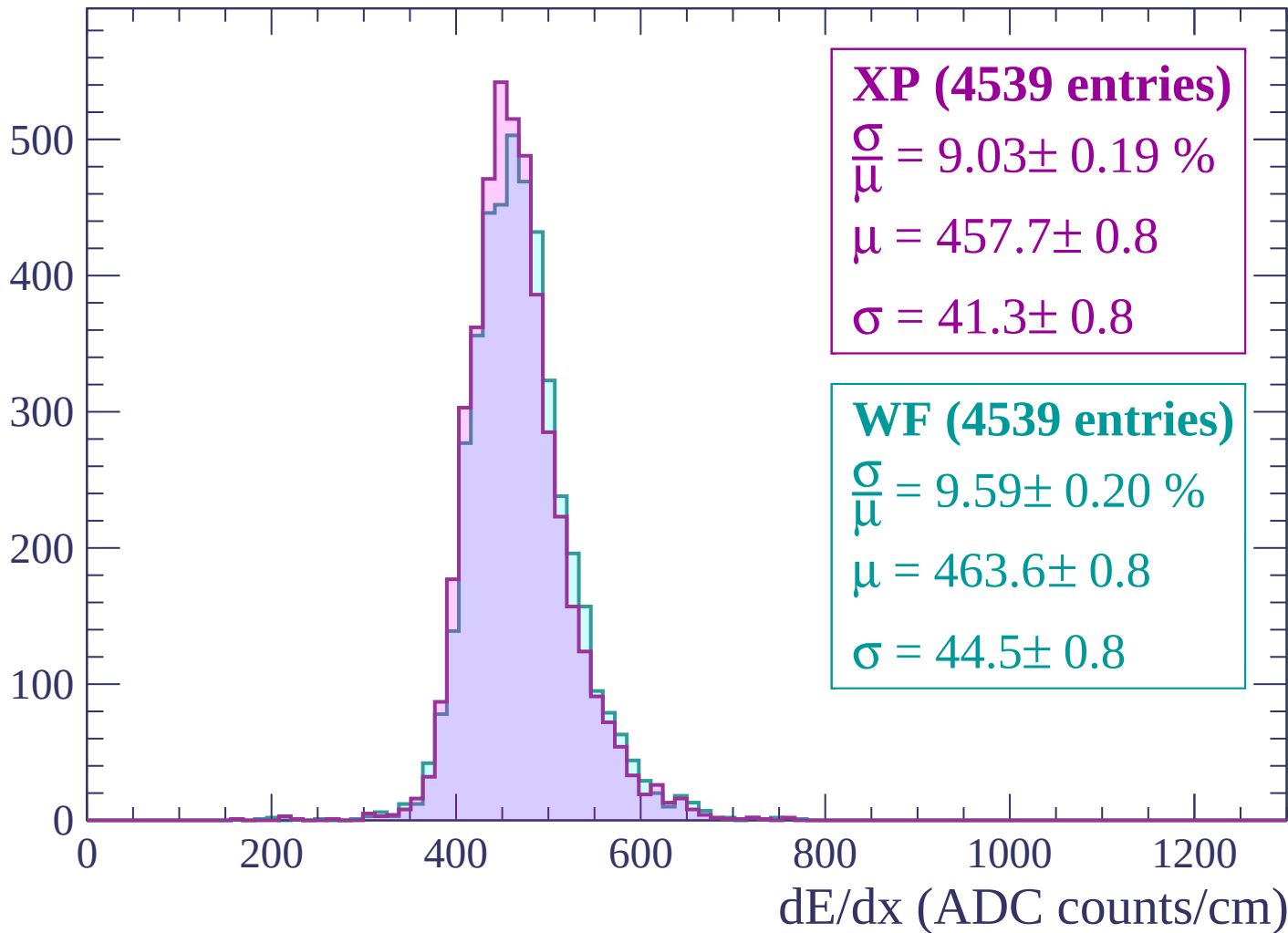
$$\mu = 462.6 \pm 0.8$$

$$\sigma = 44.7 \pm 0.9$$

dE/dx (ADC counts/cm)

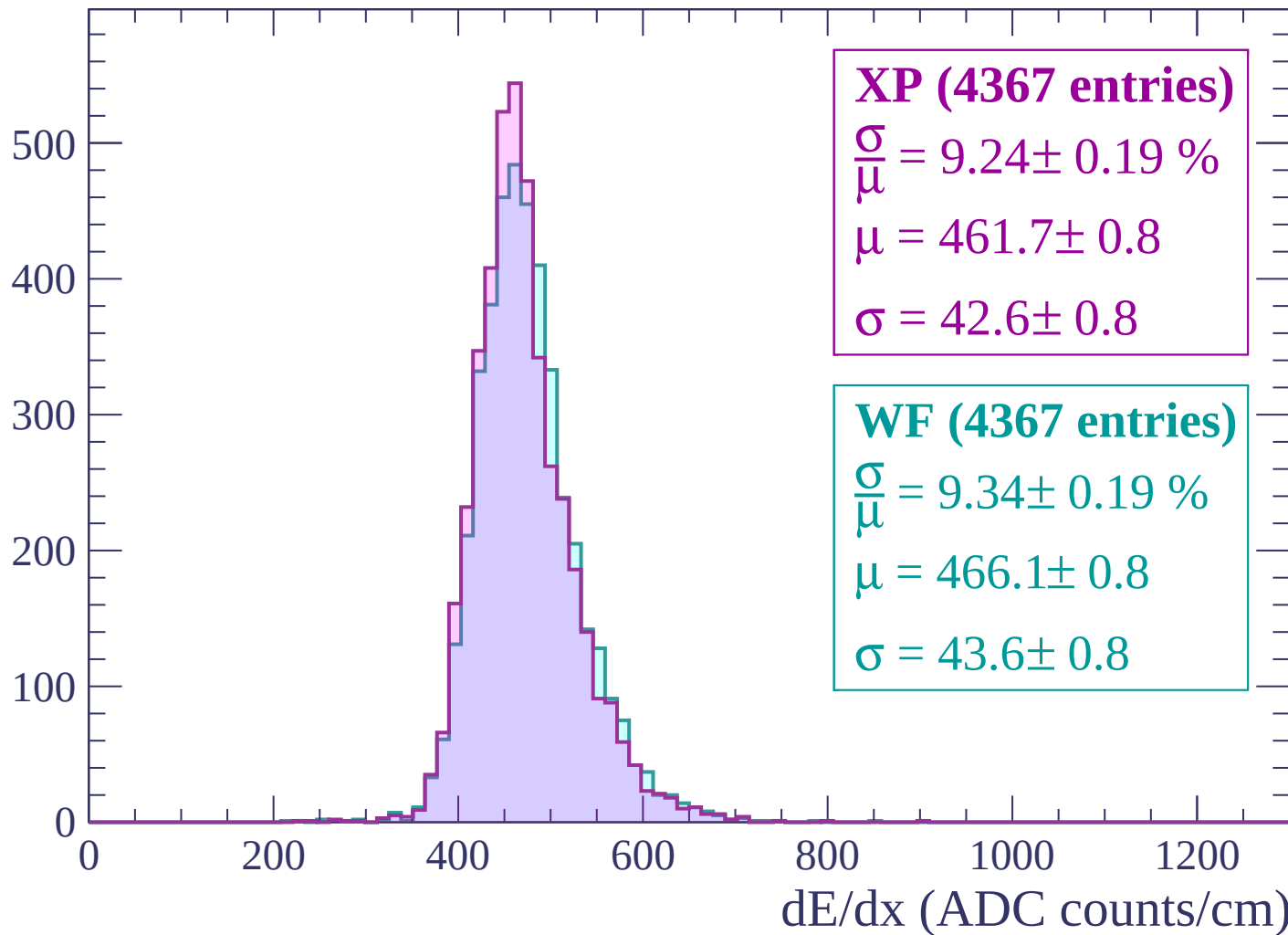
Energy loss | $1620 < p < 1680$

Count



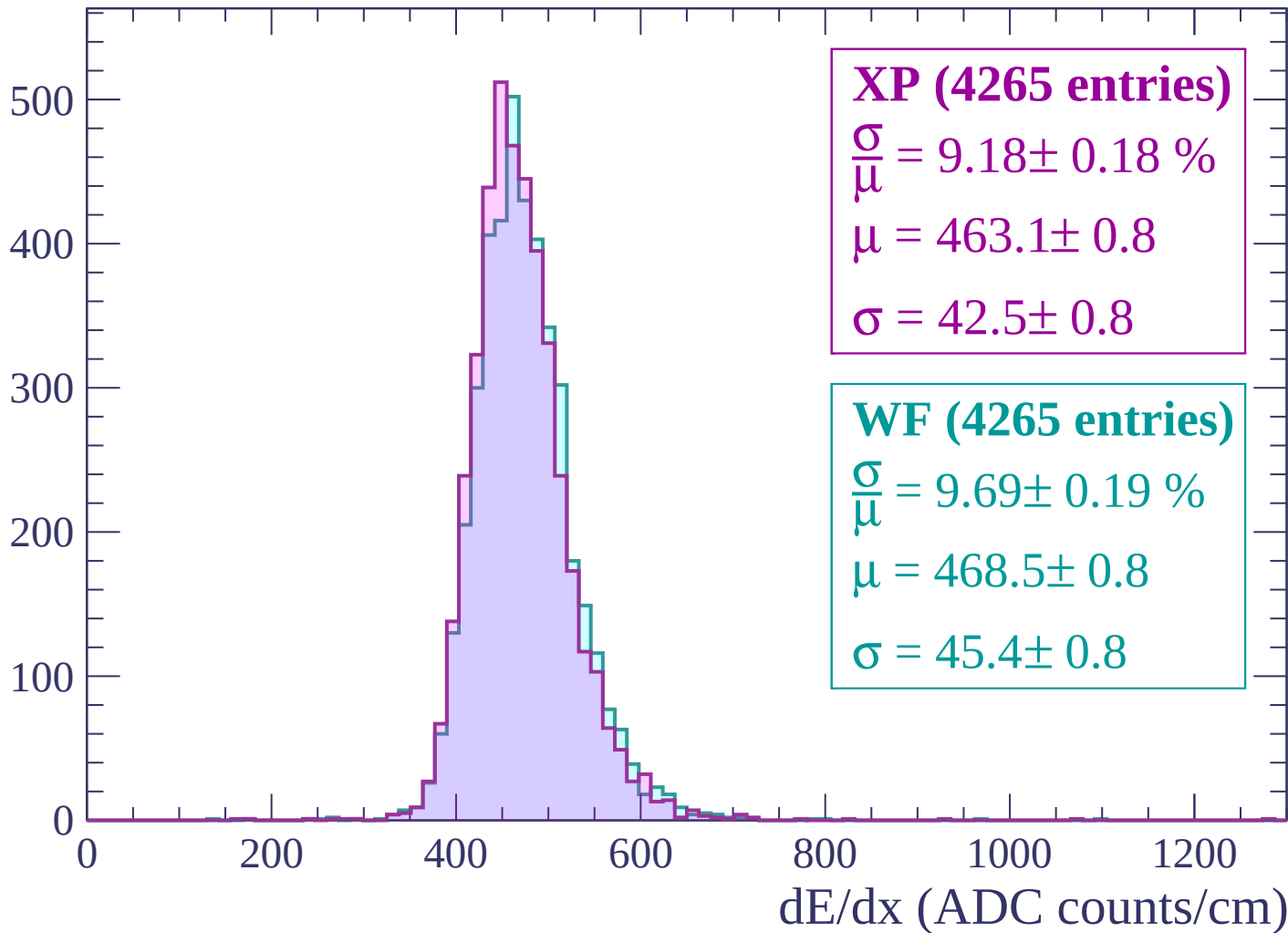
Energy loss | $1680 < p < 1740$

Count



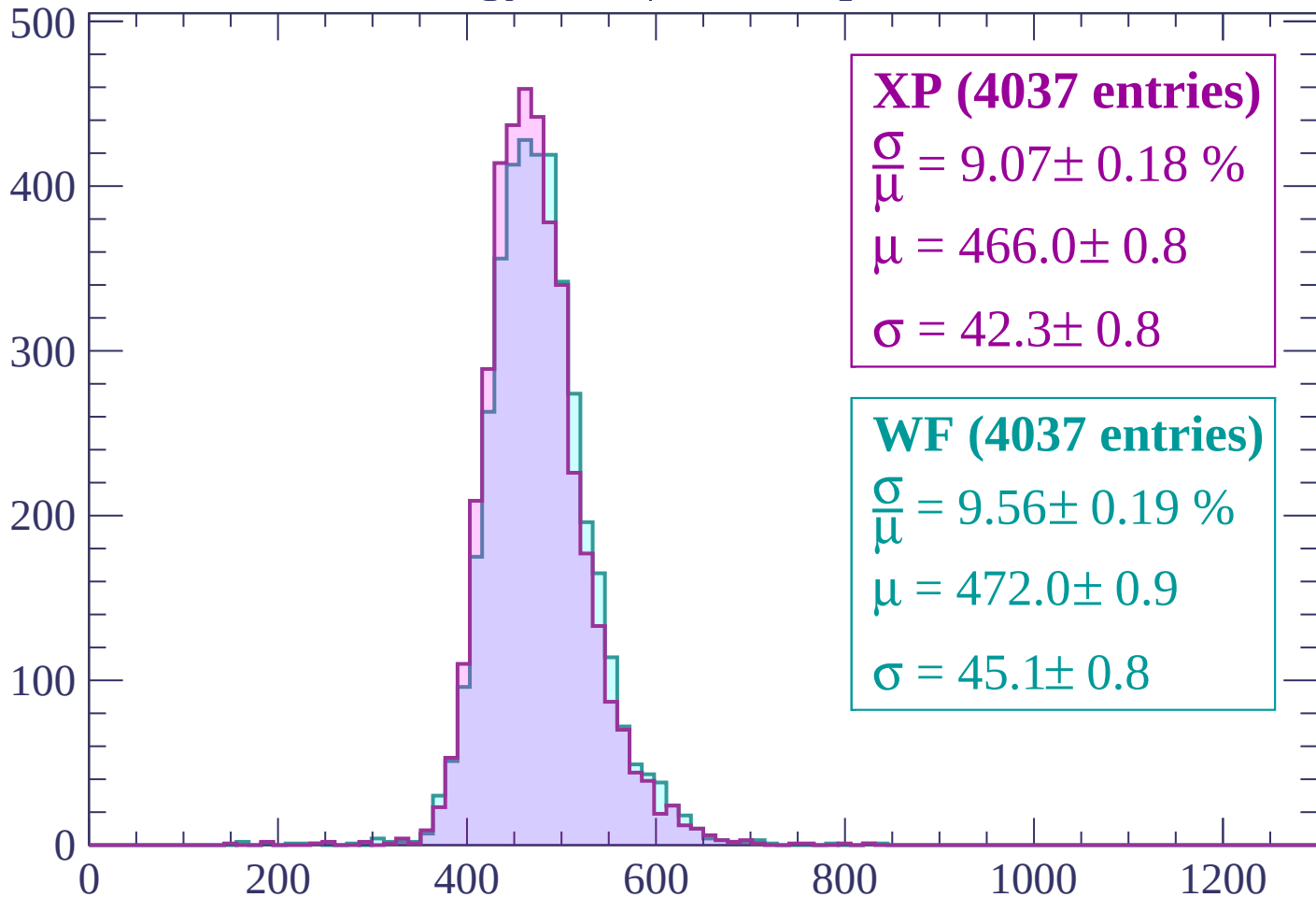
Energy loss | $1740 < p < 1800$

Count



Energy loss | $1800 < p < 1860$

Count



XP (4037 entries)

$$\frac{\sigma}{\mu} = 9.07 \pm 0.18 \%$$

$$\mu = 466.0 \pm 0.8$$

$$\sigma = 42.3 \pm 0.8$$

WF (4037 entries)

$$\frac{\sigma}{\mu} = 9.56 \pm 0.19 \%$$

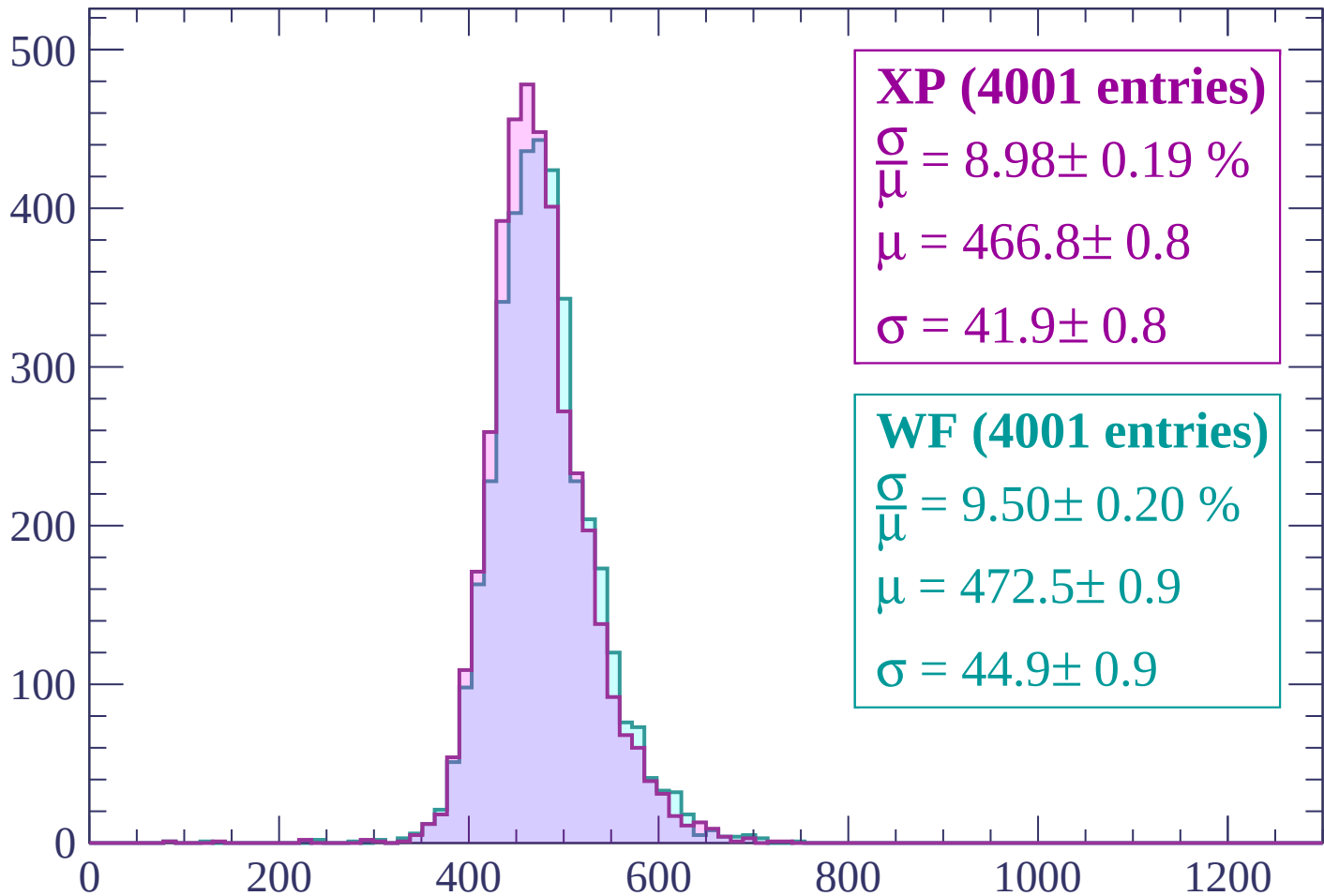
$$\mu = 472.0 \pm 0.9$$

$$\sigma = 45.1 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1860 < p < 1920$

Count



XP (4001 entries)

$$\frac{\sigma}{\mu} = 8.98 \pm 0.19 \%$$

$$\mu = 466.8 \pm 0.8$$

$$\sigma = 41.9 \pm 0.8$$

WF (4001 entries)

$$\frac{\sigma}{\mu} = 9.50 \pm 0.20 \%$$

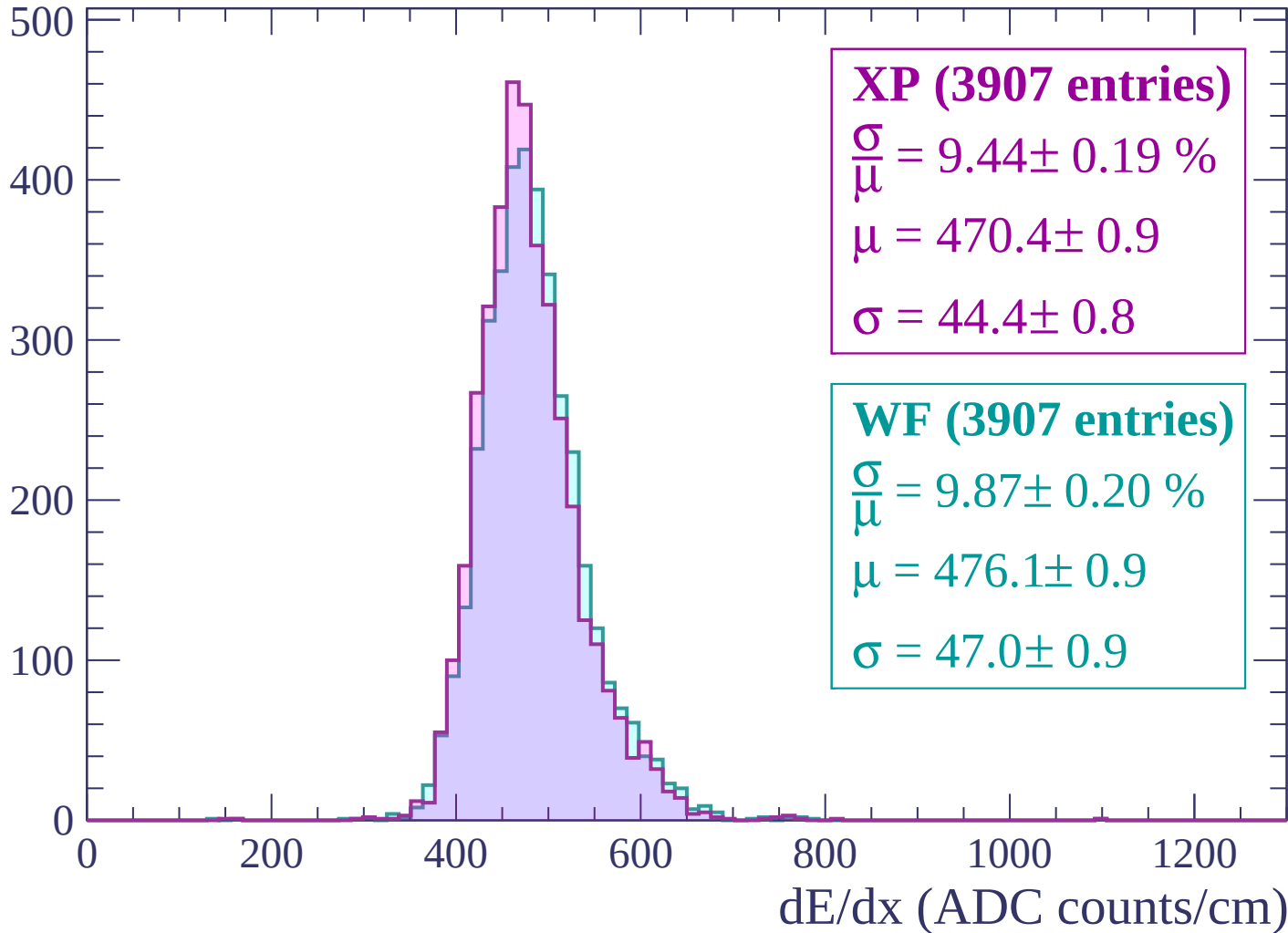
$$\mu = 472.5 \pm 0.9$$

$$\sigma = 44.9 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1920 < p < 1980$

Count



Energy loss | $1980 < p < 2040$

Count

500

400

300

200

100

0

0

200

400

600

800

1000

1200

XP (3818 entries)

$$\frac{\sigma}{\mu} = 9.14 \pm 0.23 \%$$

$$\mu = 468.9 \pm 0.9$$

$$\sigma = 42.8 \pm 1.0$$

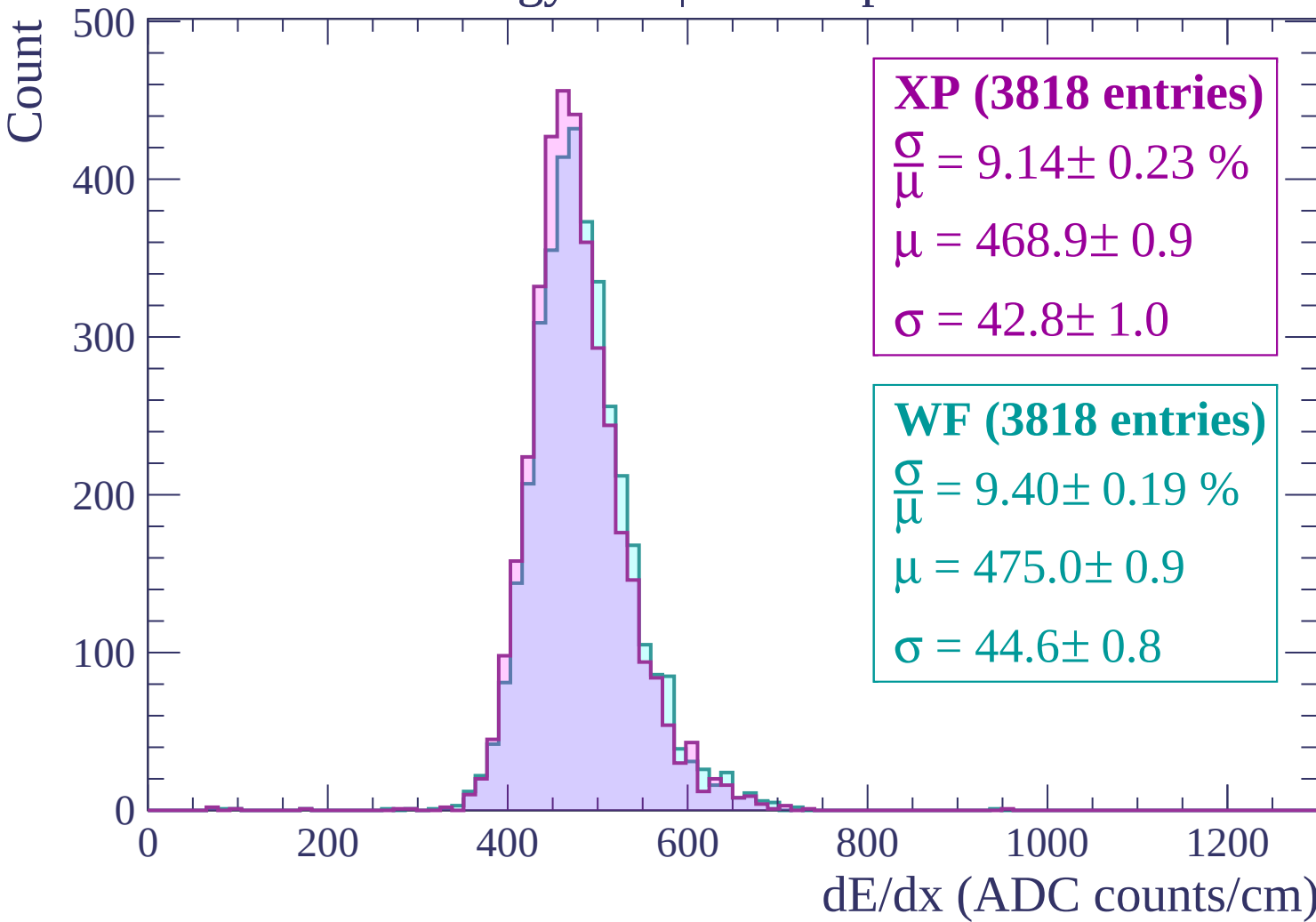
WF (3818 entries)

$$\frac{\sigma}{\mu} = 9.40 \pm 0.19 \%$$

$$\mu = 475.0 \pm 0.9$$

$$\sigma = 44.6 \pm 0.8$$

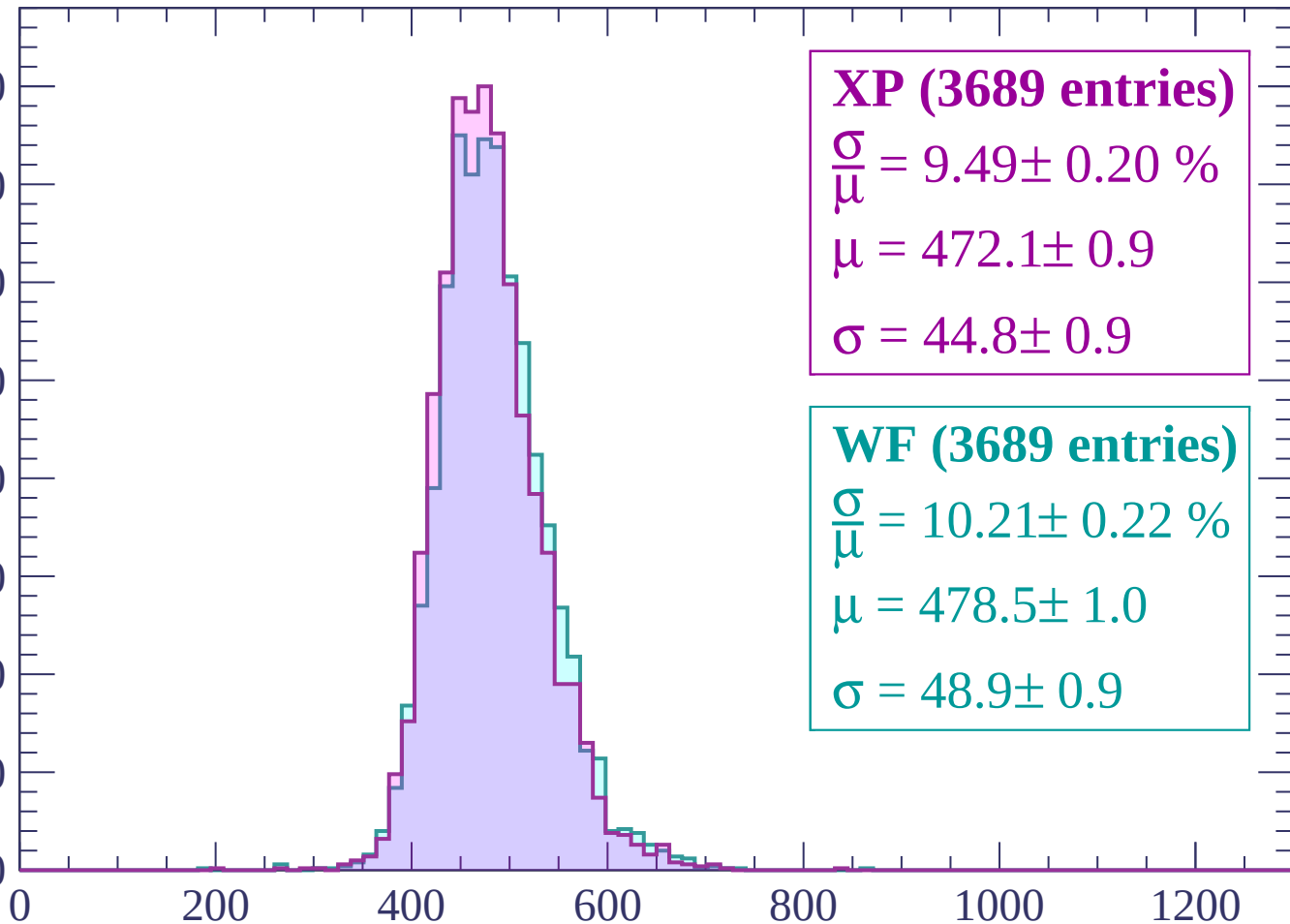
dE/dx (ADC counts/cm)



Energy loss | $2040 < p < 2100$

Count

400
350
300
250
200
150
100
50
0



XP (3689 entries)

$$\frac{\sigma}{\mu} = 9.49 \pm 0.20 \%$$

$$\mu = 472.1 \pm 0.9$$

$$\sigma = 44.8 \pm 0.9$$

WF (3689 entries)

$$\frac{\sigma}{\mu} = 10.21 \pm 0.22 \%$$

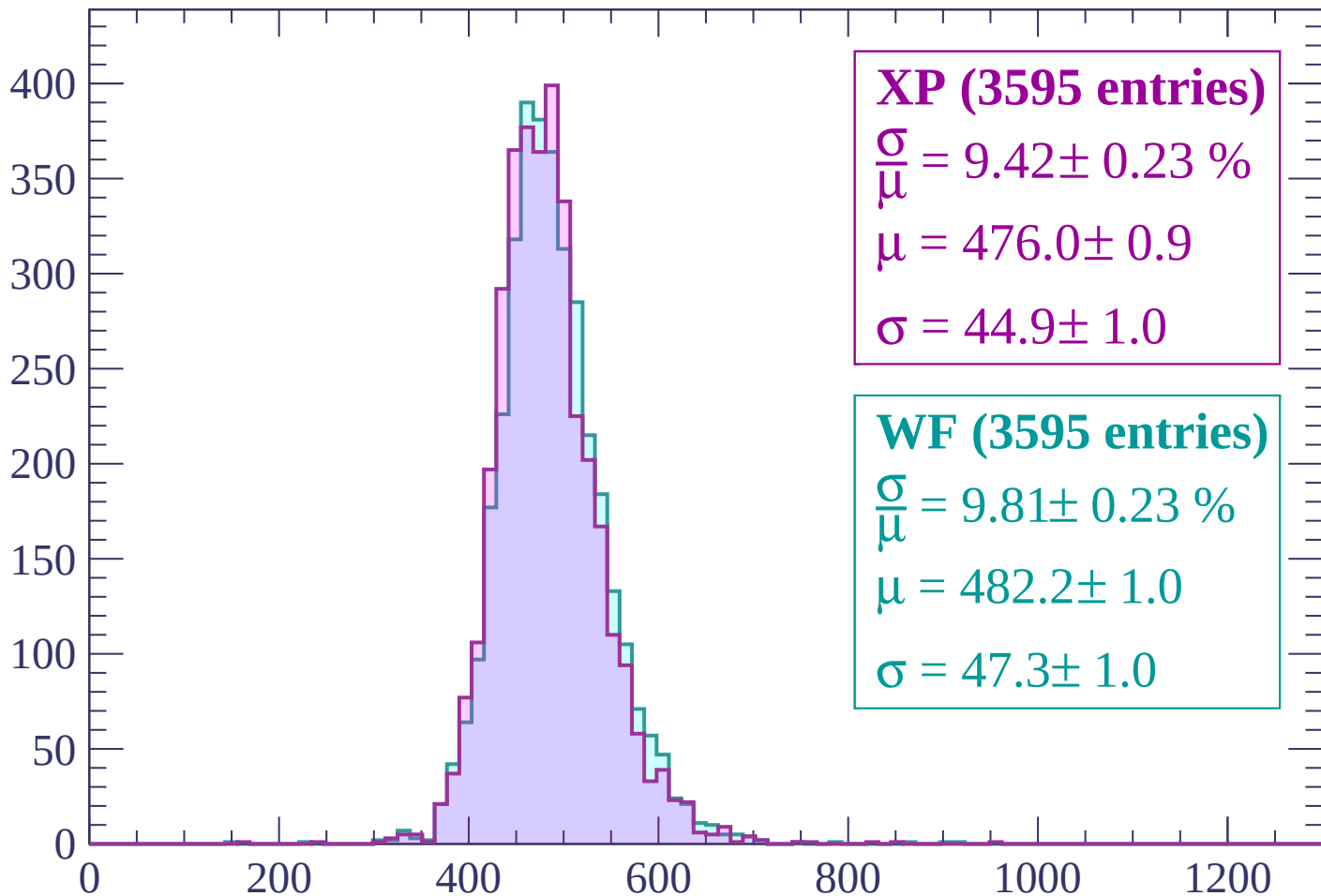
$$\mu = 478.5 \pm 1.0$$

$$\sigma = 48.9 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $2100 < p < 2160$

Count



XP (3595 entries)

$$\frac{\sigma}{\mu} = 9.42 \pm 0.23 \%$$

$$\mu = 476.0 \pm 0.9$$

$$\sigma = 44.9 \pm 1.0$$

WF (3595 entries)

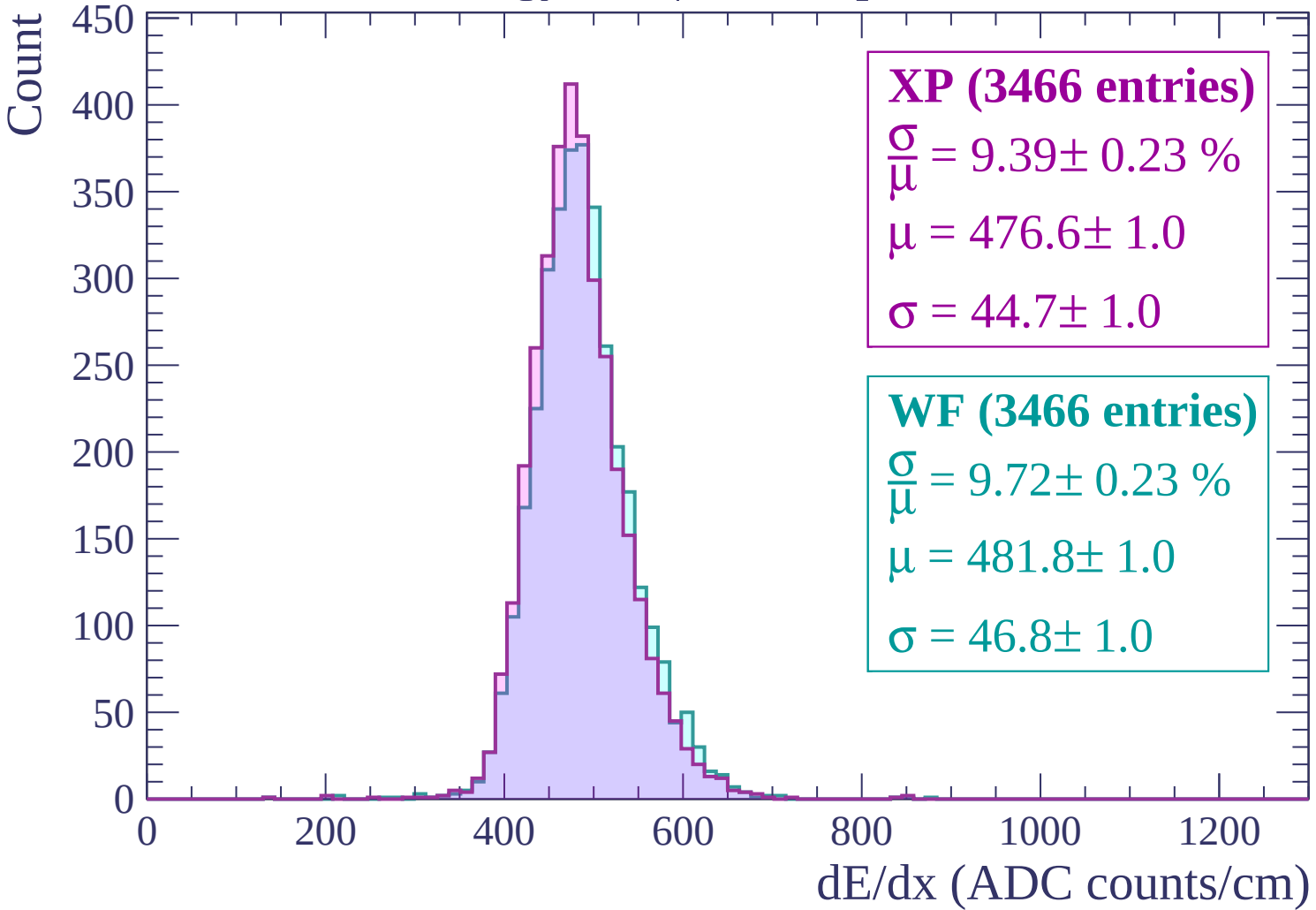
$$\frac{\sigma}{\mu} = 9.81 \pm 0.23 \%$$

$$\mu = 482.2 \pm 1.0$$

$$\sigma = 47.3 \pm 1.0$$

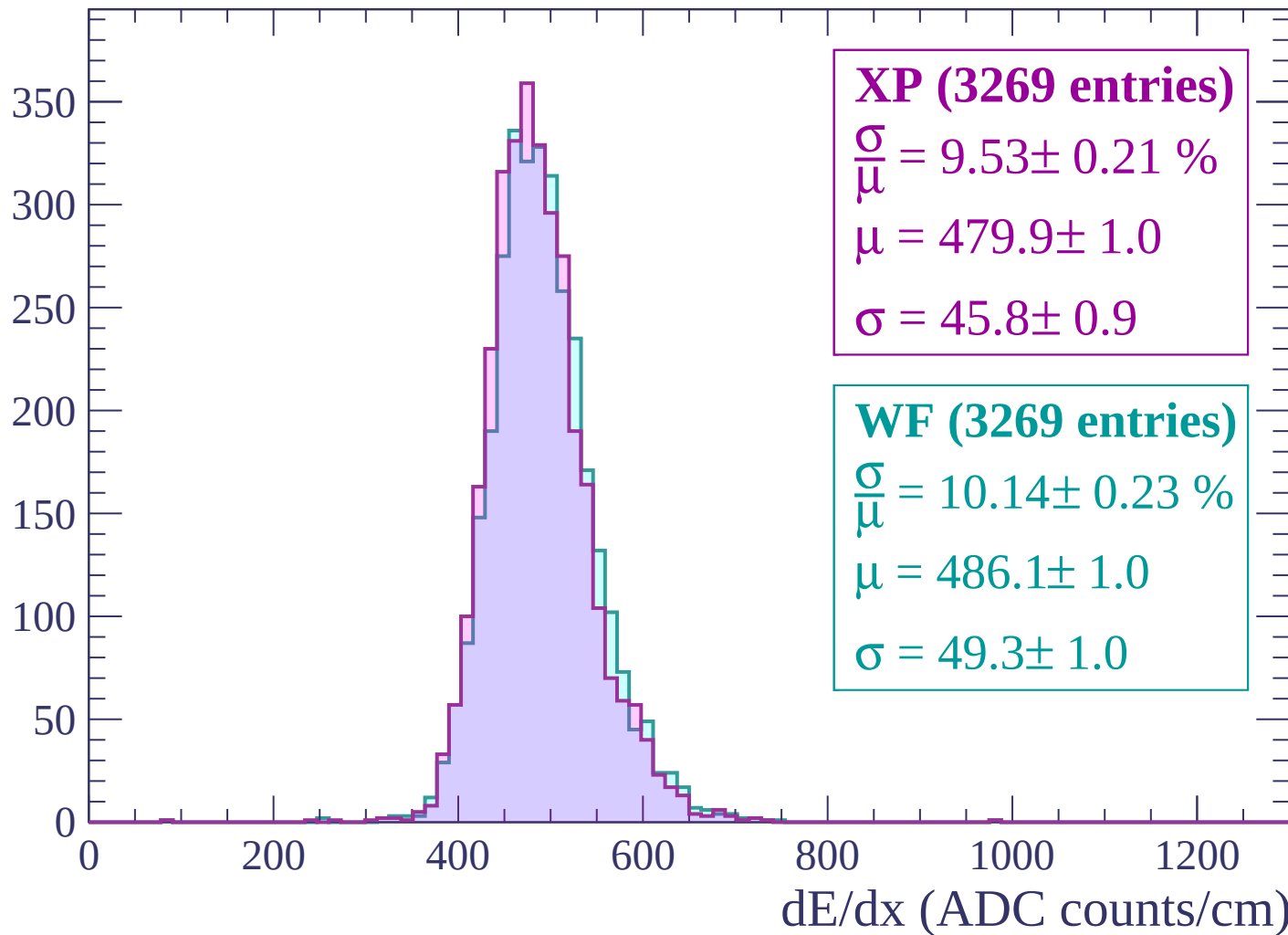
dE/dx (ADC counts/cm)

Energy loss | $2160 < p < 2220$



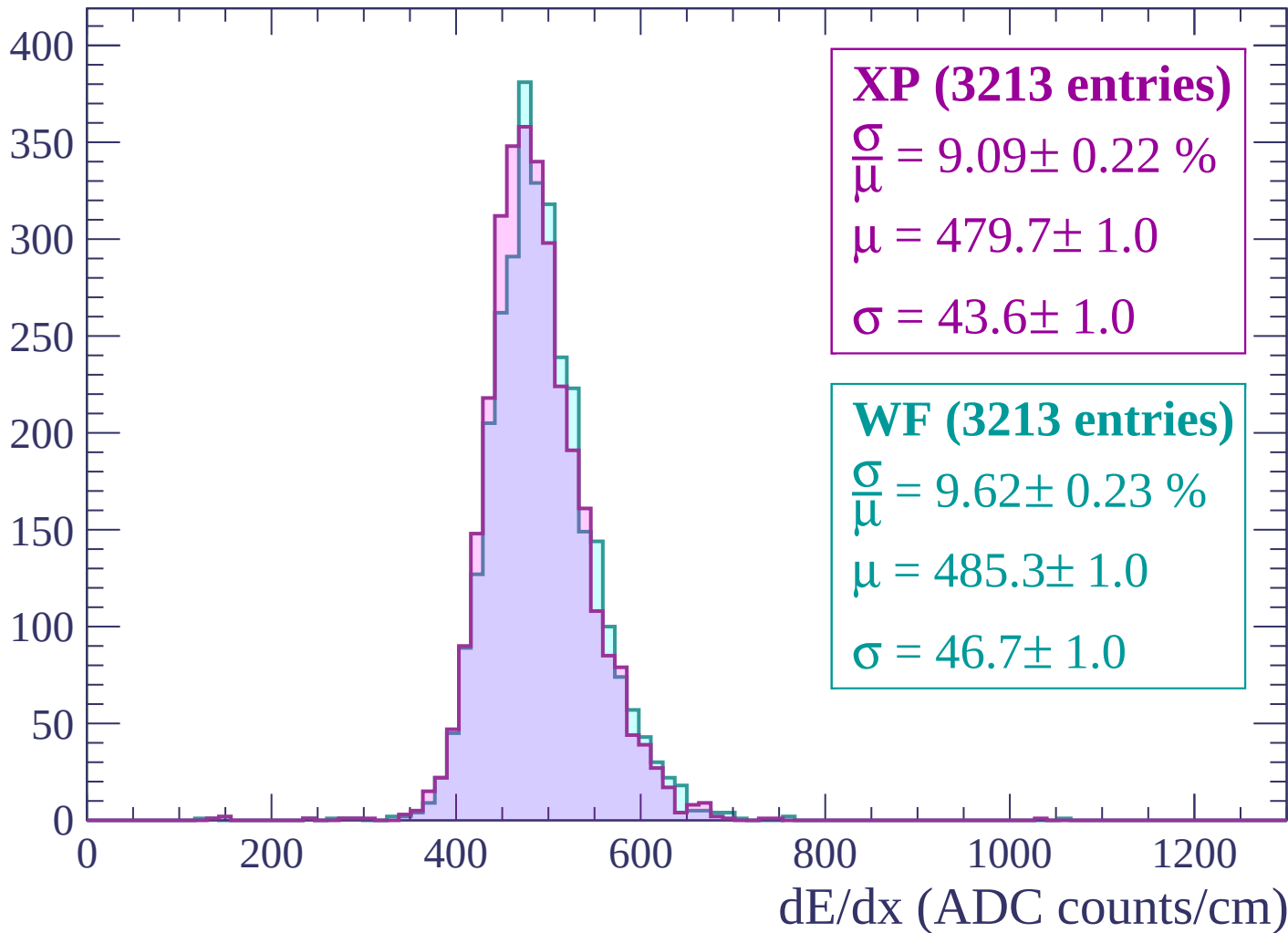
Energy loss | $2220 < p < 2280$

Count



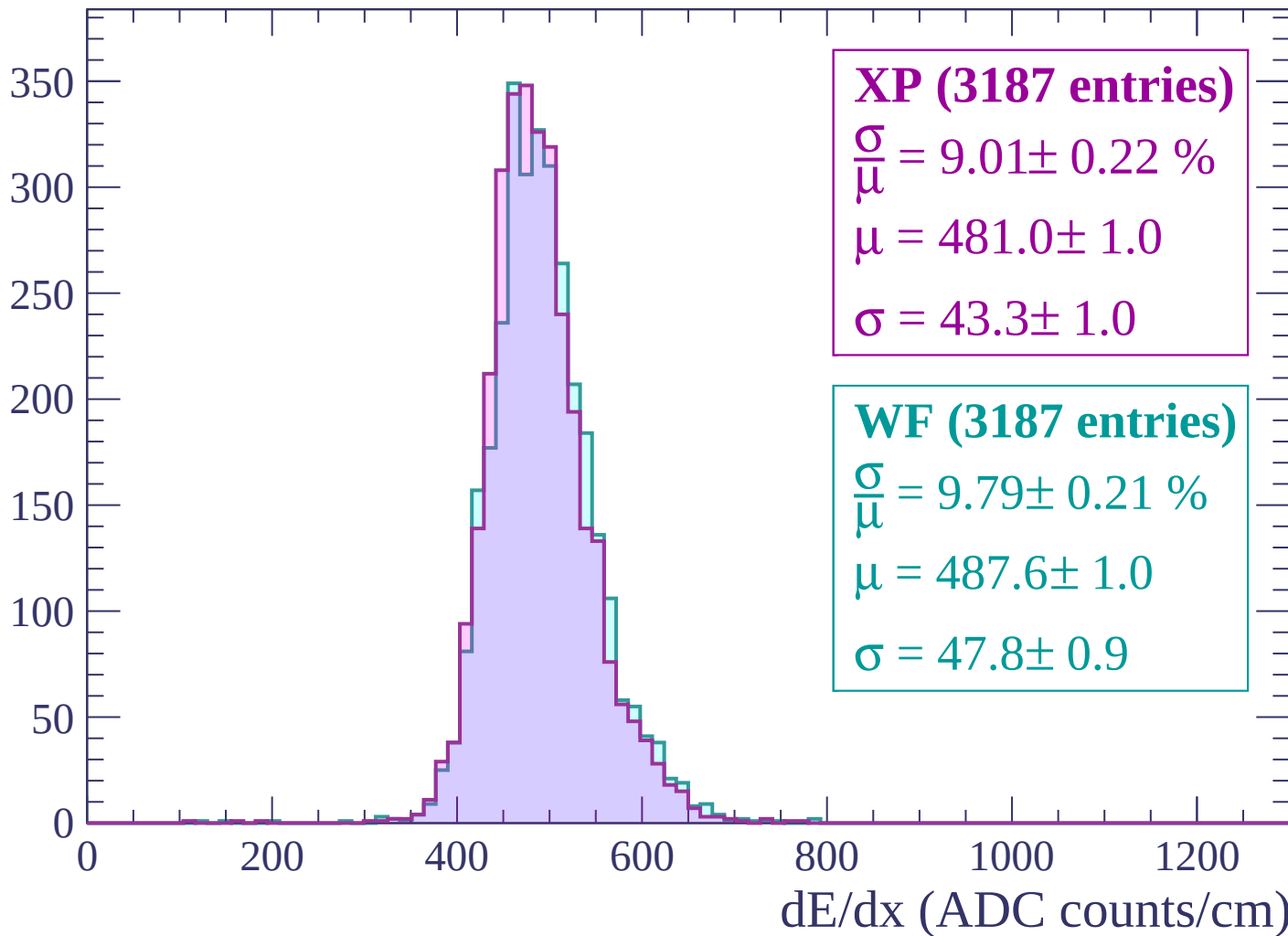
Energy loss | $2280 < p < 2340$

Count



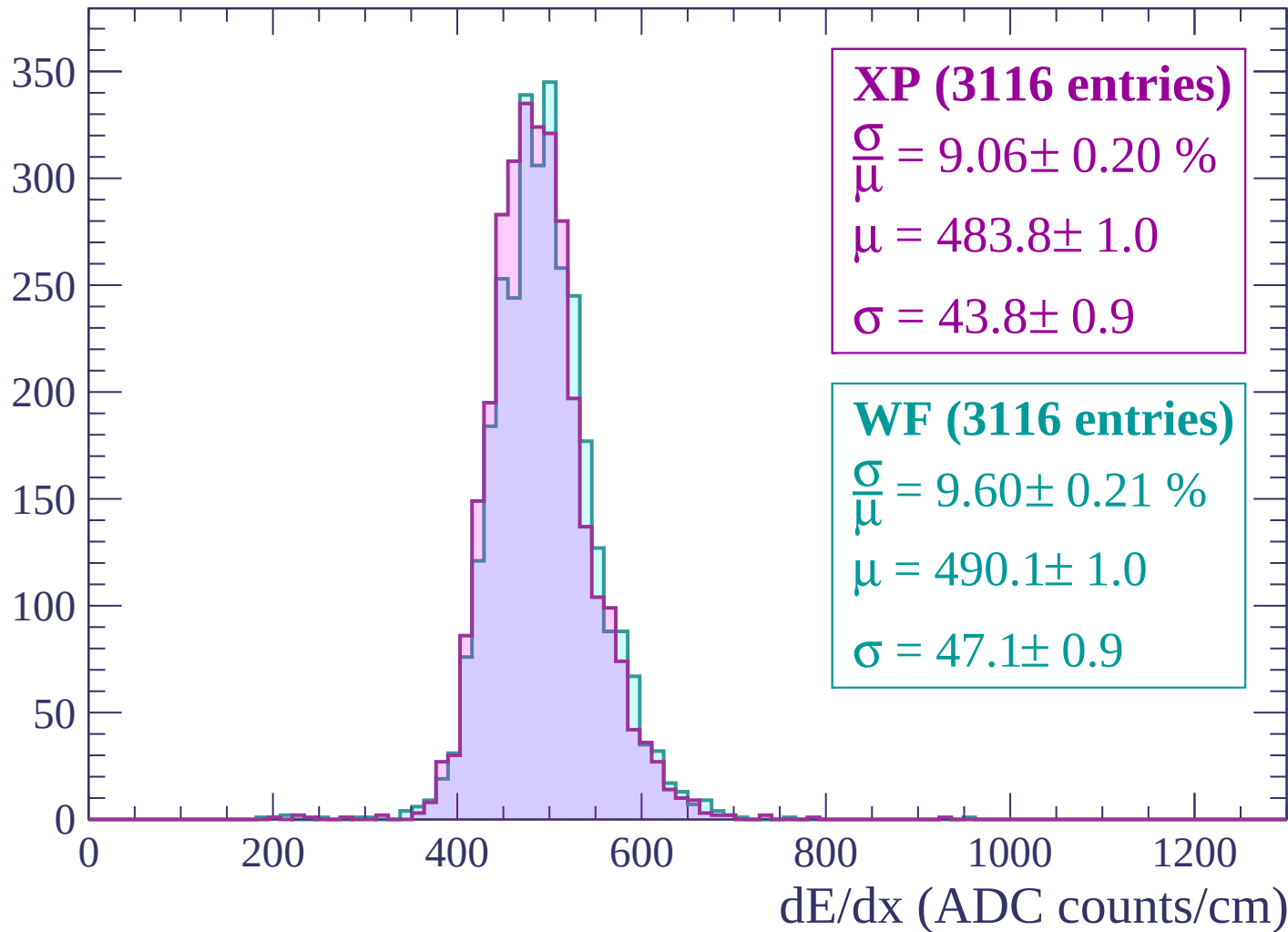
Energy loss | $2340 < p < 2400$

Count



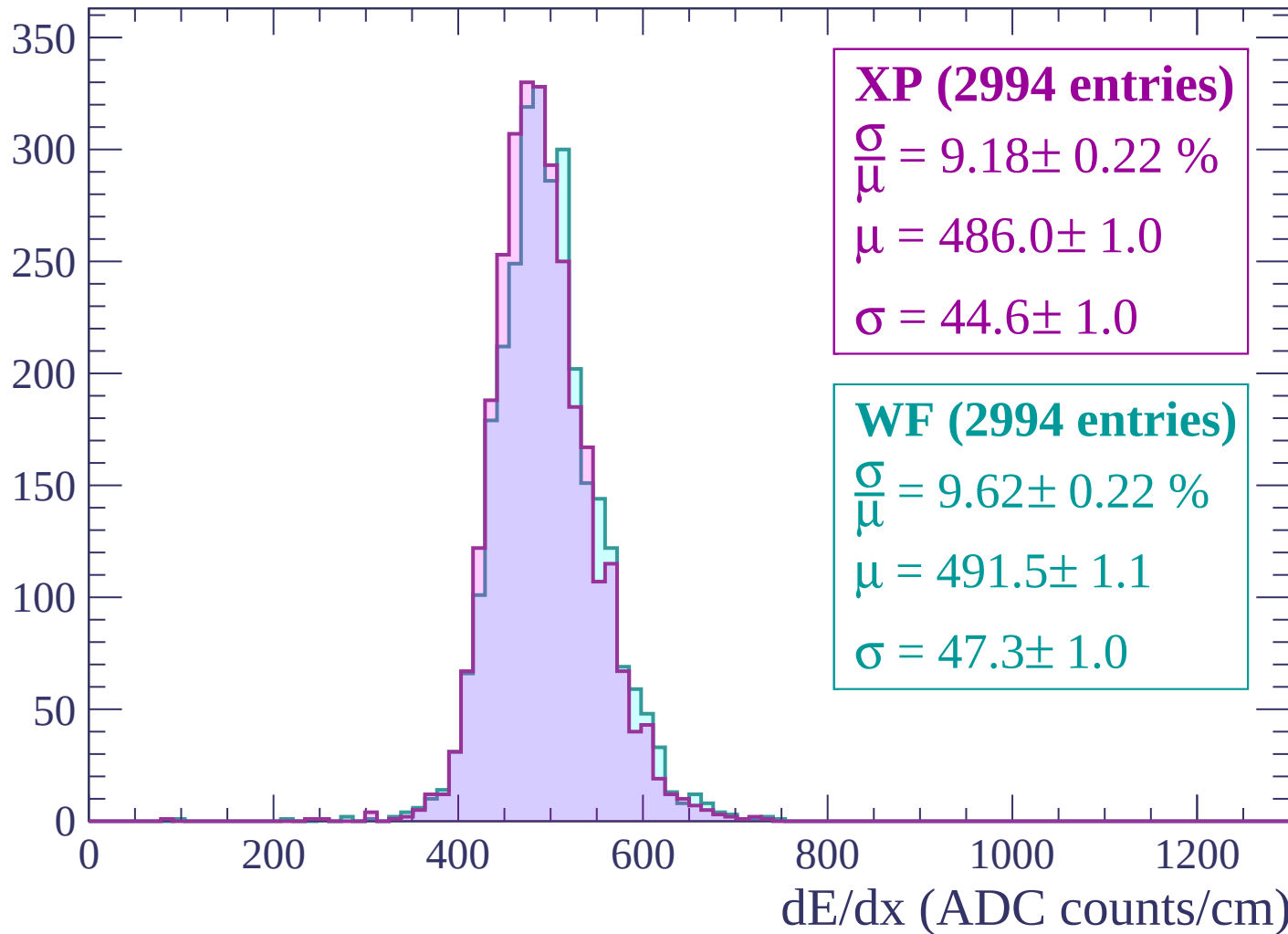
Energy loss | $2400 < p < 2460$

Count



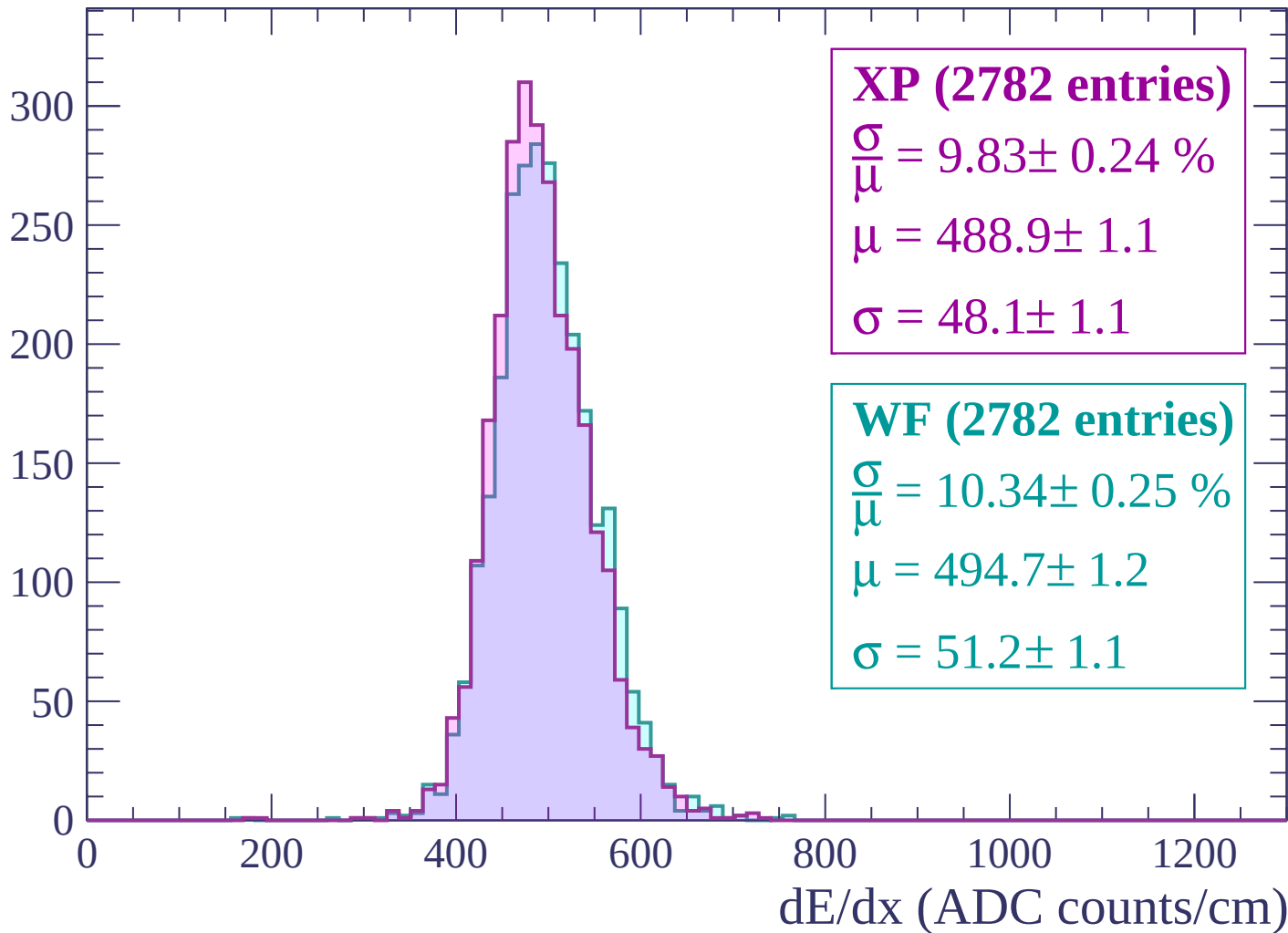
Energy loss | $2460 < p < 2520$

Count



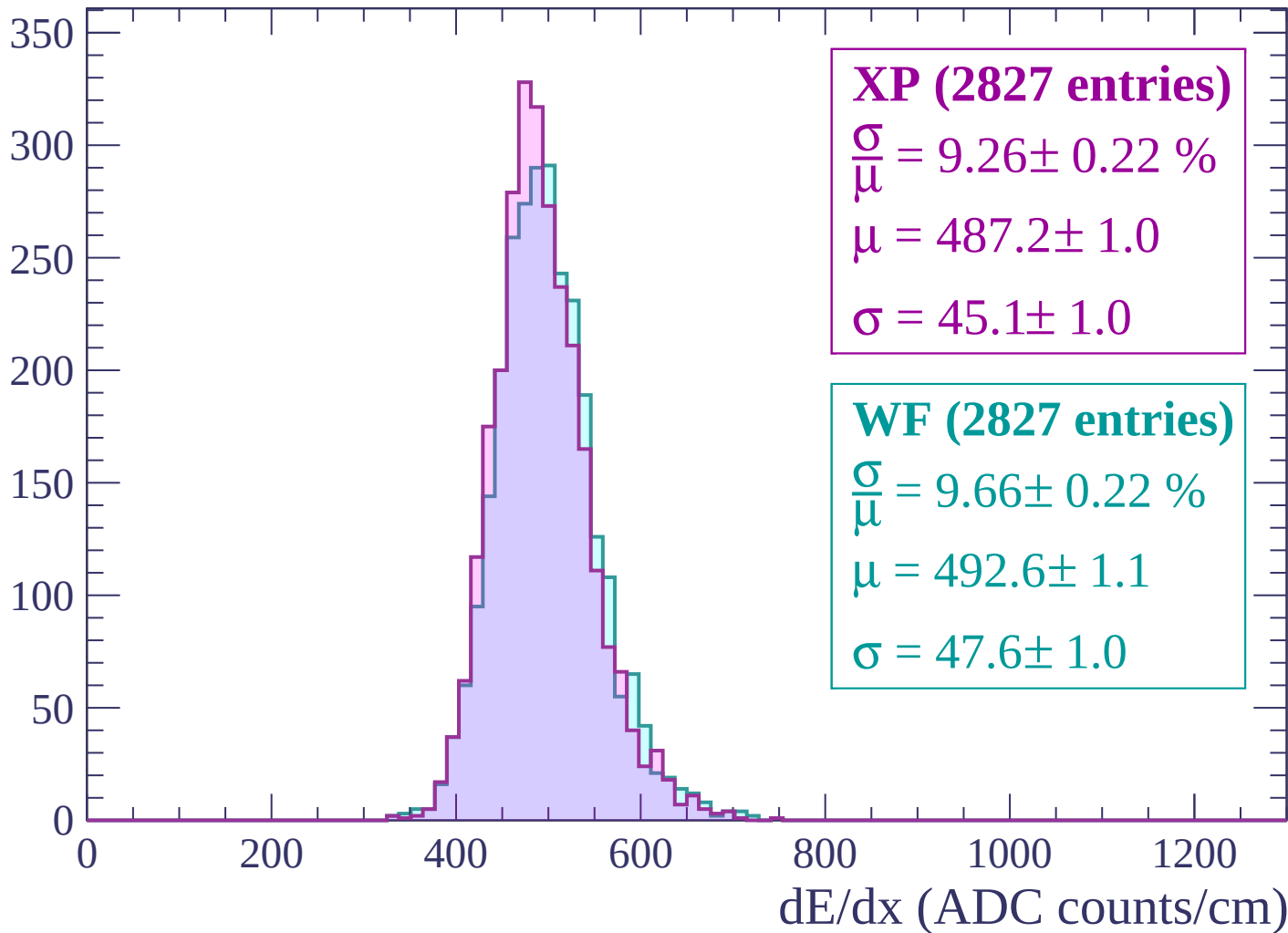
Energy loss | $2520 < p < 2580$

Count



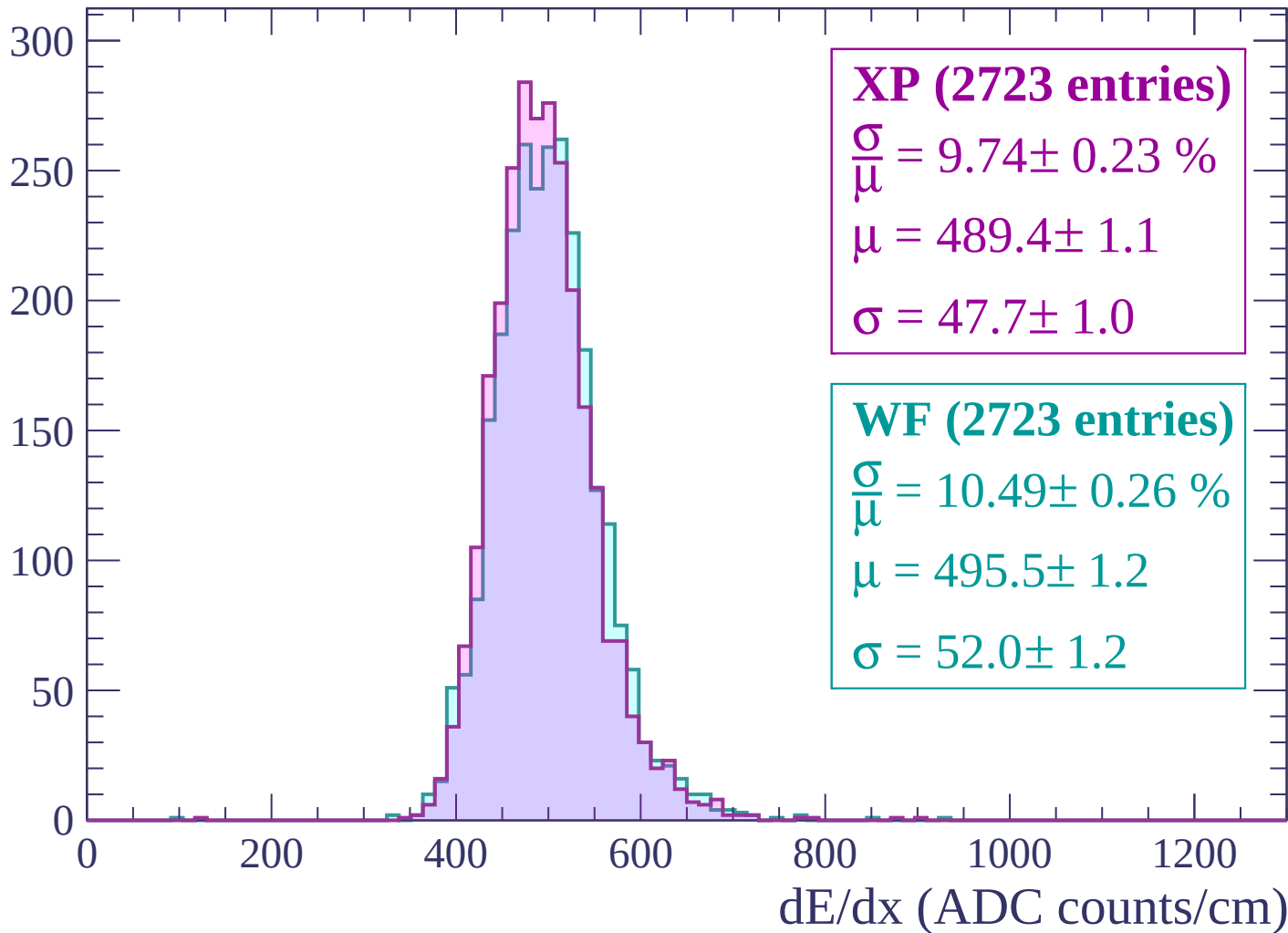
Energy loss | $2580 < p < 2640$

Count



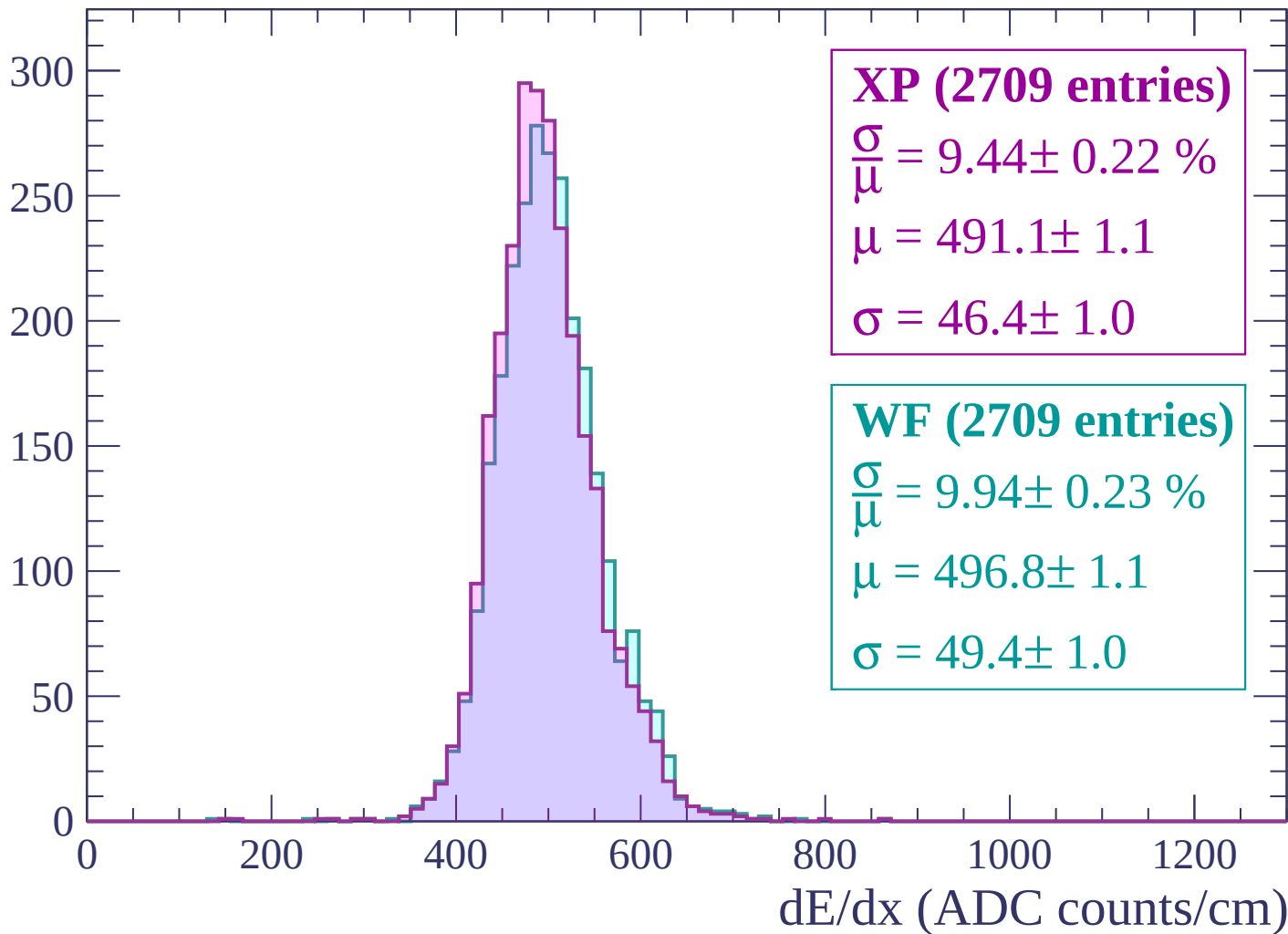
Energy loss | $2640 < p < 2700$

Count



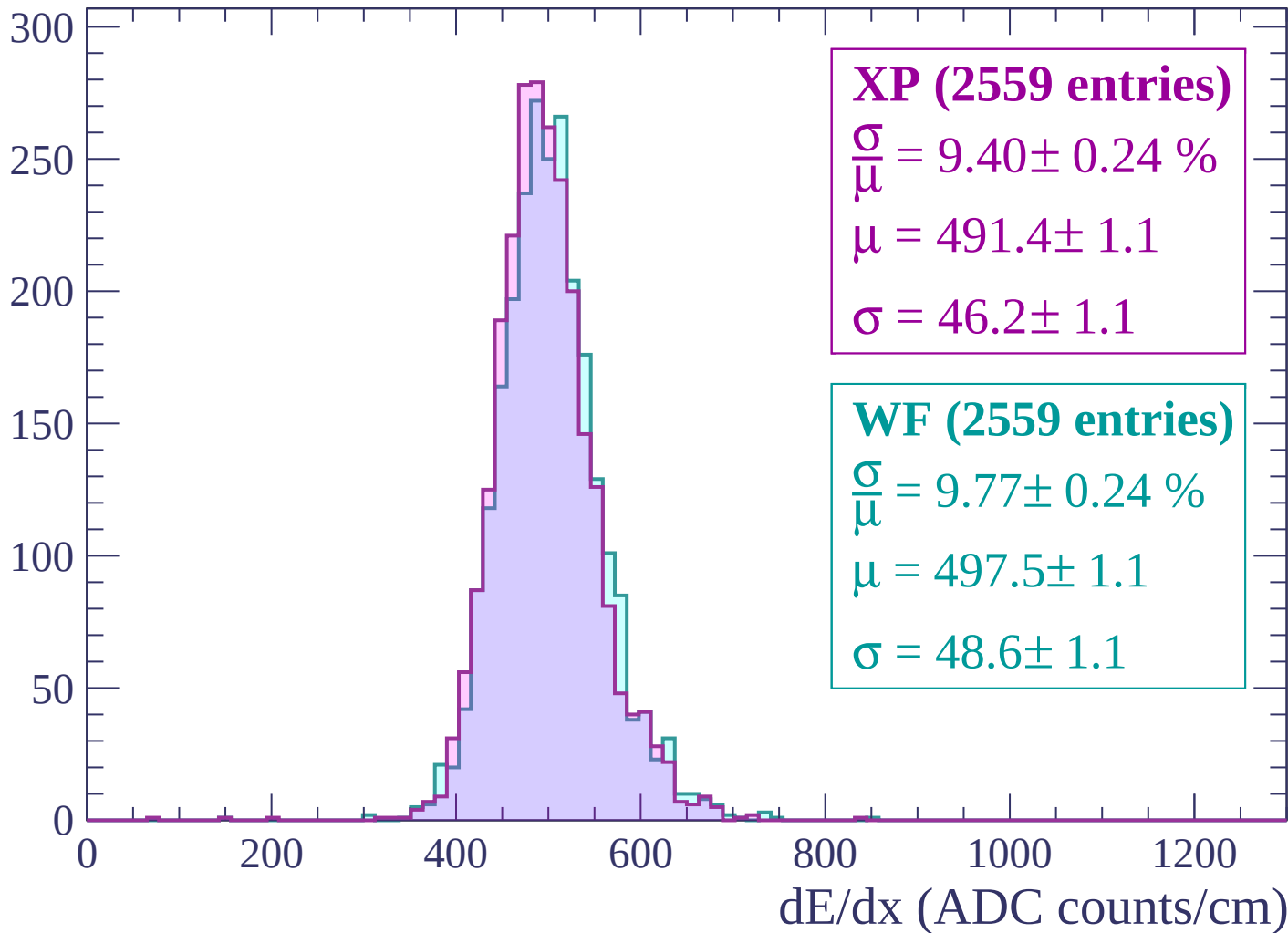
Energy loss | $2700 < p < 2760$

Count



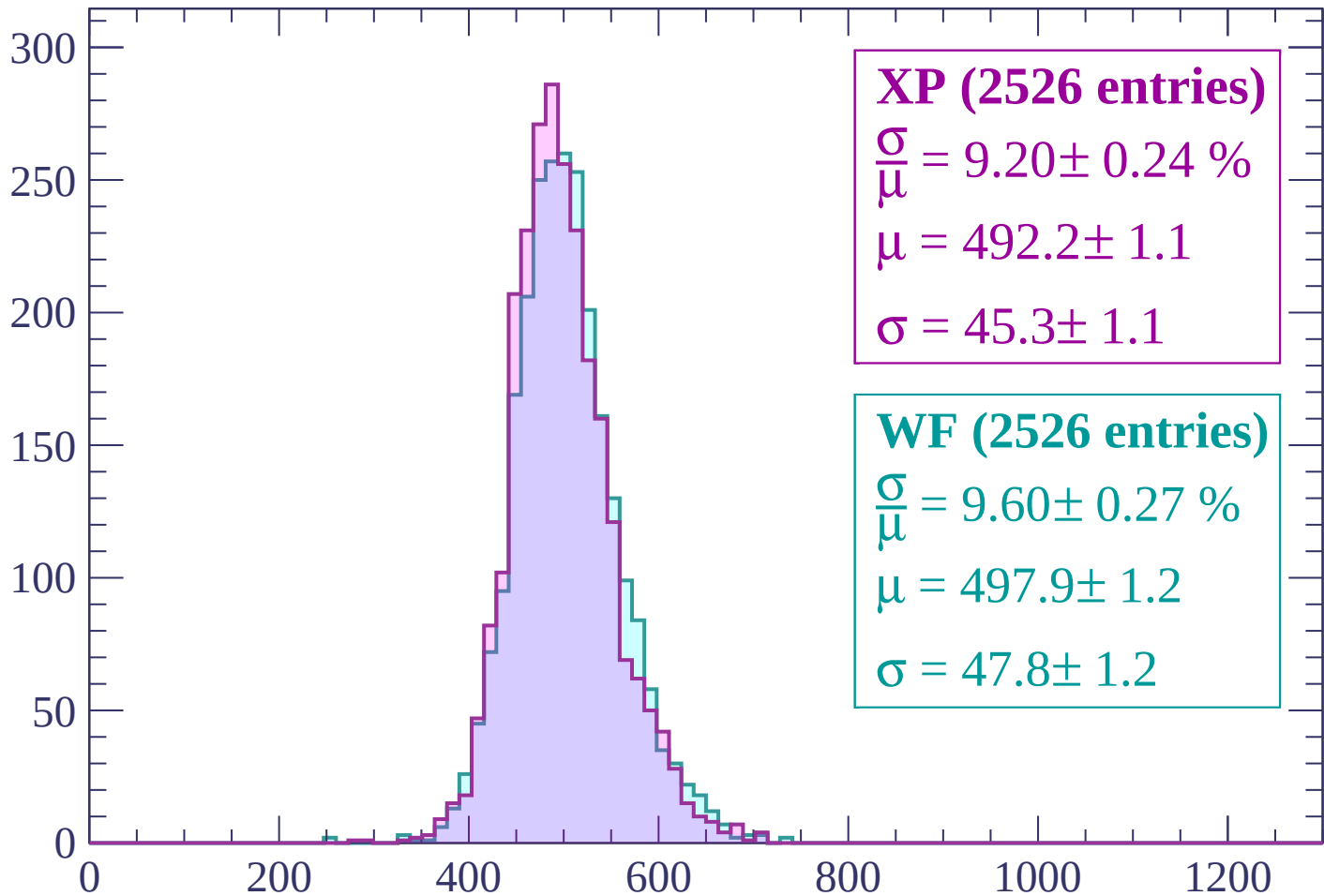
Energy loss | $2760 < p < 2820$

Count



Energy loss | $2820 < p < 2880$

Count



XP (2526 entries)

$\frac{\sigma}{\mu} = 9.20 \pm 0.24 \%$

$\mu = 492.2 \pm 1.1$

$\sigma = 45.3 \pm 1.1$

WF (2526 entries)

$\frac{\sigma}{\mu} = 9.60 \pm 0.27 \%$

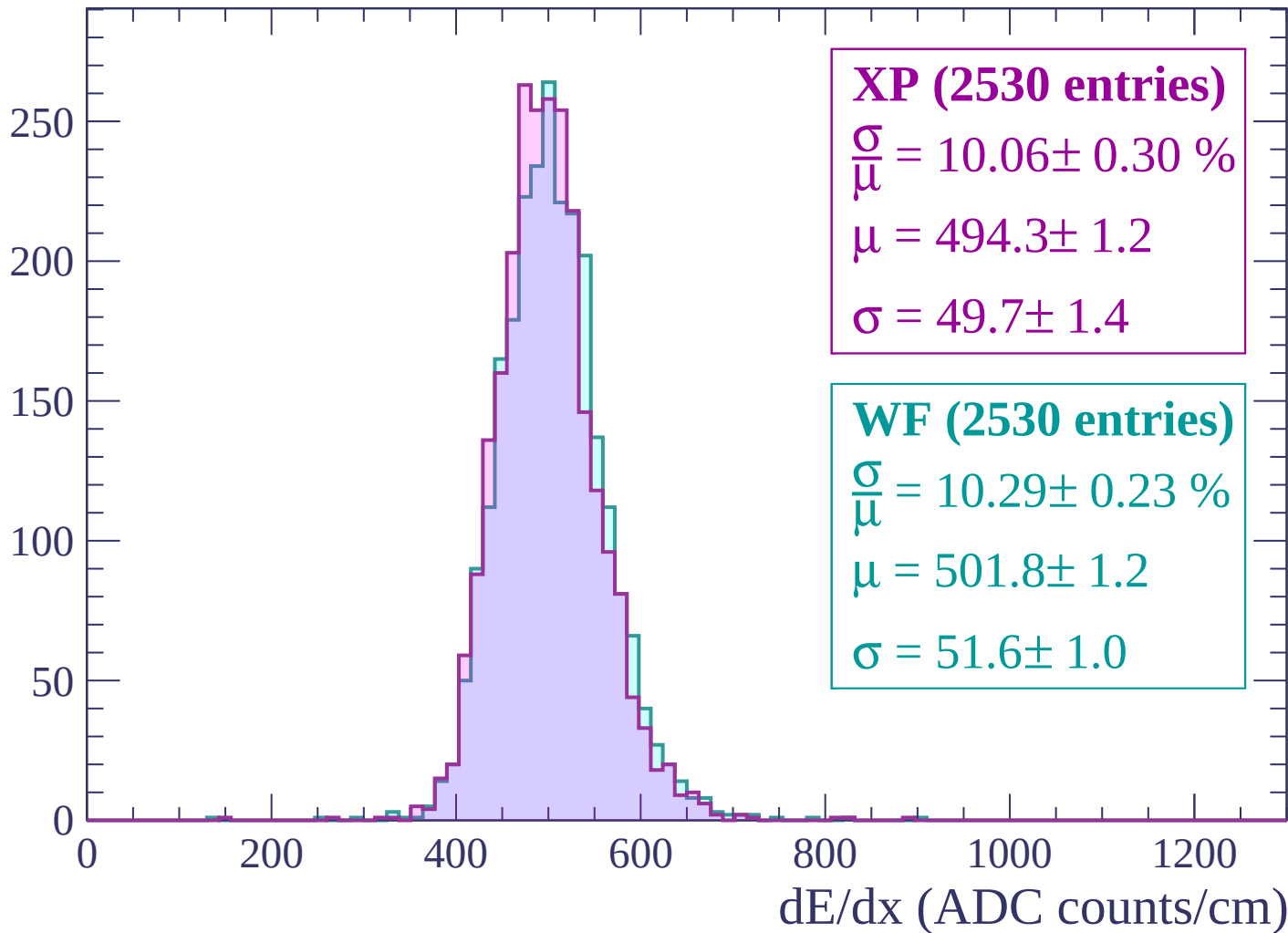
$\mu = 497.9 \pm 1.2$

$\sigma = 47.8 \pm 1.2$

dE/dx (ADC counts/cm)

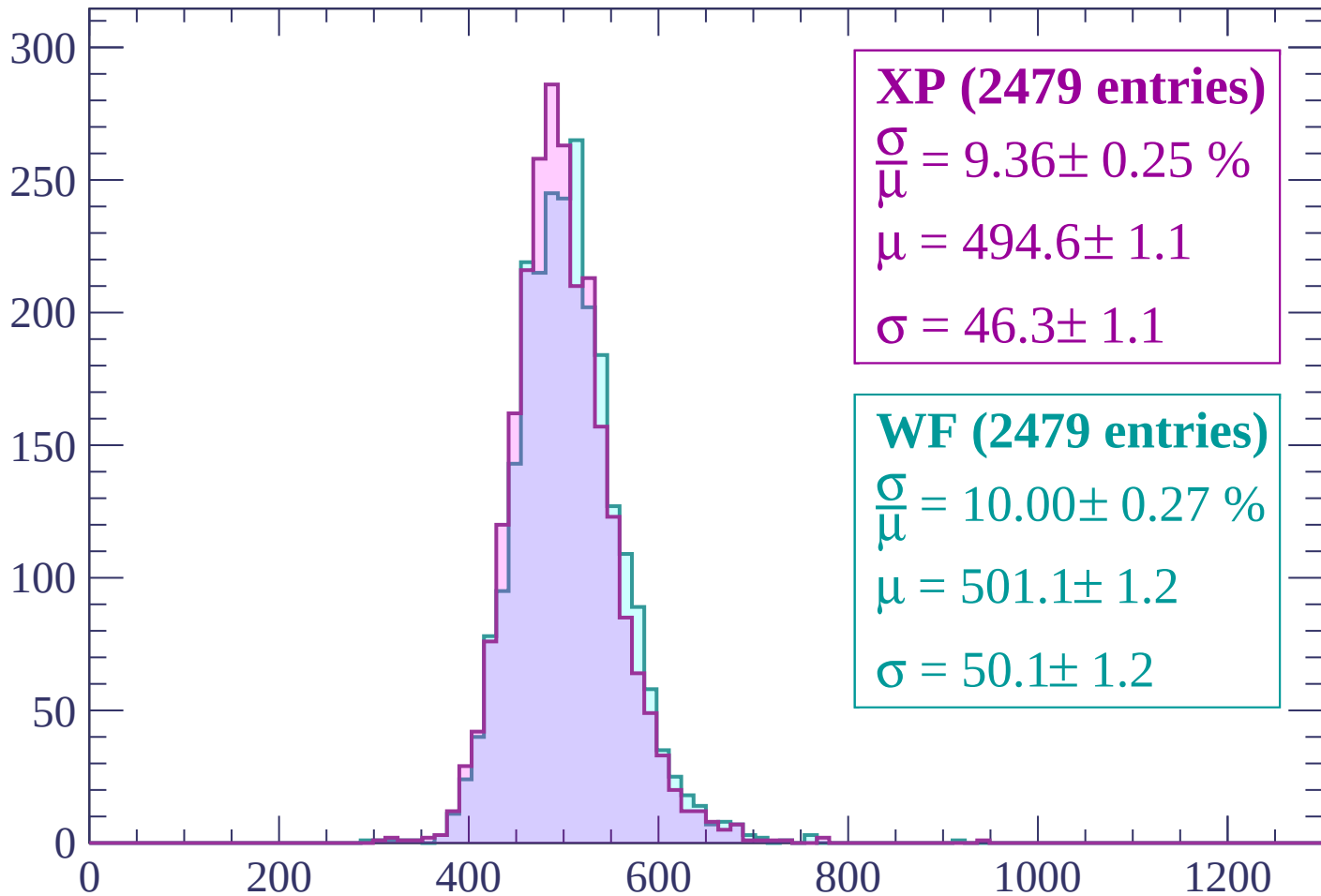
Energy loss | $2880 < p < 2940$

Count



Energy loss | $2940 < p < 3000$

Count



XP (2479 entries)

$$\frac{\sigma}{\mu} = 9.36 \pm 0.25 \%$$

$$\mu = 494.6 \pm 1.1$$

$$\sigma = 46.3 \pm 1.1$$

WF (2479 entries)

$$\frac{\sigma}{\mu} = 10.00 \pm 0.27 \%$$

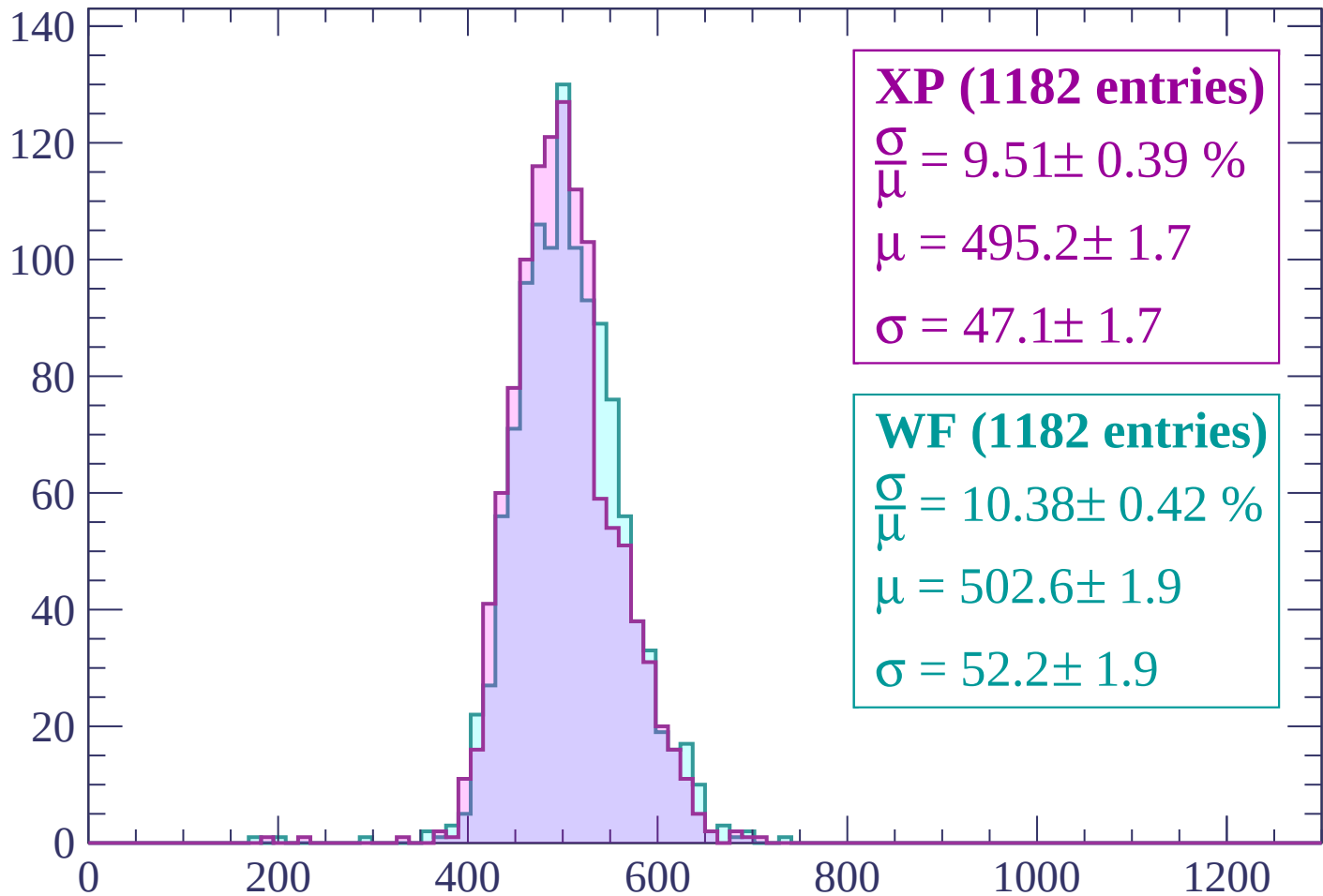
$$\mu = 501.1 \pm 1.2$$

$$\sigma = 50.1 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $3000 < p < 3060$

Count



XP (1182 entries)

$$\frac{\sigma}{\mu} = 9.51 \pm 0.39 \%$$

$$\mu = 495.2 \pm 1.7$$

$$\sigma = 47.1 \pm 1.7$$

WF (1182 entries)

$$\frac{\sigma}{\mu} = 10.38 \pm 0.42 \%$$

$$\mu = 502.6 \pm 1.9$$

$$\sigma = 52.2 \pm 1.9$$

dE/dx (ADC counts/cm)

